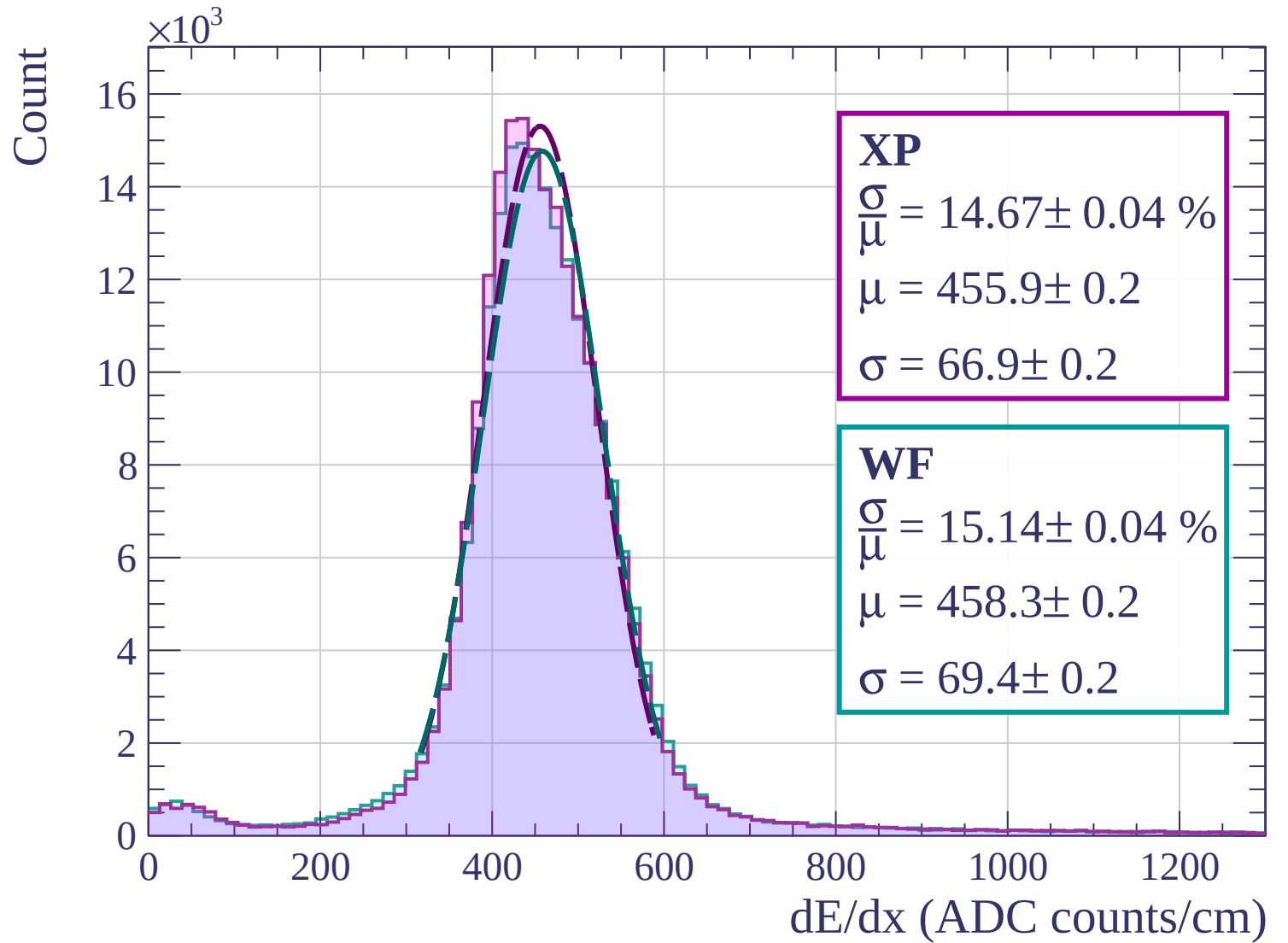
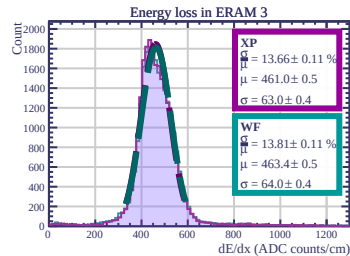
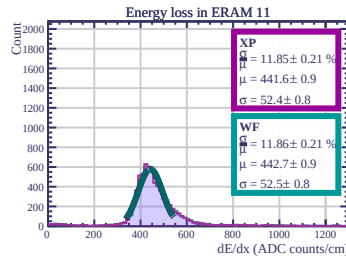
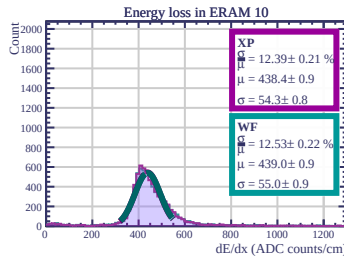
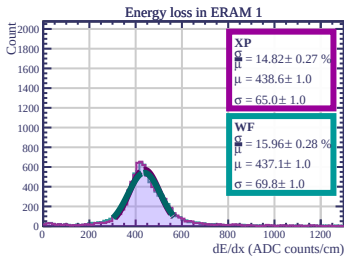
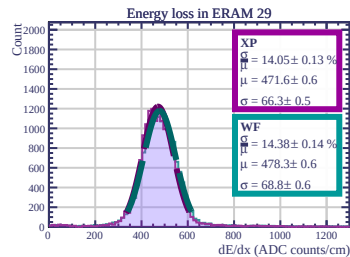
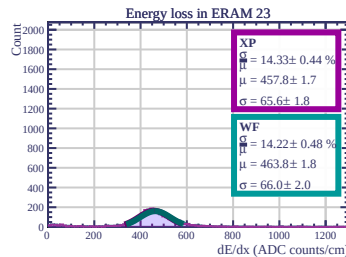
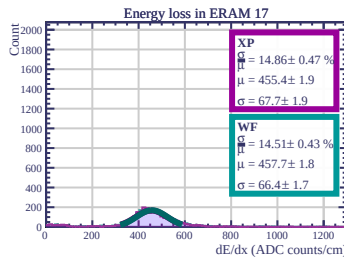
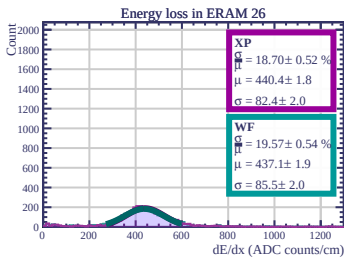
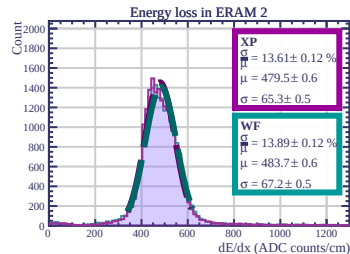
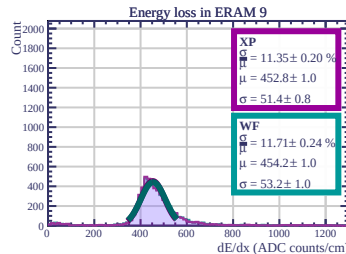
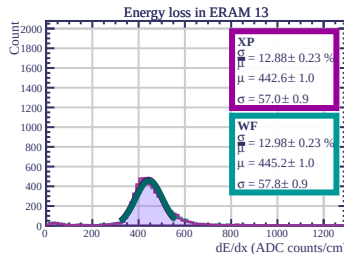
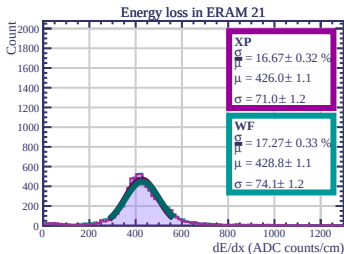
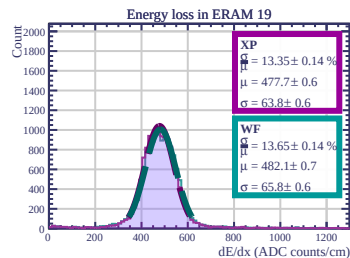
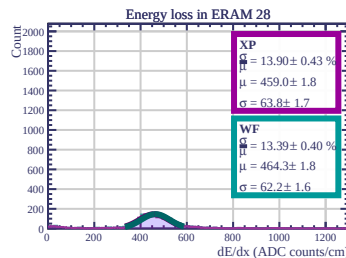
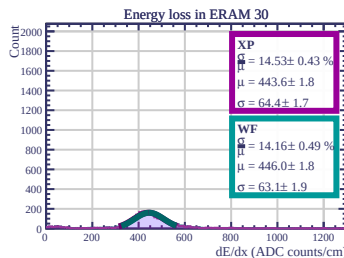
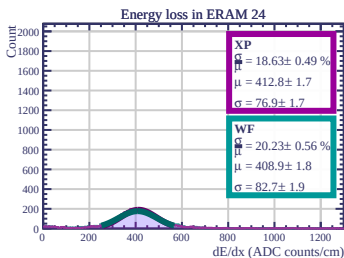
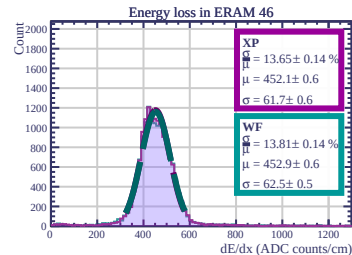
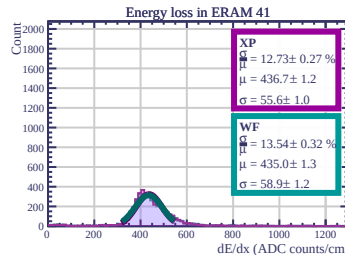
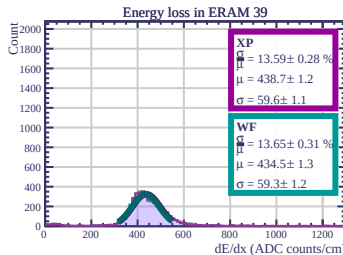
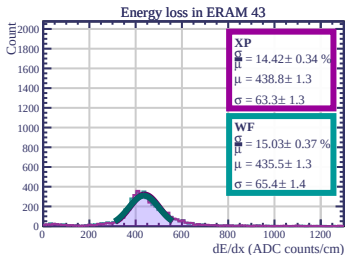
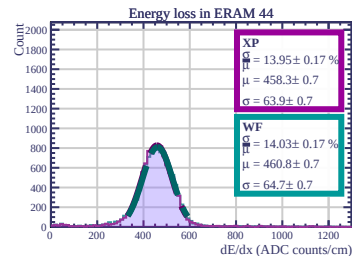
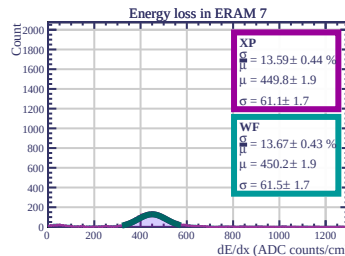
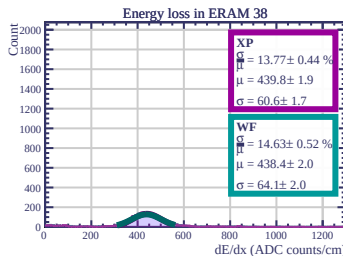
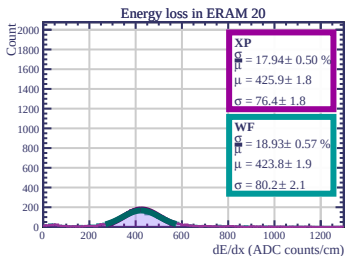
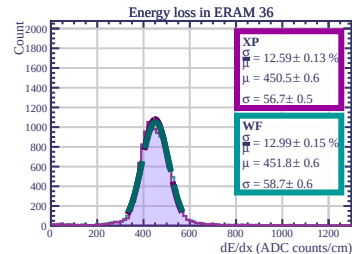
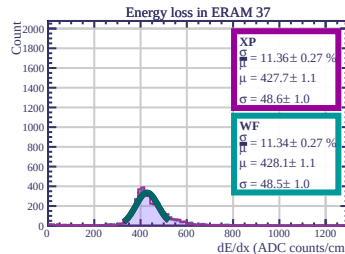
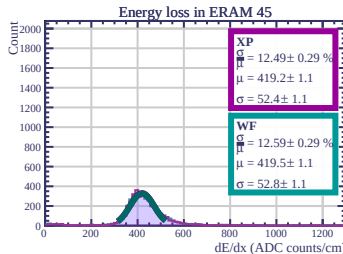
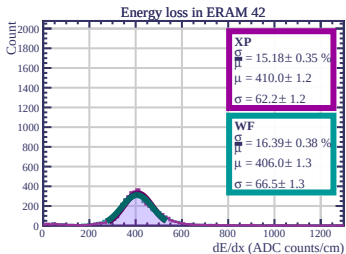
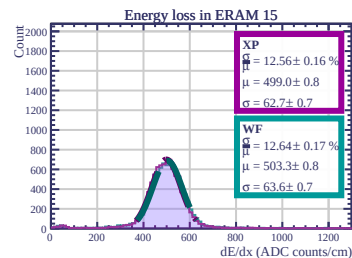
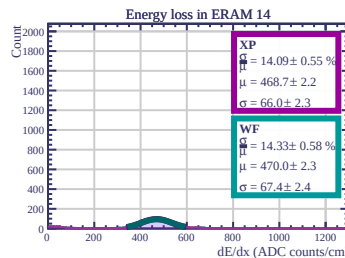
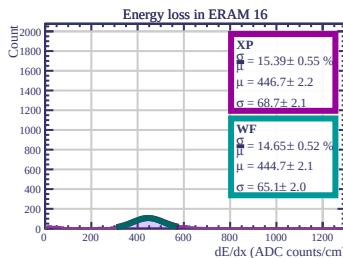
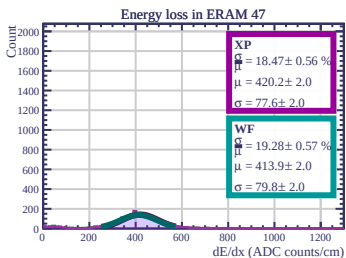
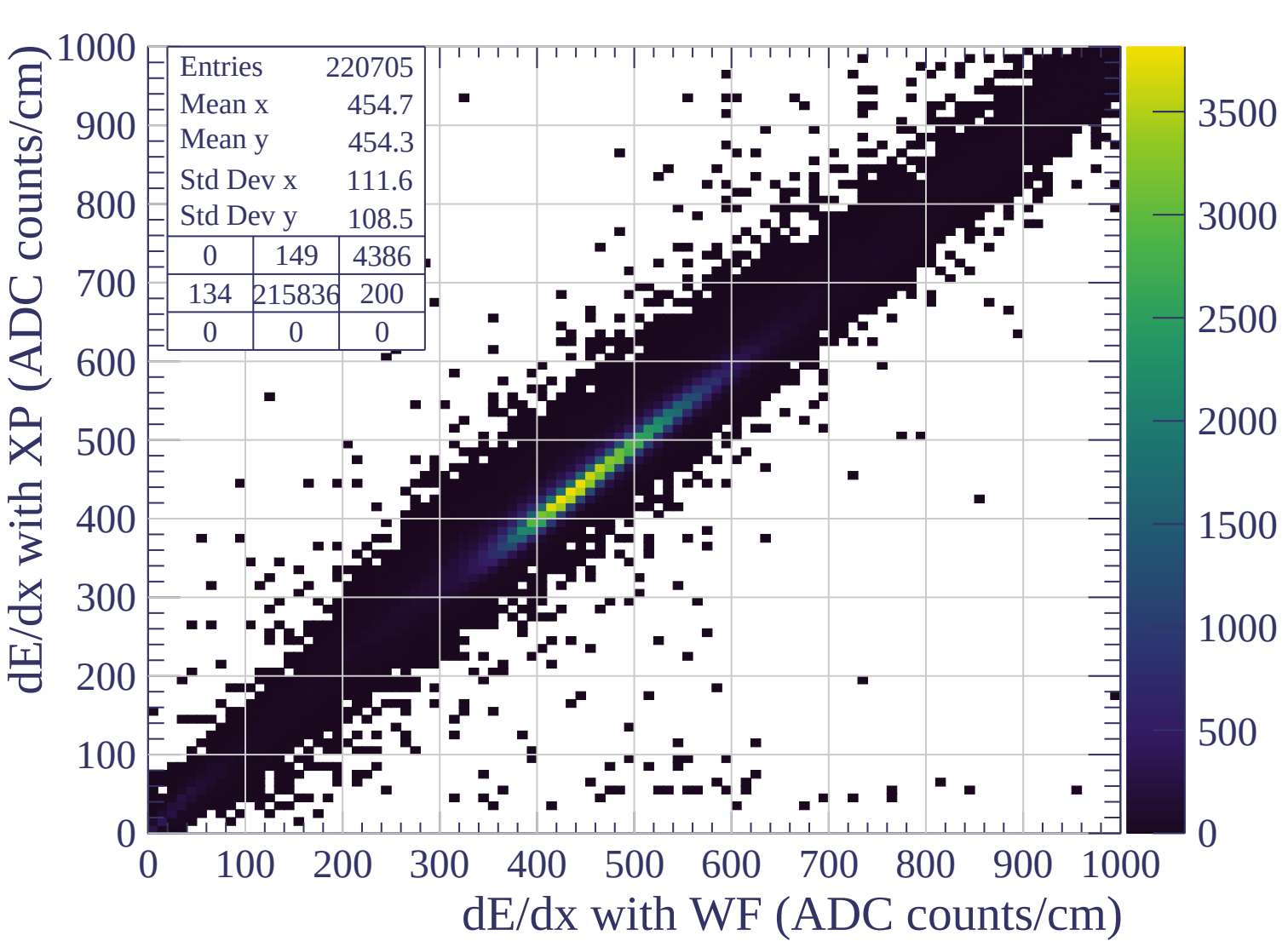
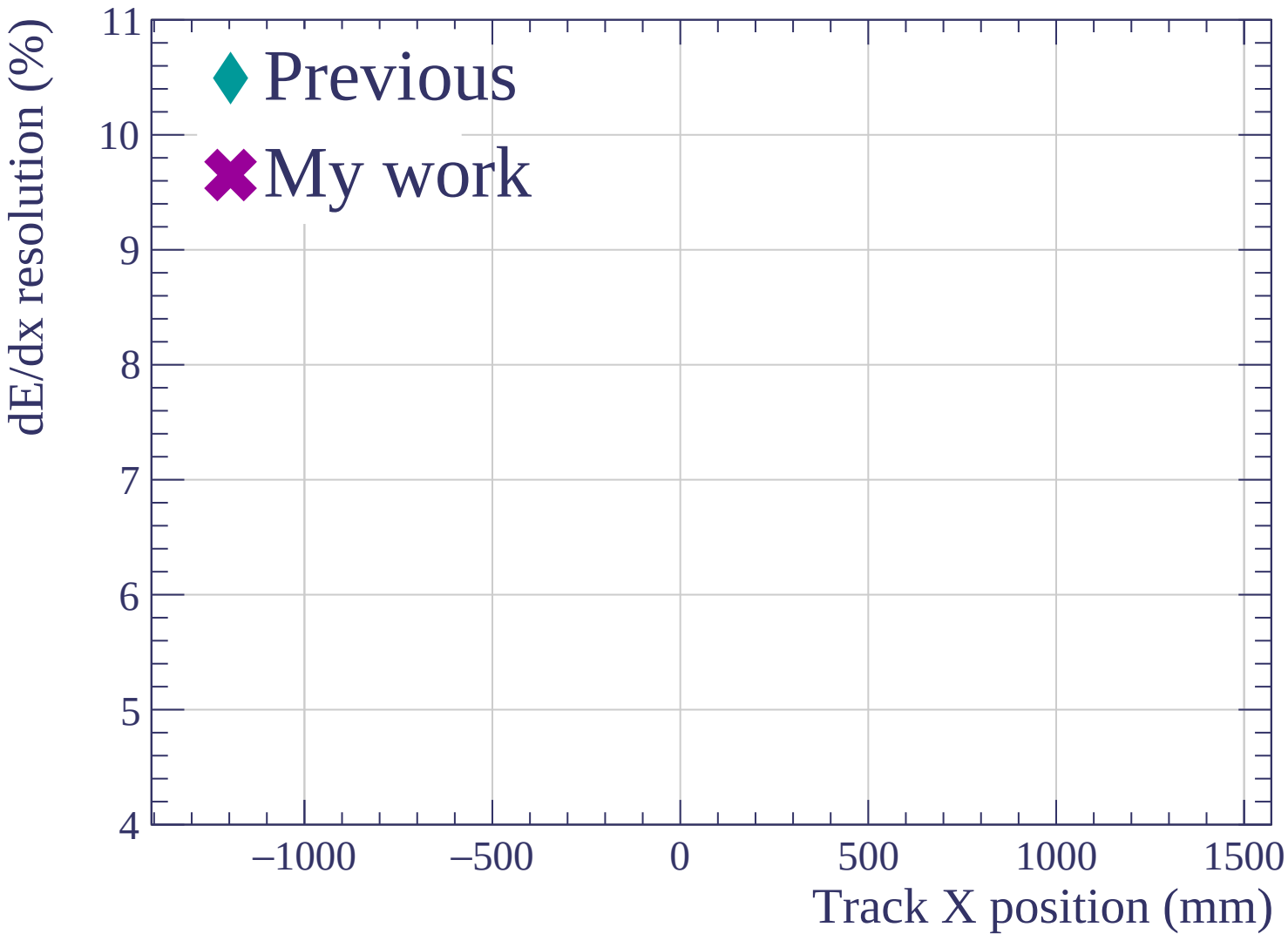


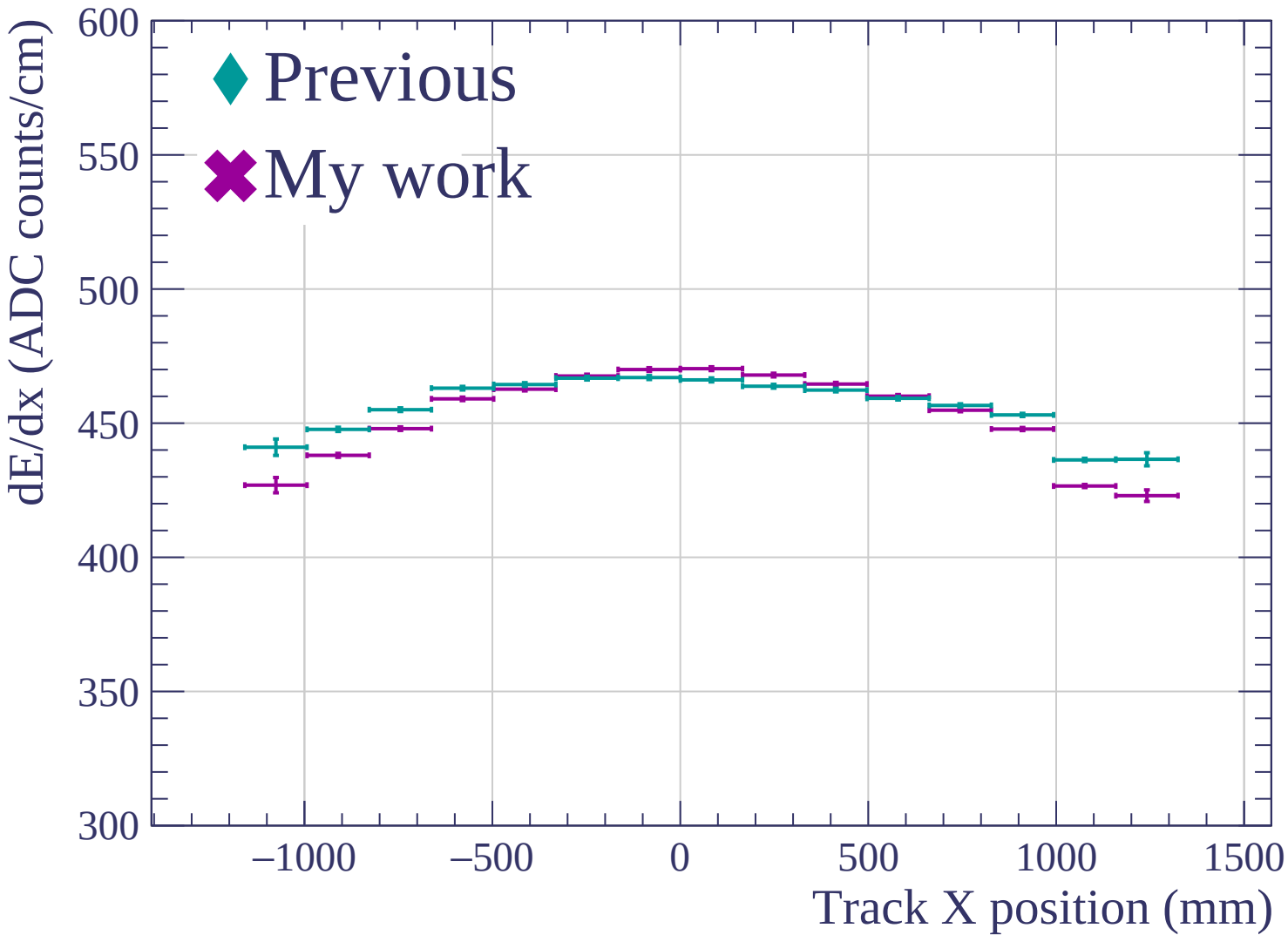


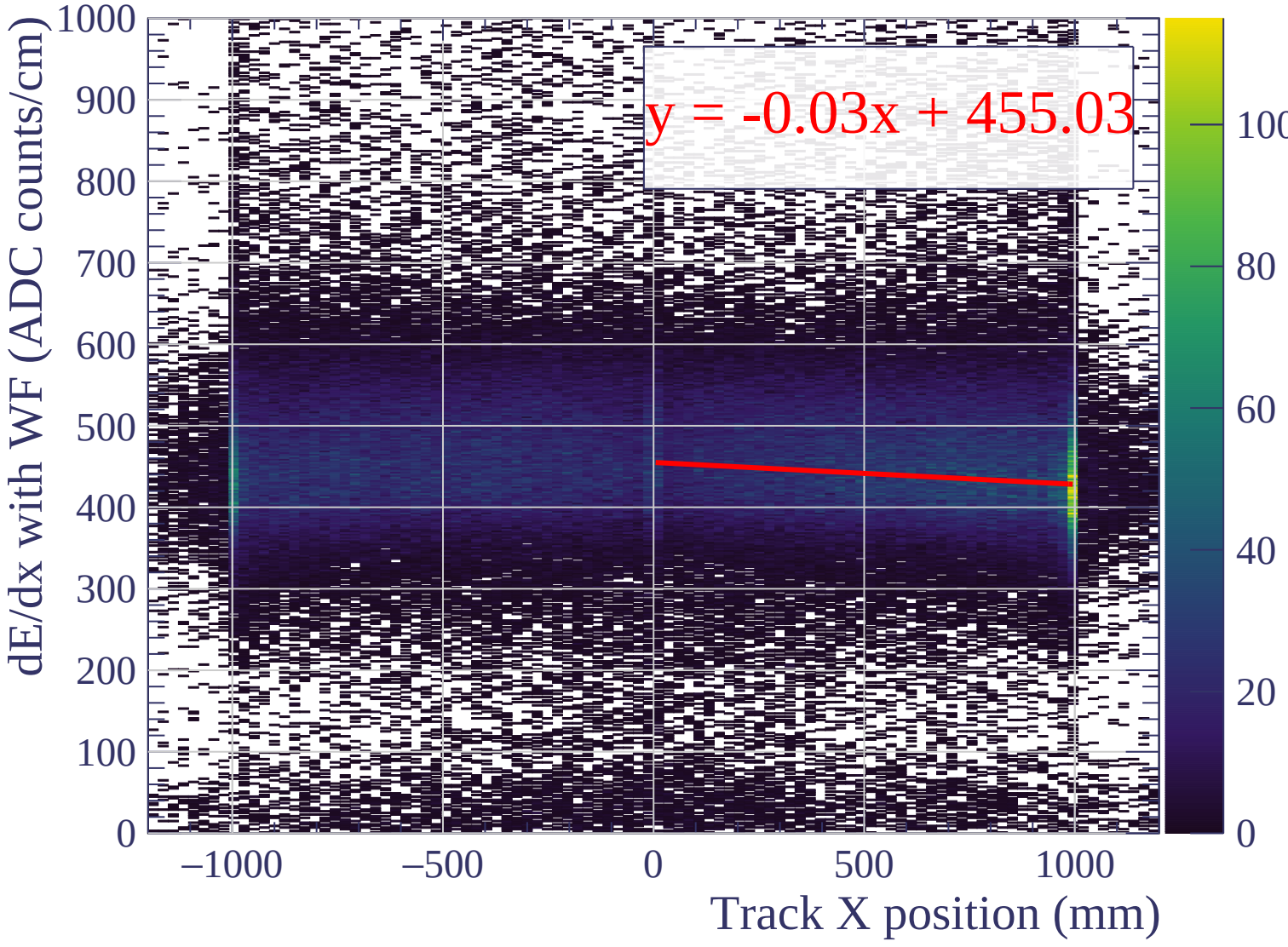


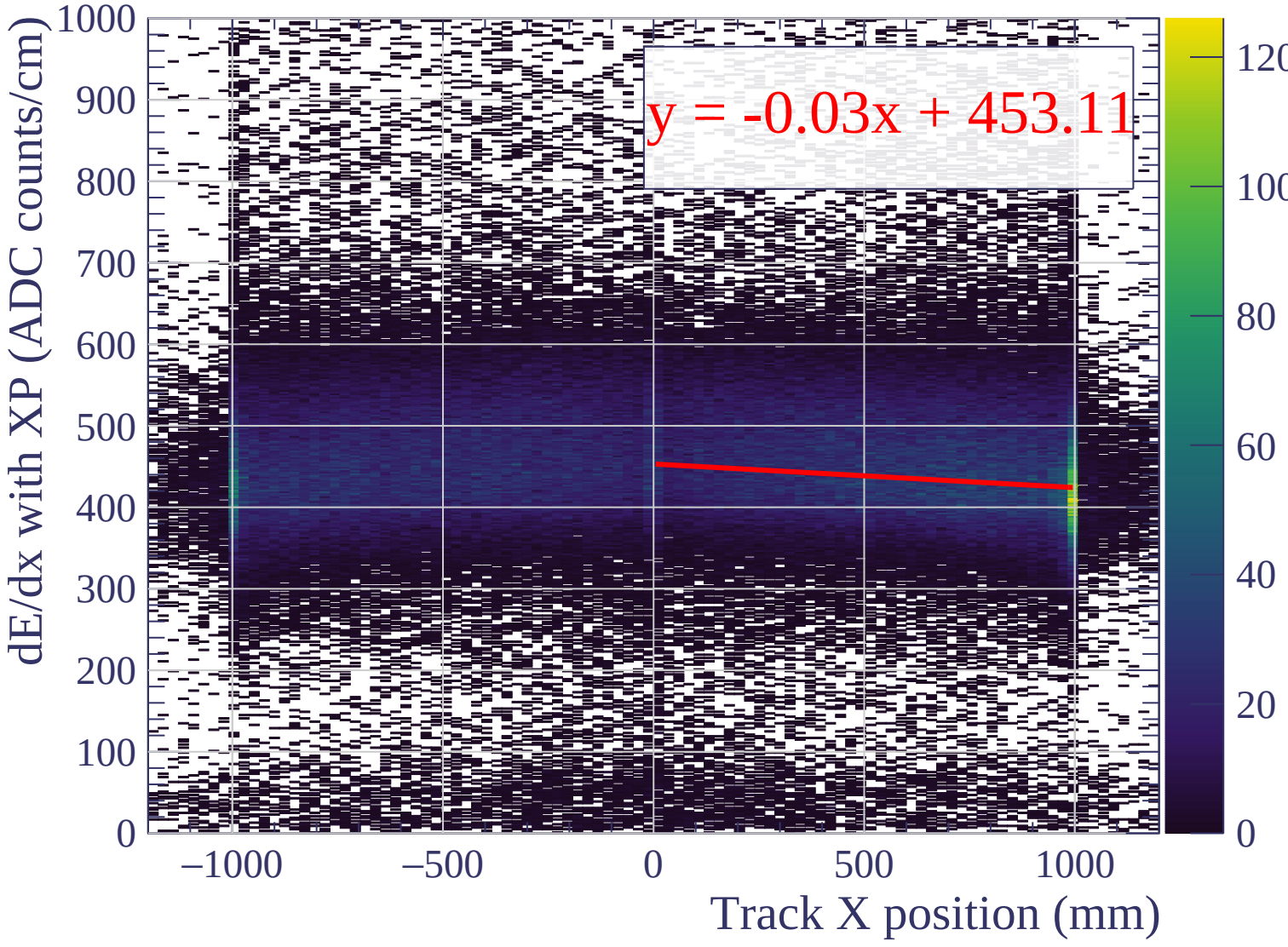


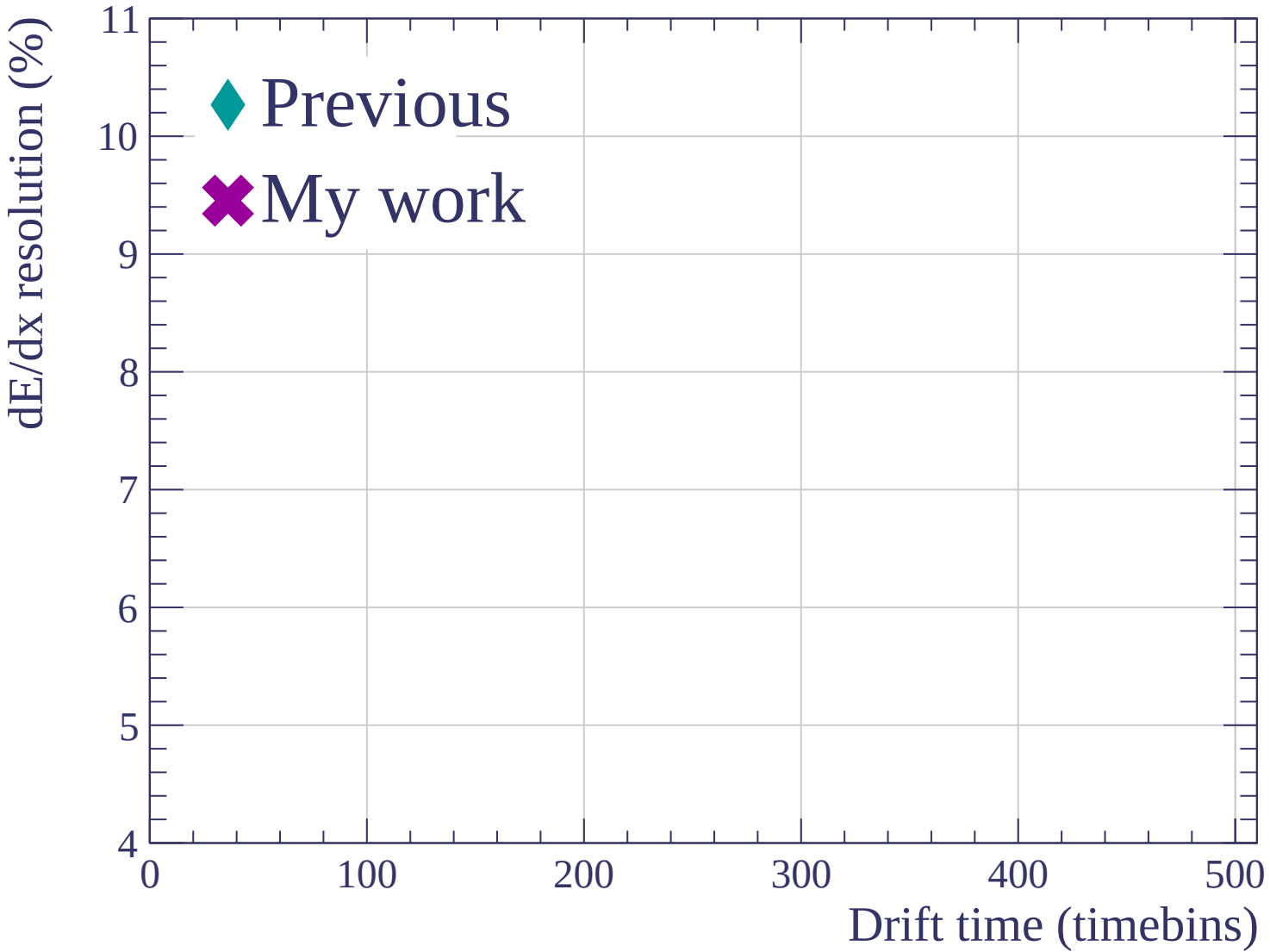


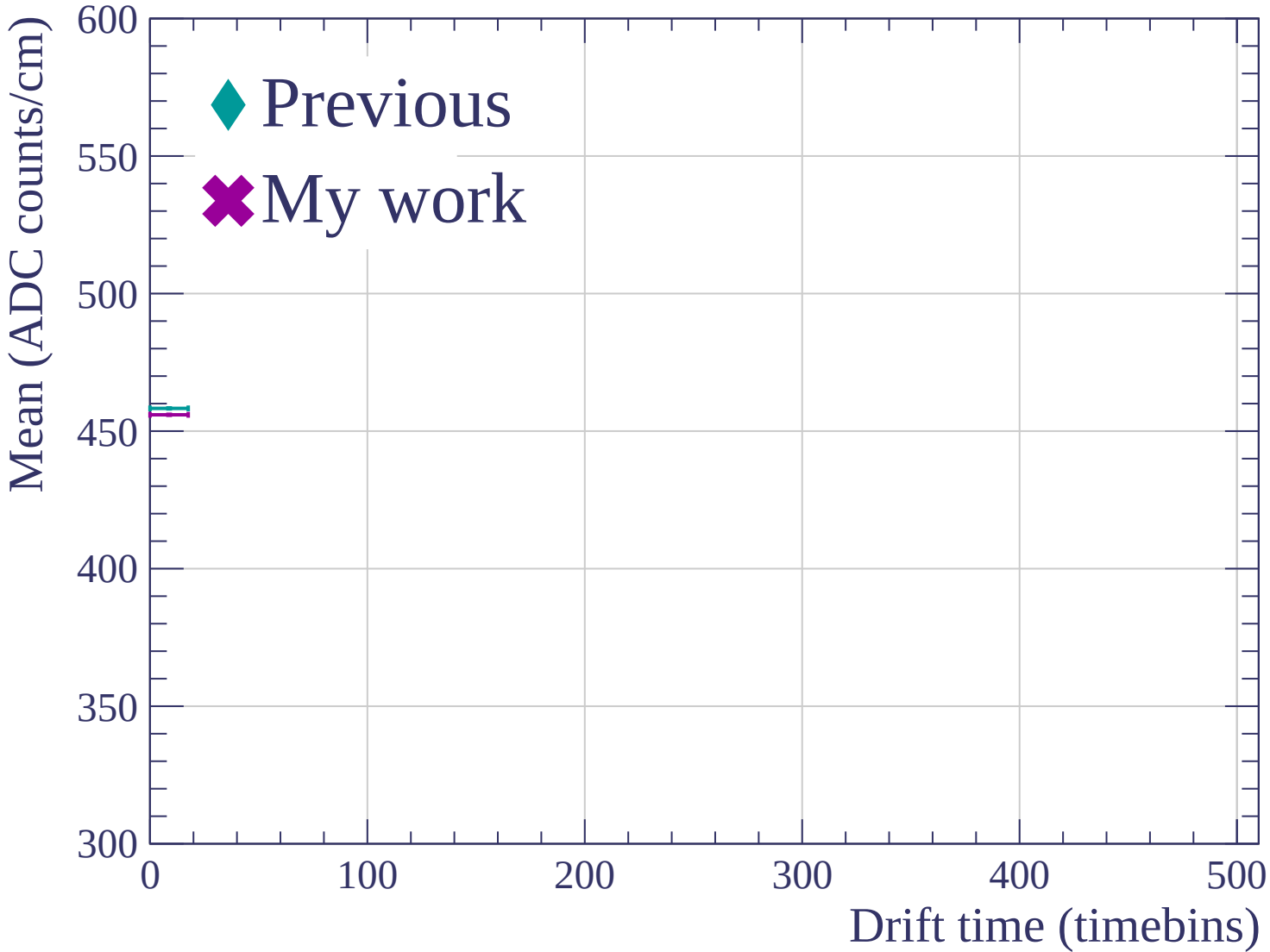


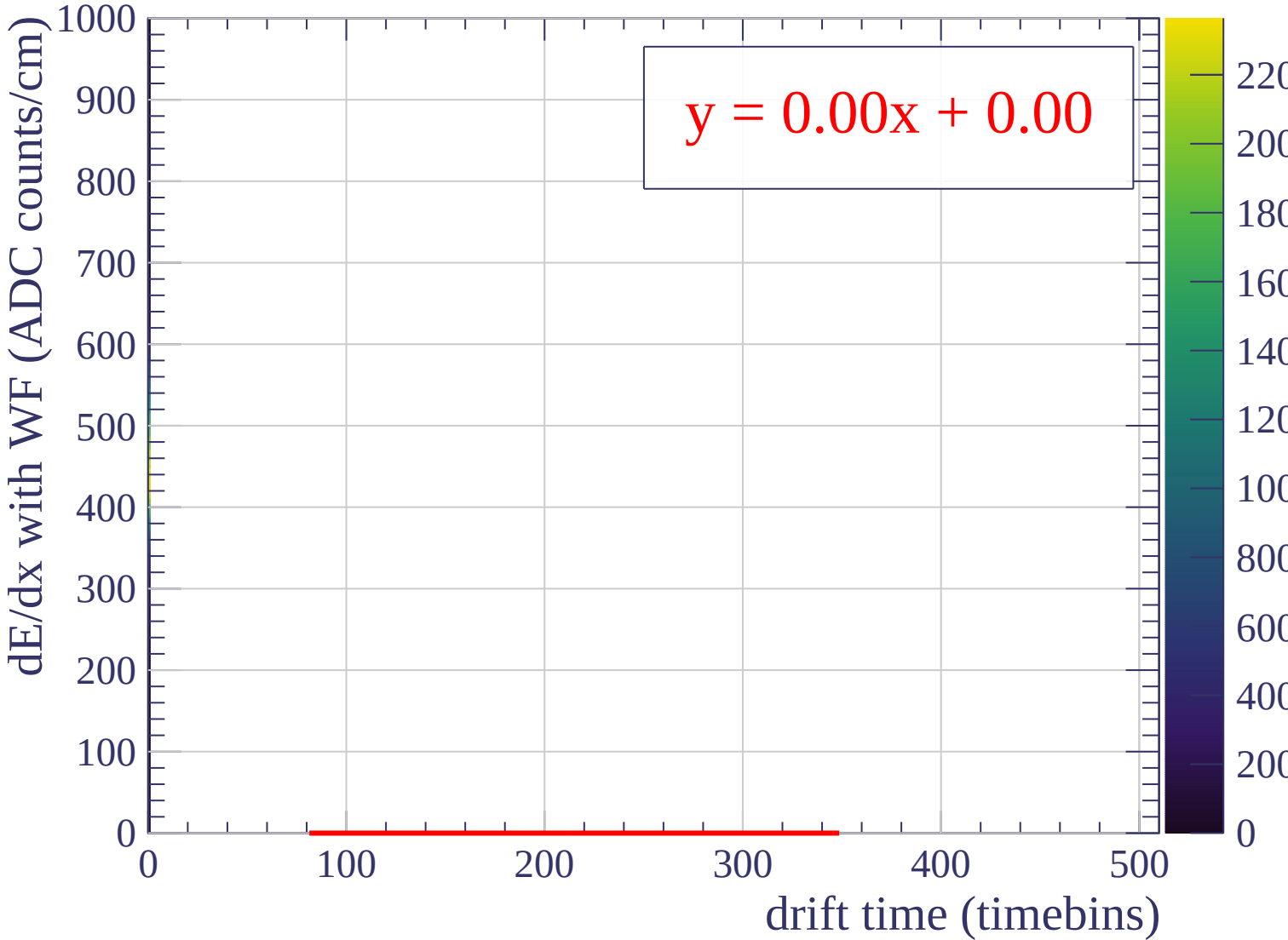


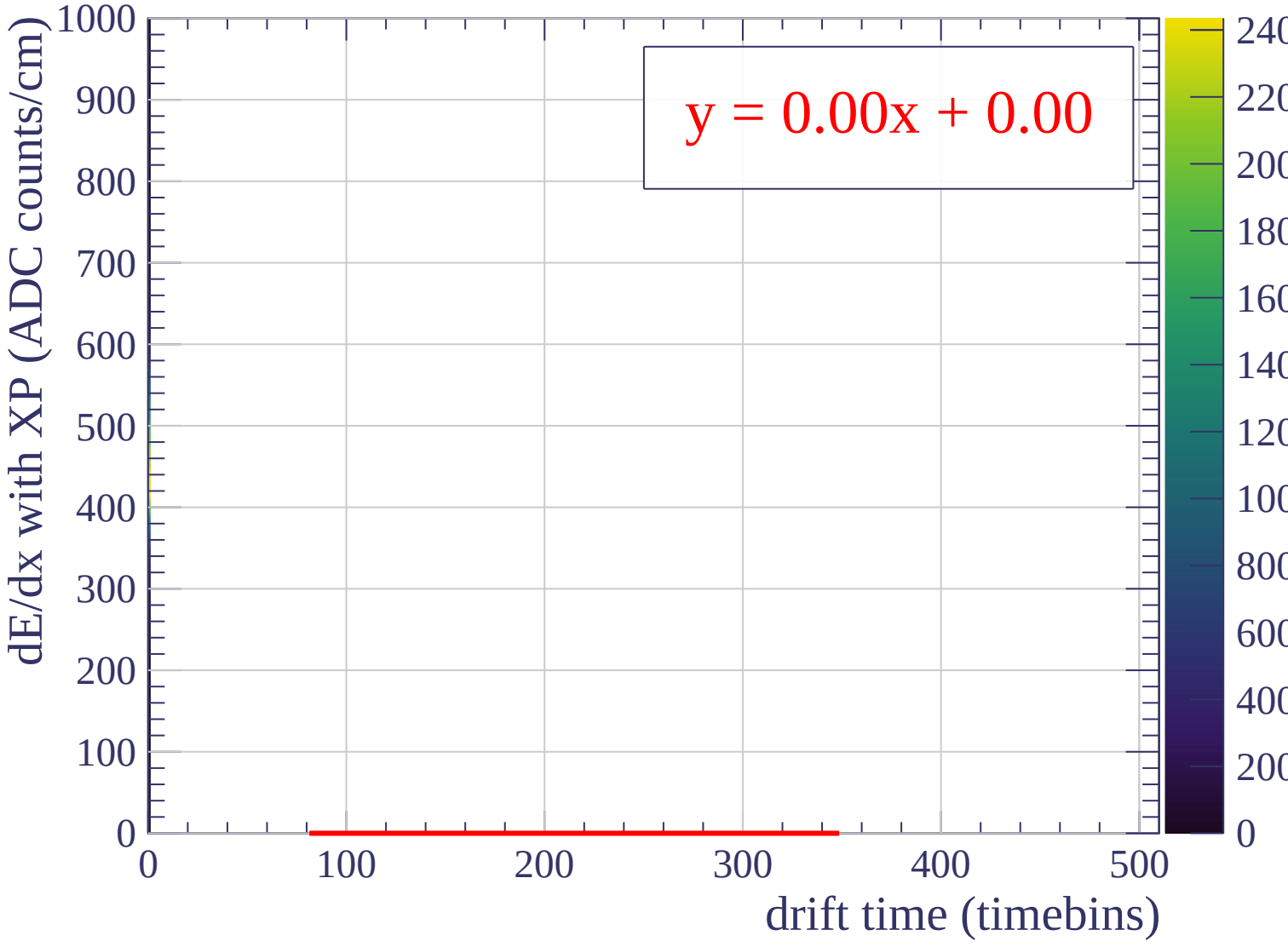


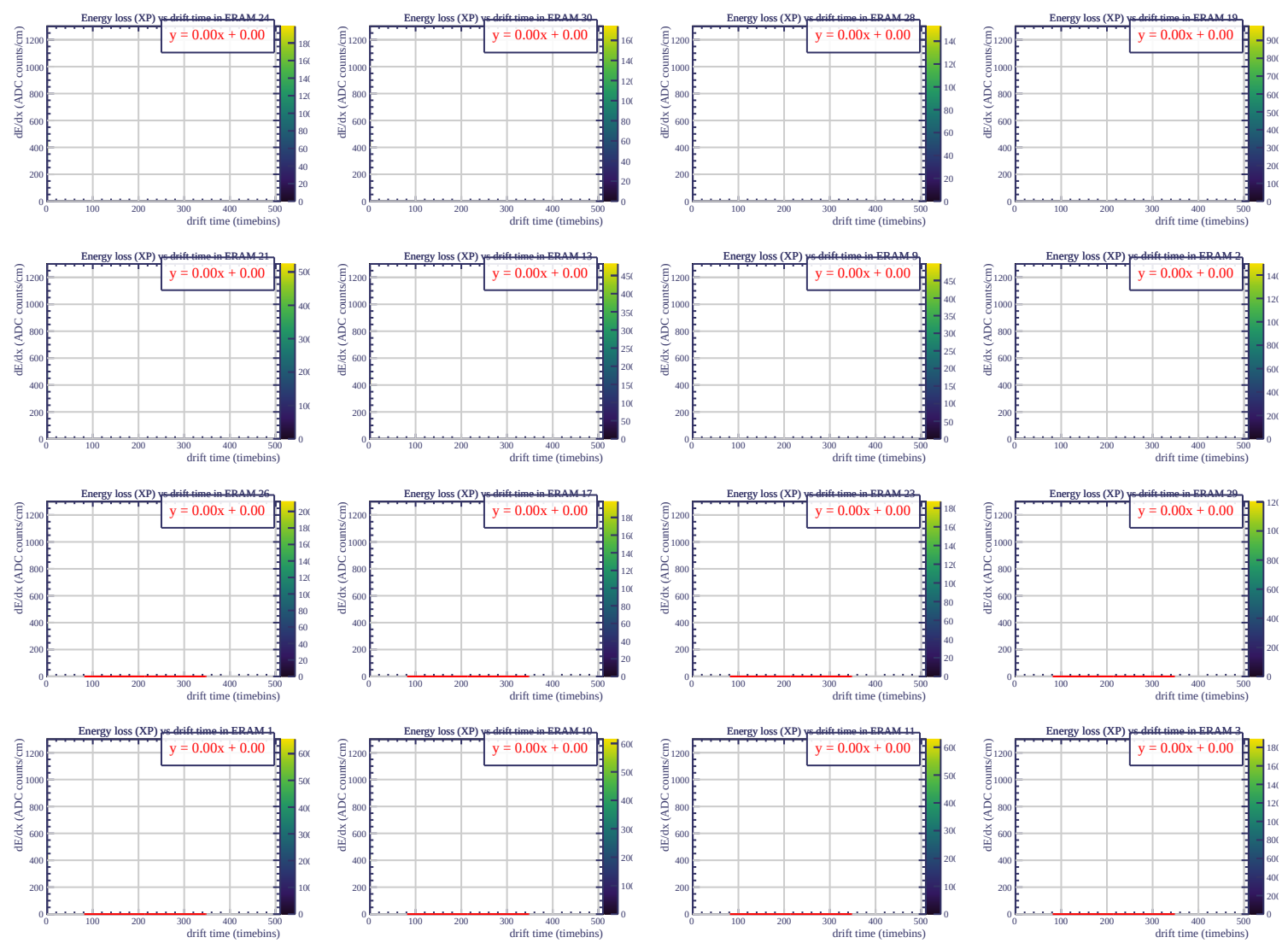


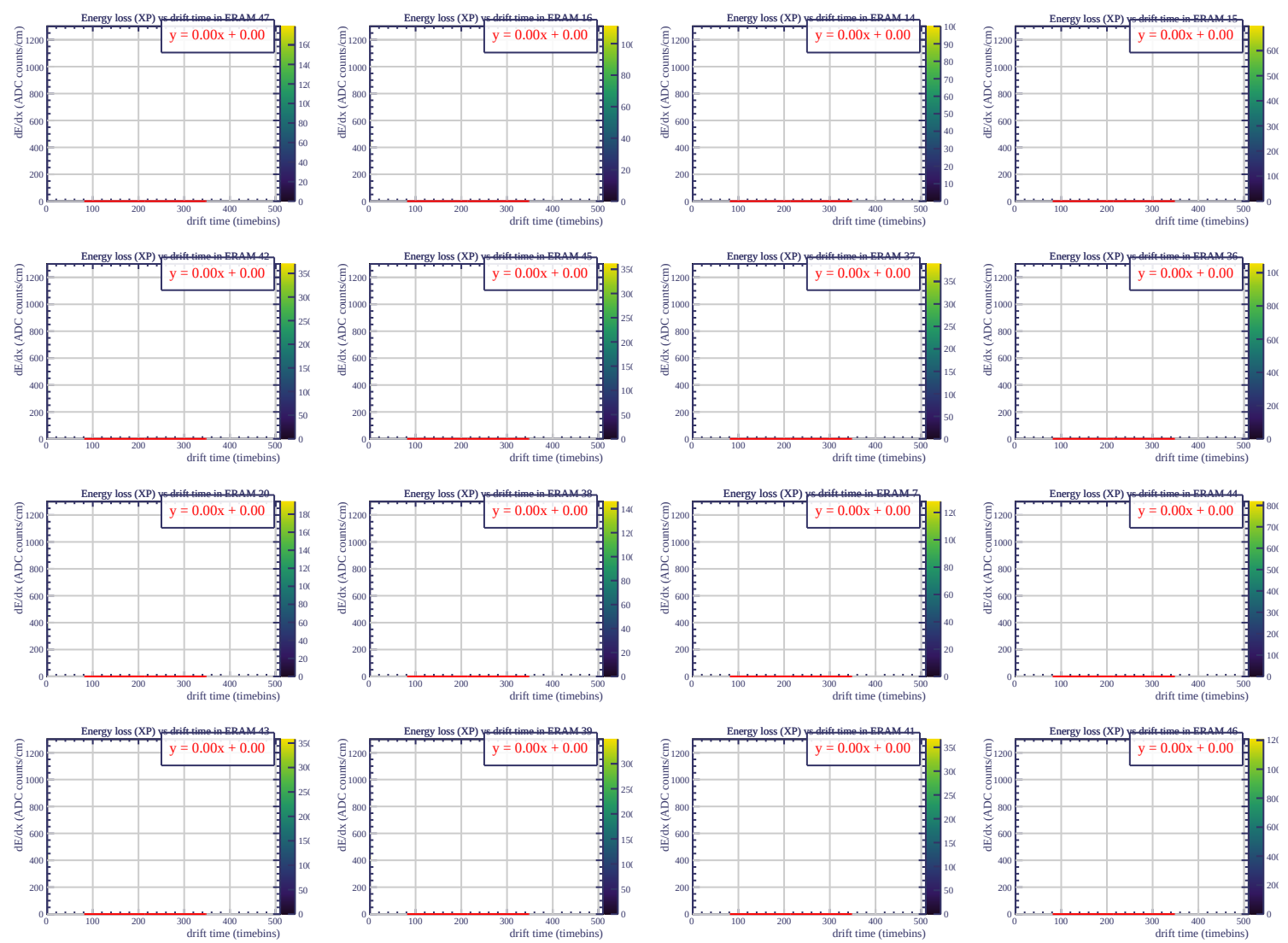


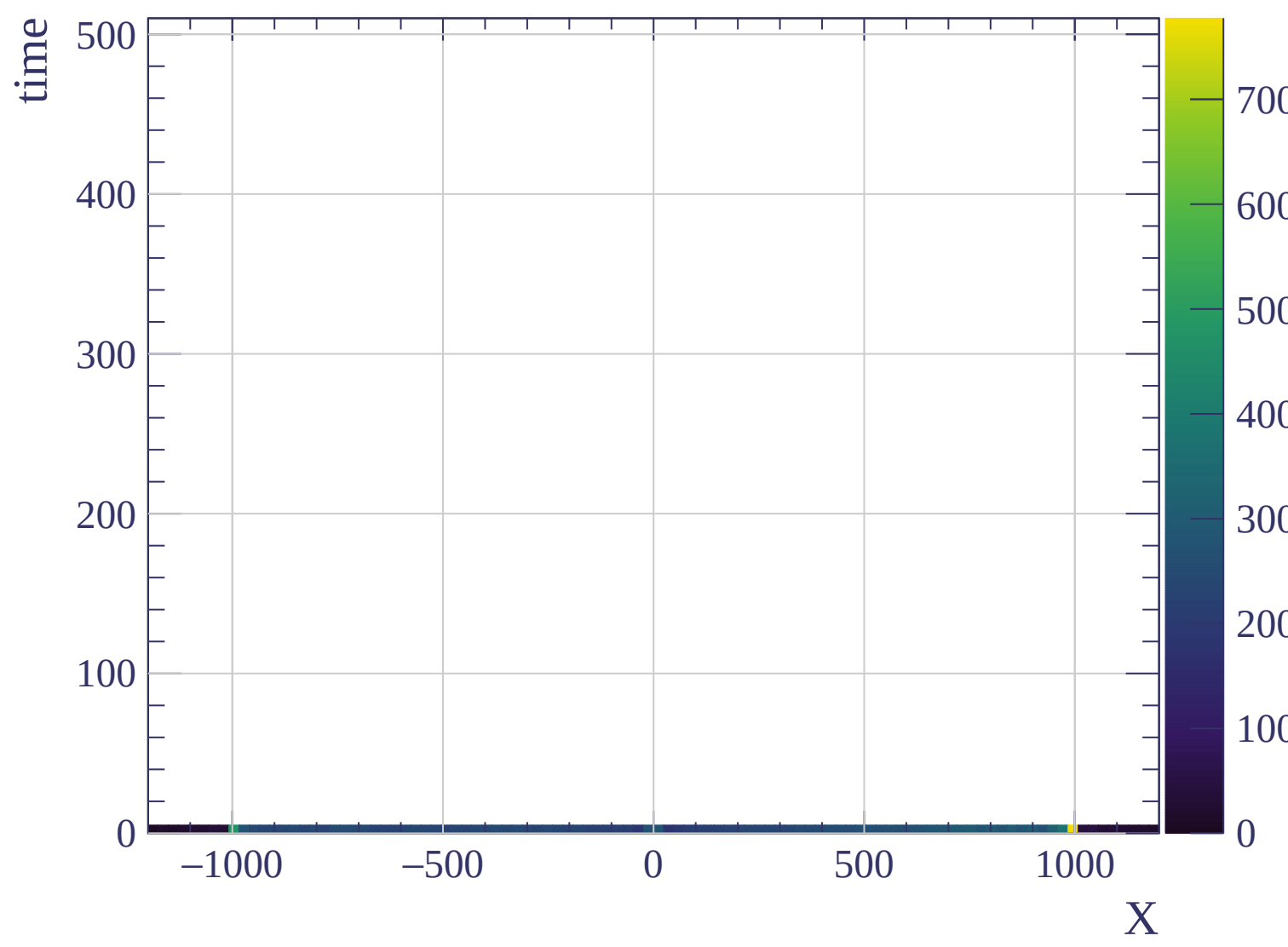


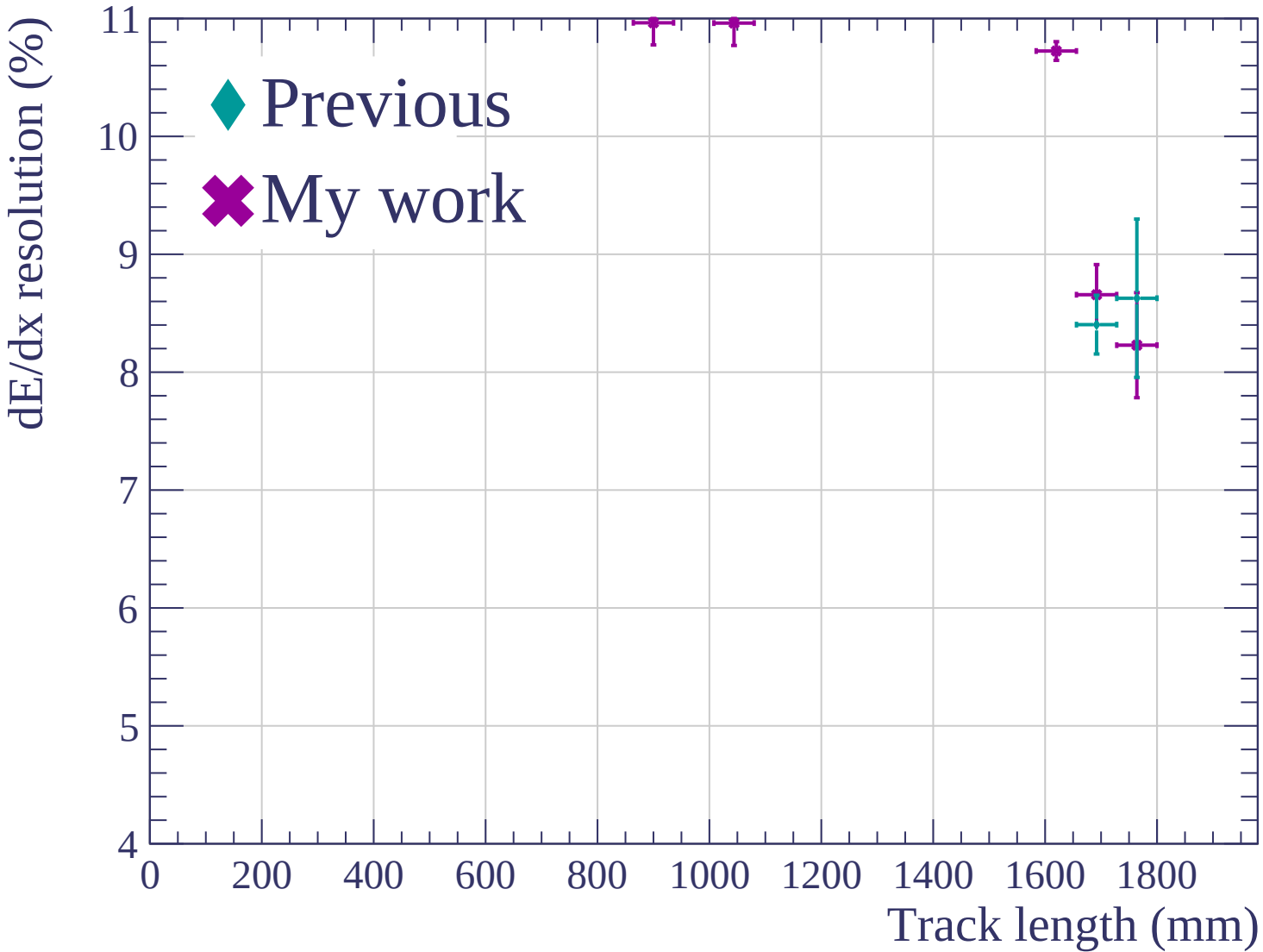


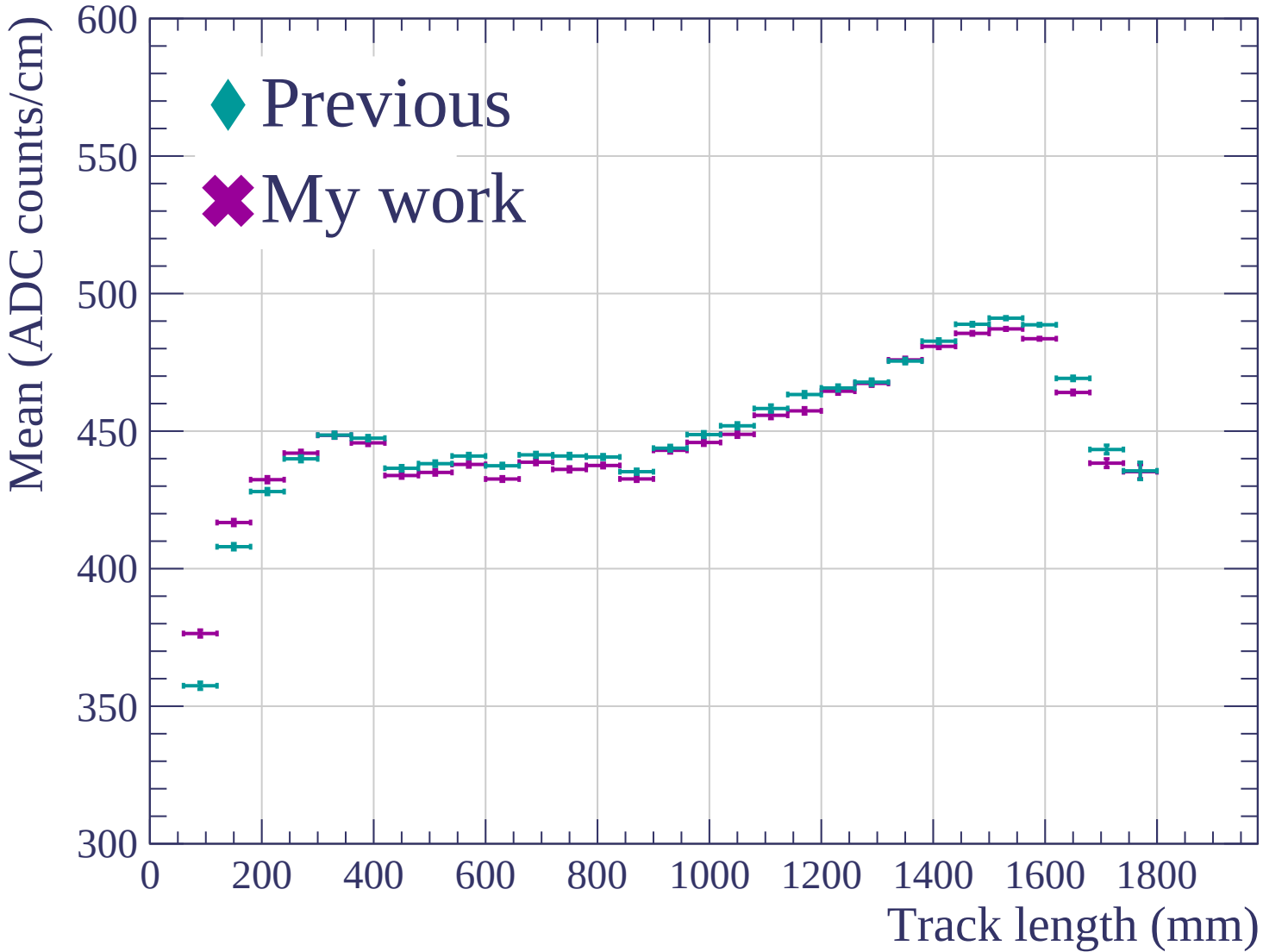


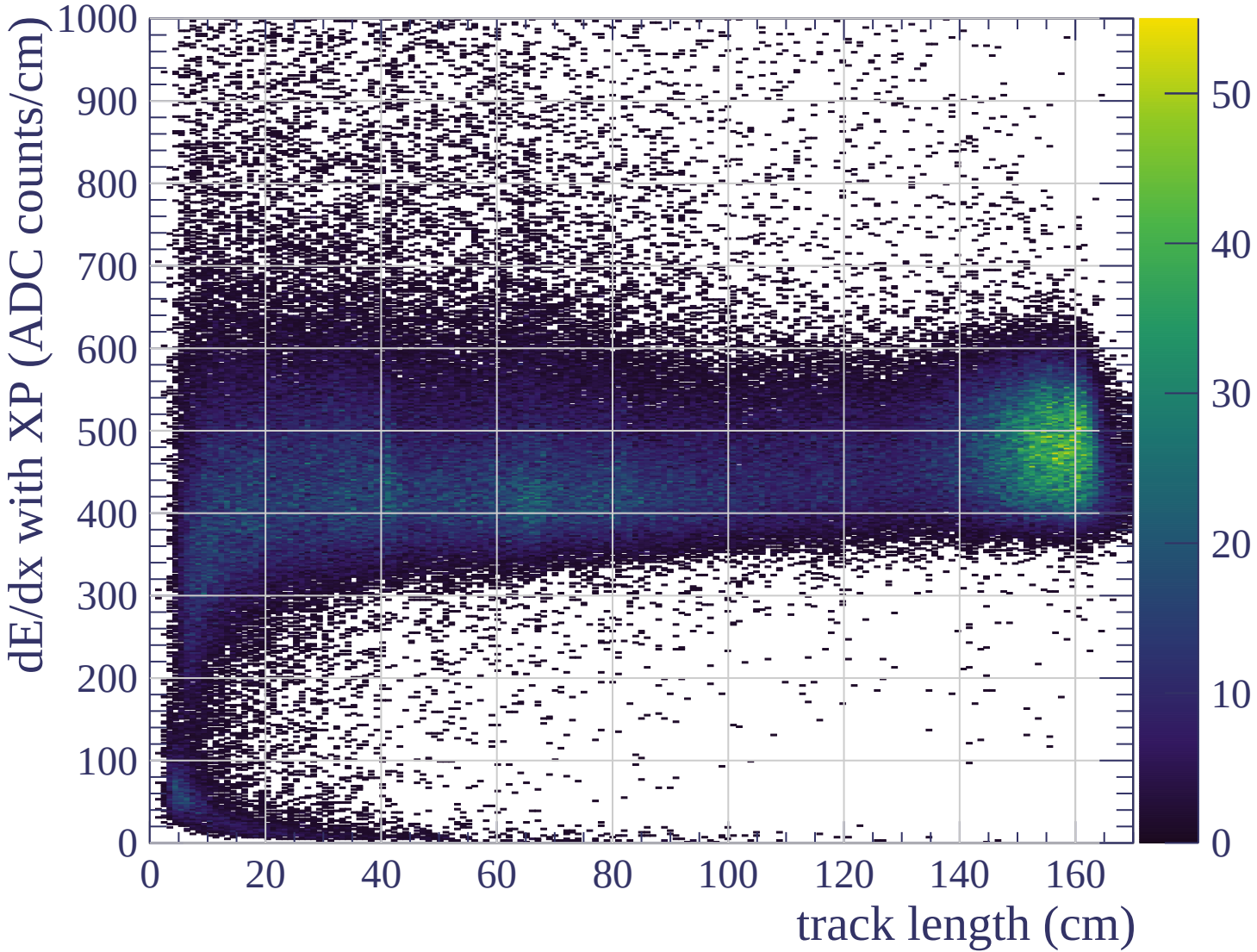


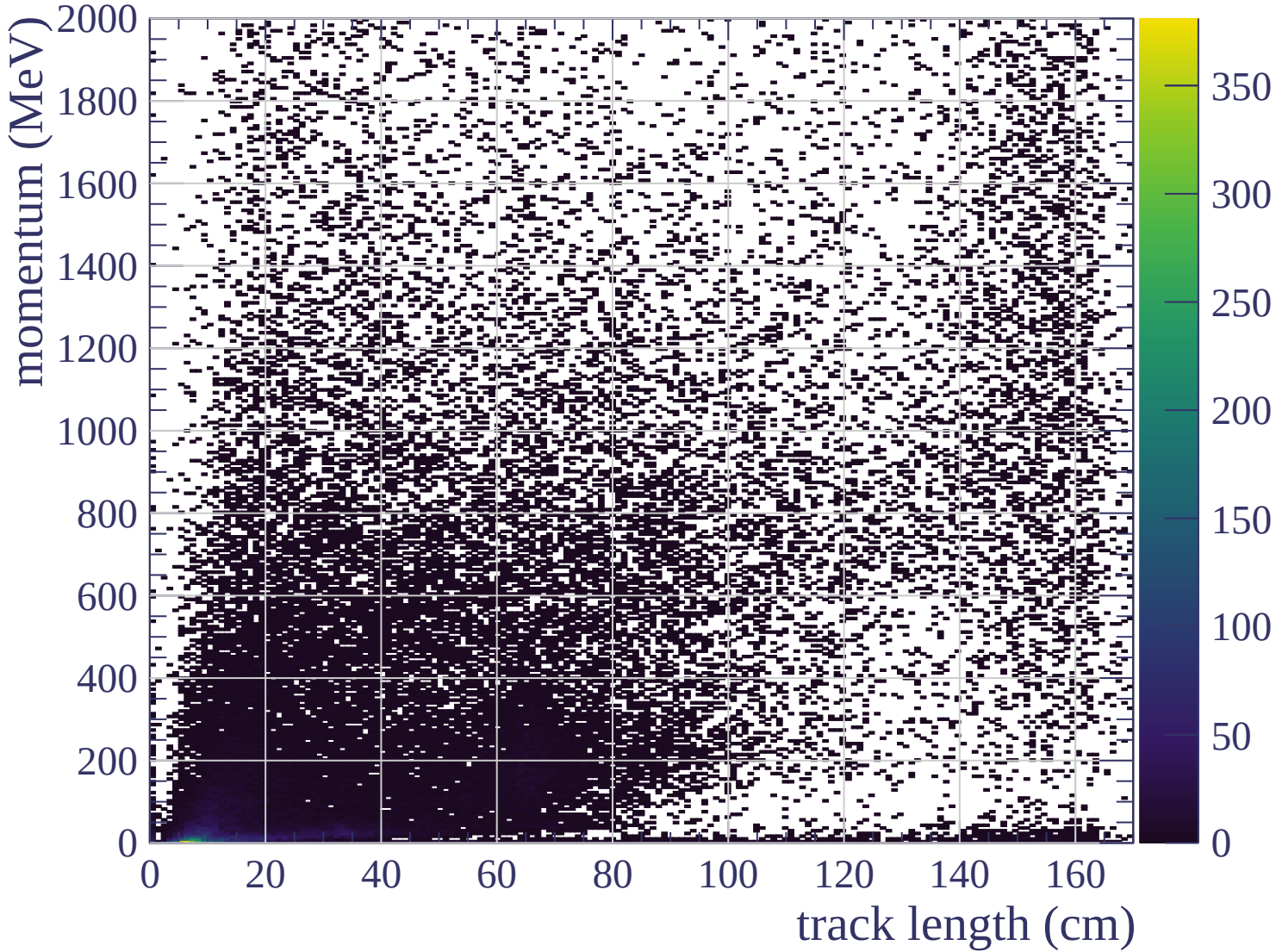


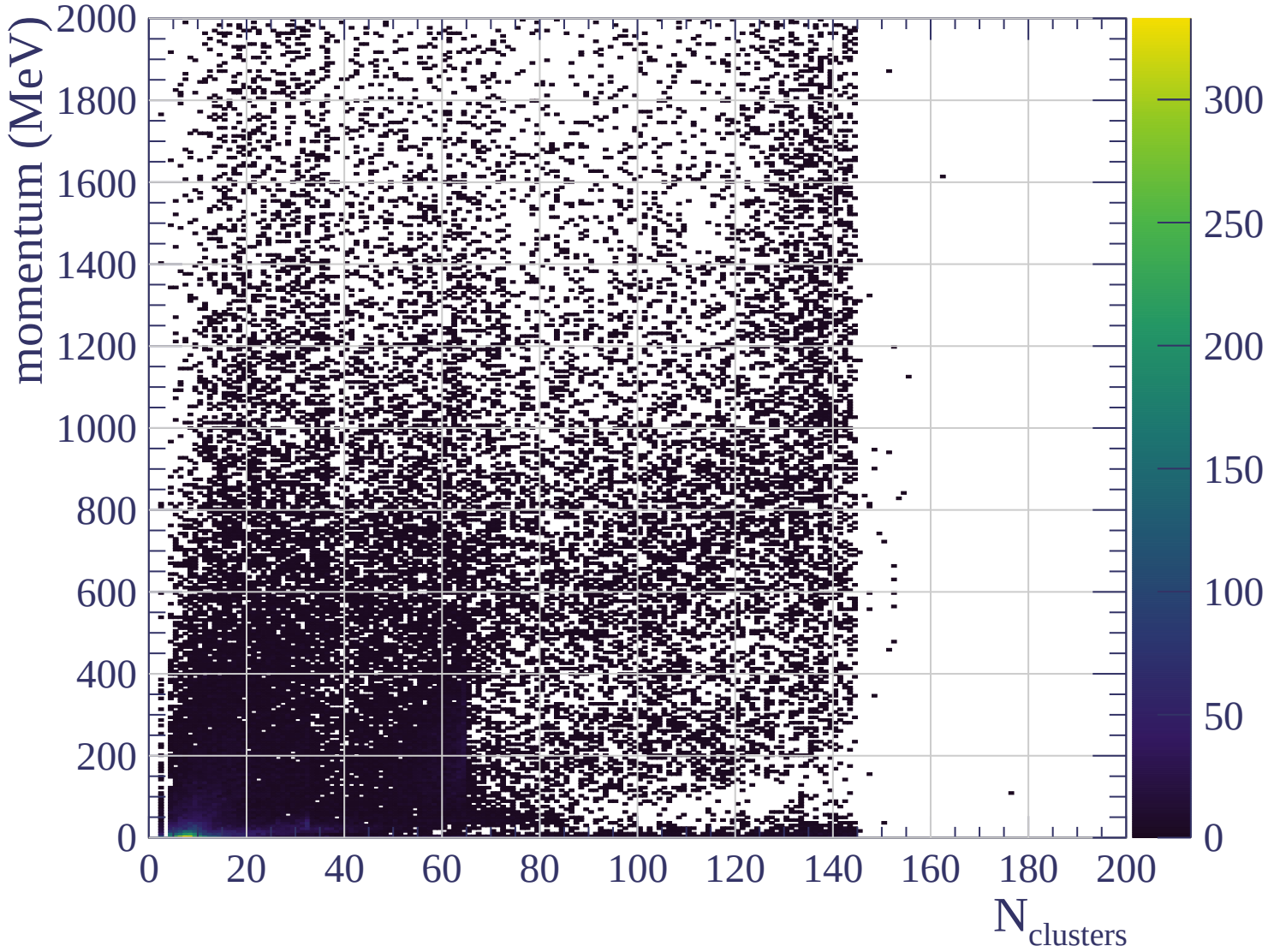


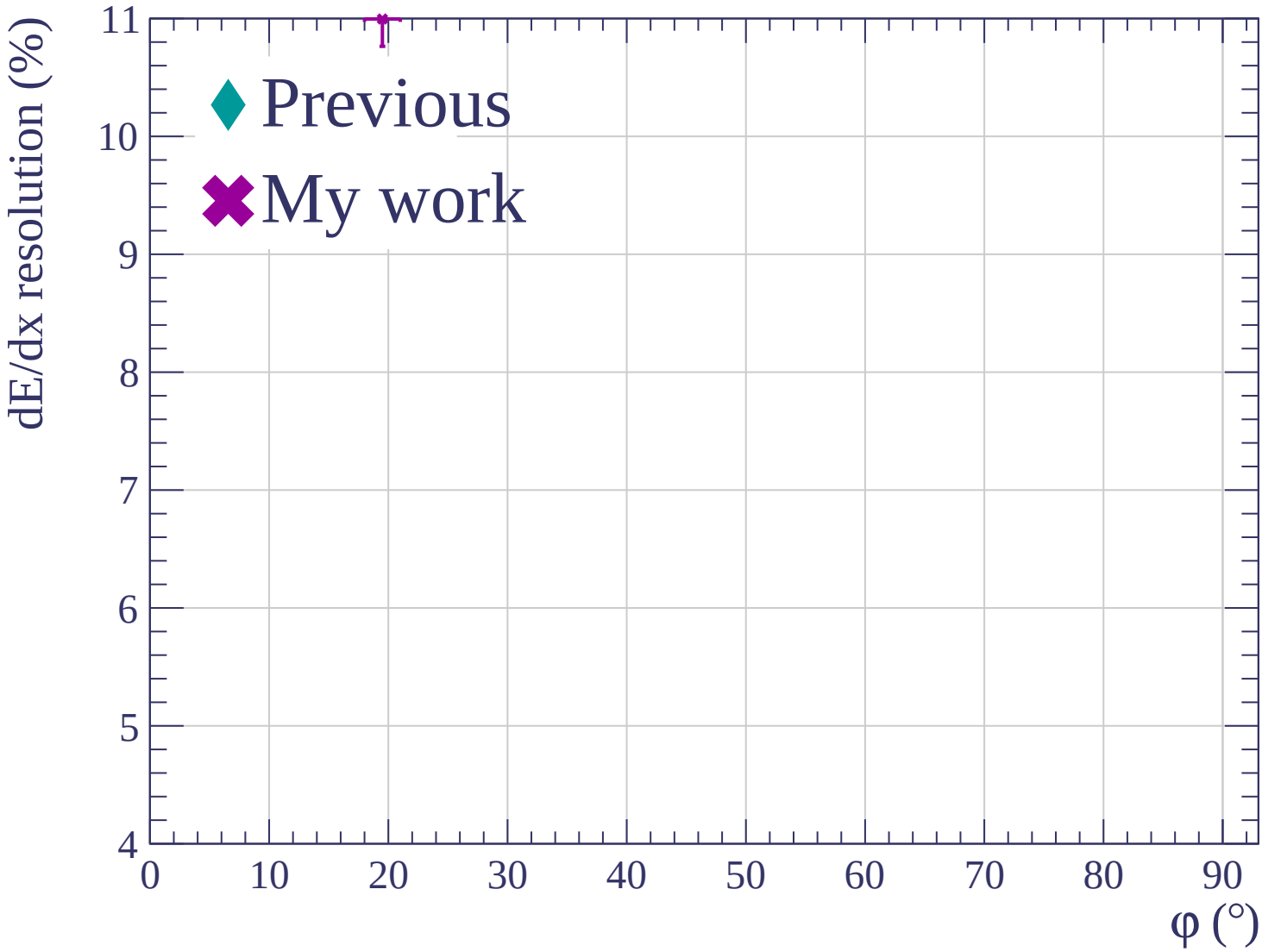


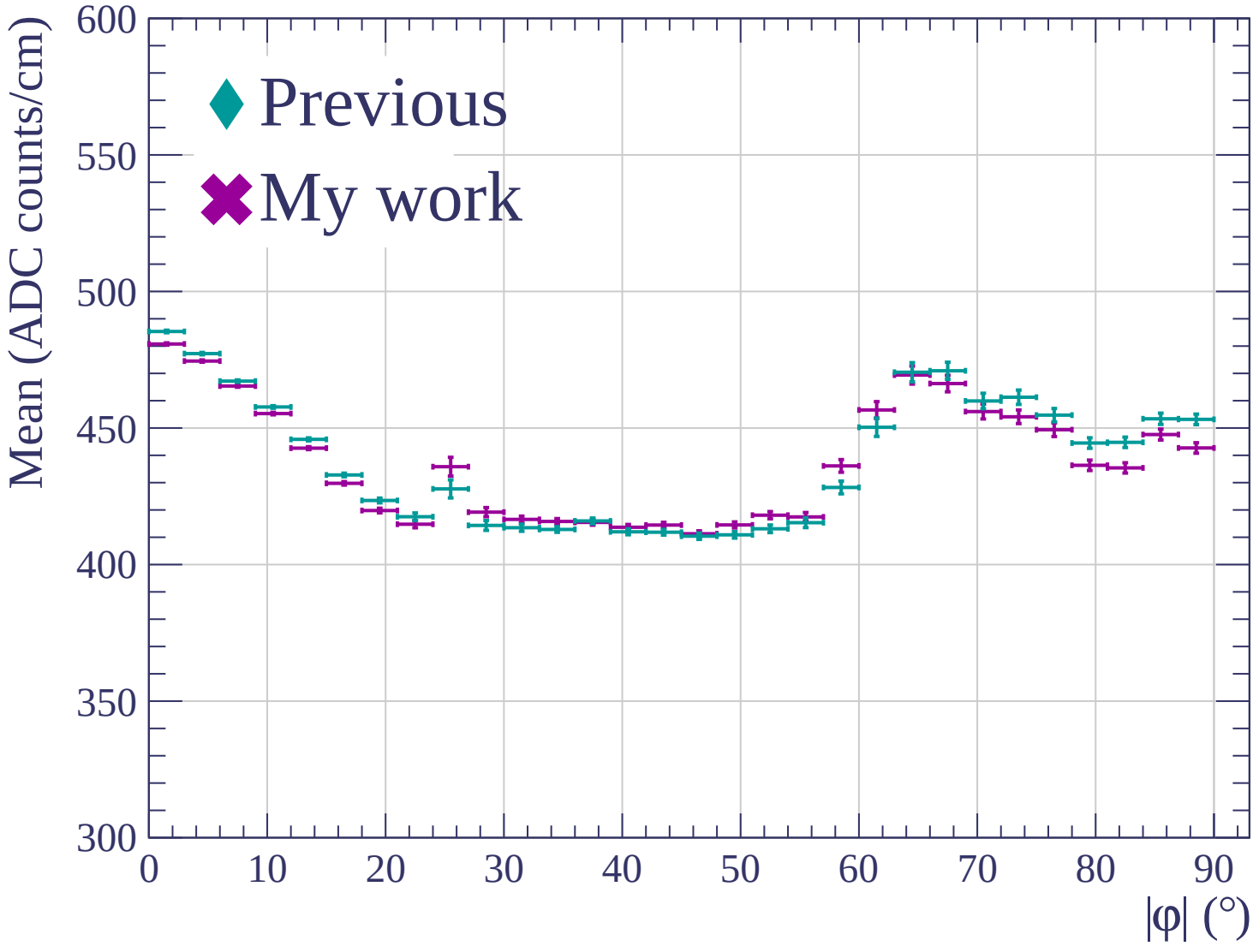


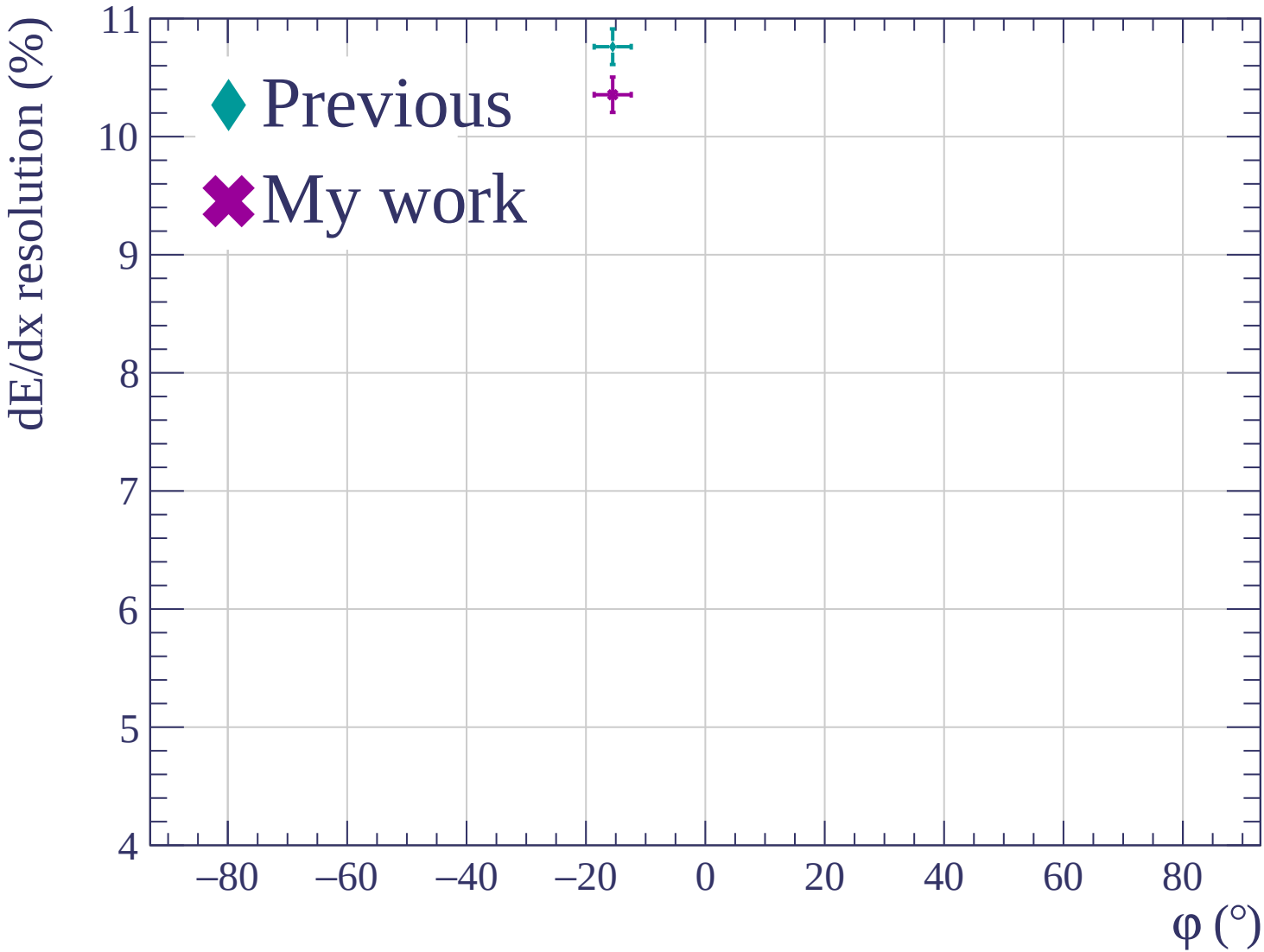


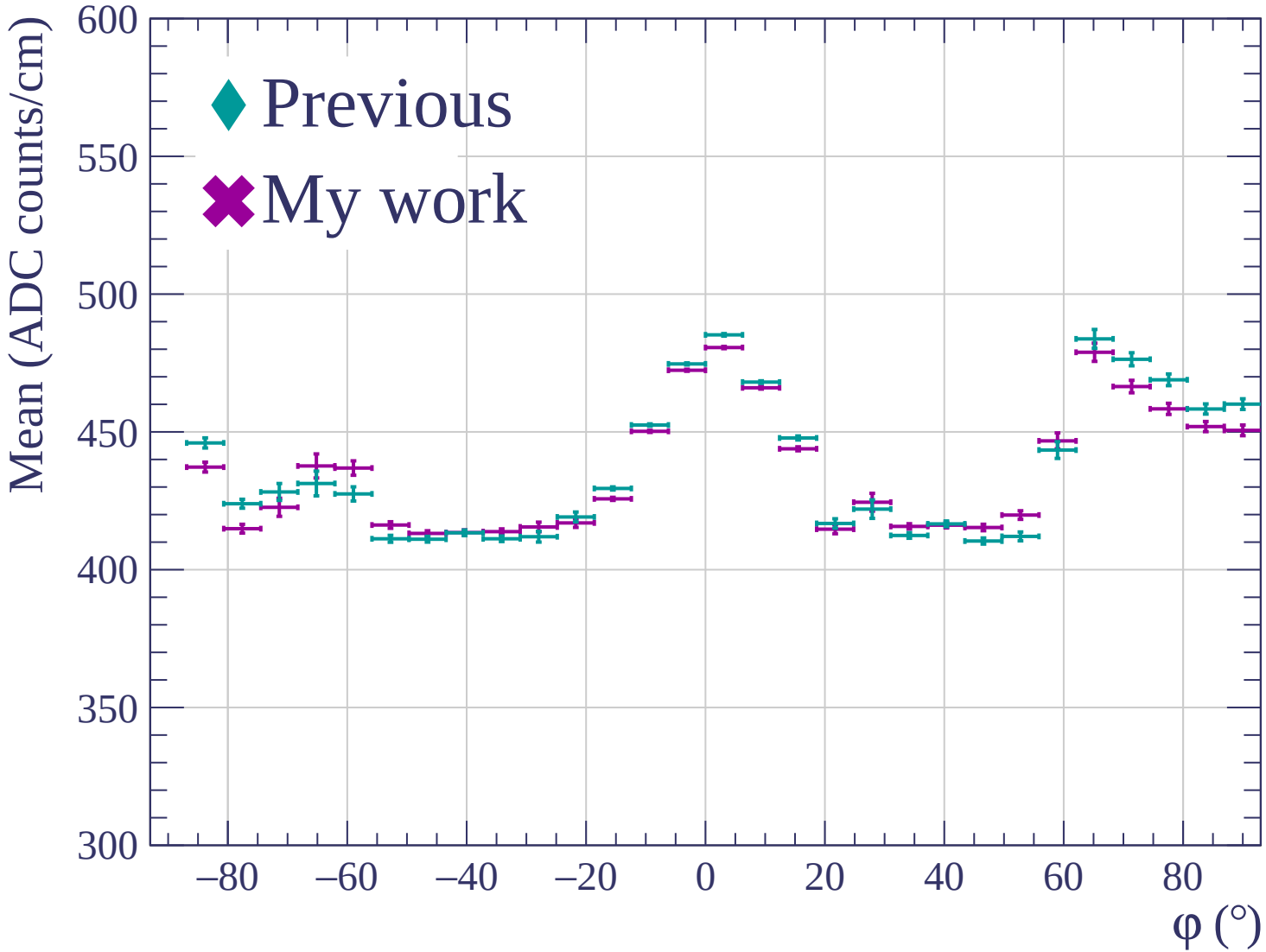


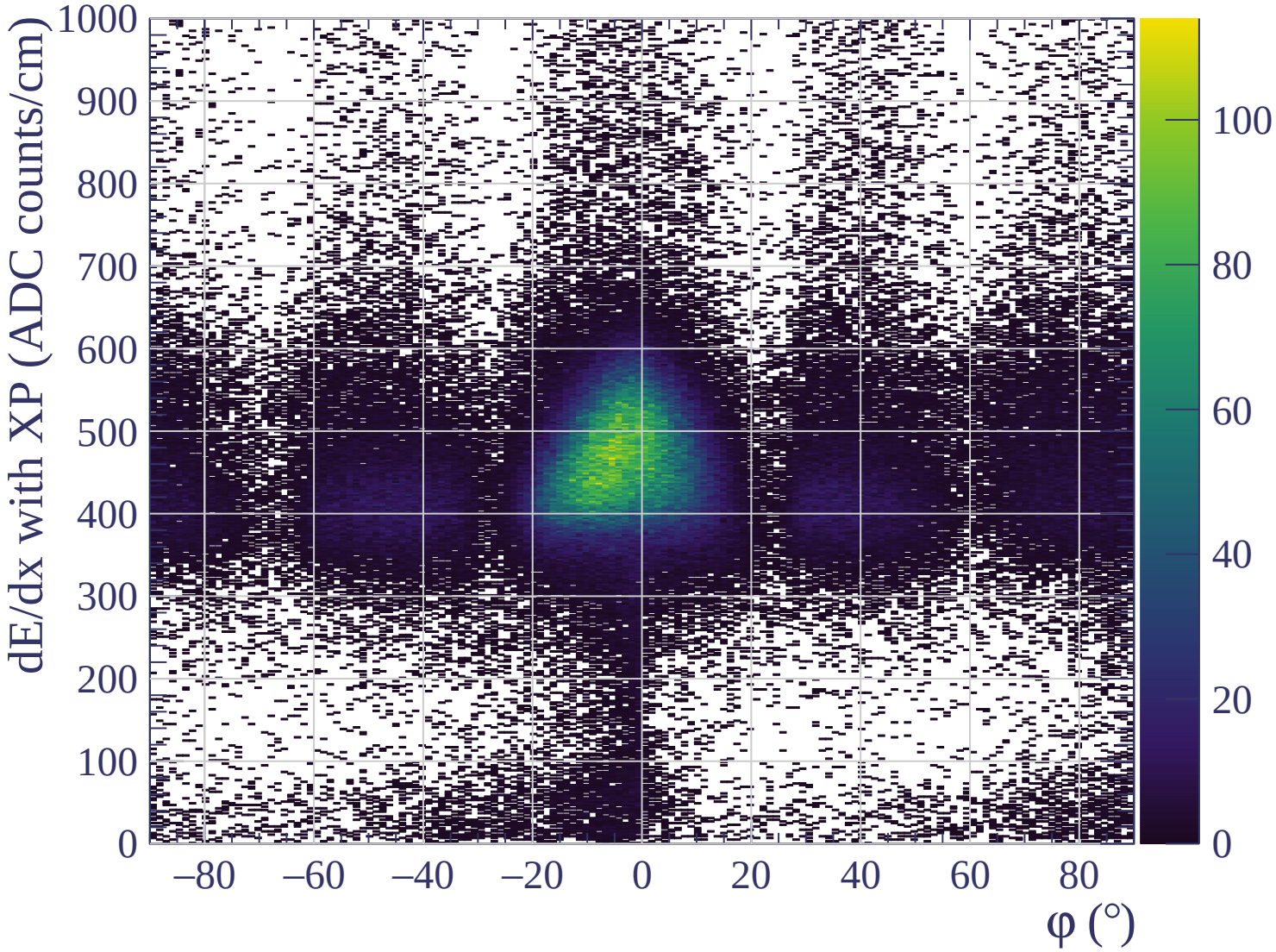


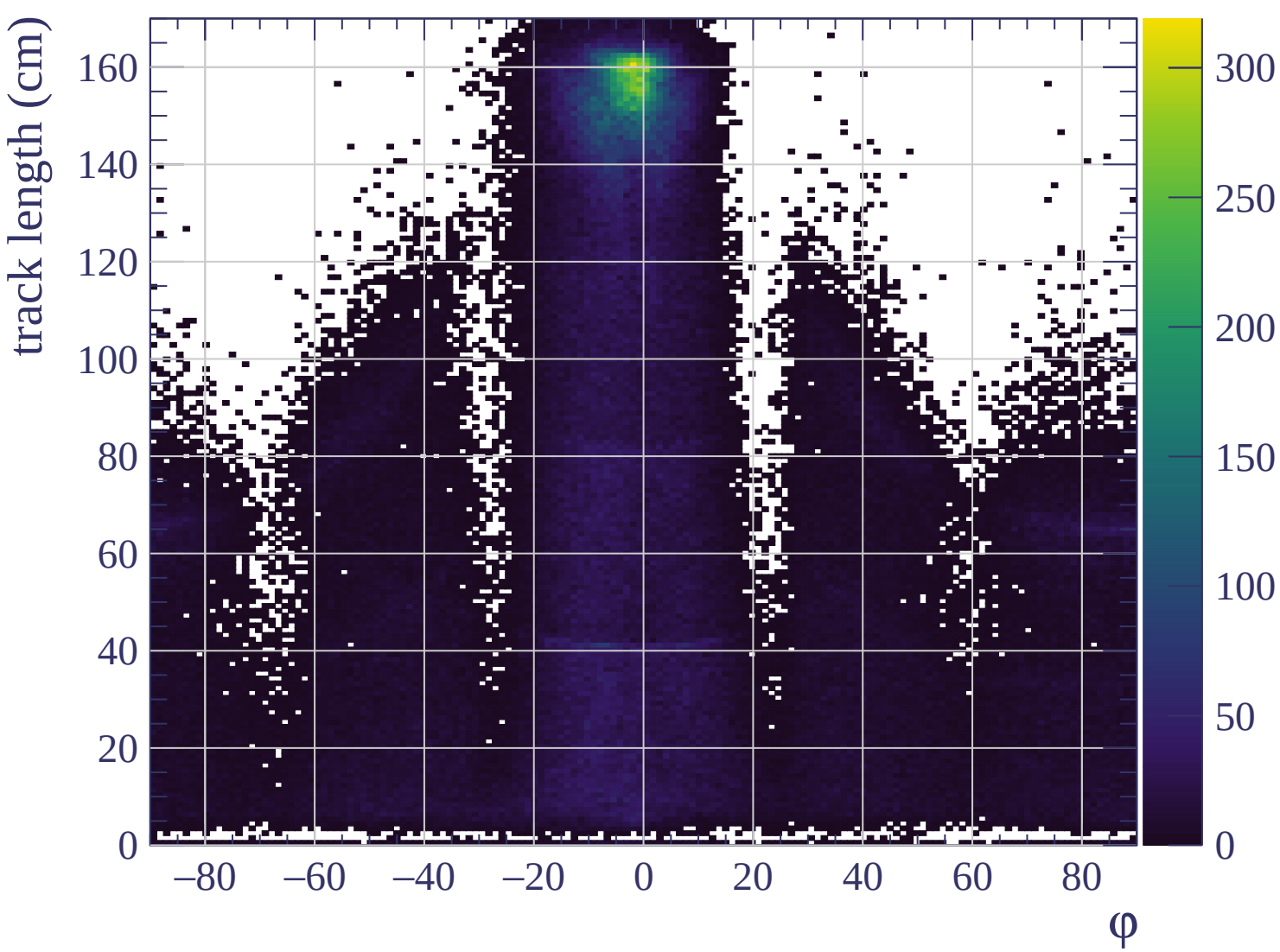


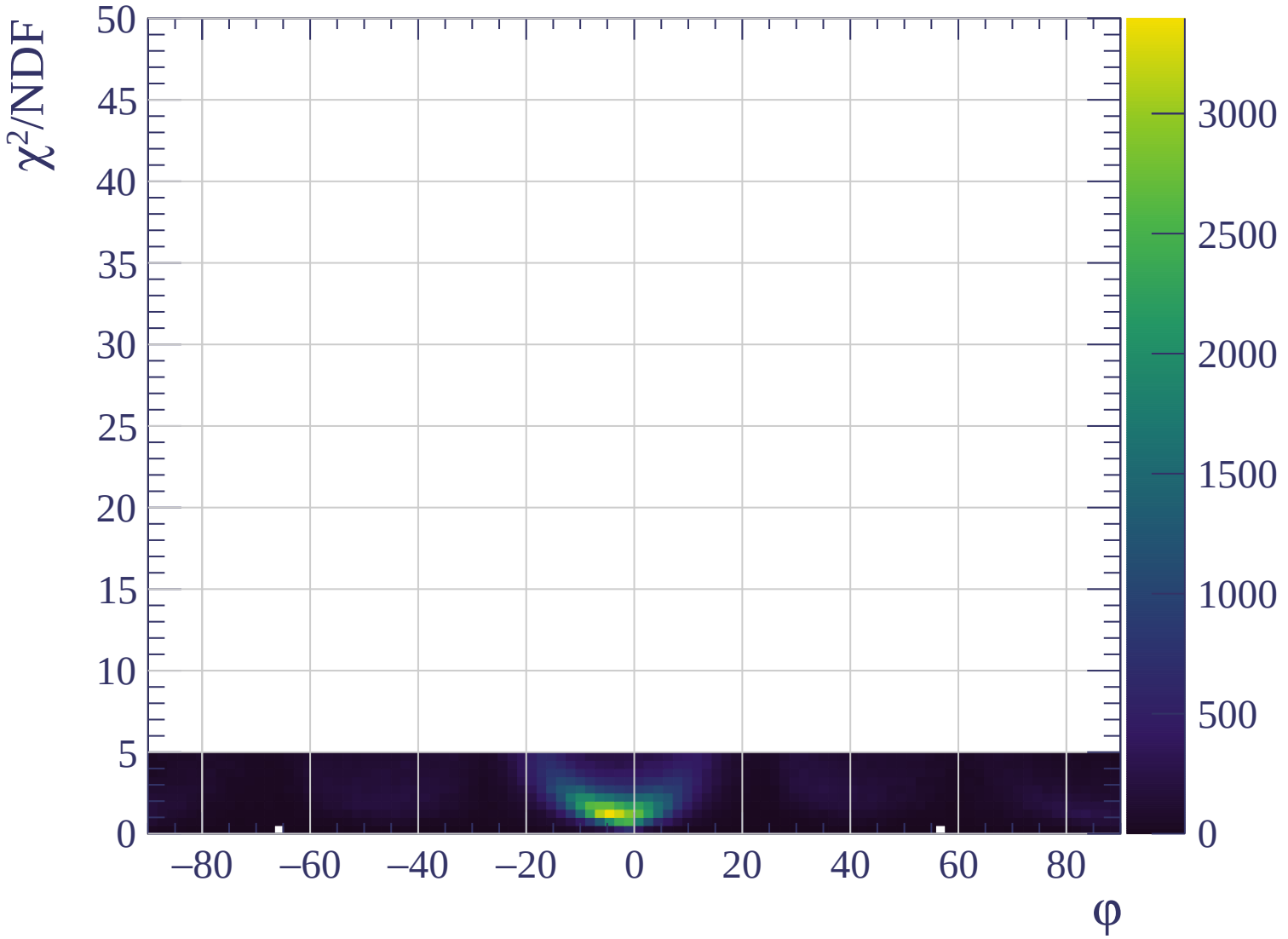


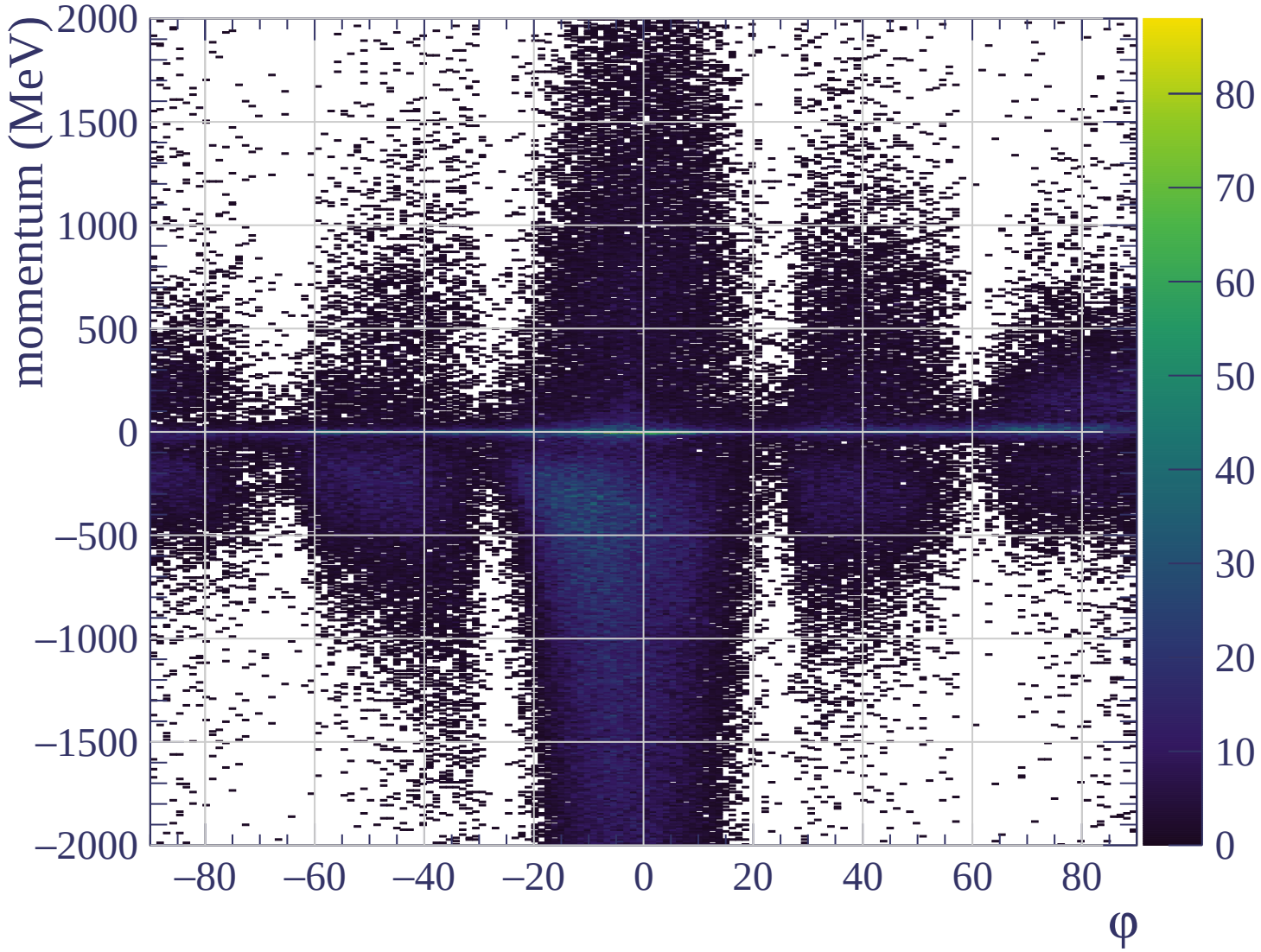


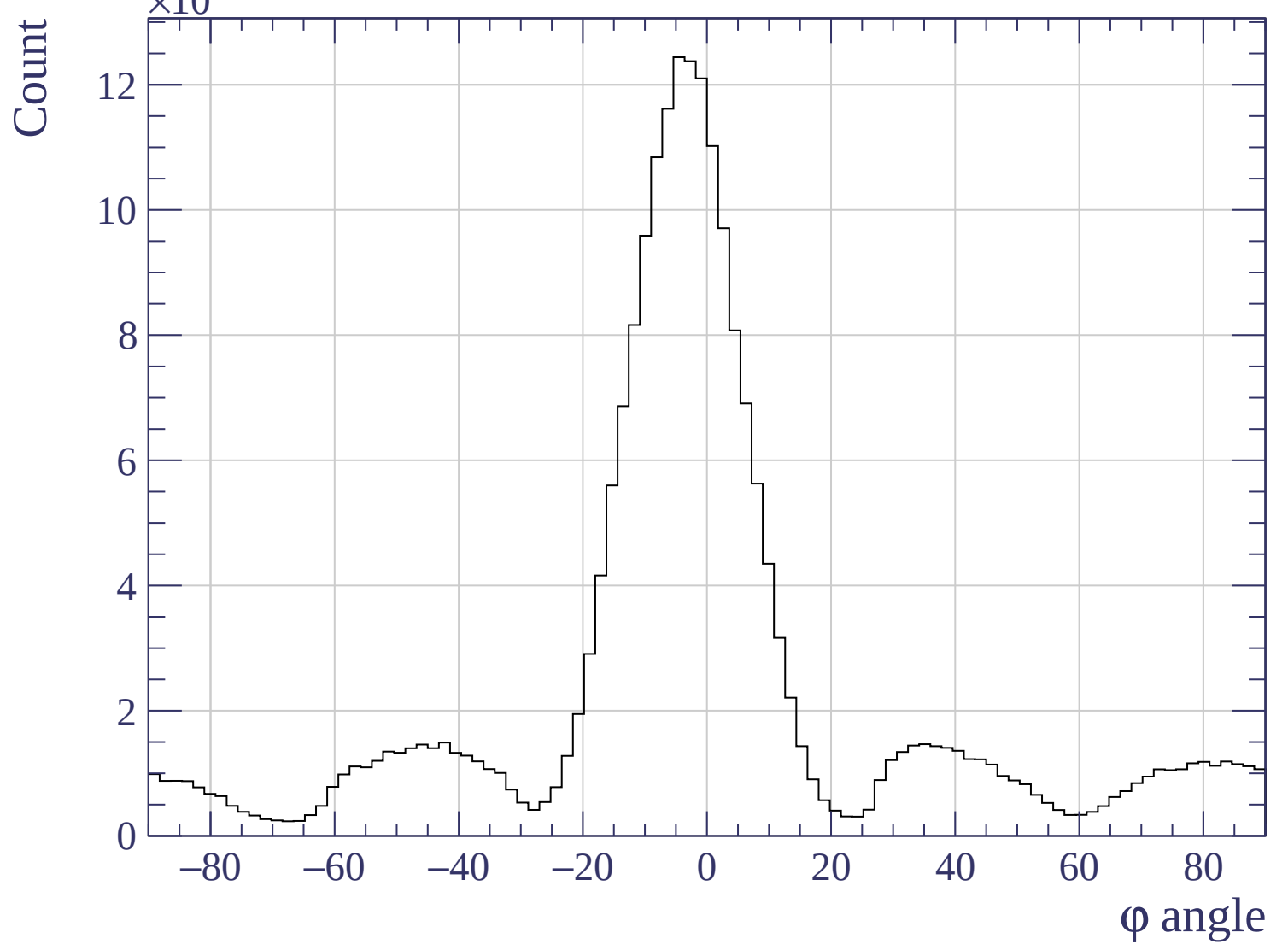


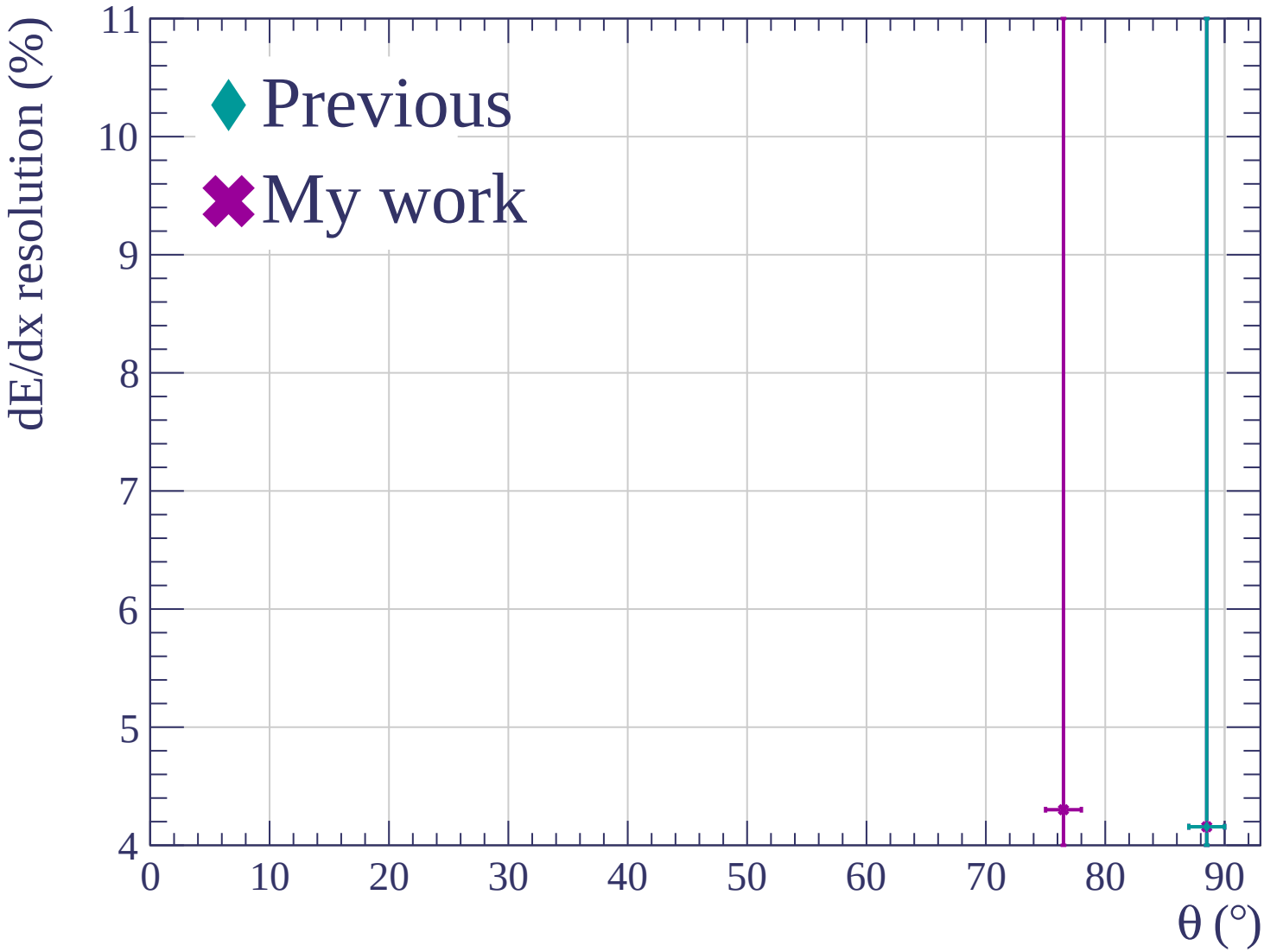


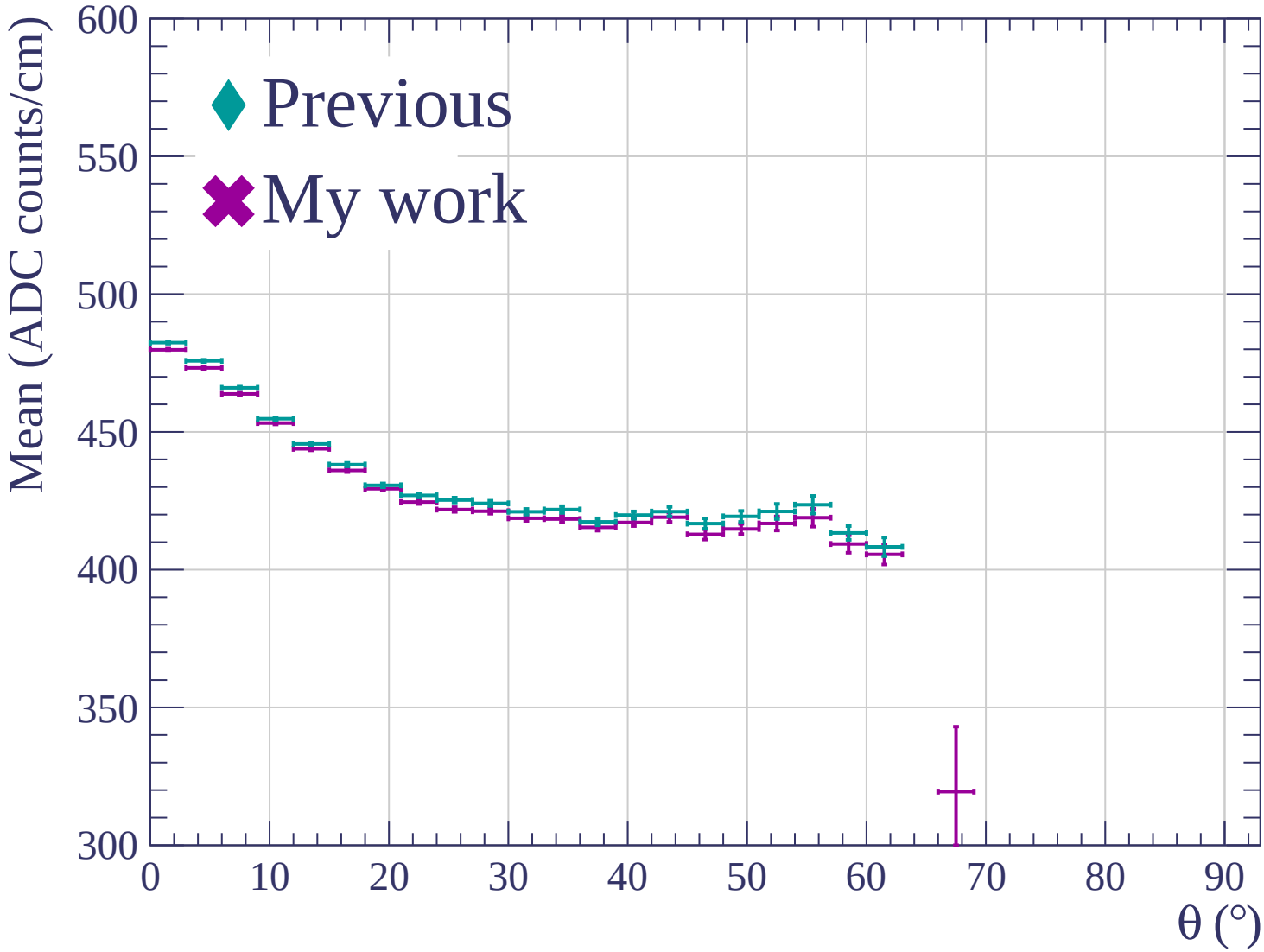


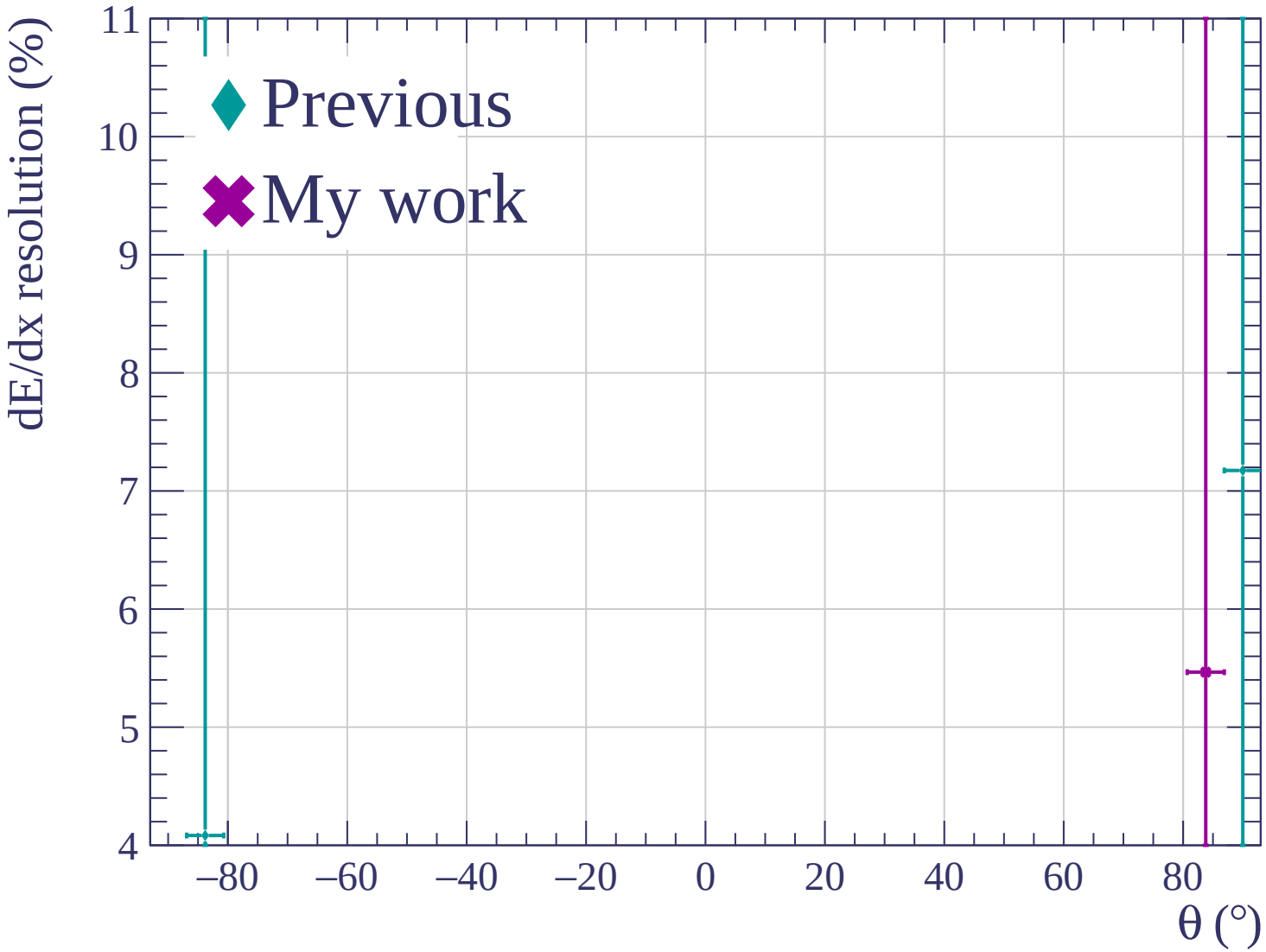


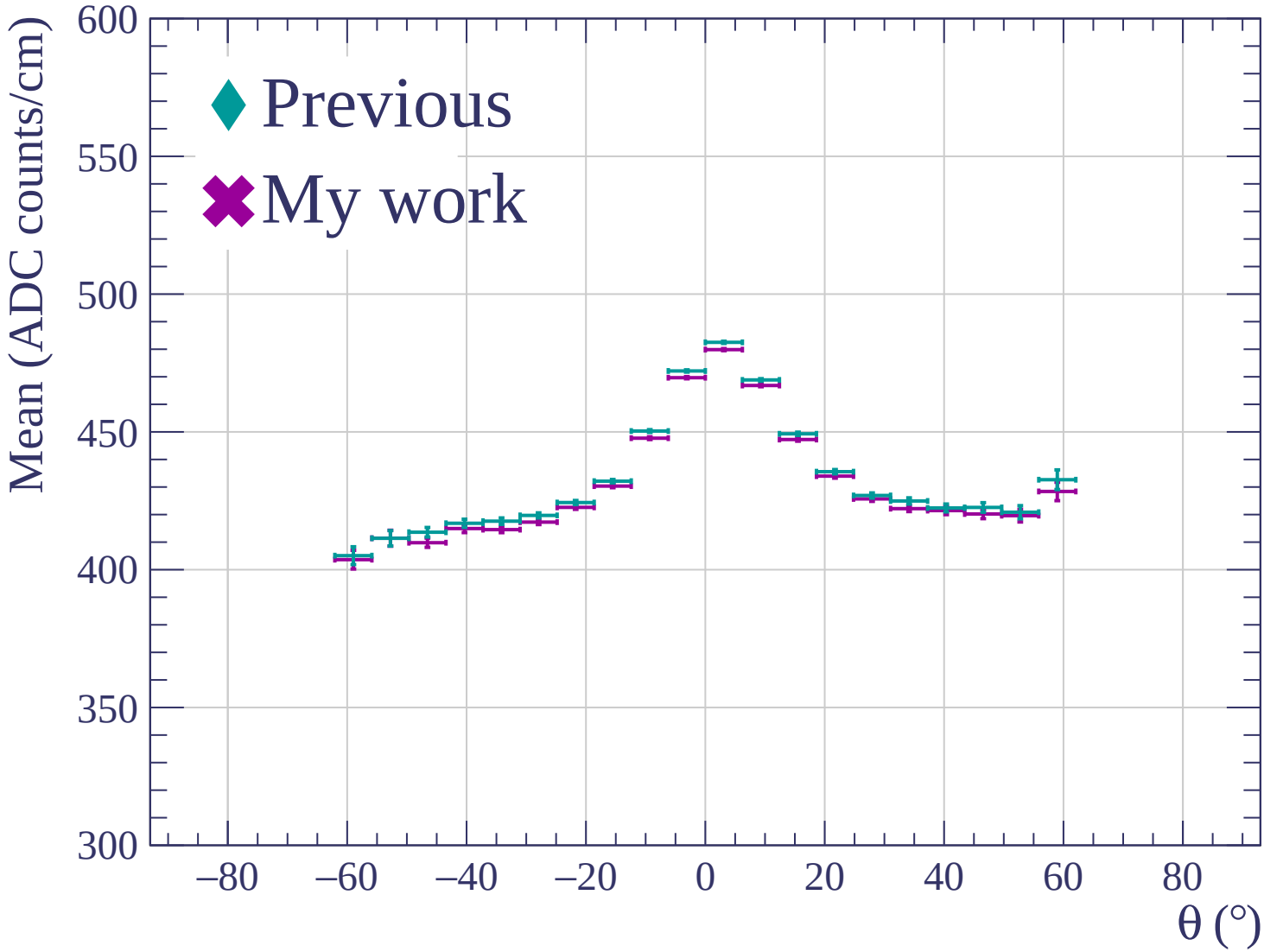


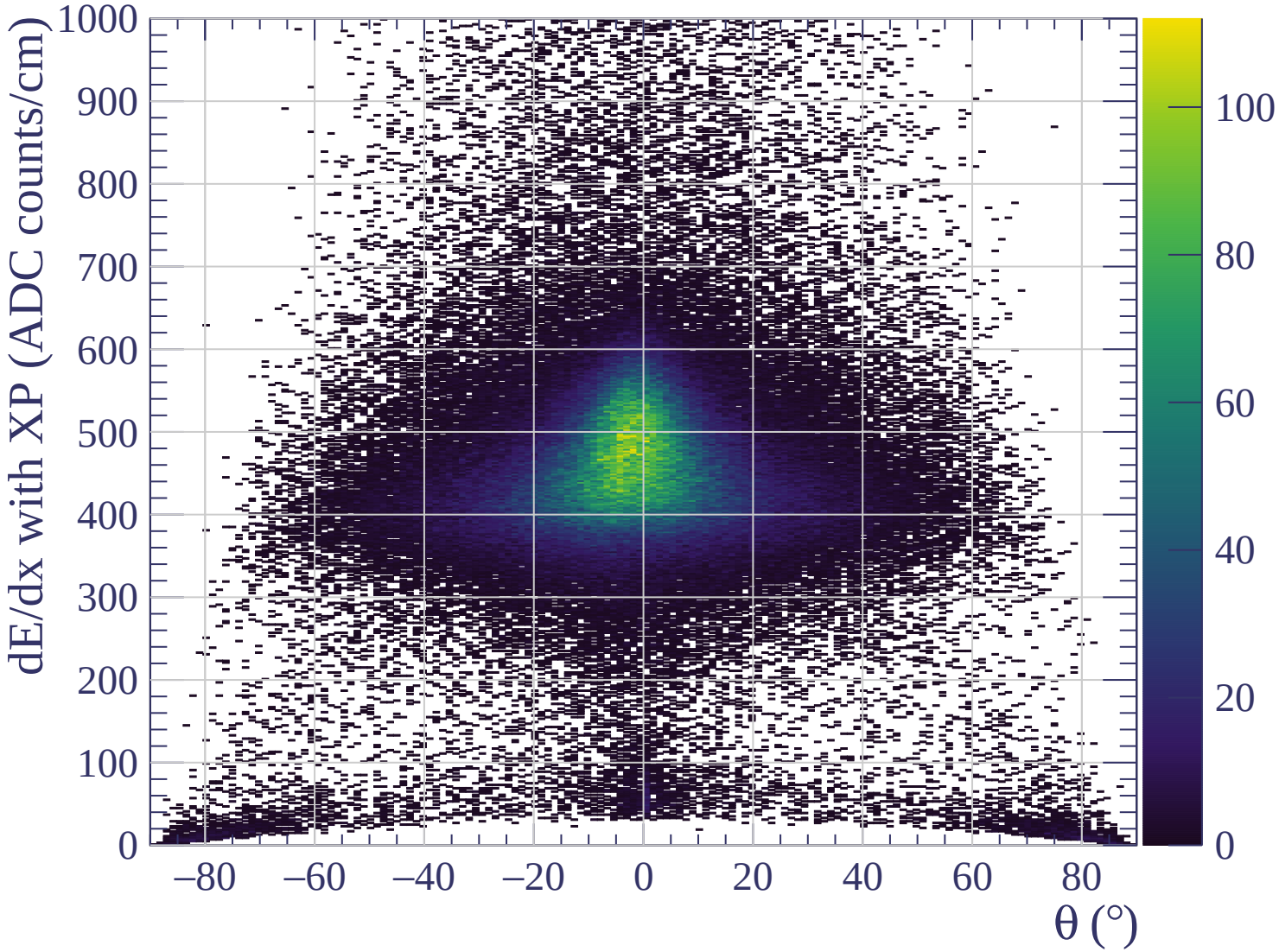


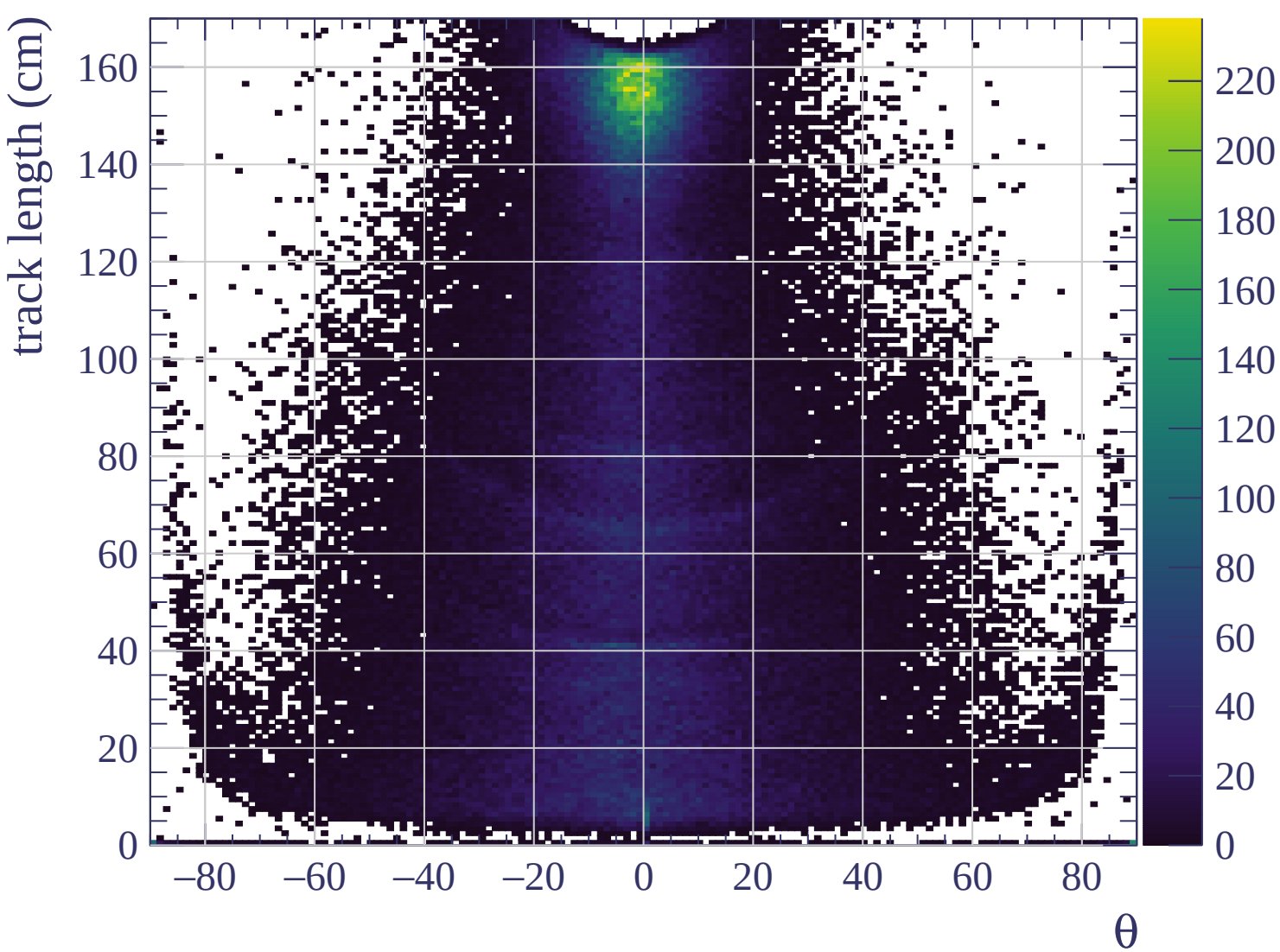


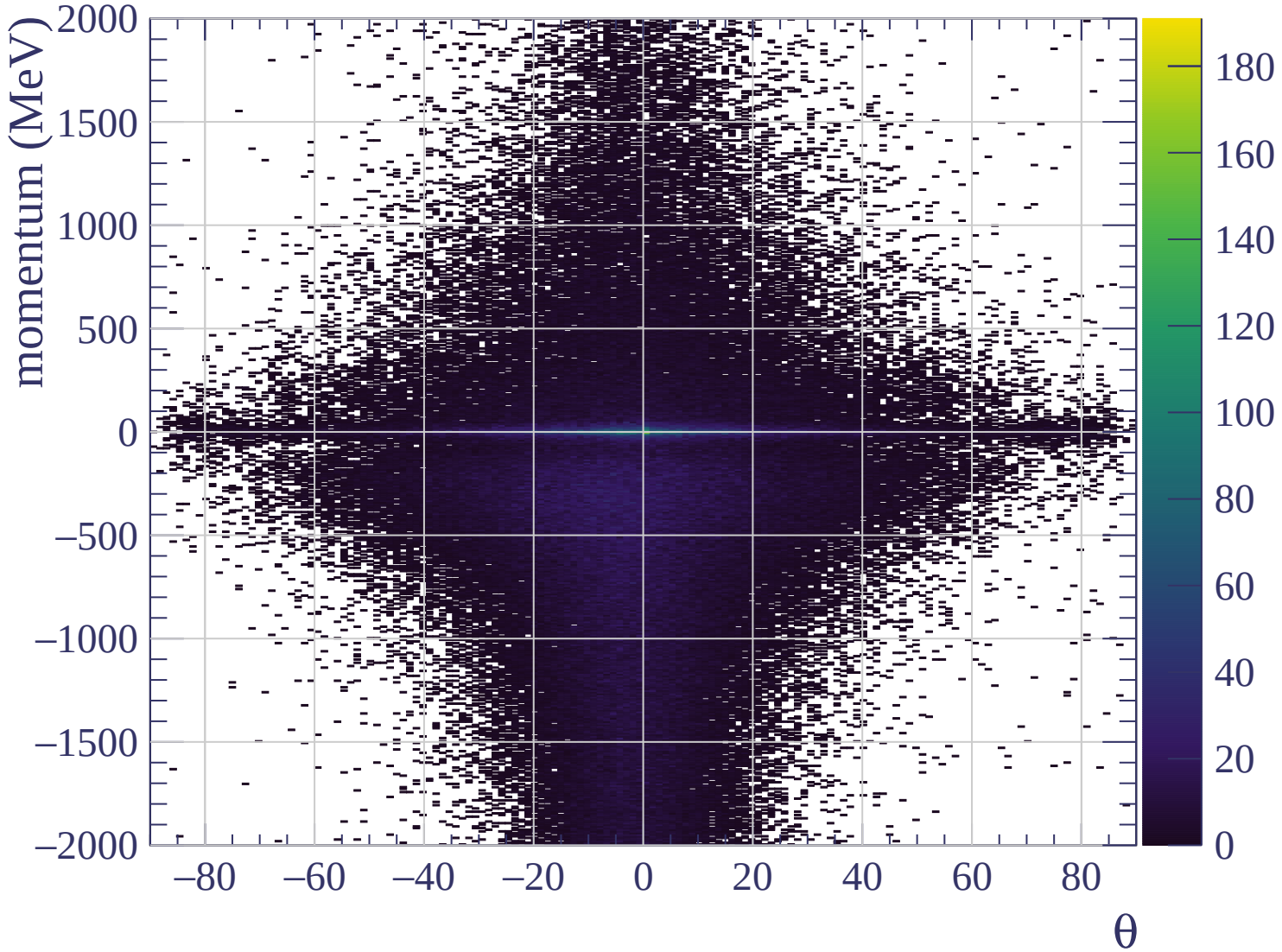


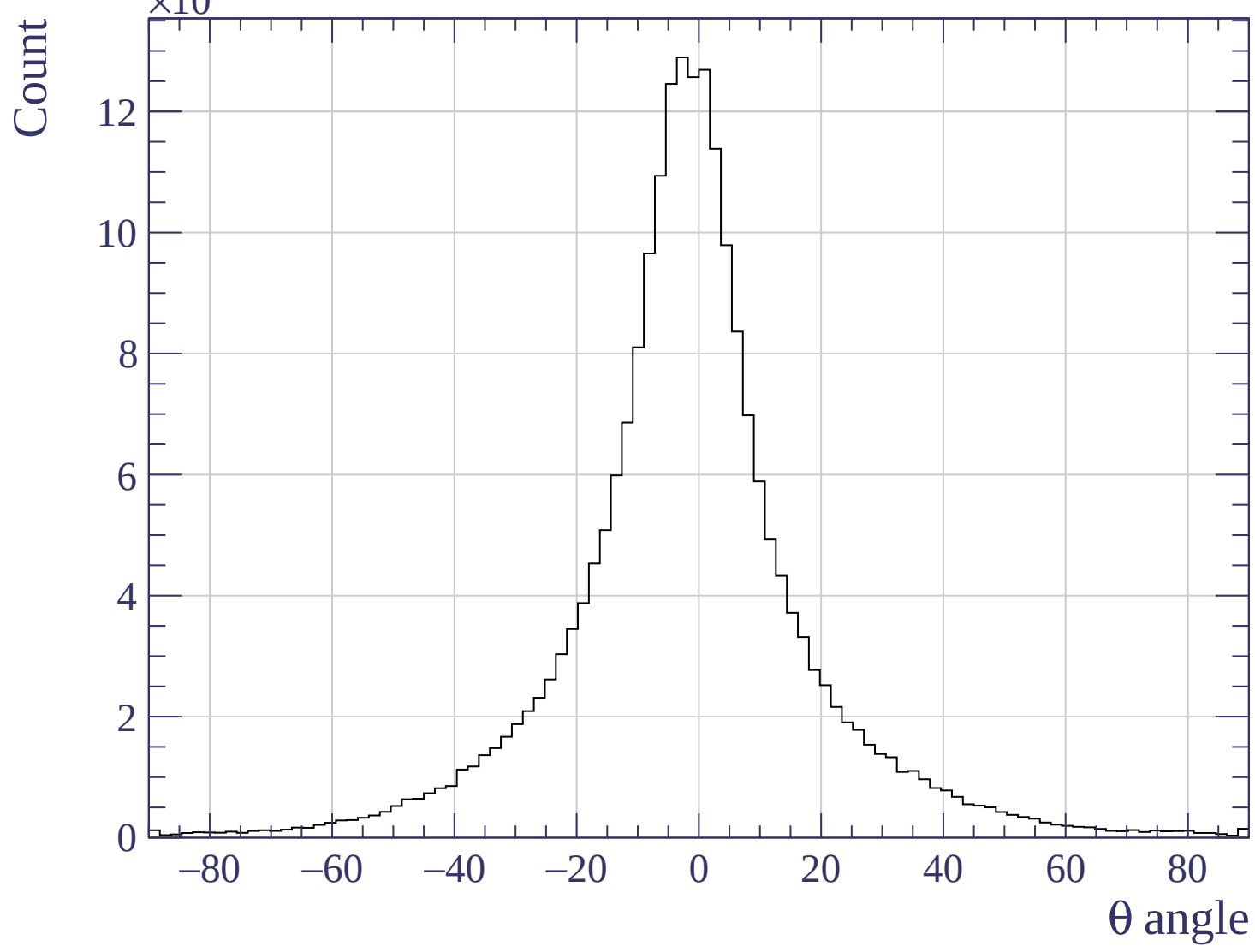


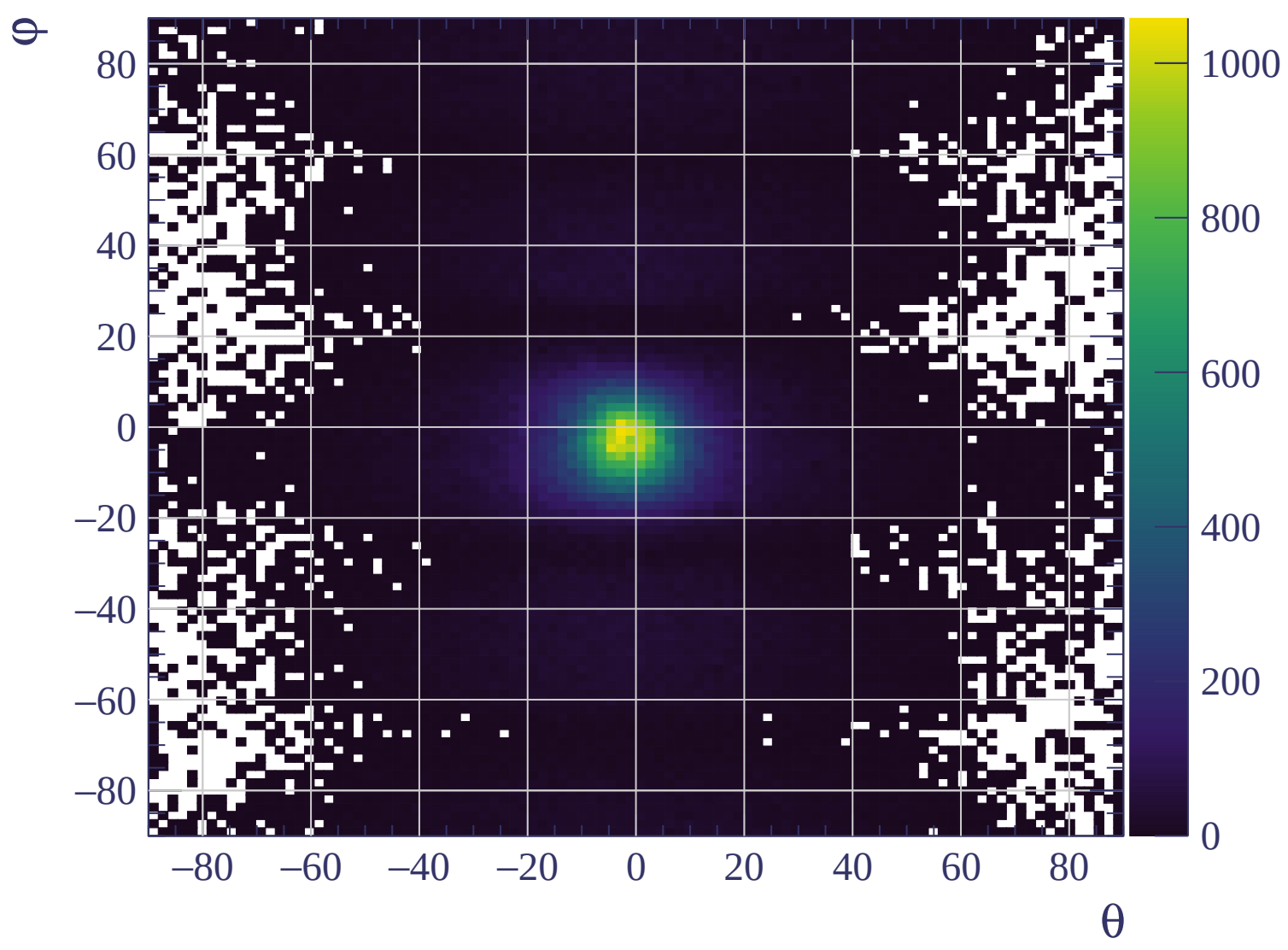


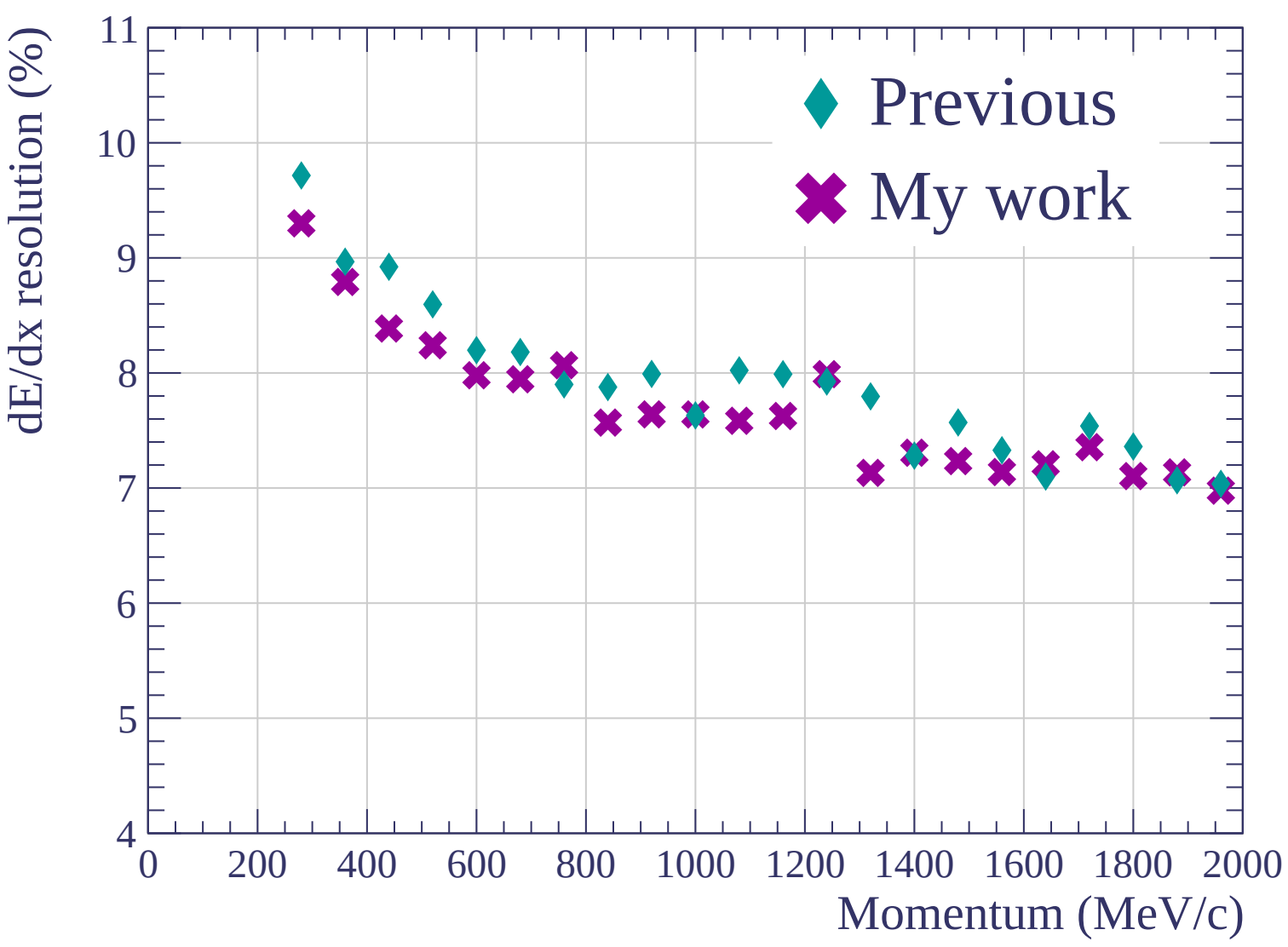


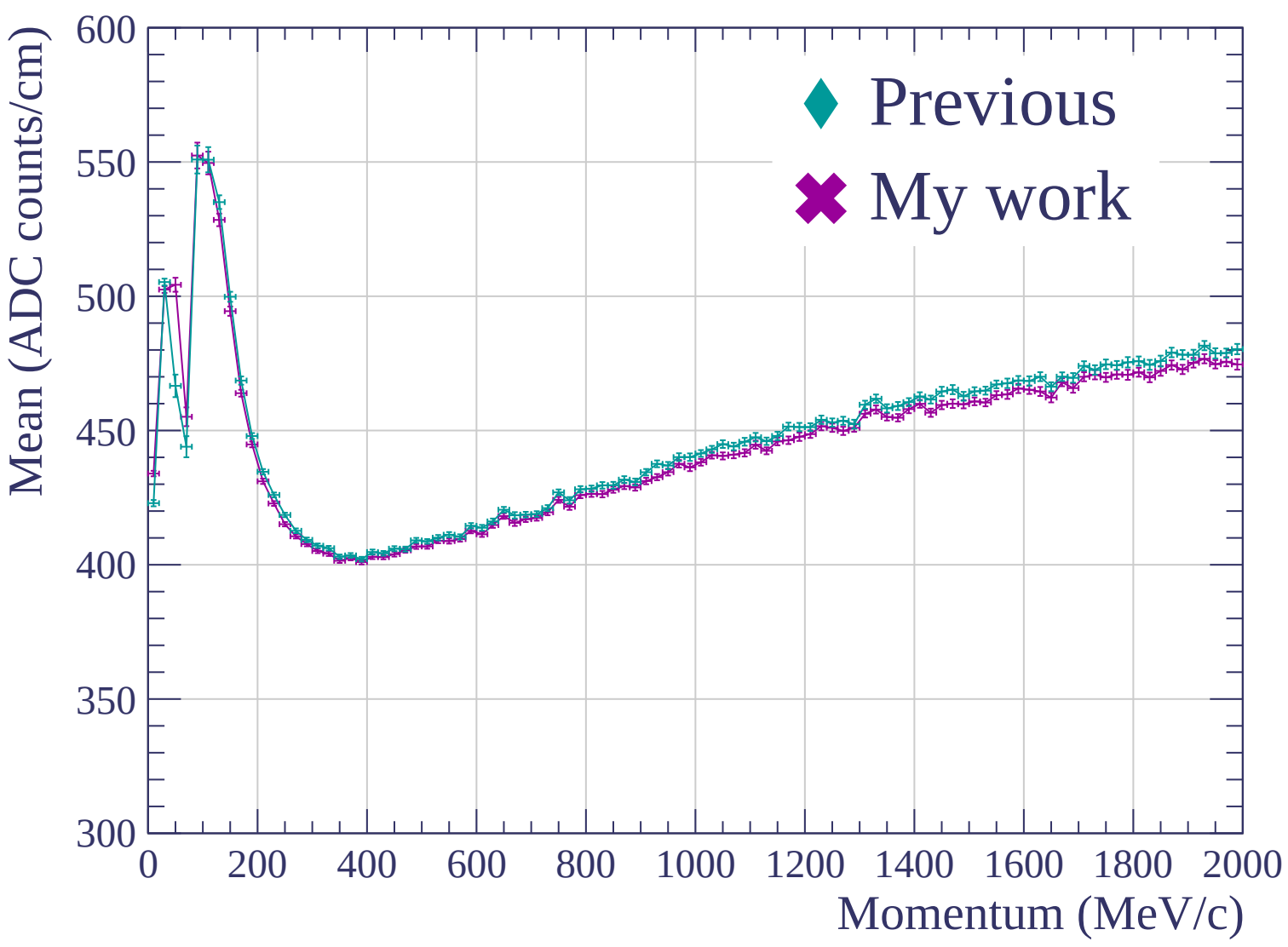


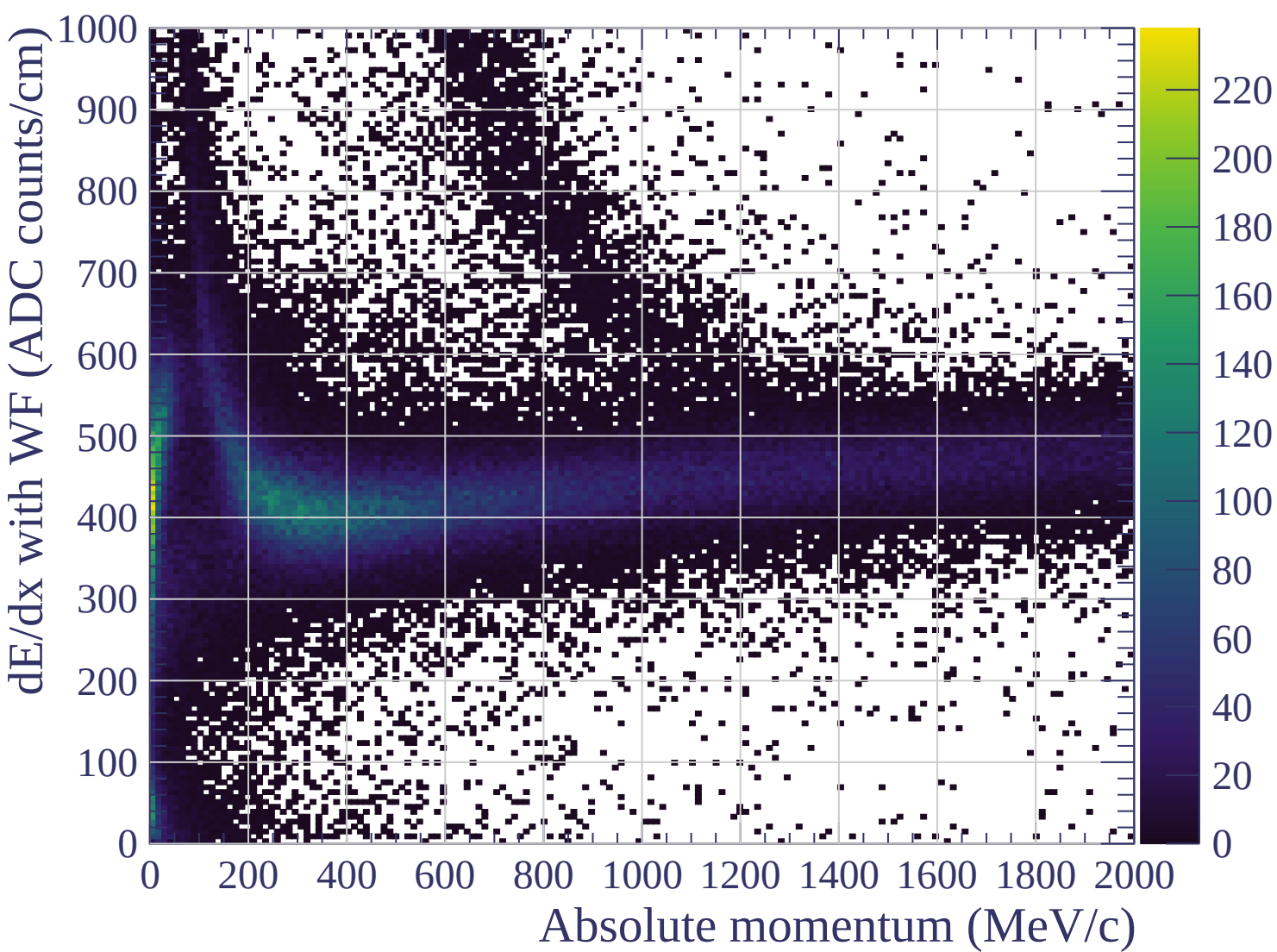


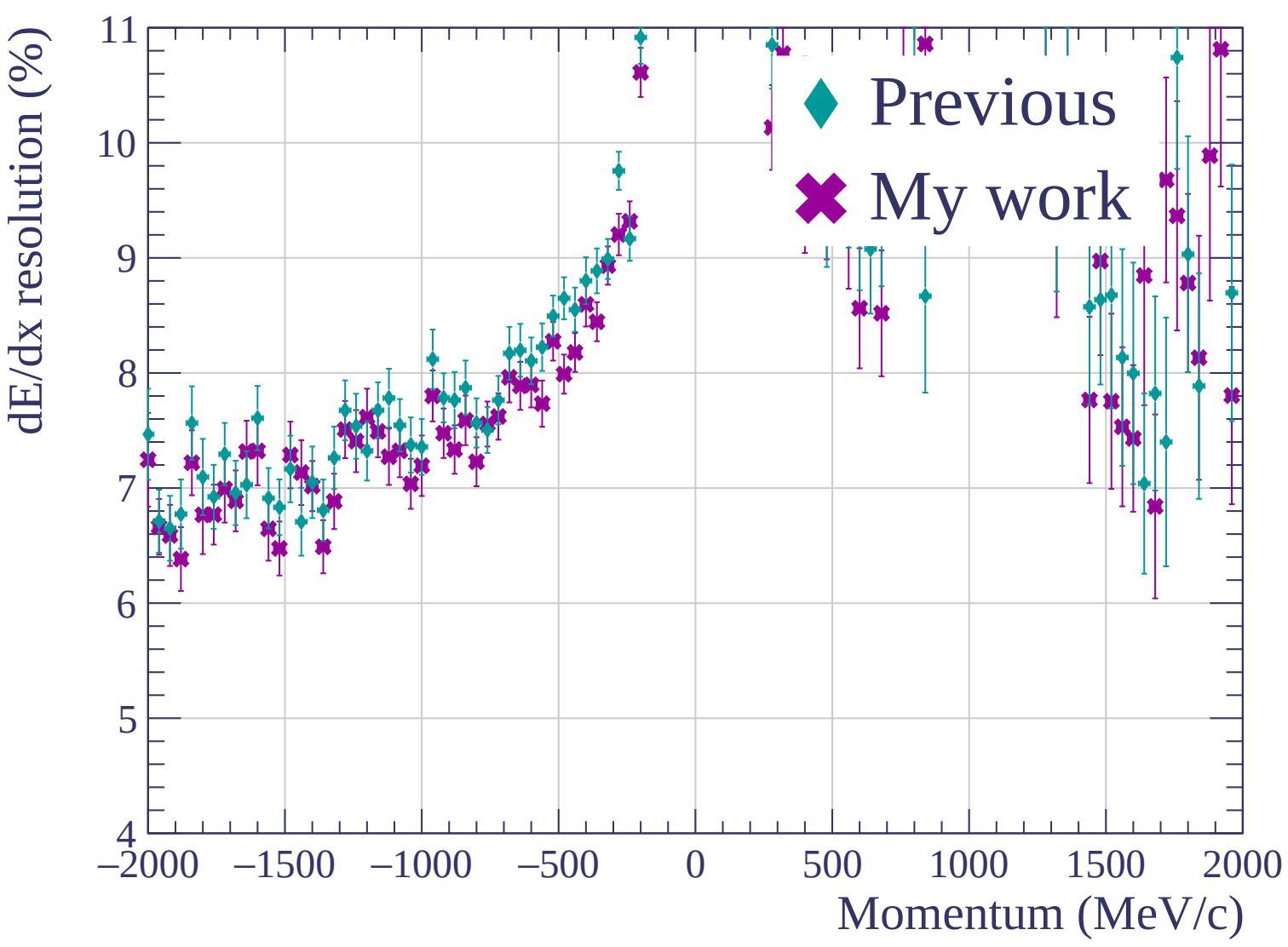


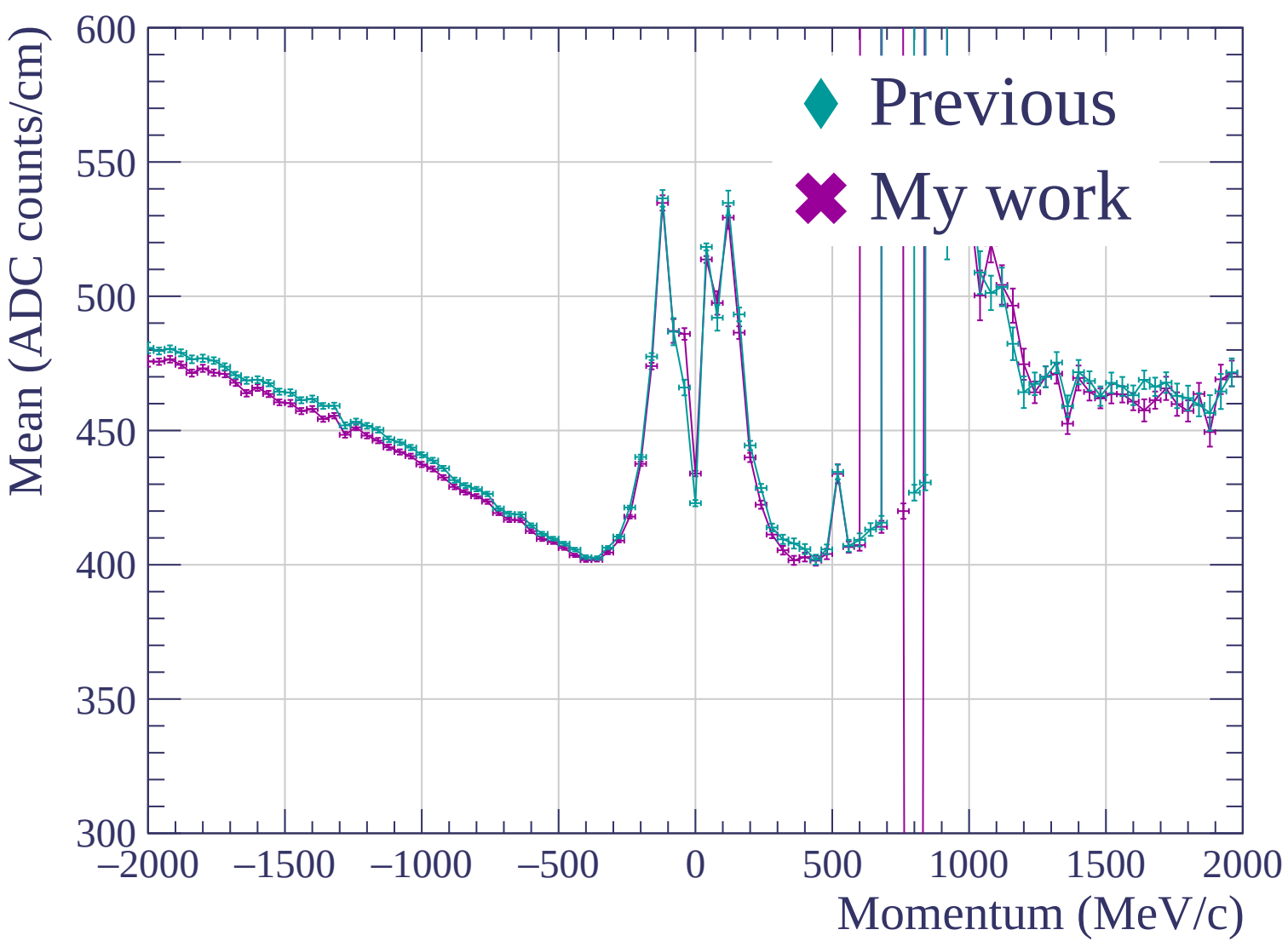


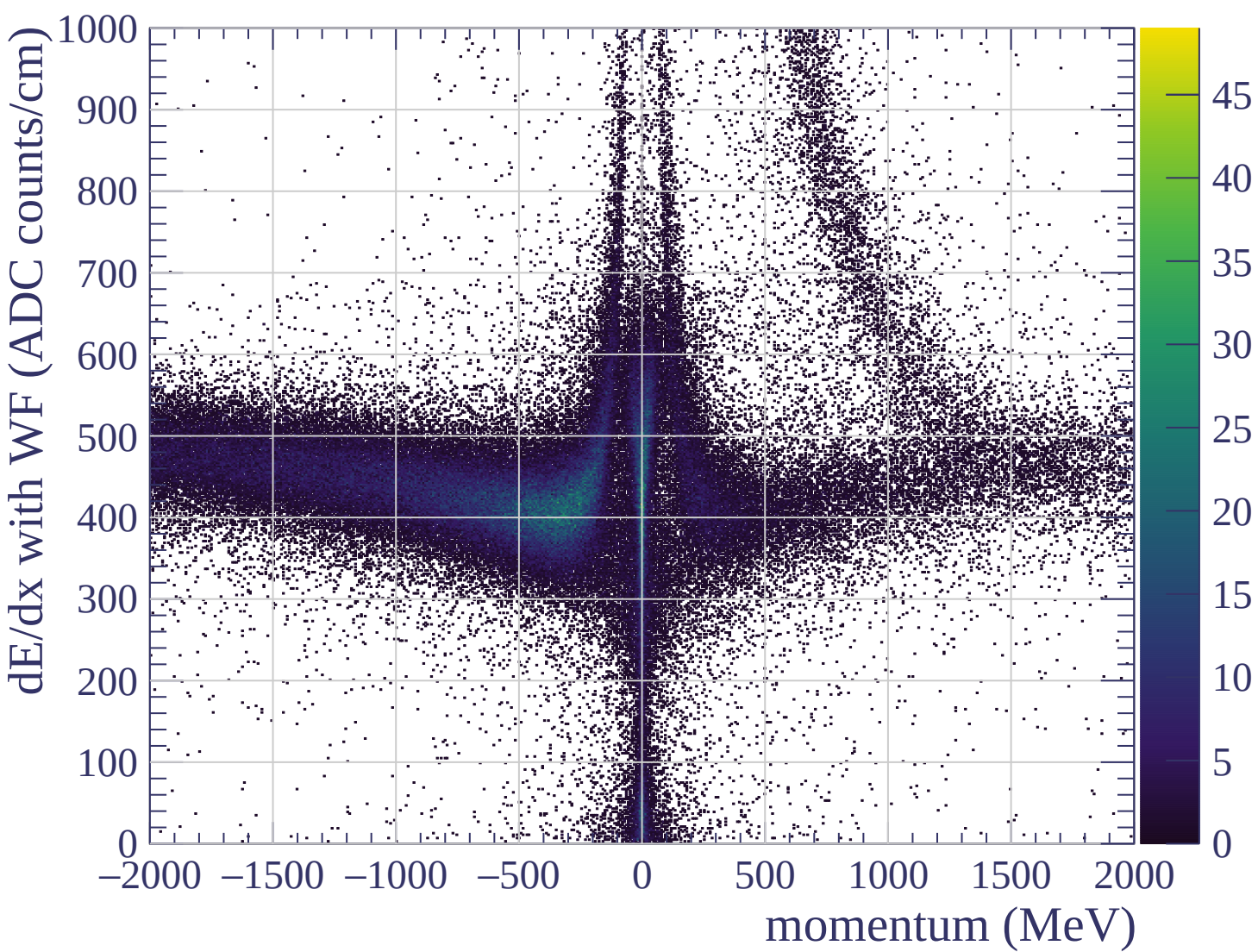


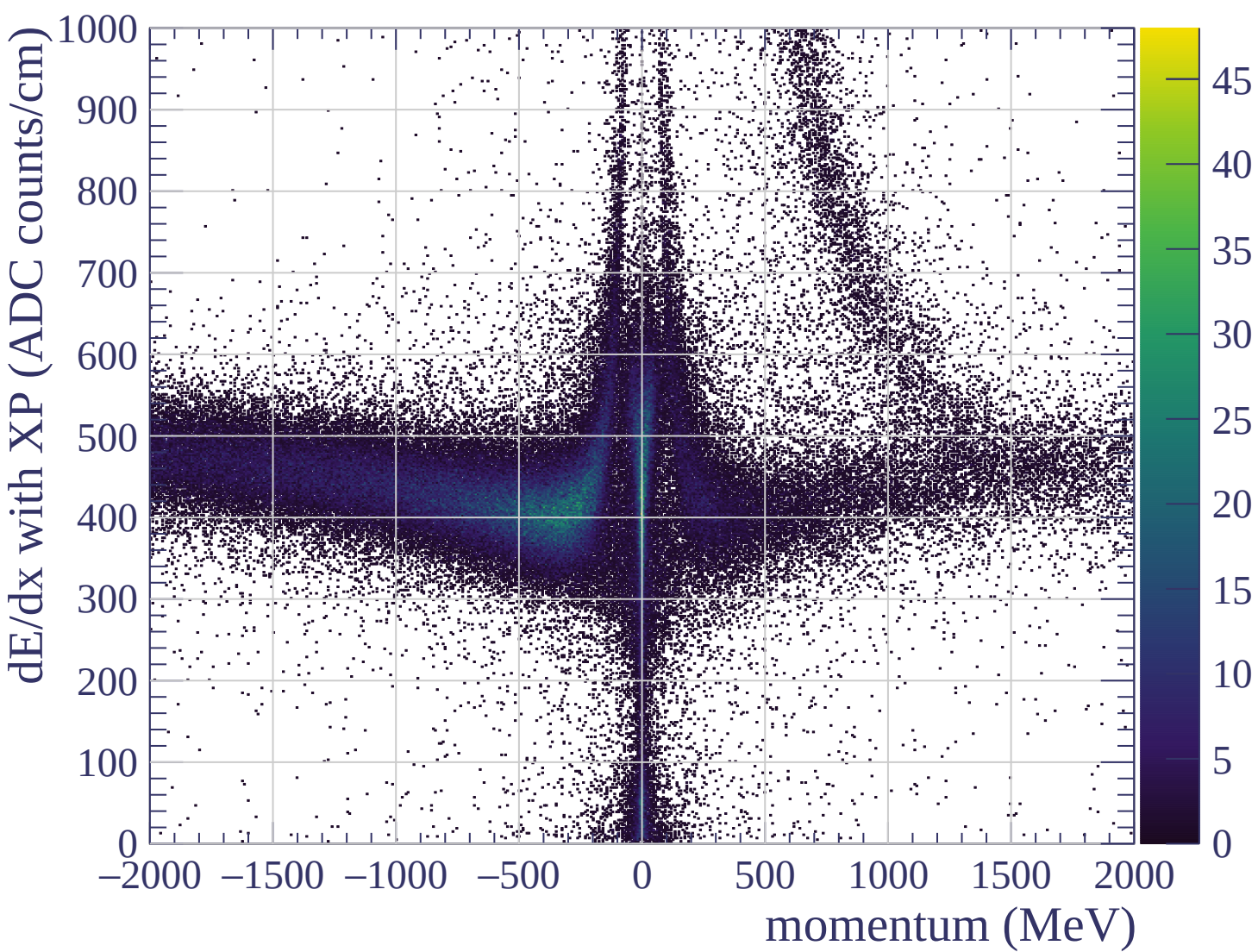


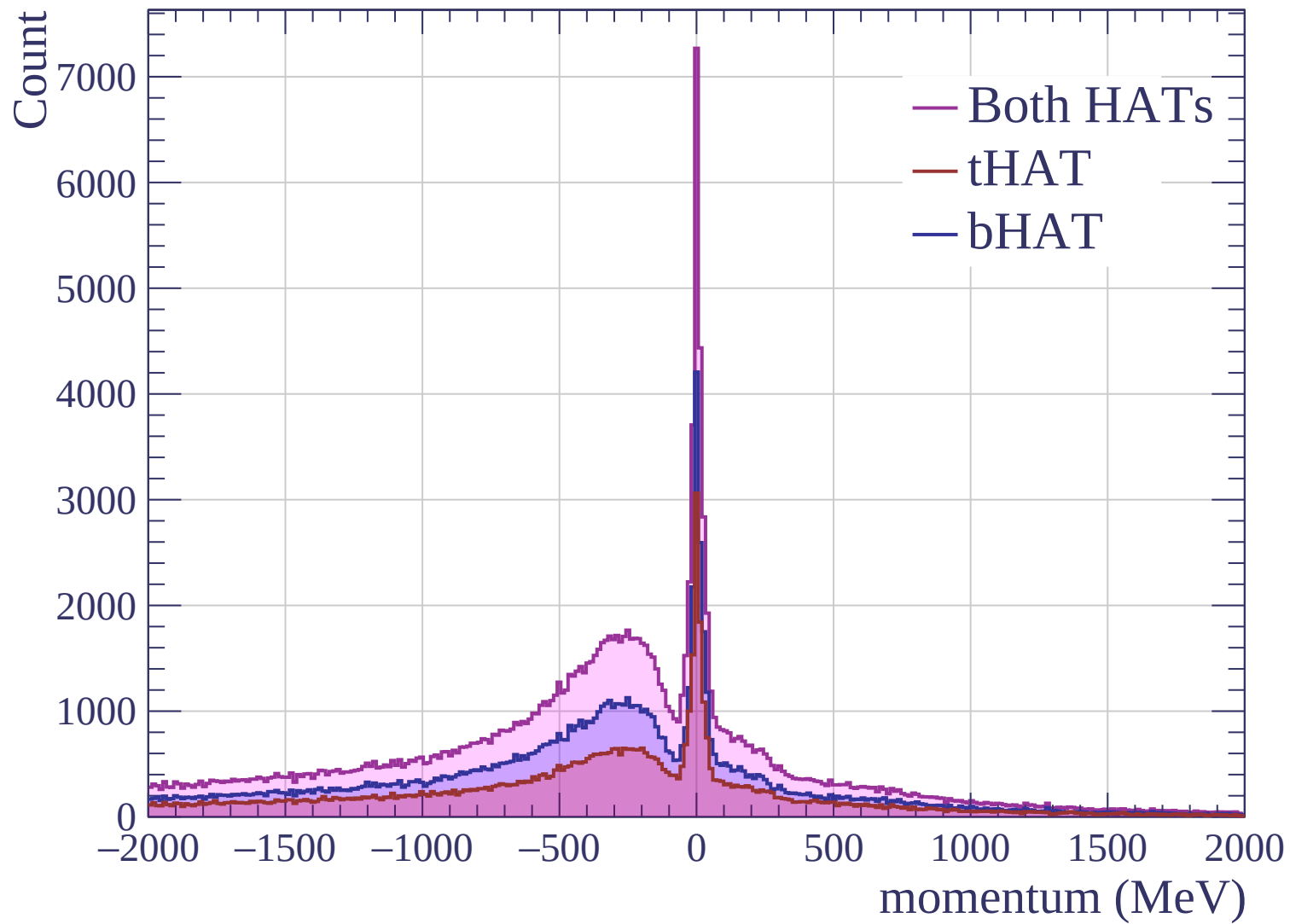


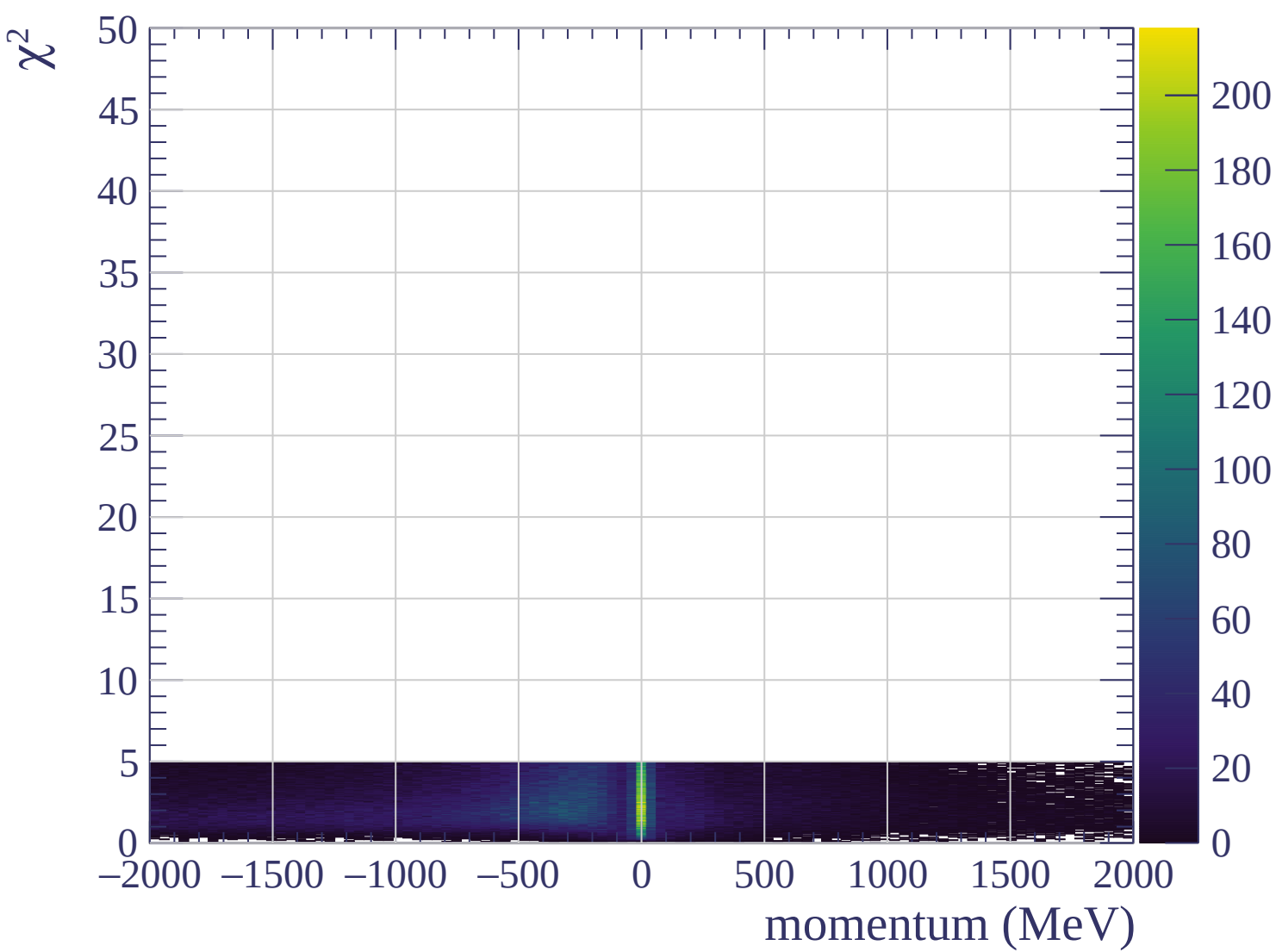




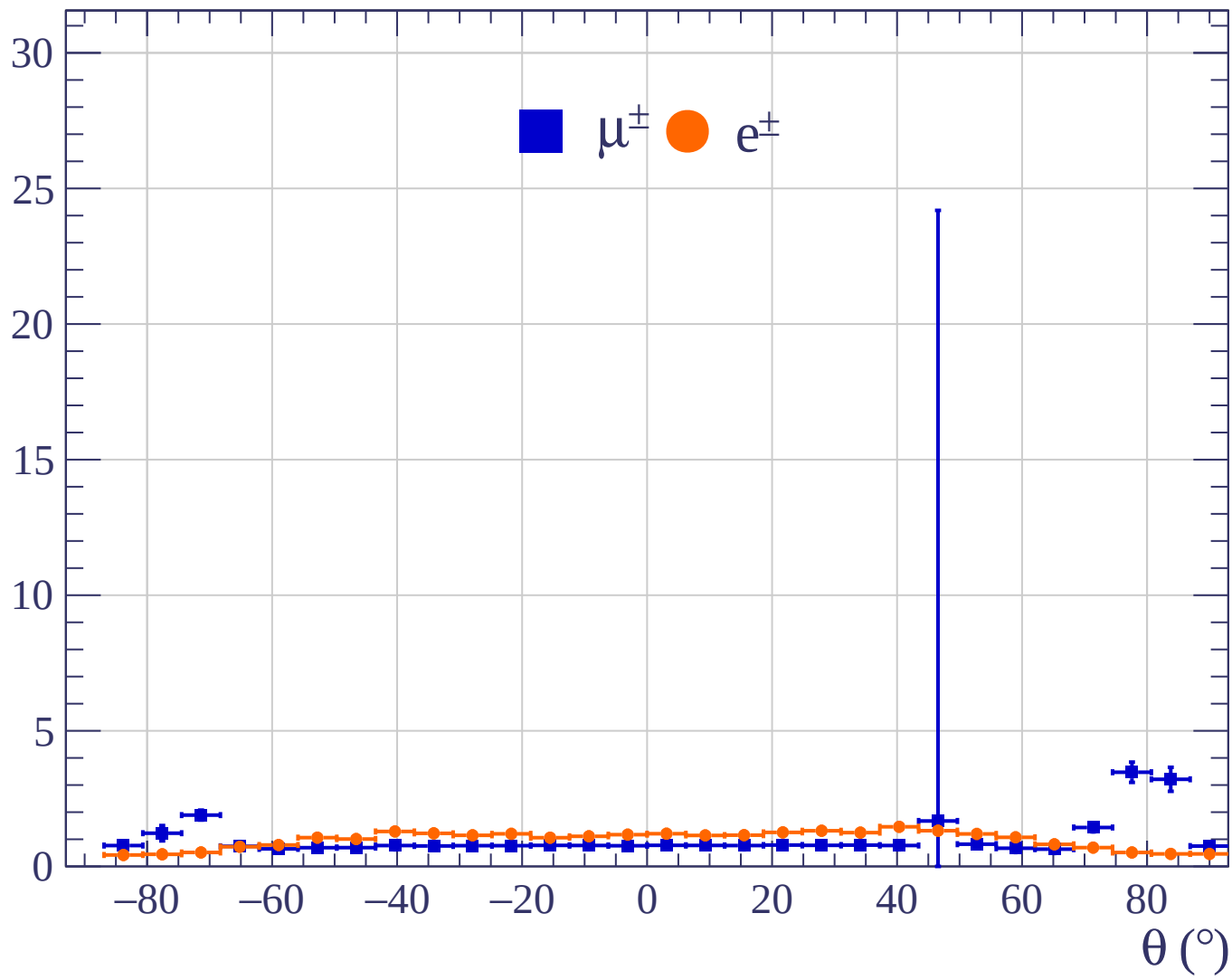


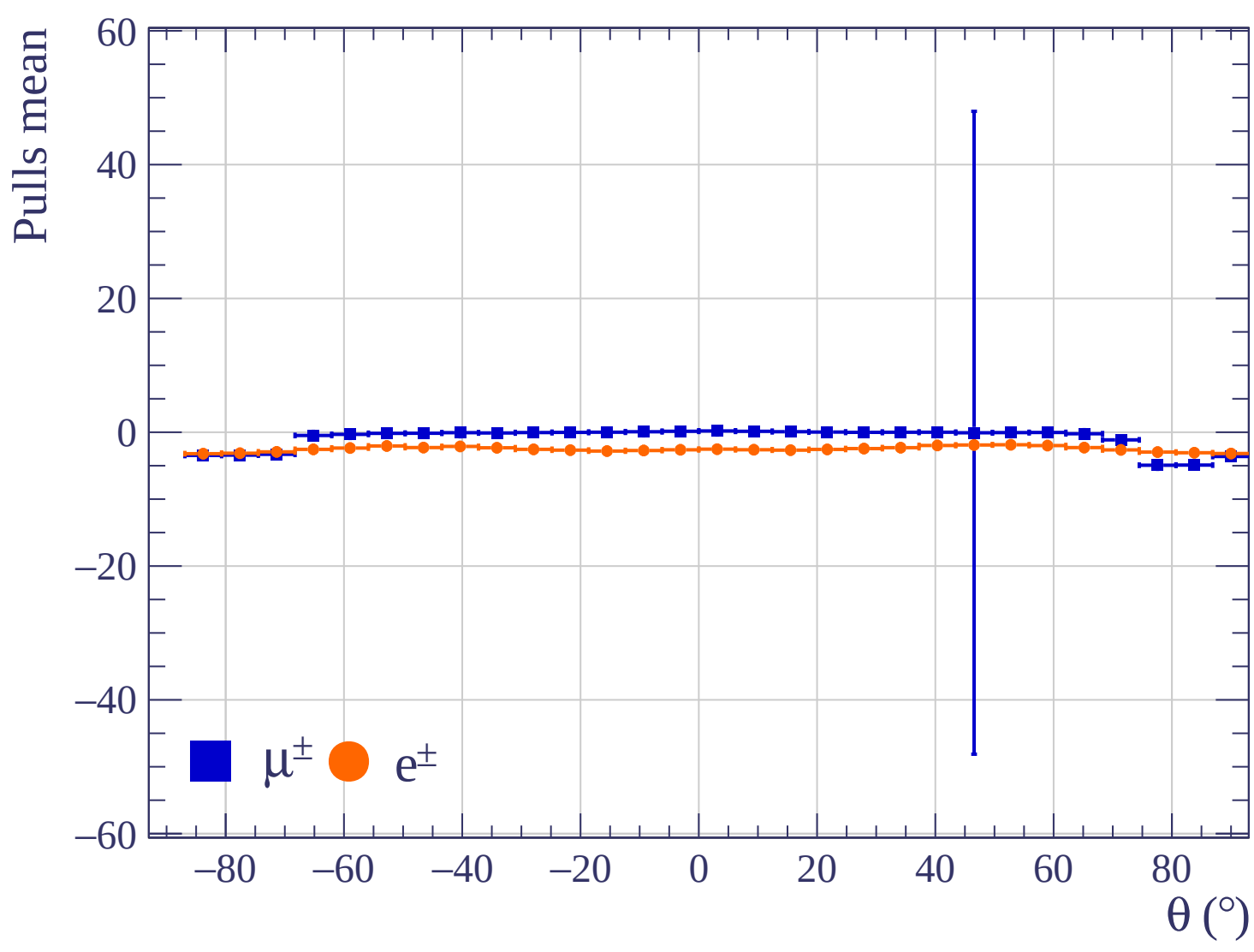




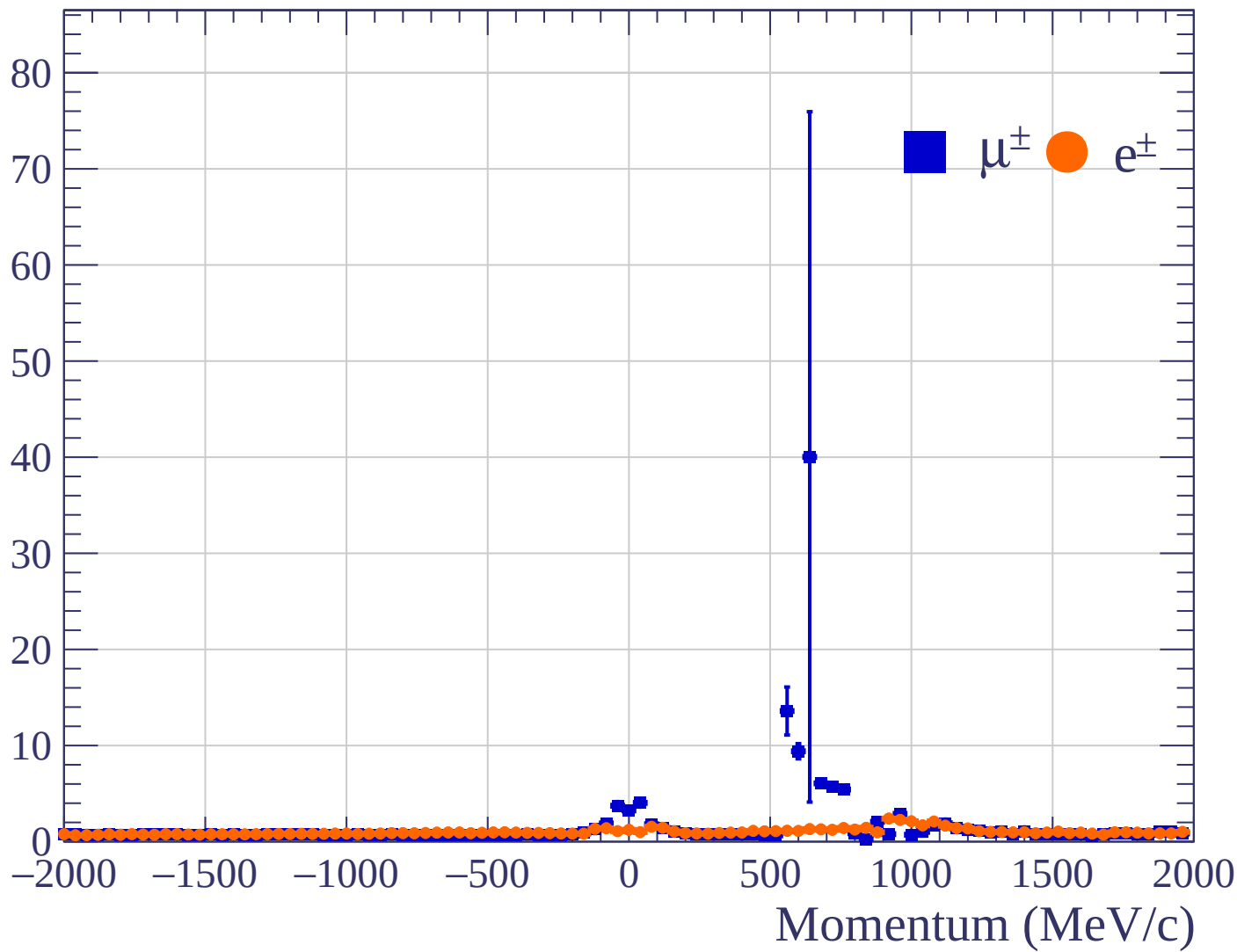


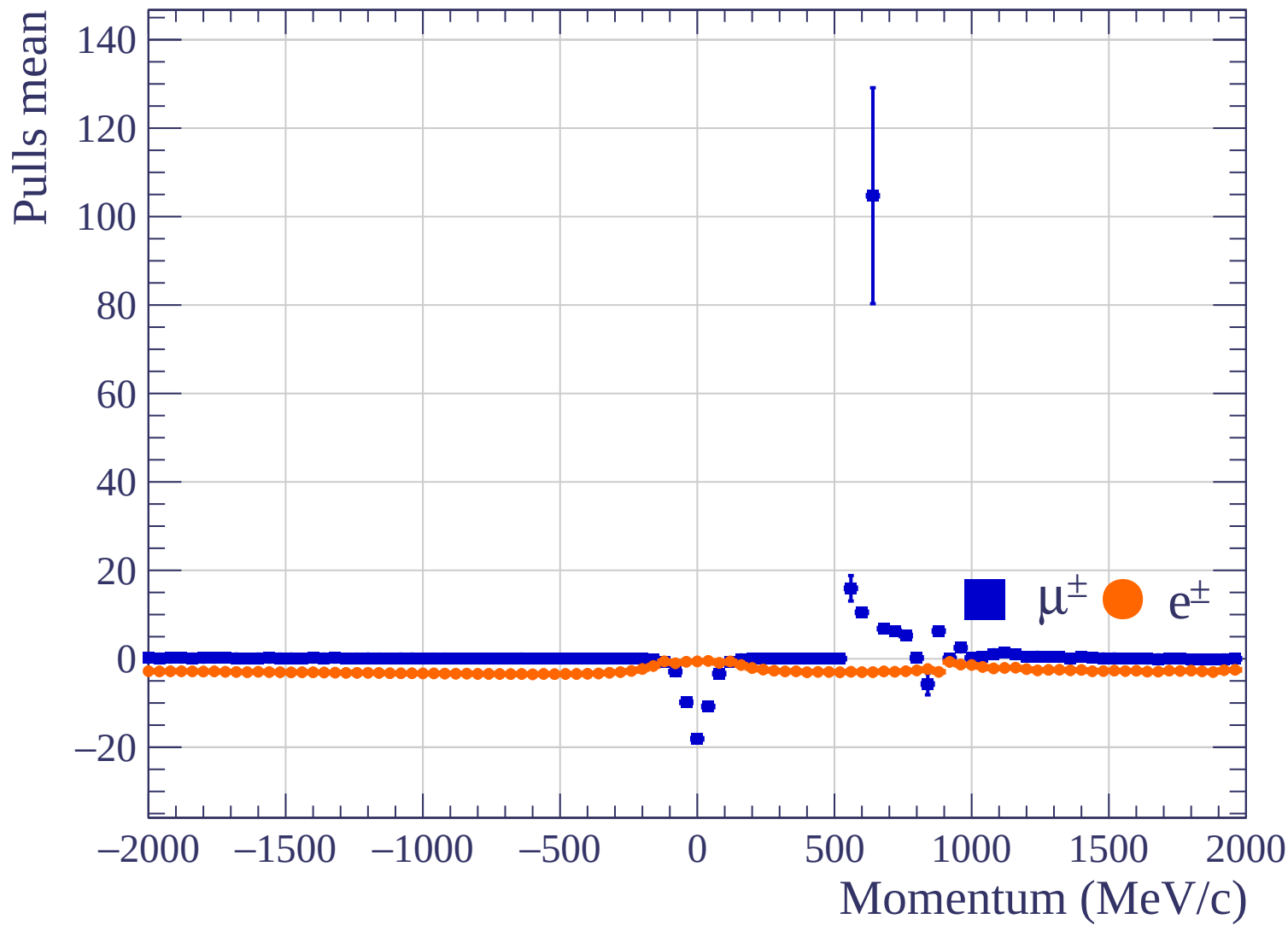
Pulls standard deviation

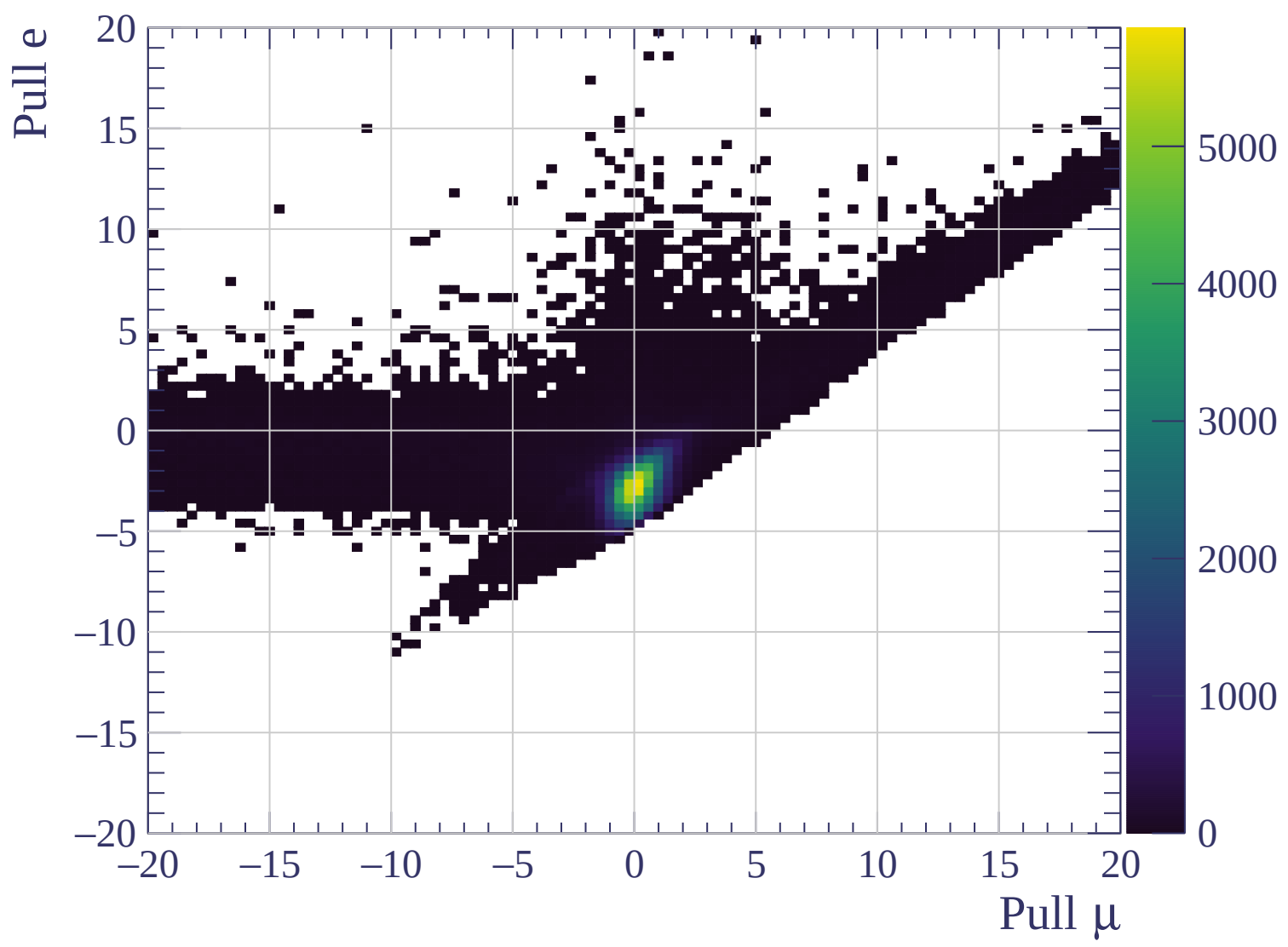


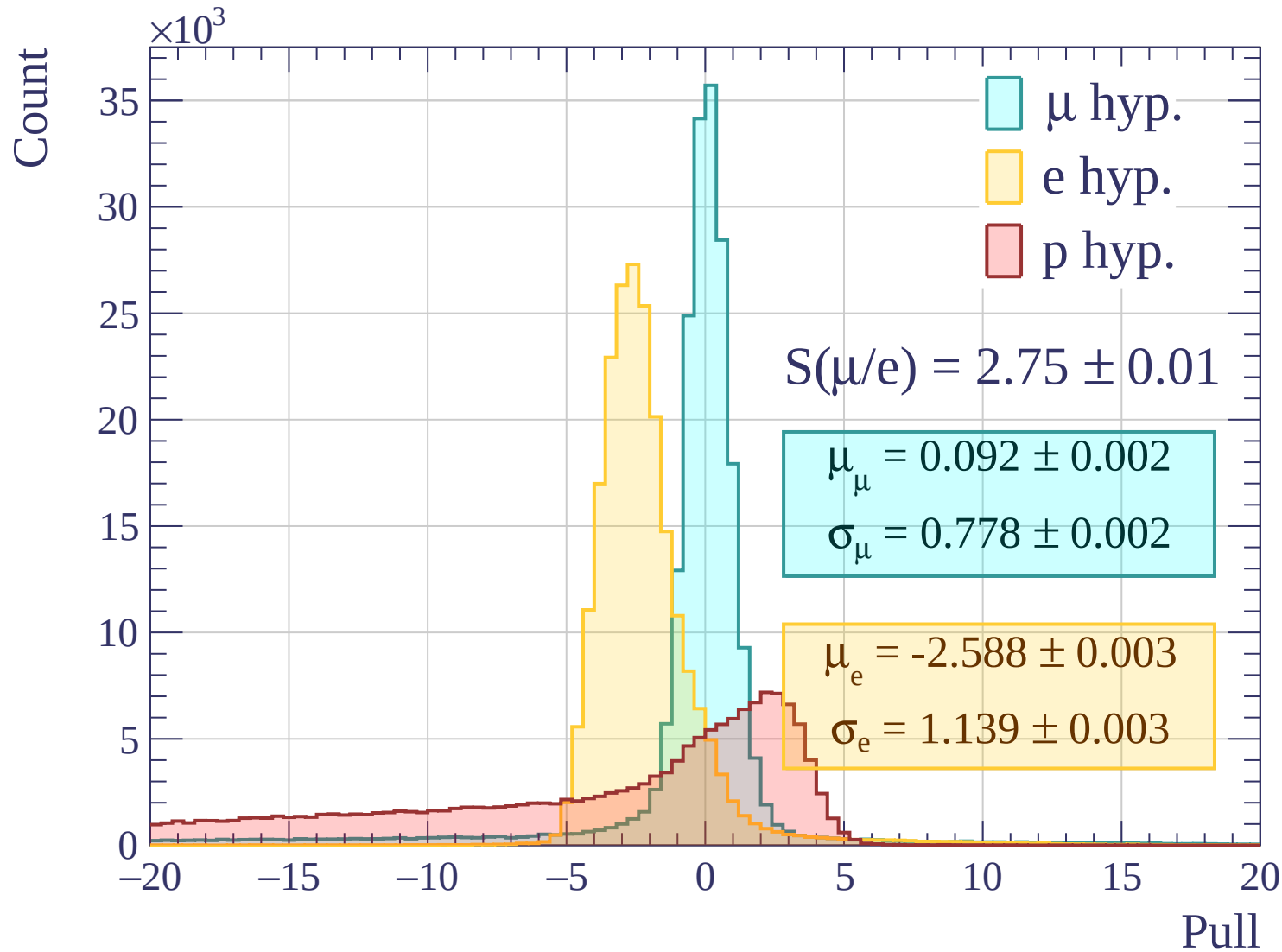


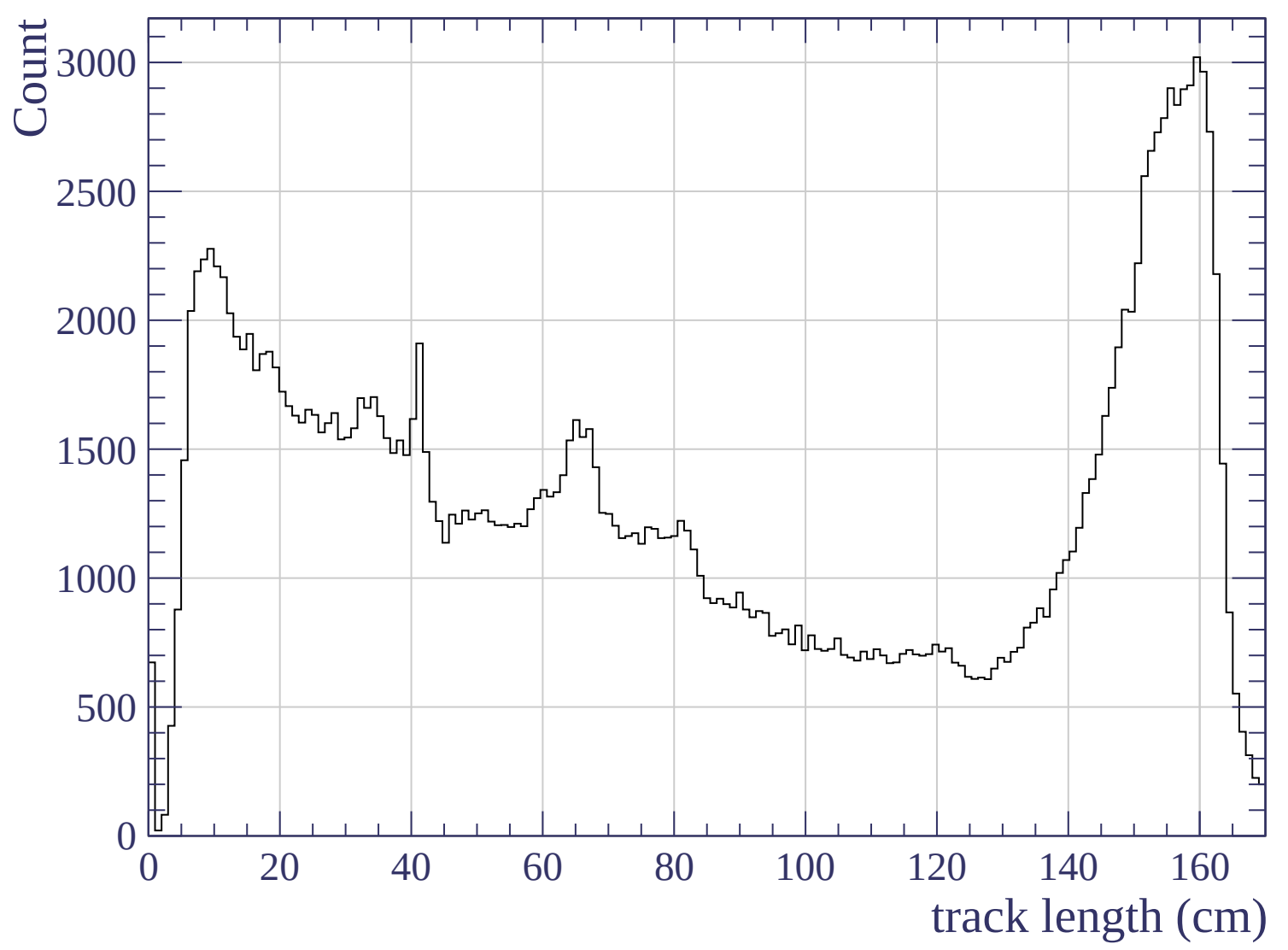
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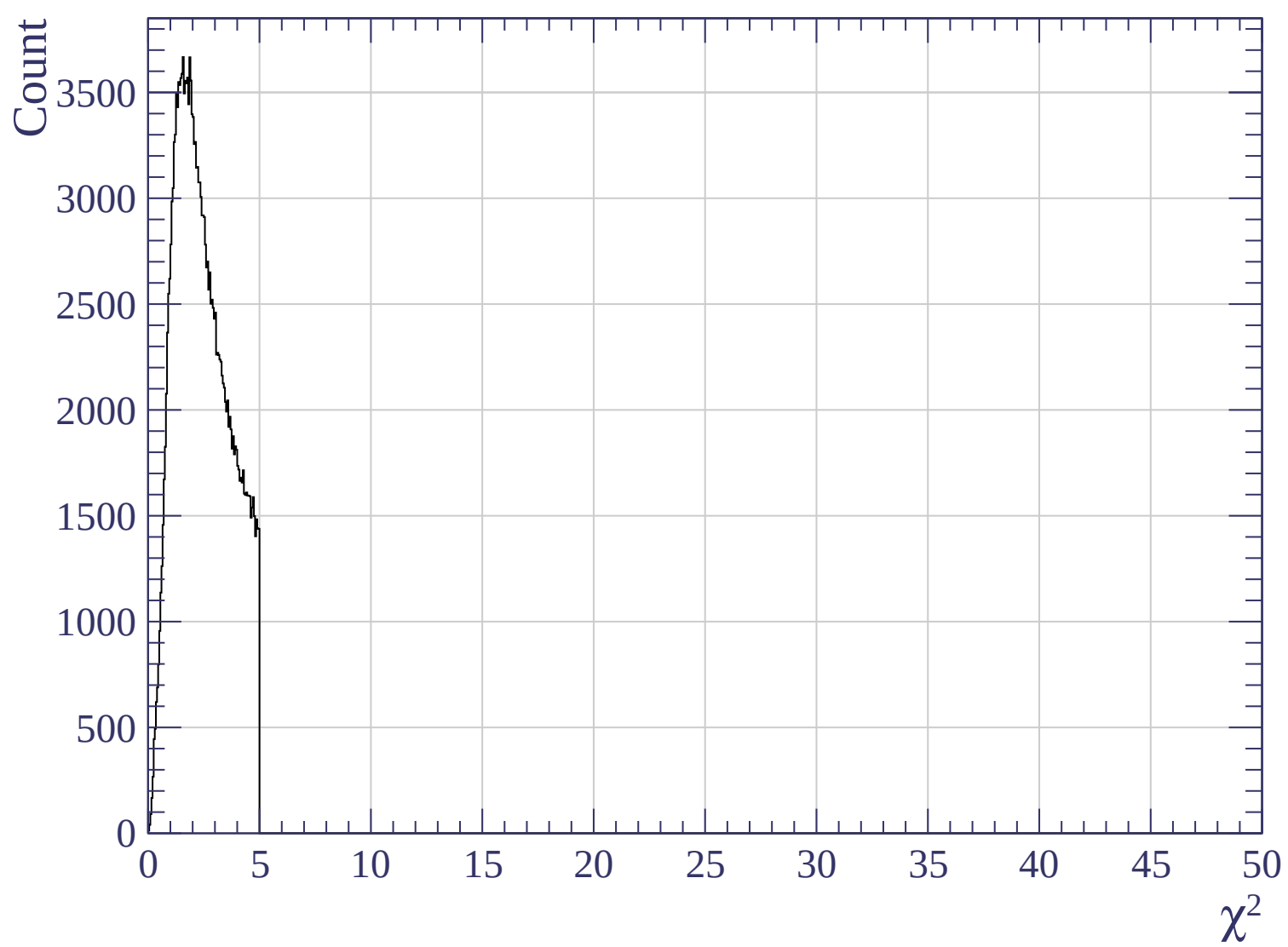


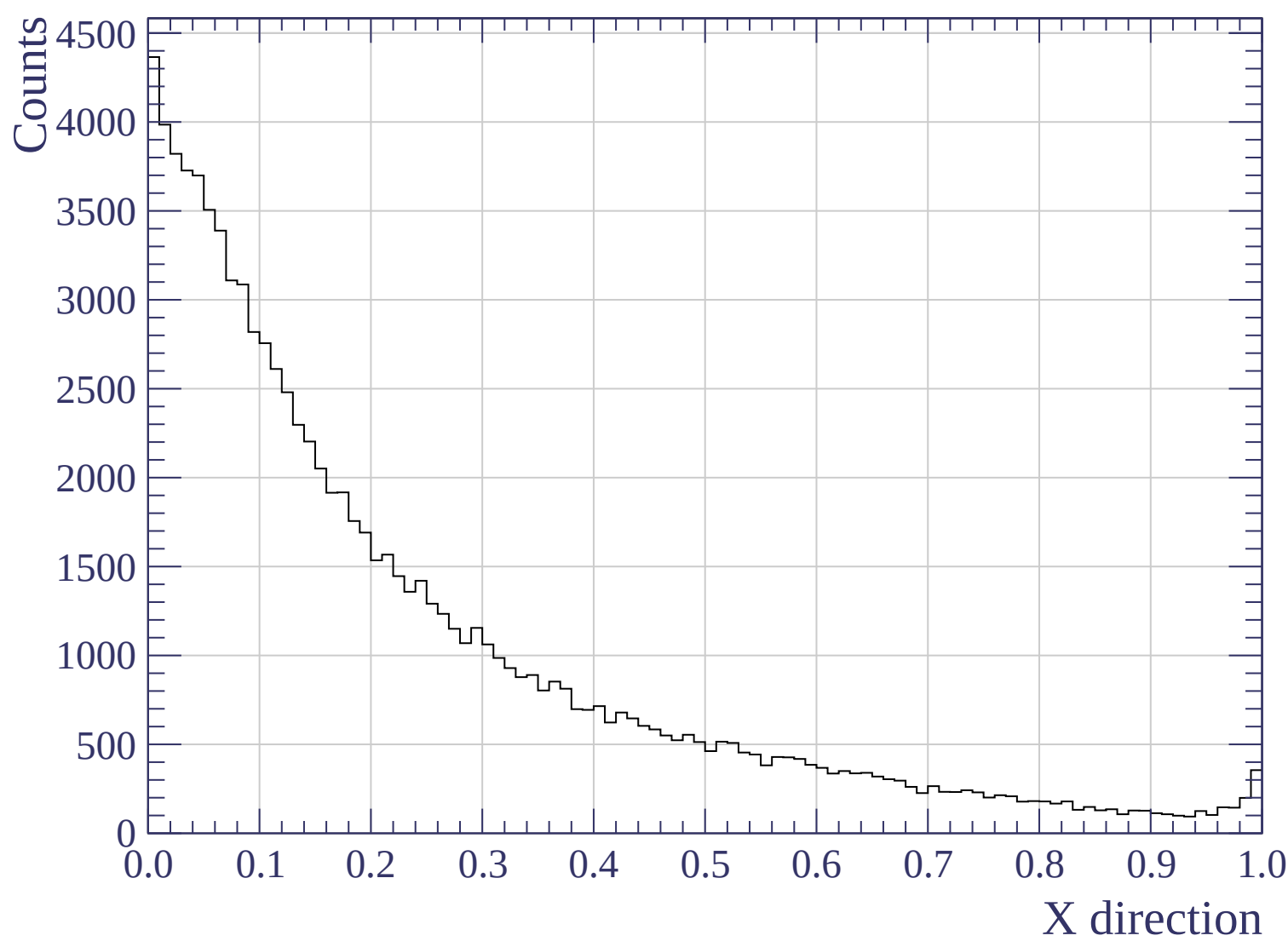


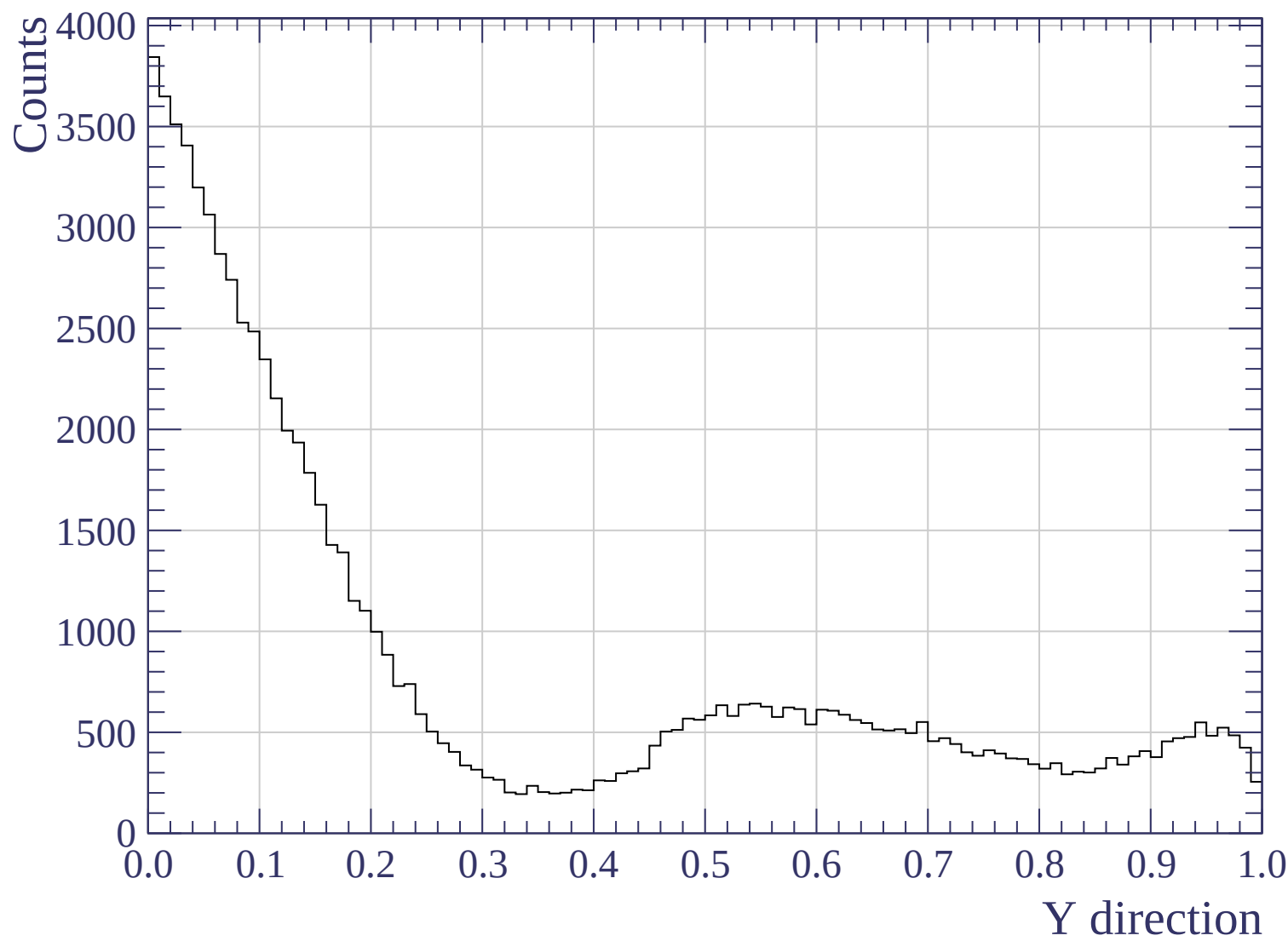


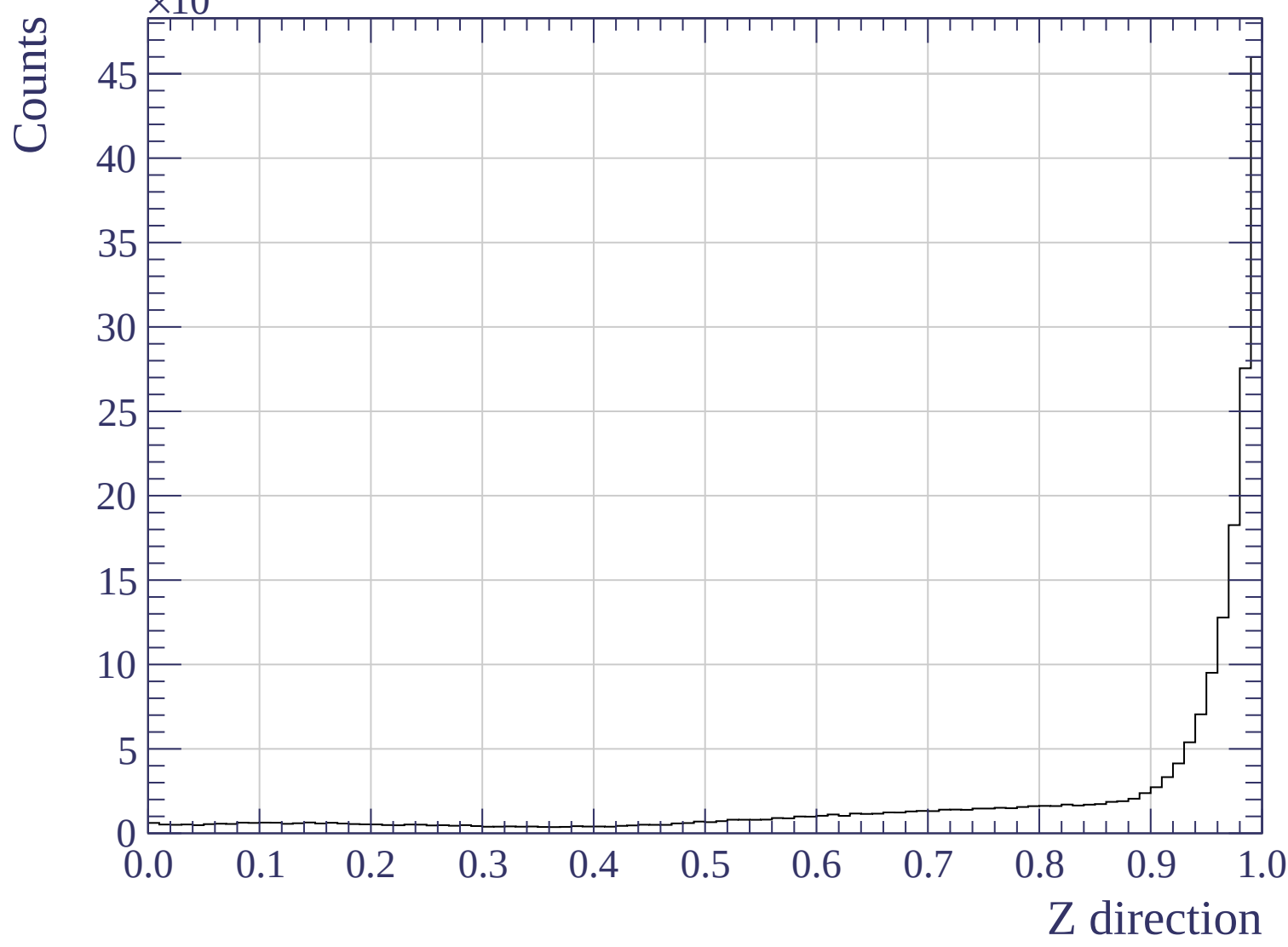


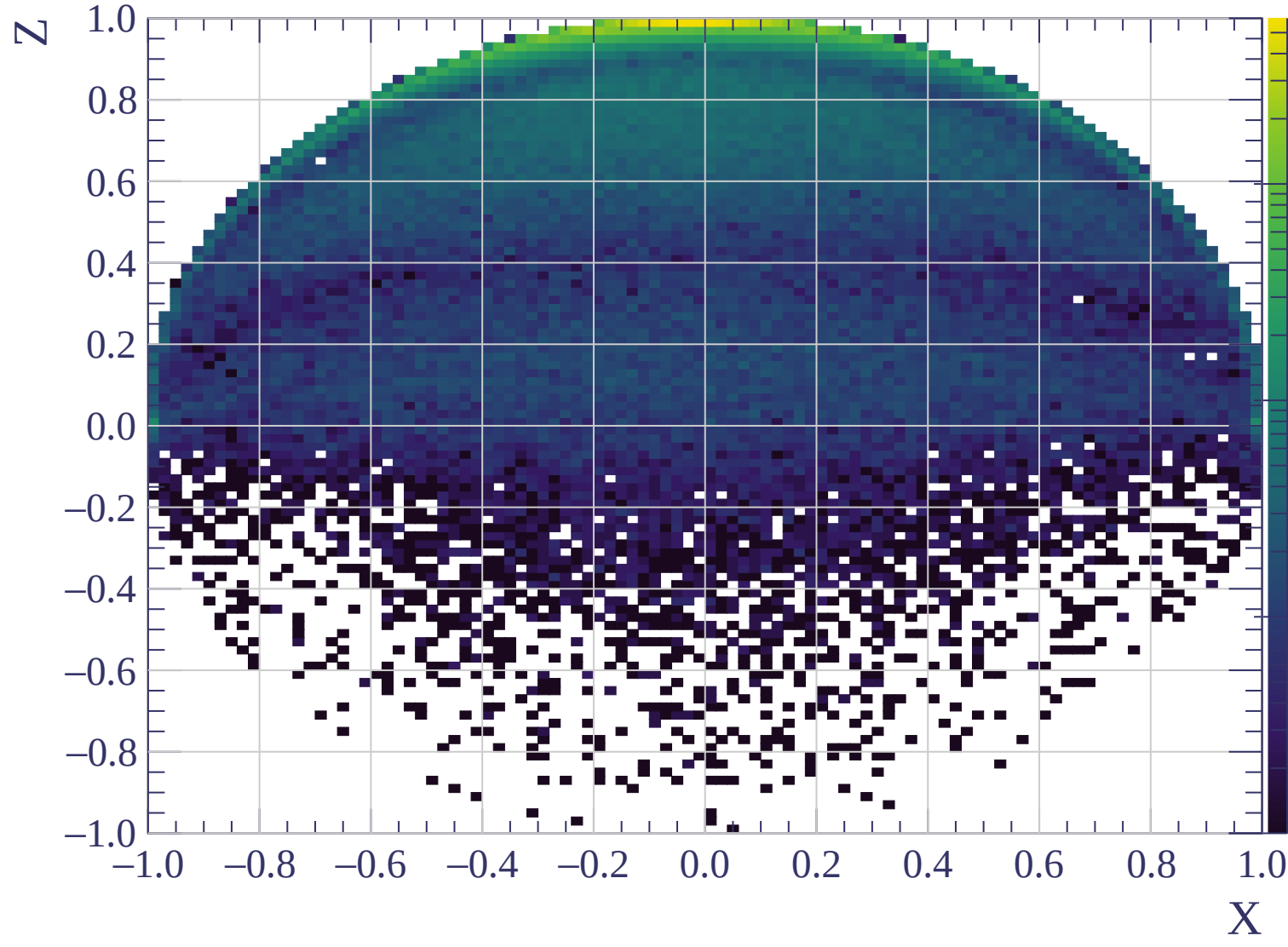


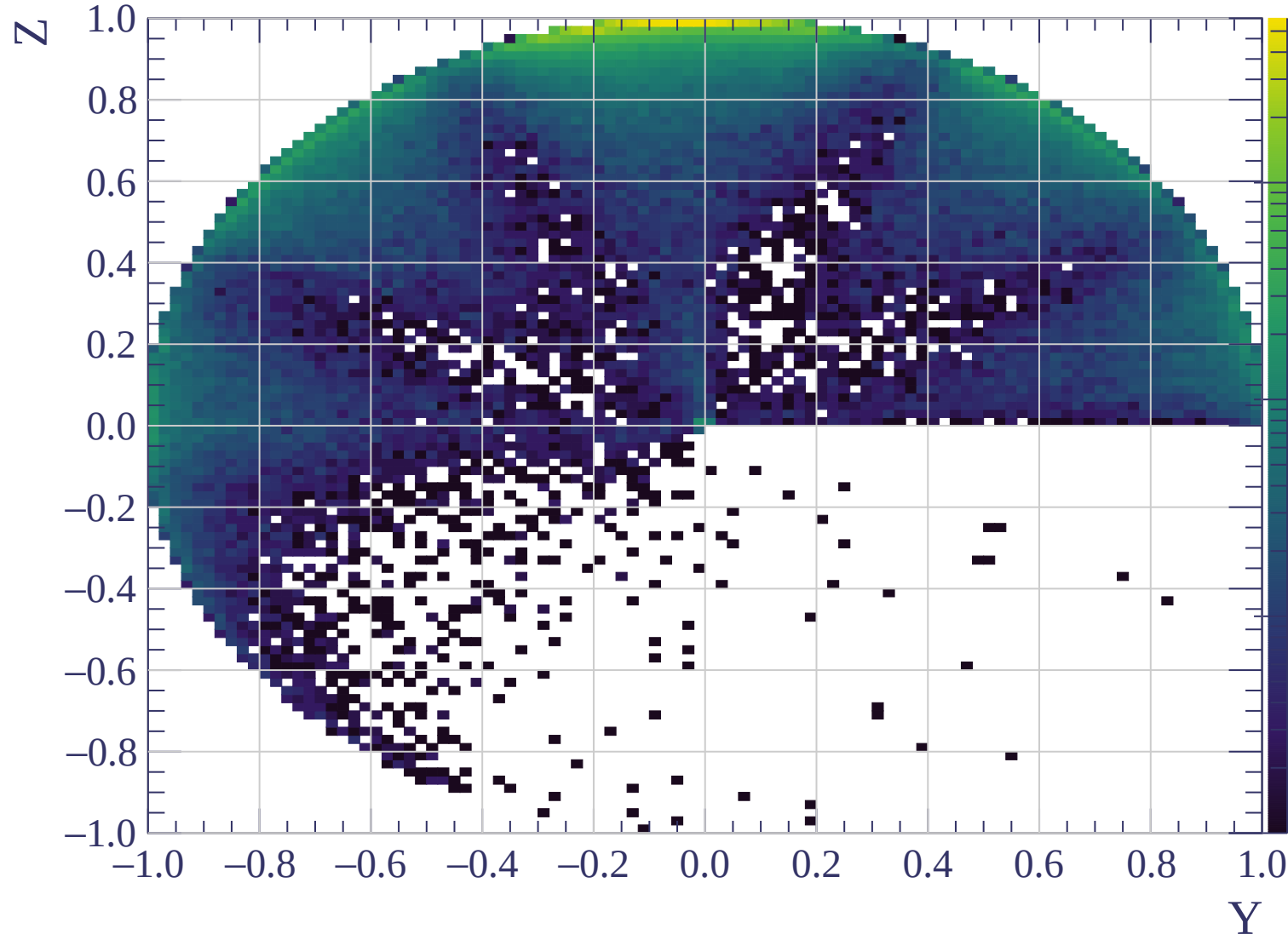


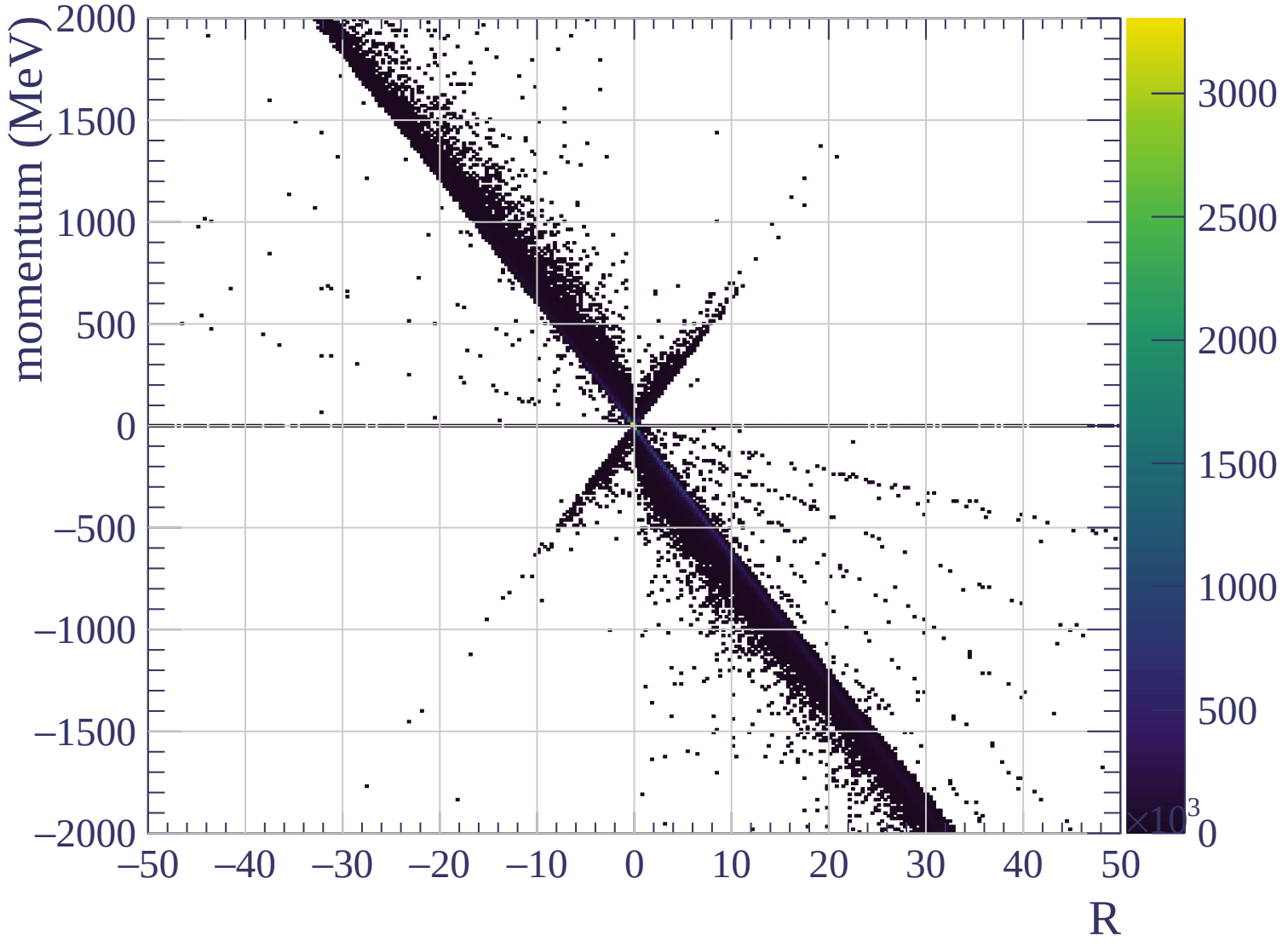


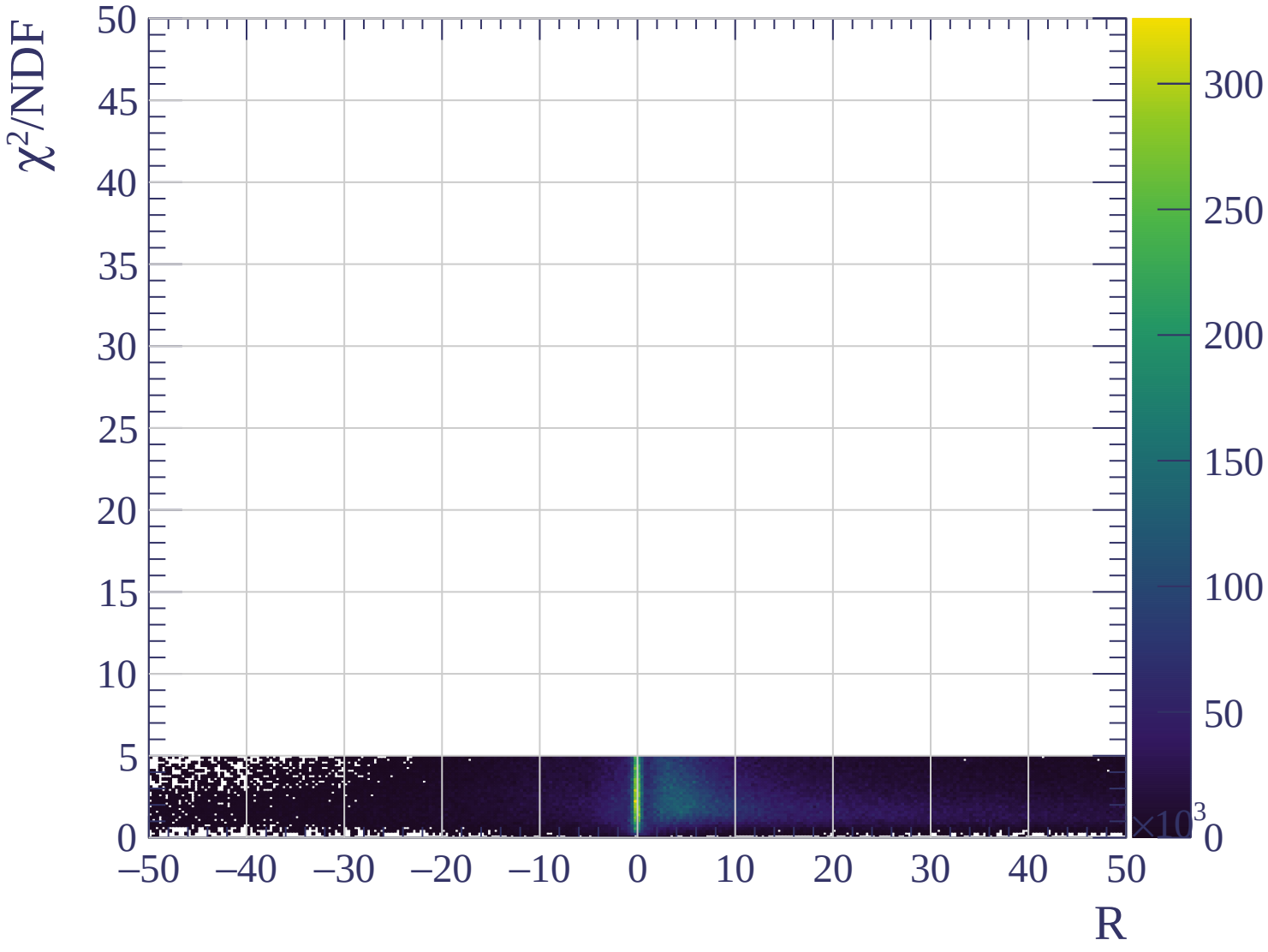






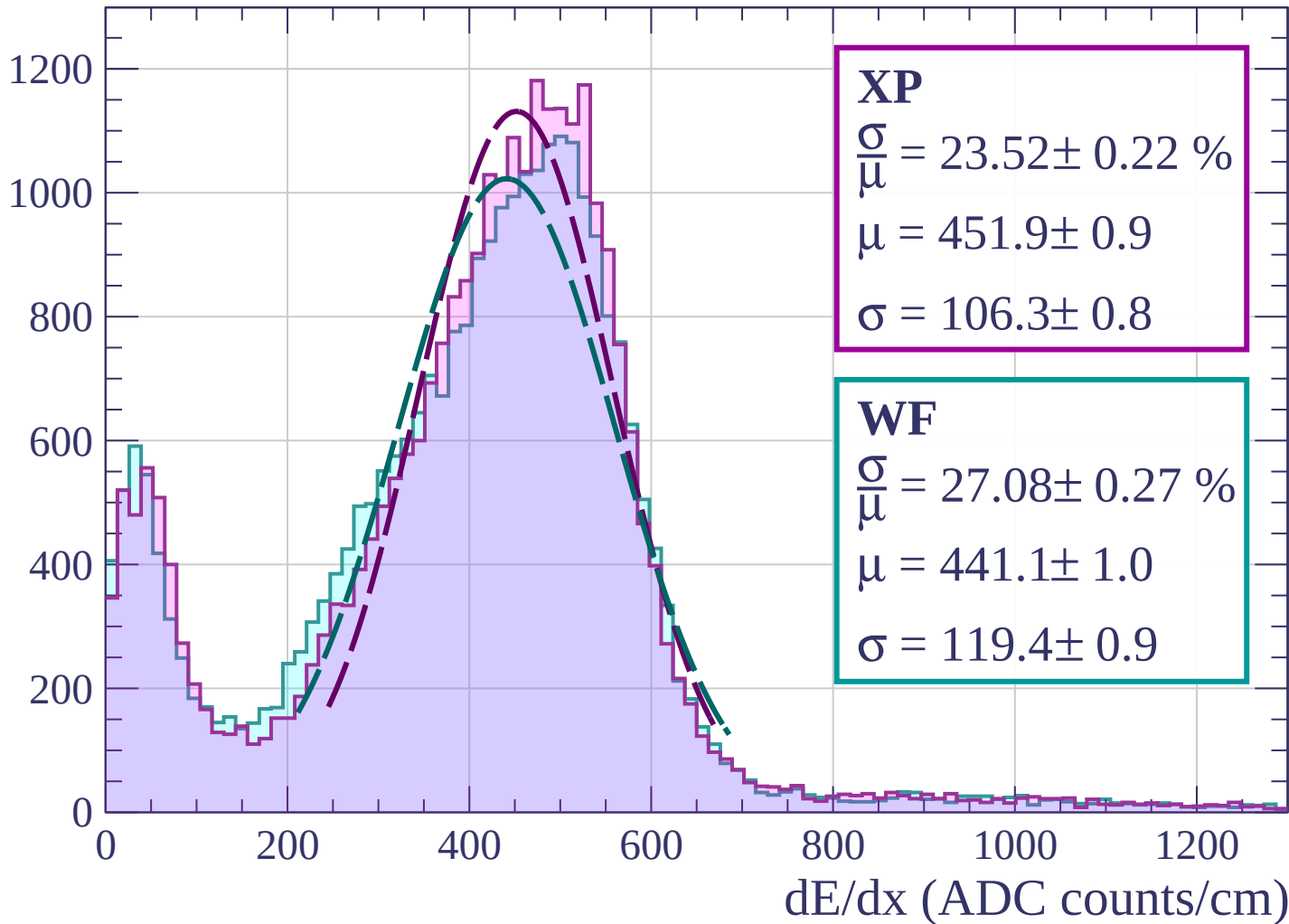




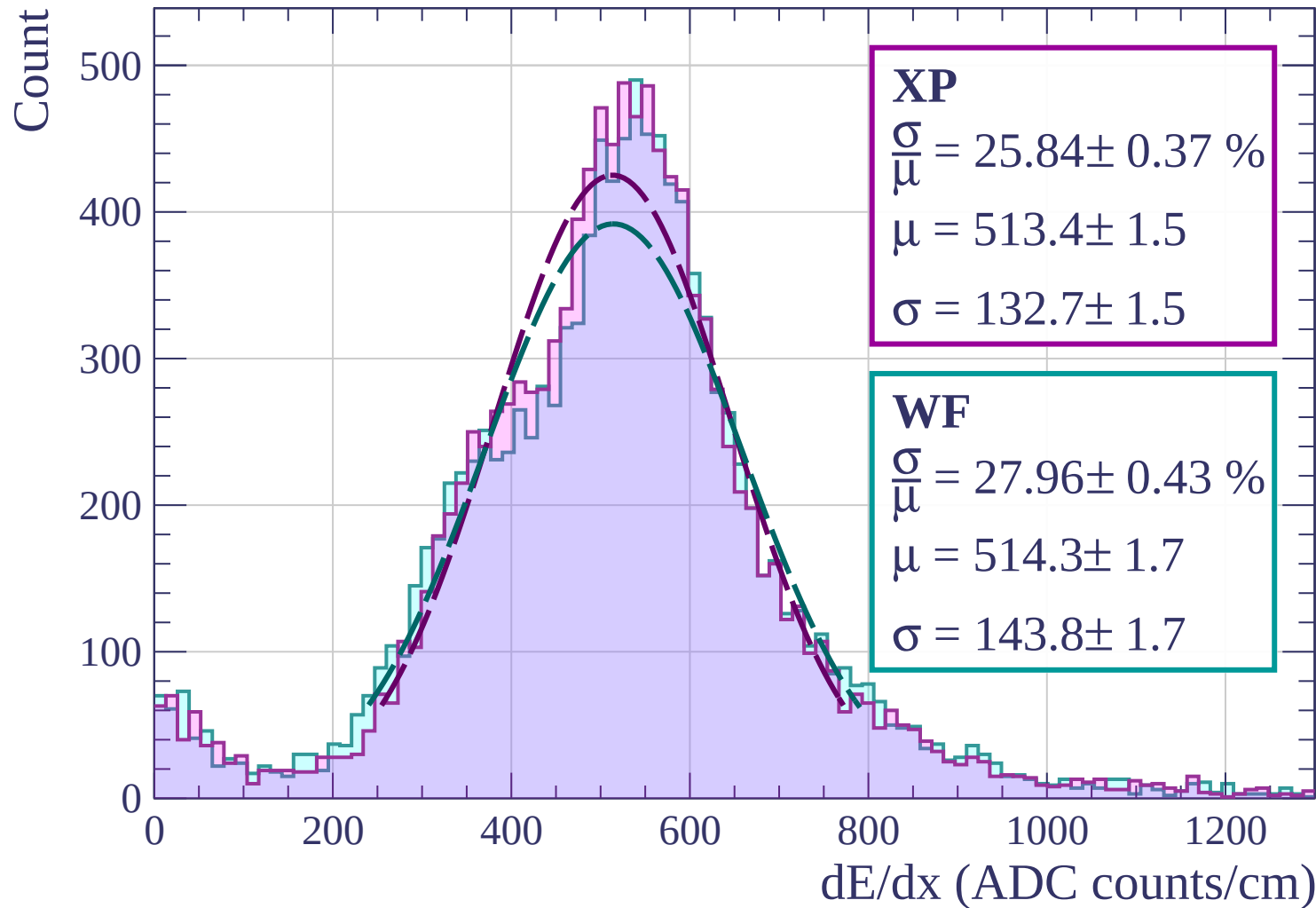


Energy loss | $0 < p < 80$

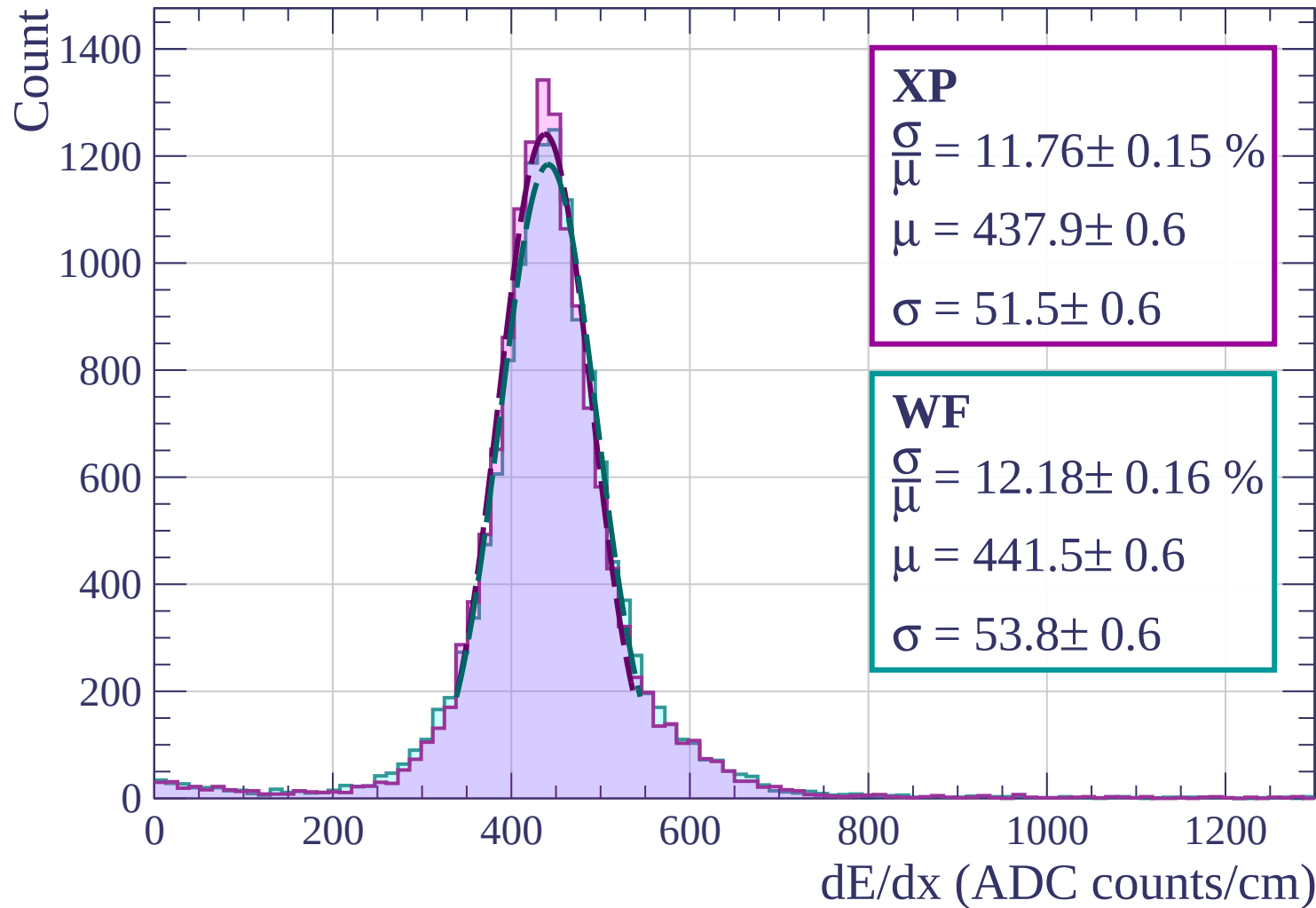
Count



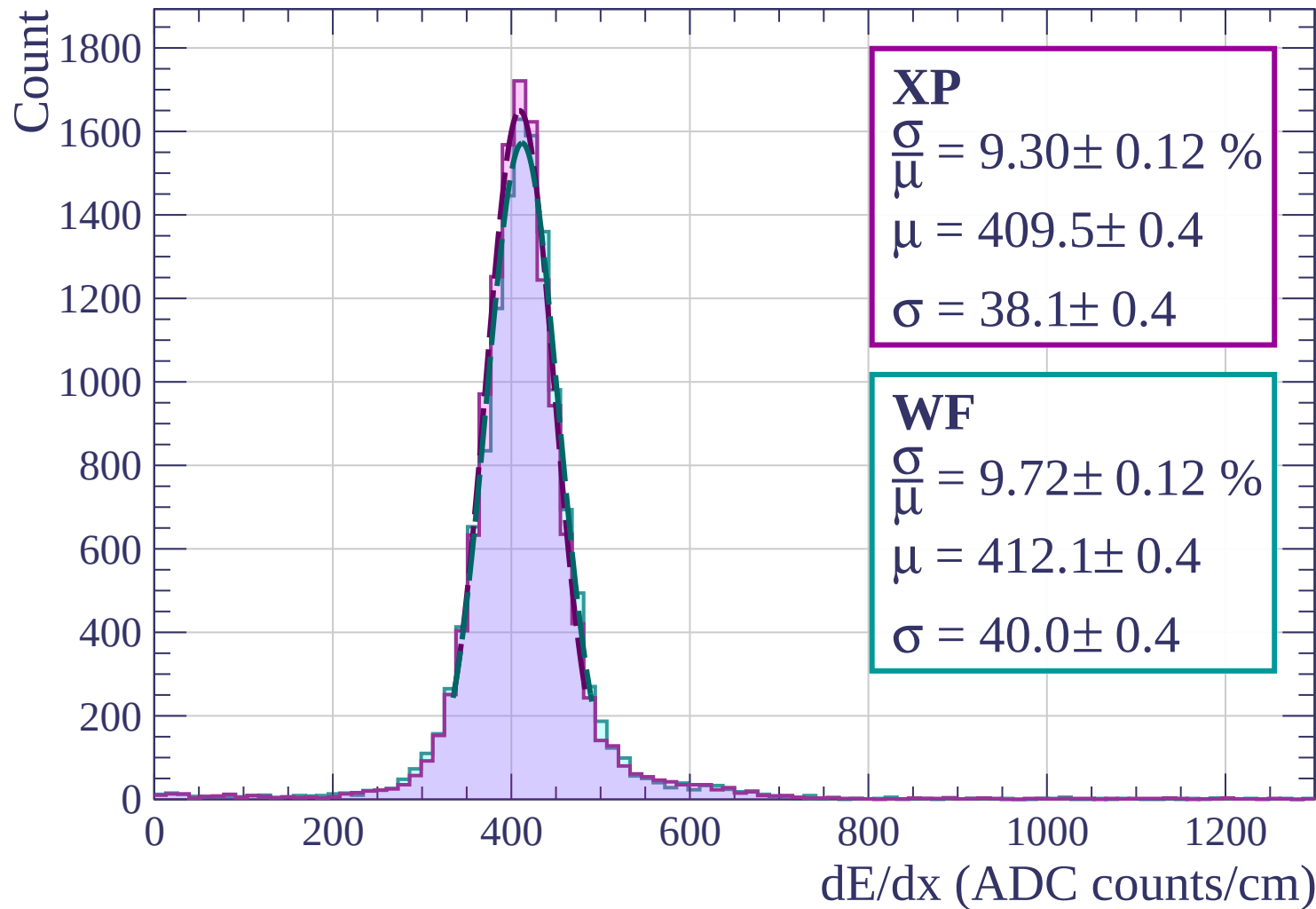
Energy loss | $80 < p < 160$



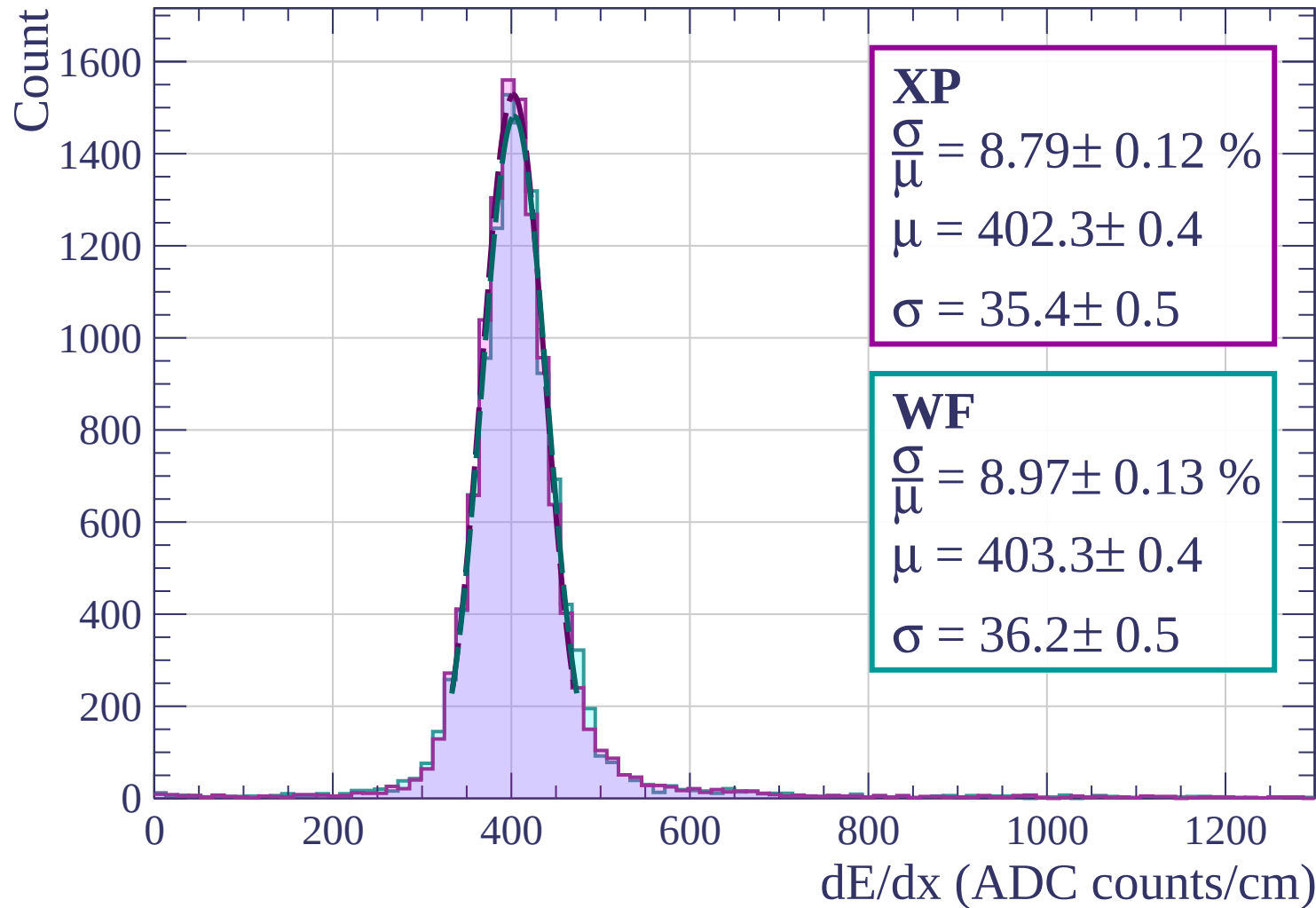
Energy loss | $160 < p < 240$



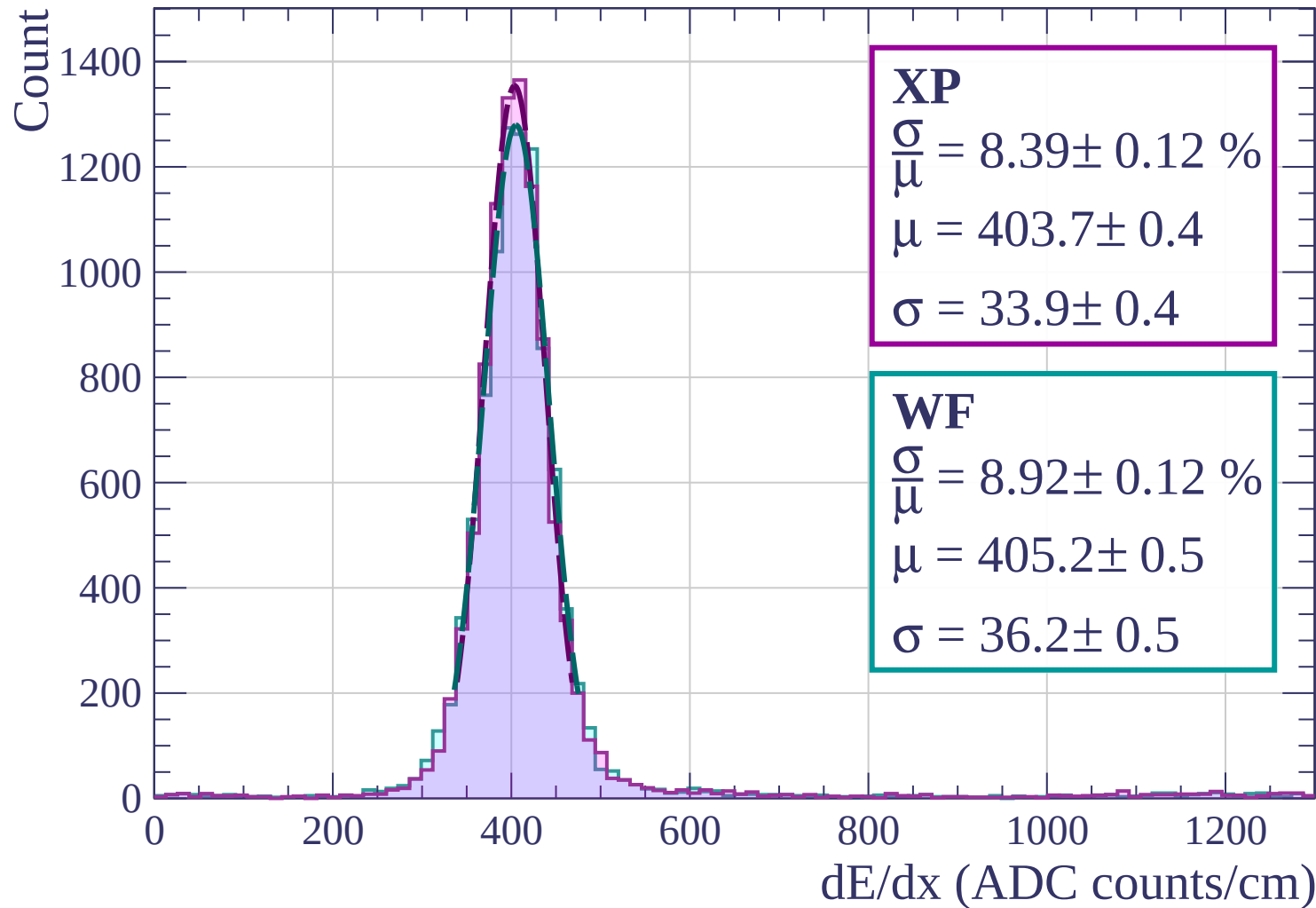
Energy loss | $240 < p < 320$



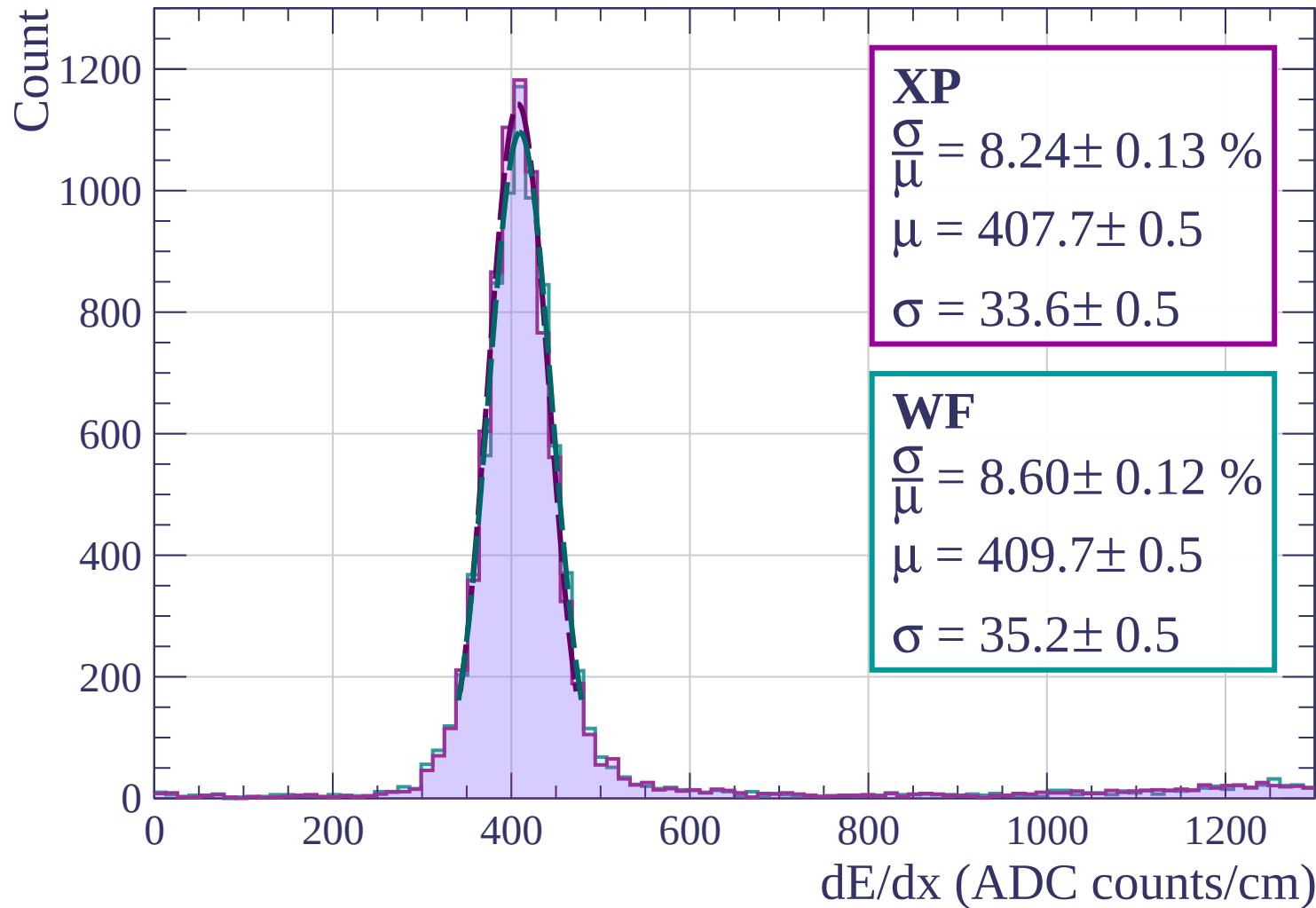
Energy loss | $320 < p < 400$



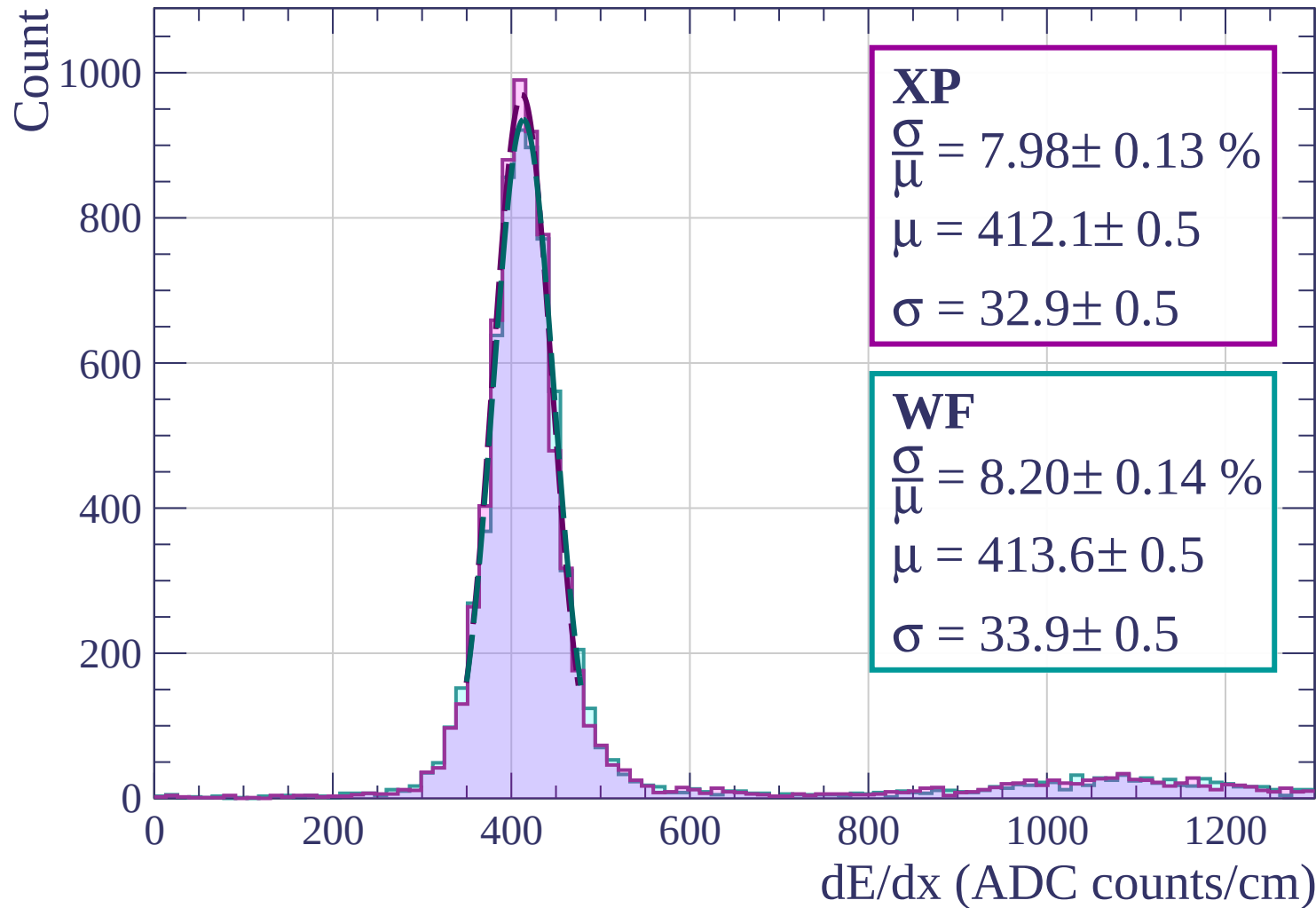
Energy loss | $400 < p < 480$



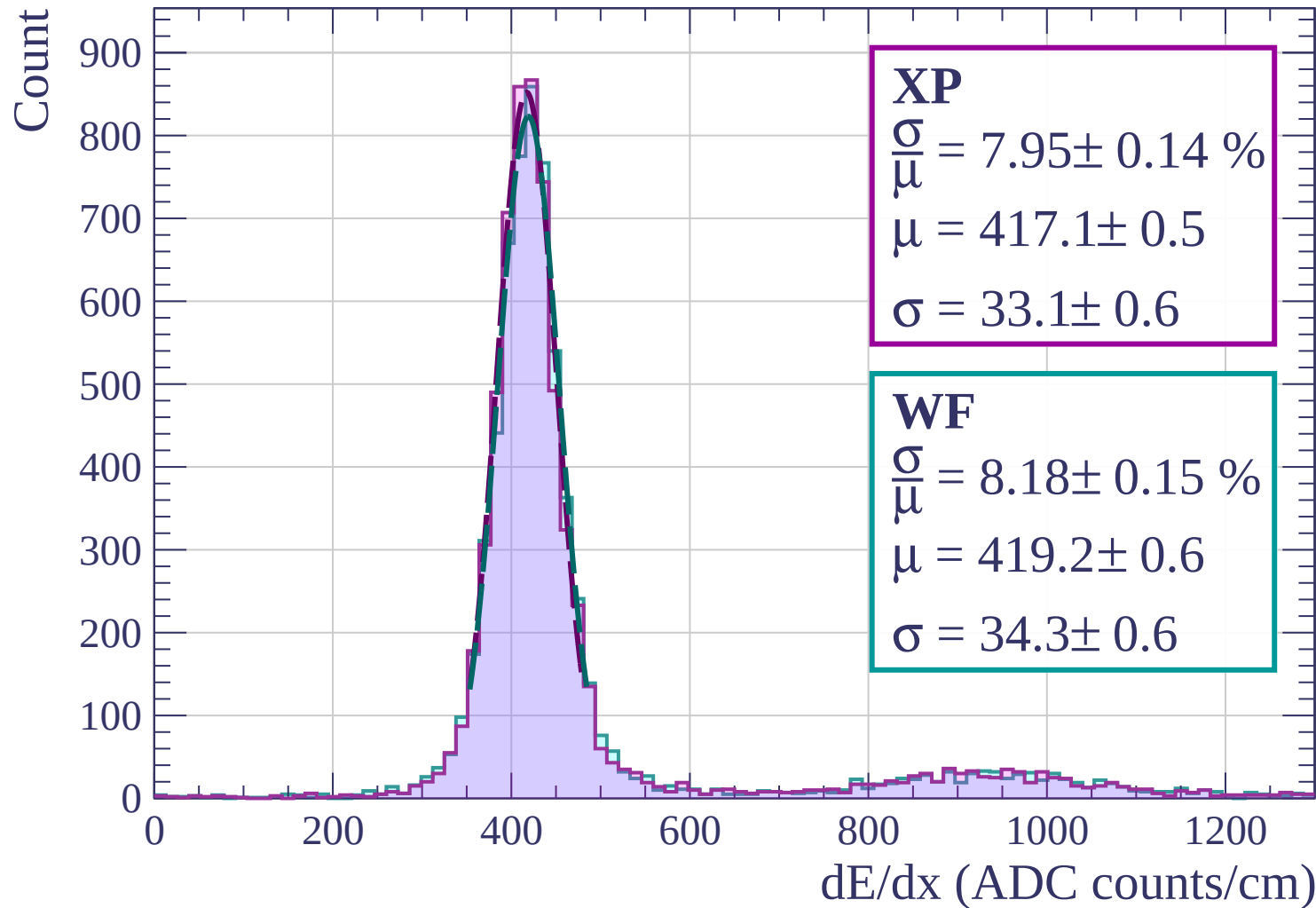
Energy loss | $480 < p < 560$



Energy loss | $560 < p < 640$

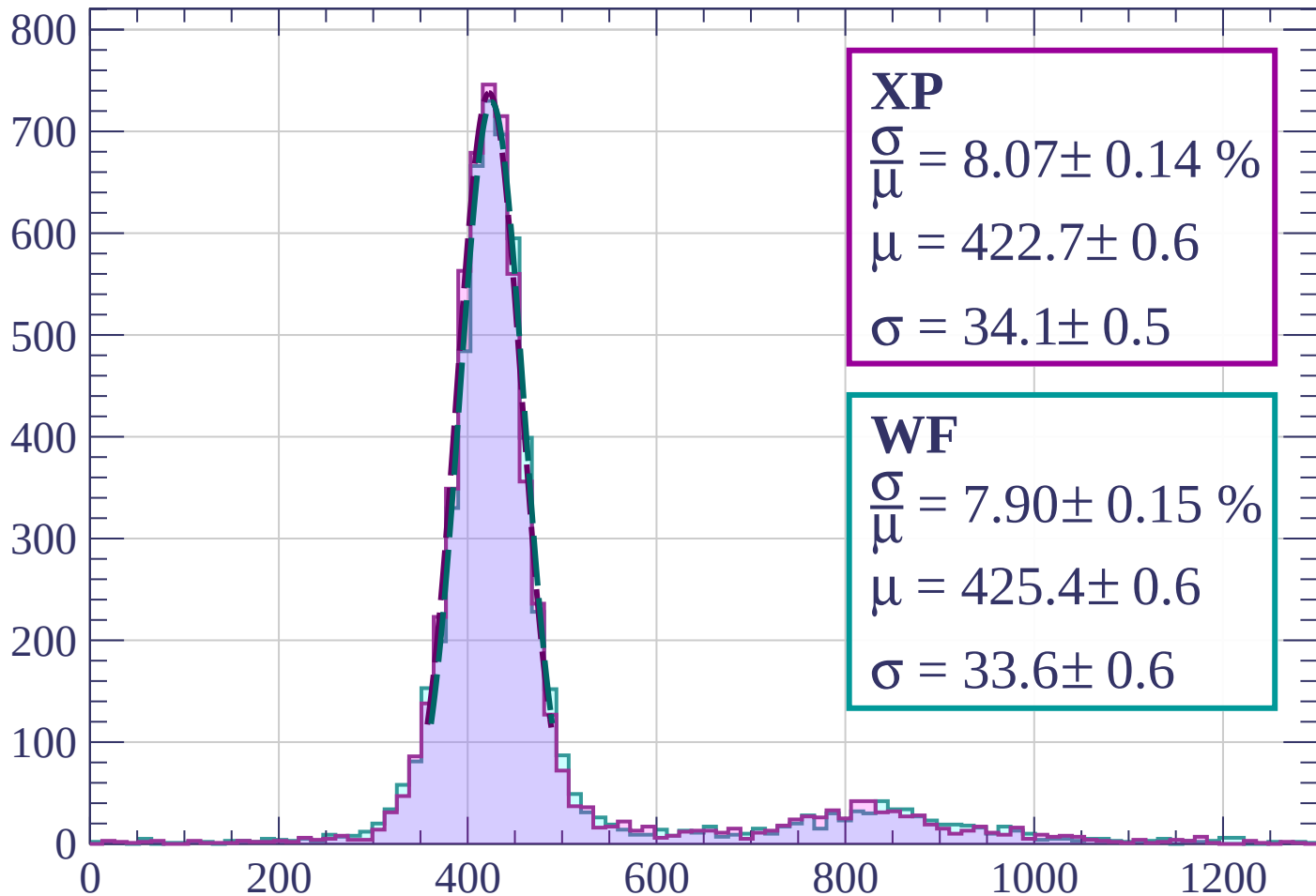


Energy loss | $640 < p < 720$



Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 8.07 \pm 0.14 \%$$

$$\mu = 422.7 \pm 0.6$$

$$\sigma = 34.1 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 7.90 \pm 0.15 \%$$

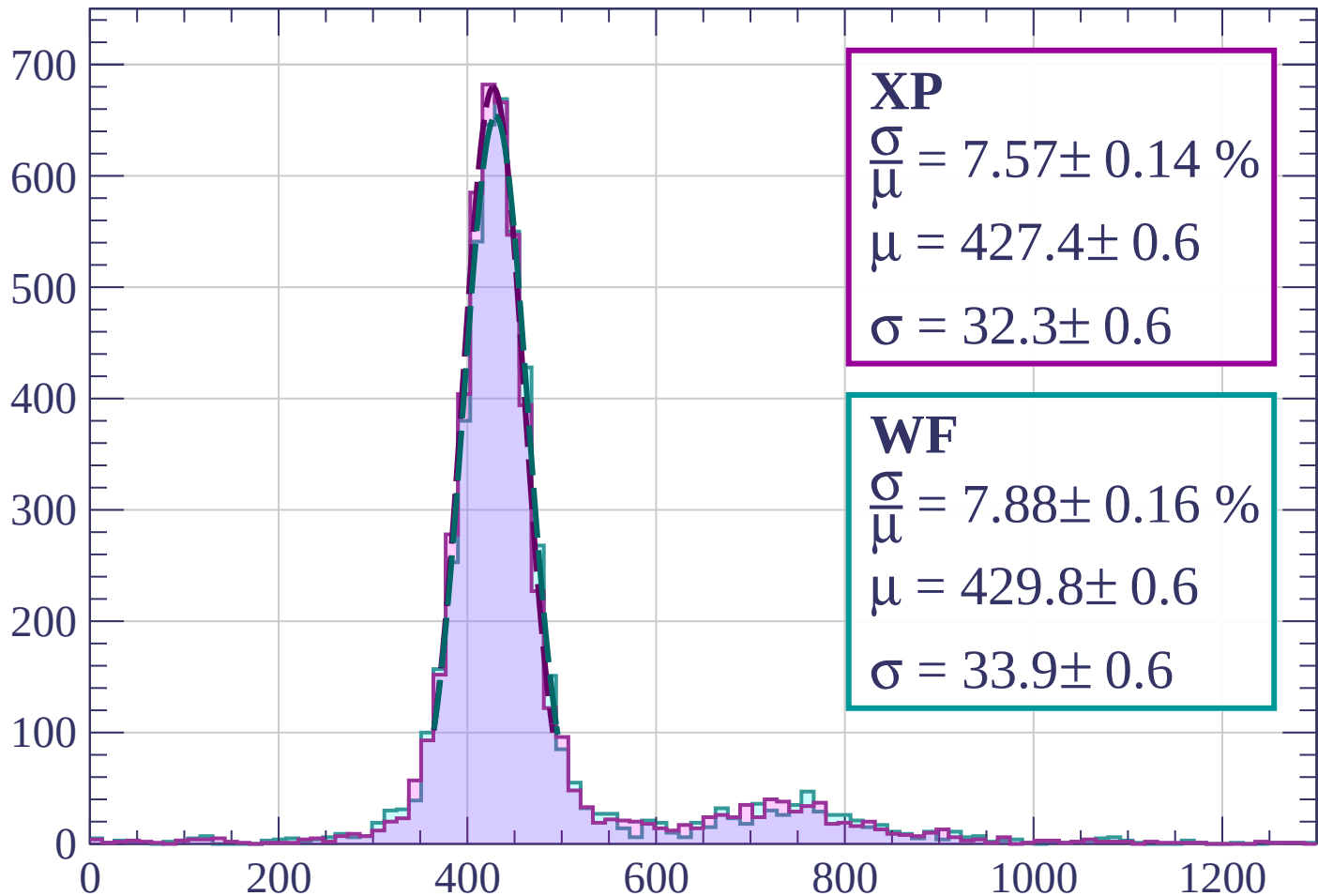
$$\mu = 425.4 \pm 0.6$$

$$\sigma = 33.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 880$

Count



XP

$$\frac{\sigma}{\mu} = 7.57 \pm 0.14 \%$$

$$\mu = 427.4 \pm 0.6$$

$$\sigma = 32.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 7.88 \pm 0.16 \%$$

$$\mu = 429.8 \pm 0.6$$

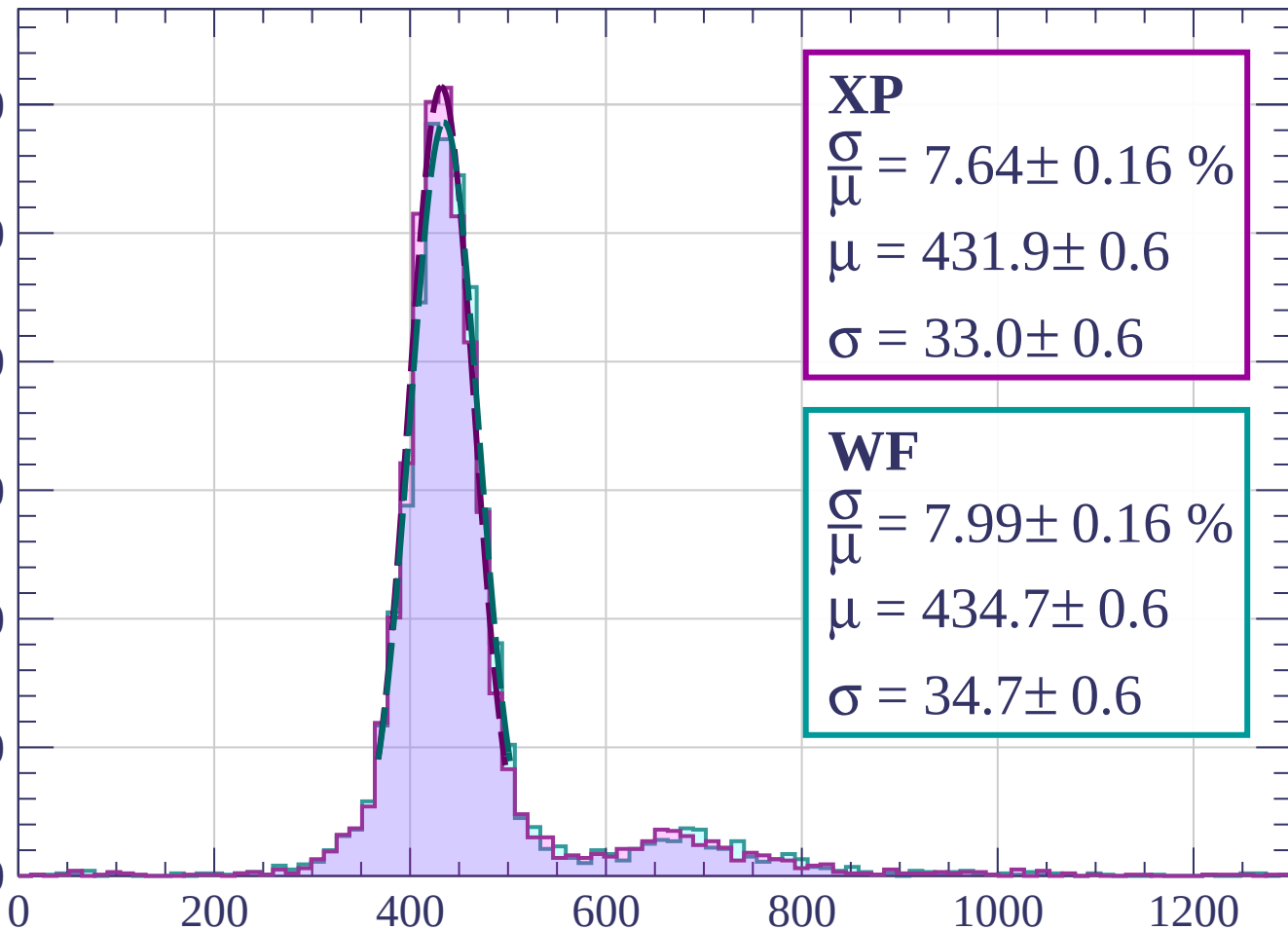
$$\sigma = 33.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $880 < p < 960$

Count

600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 7.64 \pm 0.16 \%$$

$$\mu = 431.9 \pm 0.6$$

$$\sigma = 33.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 7.99 \pm 0.16 \%$$

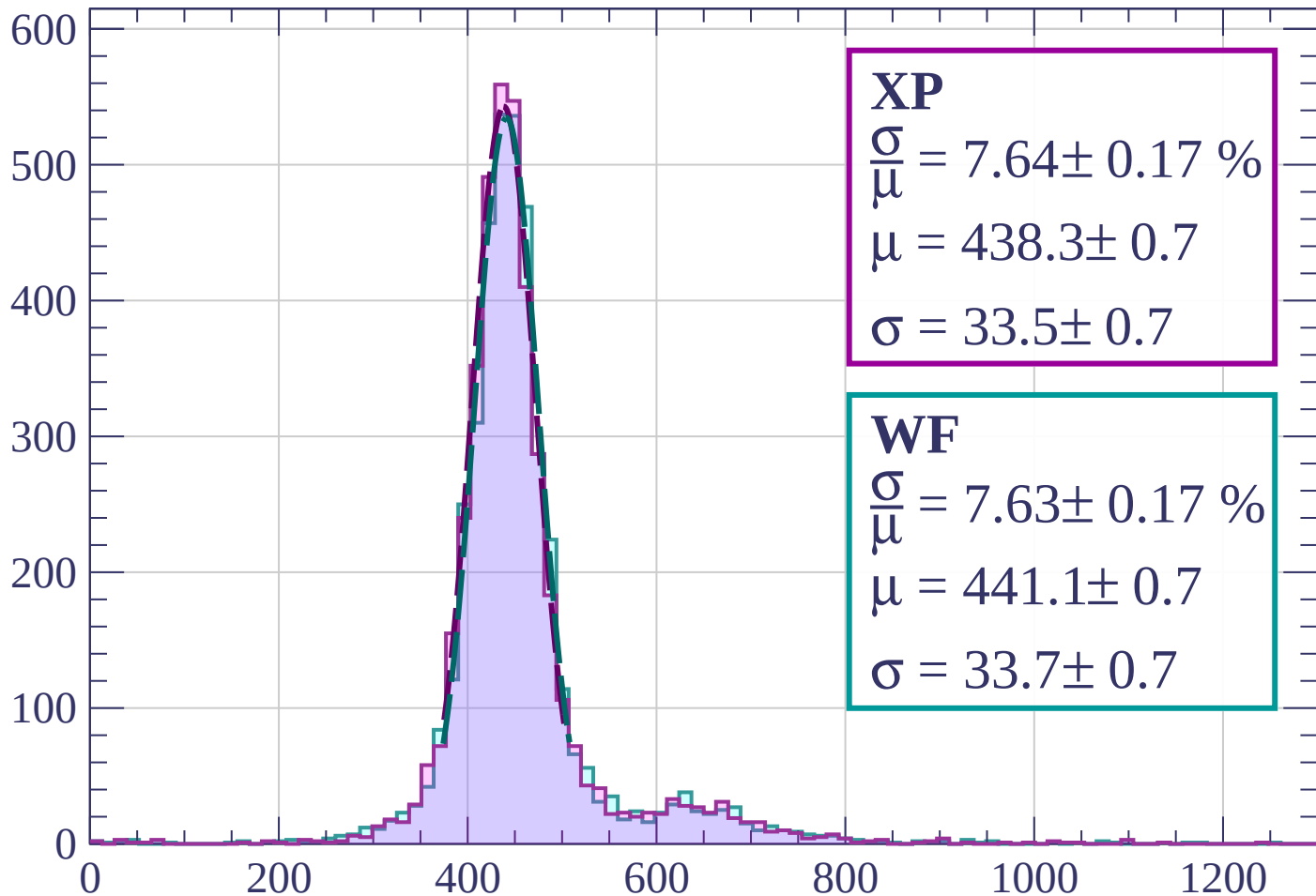
$$\mu = 434.7 \pm 0.6$$

$$\sigma = 34.7 \pm 0.6$$

dE/dx (ADC counts/cm)

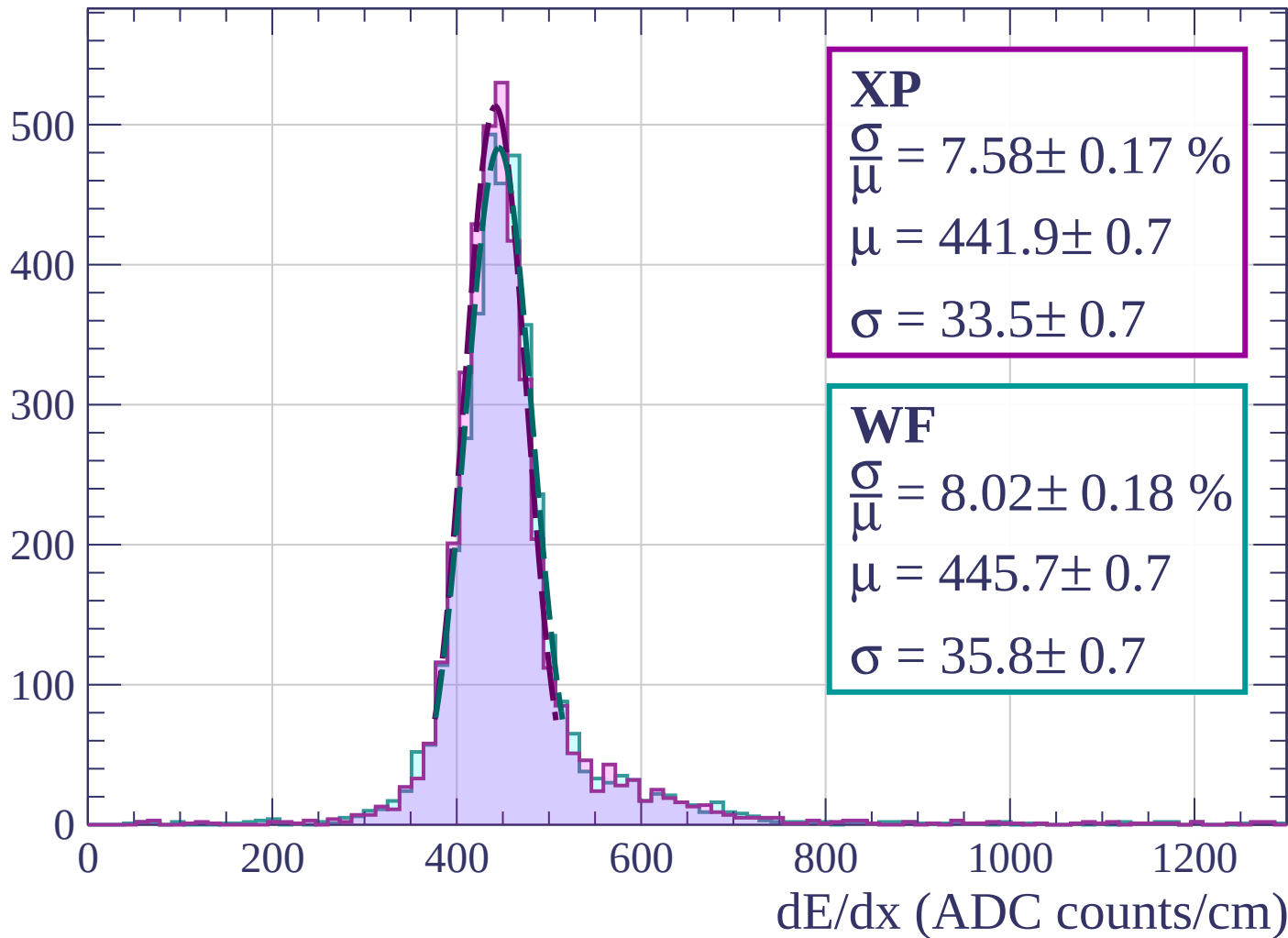
Energy loss | $960 < p < 1040$

Count



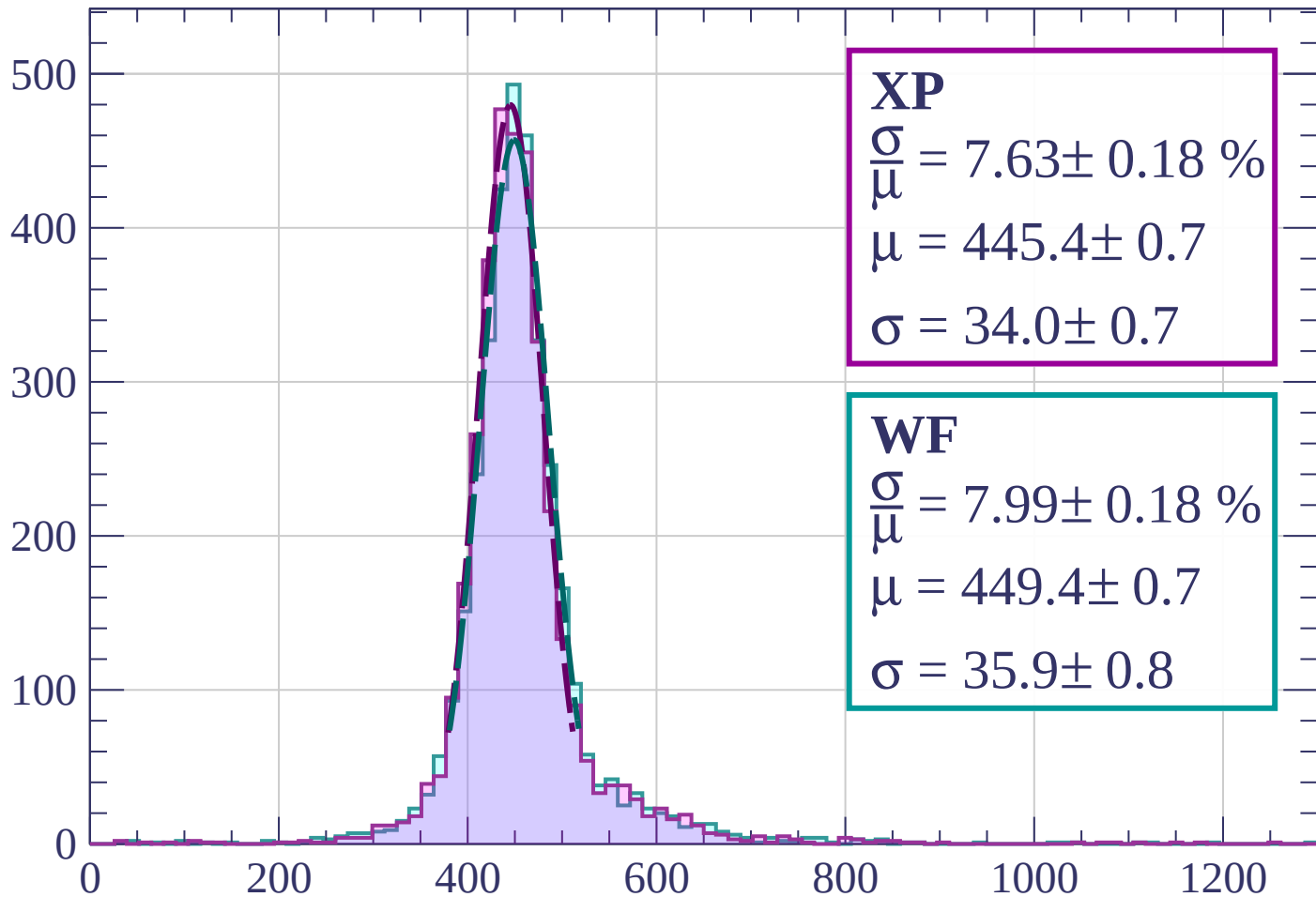
Energy loss | $1040 < p < 1120$

Count



Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 7.63 \pm 0.18 \%$$

$$\mu = 445.4 \pm 0.7$$

$$\sigma = 34.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.99 \pm 0.18 \%$$

$$\mu = 449.4 \pm 0.7$$

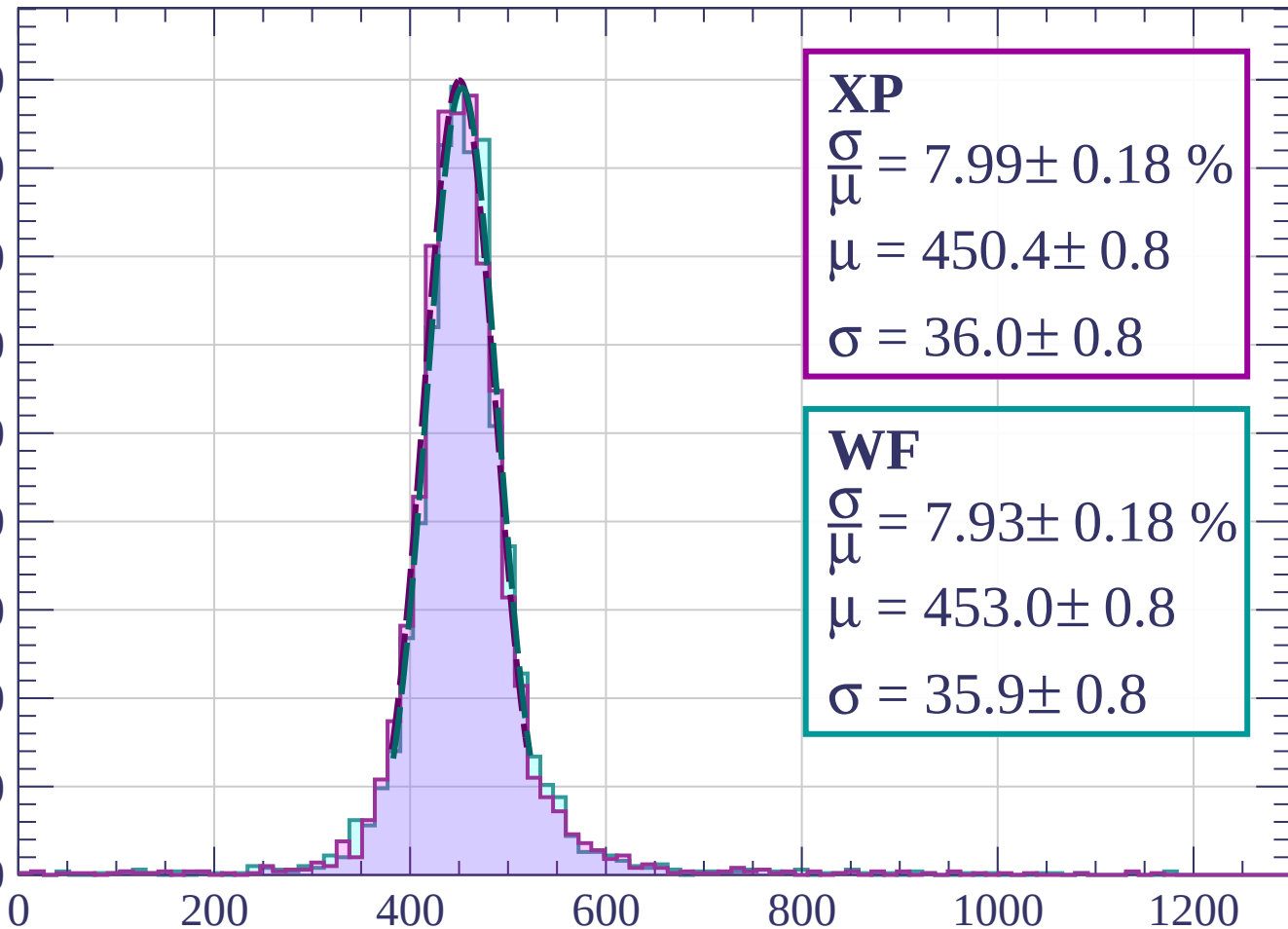
$$\sigma = 35.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1280$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 7.99 \pm 0.18 \%$$

$$\mu = 450.4 \pm 0.8$$

$$\sigma = 36.0 \pm 0.8$$

WF

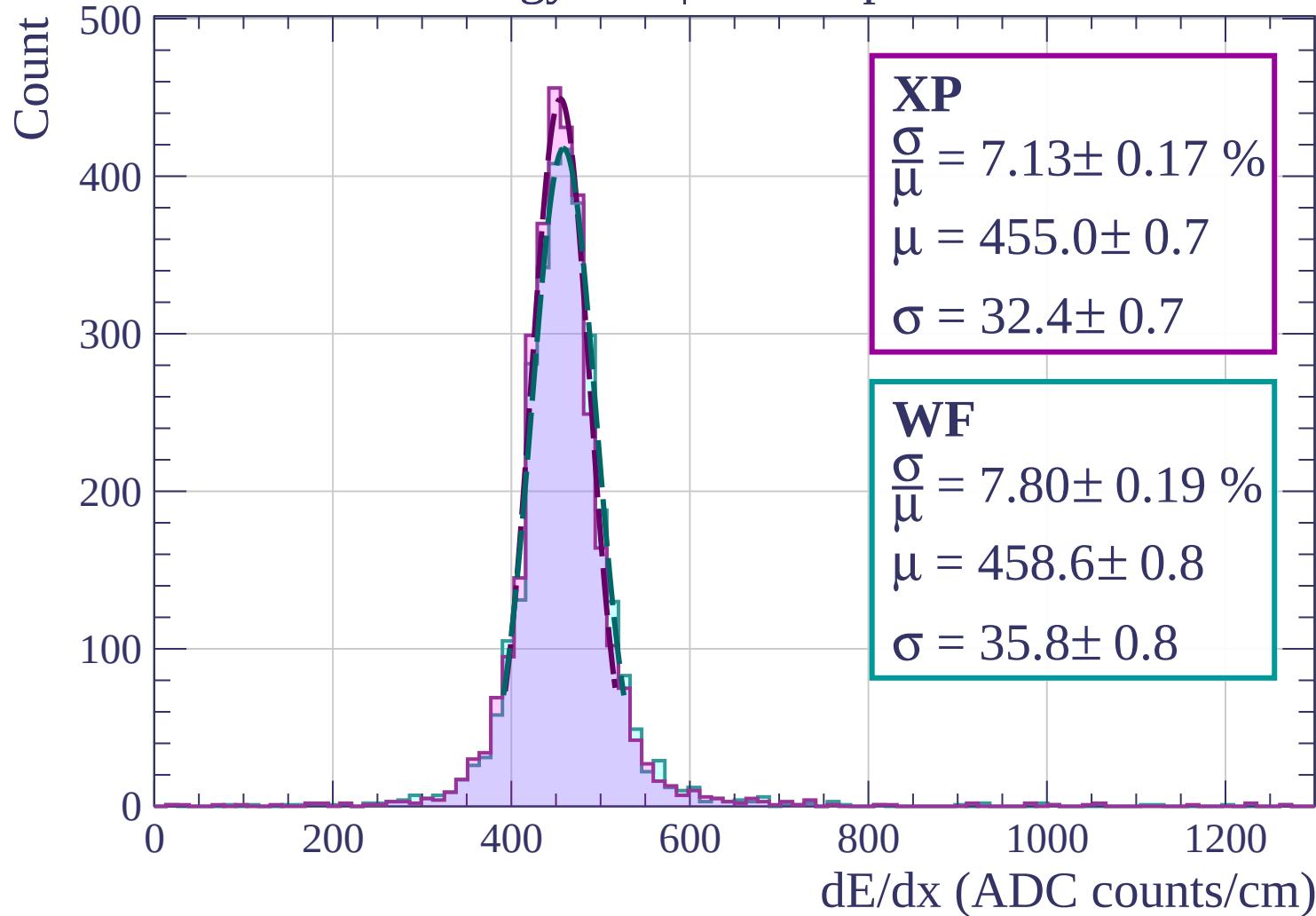
$$\frac{\sigma}{\mu} = 7.93 \pm 0.18 \%$$

$$\mu = 453.0 \pm 0.8$$

$$\sigma = 35.9 \pm 0.8$$

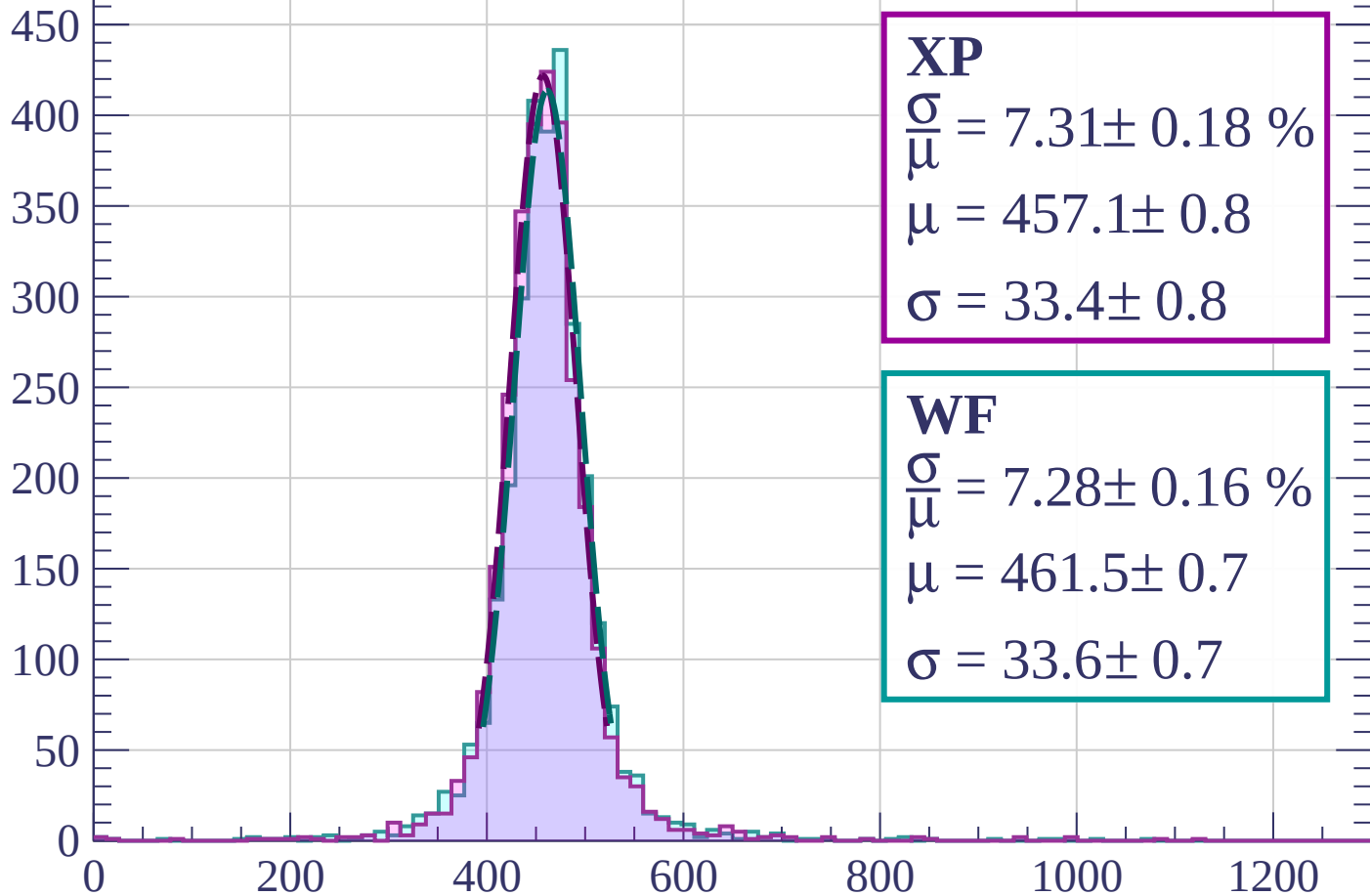
dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1360$



Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 7.31 \pm 0.18 \%$$

$$\mu = 457.1 \pm 0.8$$

$$\sigma = 33.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.28 \pm 0.16 \%$$

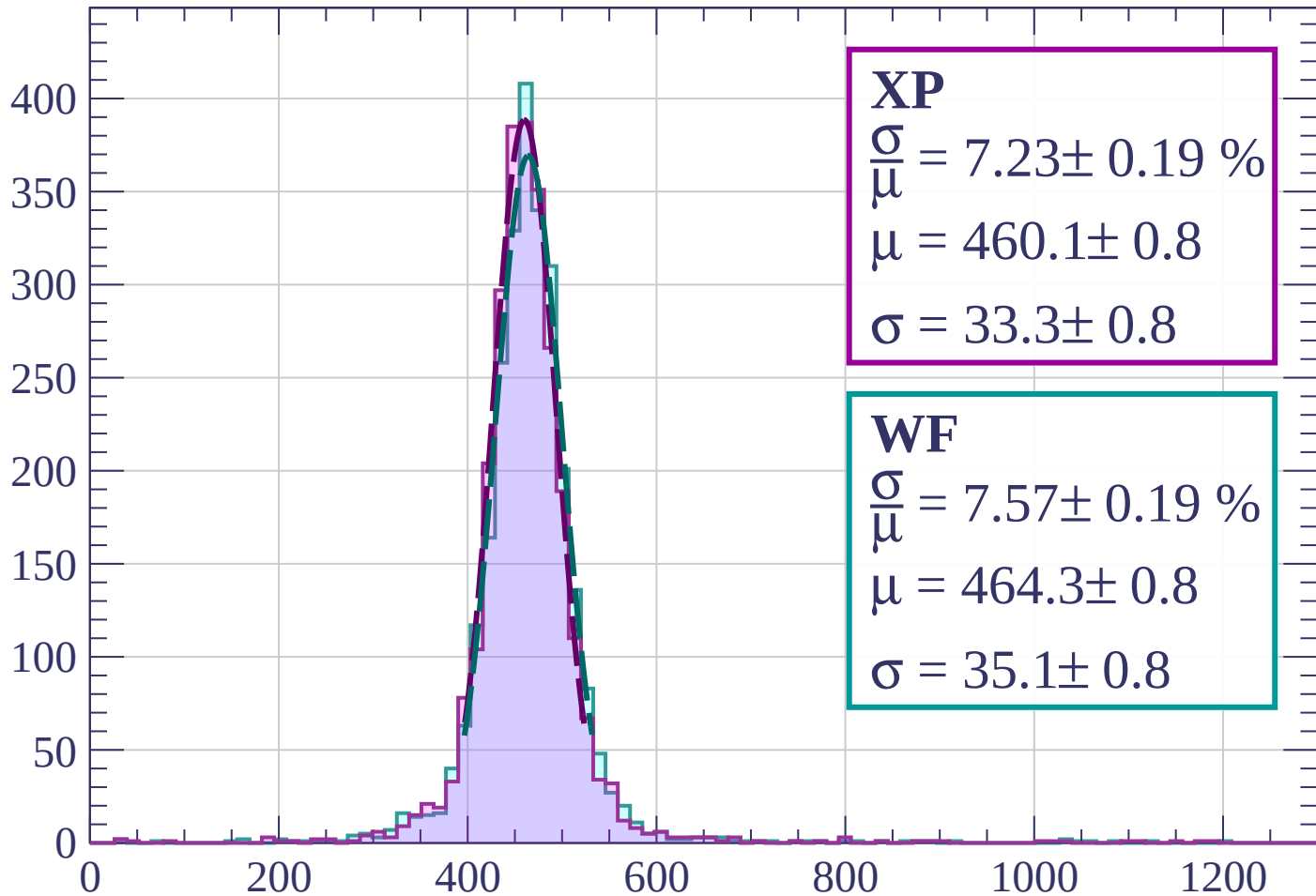
$$\mu = 461.5 \pm 0.7$$

$$\sigma = 33.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 7.23 \pm 0.19 \%$$

$$\mu = 460.1 \pm 0.8$$

$$\sigma = 33.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.57 \pm 0.19 \%$$

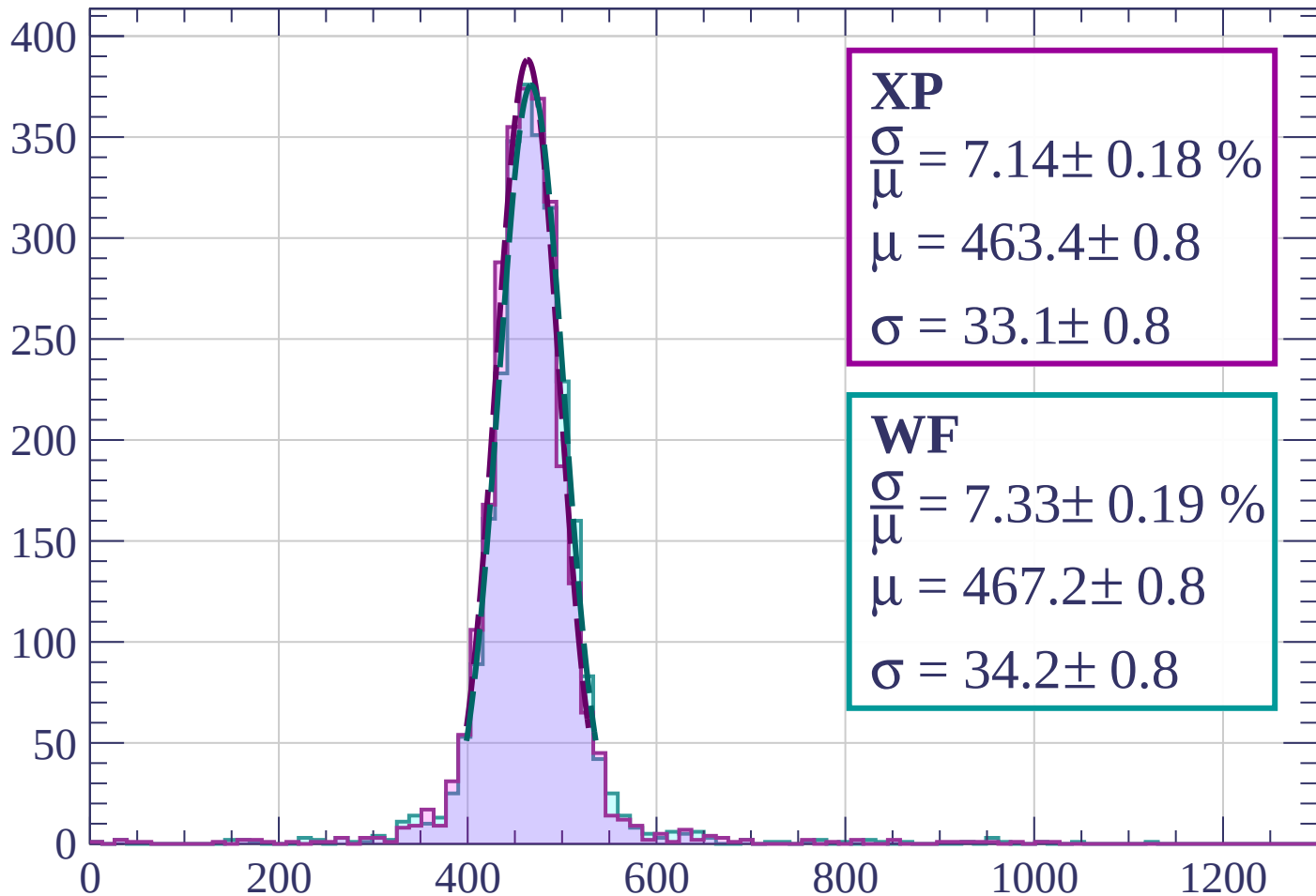
$$\mu = 464.3 \pm 0.8$$

$$\sigma = 35.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 7.14 \pm 0.18 \%$$

$$\mu = 463.4 \pm 0.8$$

$$\sigma = 33.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.33 \pm 0.19 \%$$

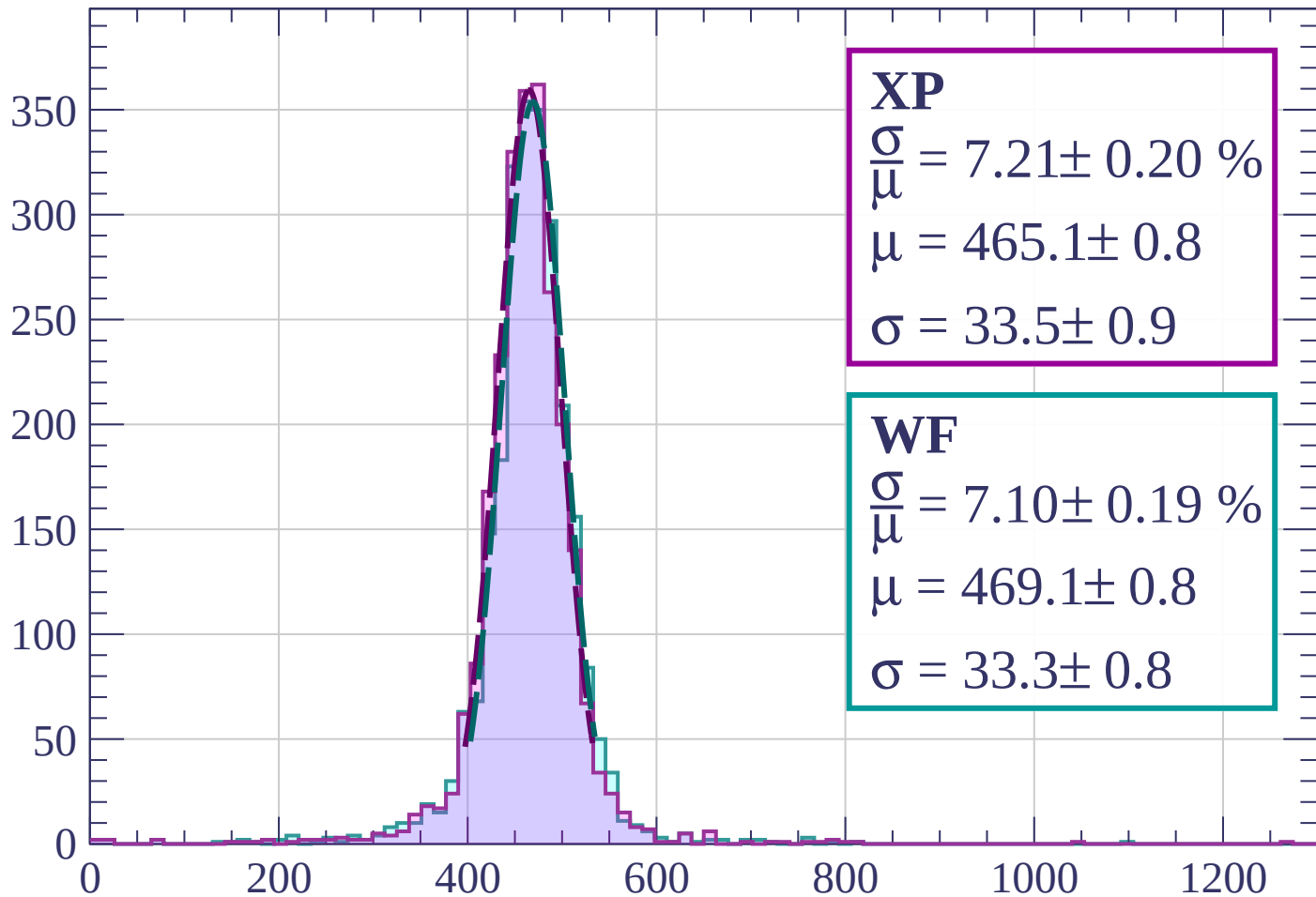
$$\mu = 467.2 \pm 0.8$$

$$\sigma = 34.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 7.21 \pm 0.20 \%$$

$$\mu = 465.1 \pm 0.8$$

$$\sigma = 33.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.10 \pm 0.19 \%$$

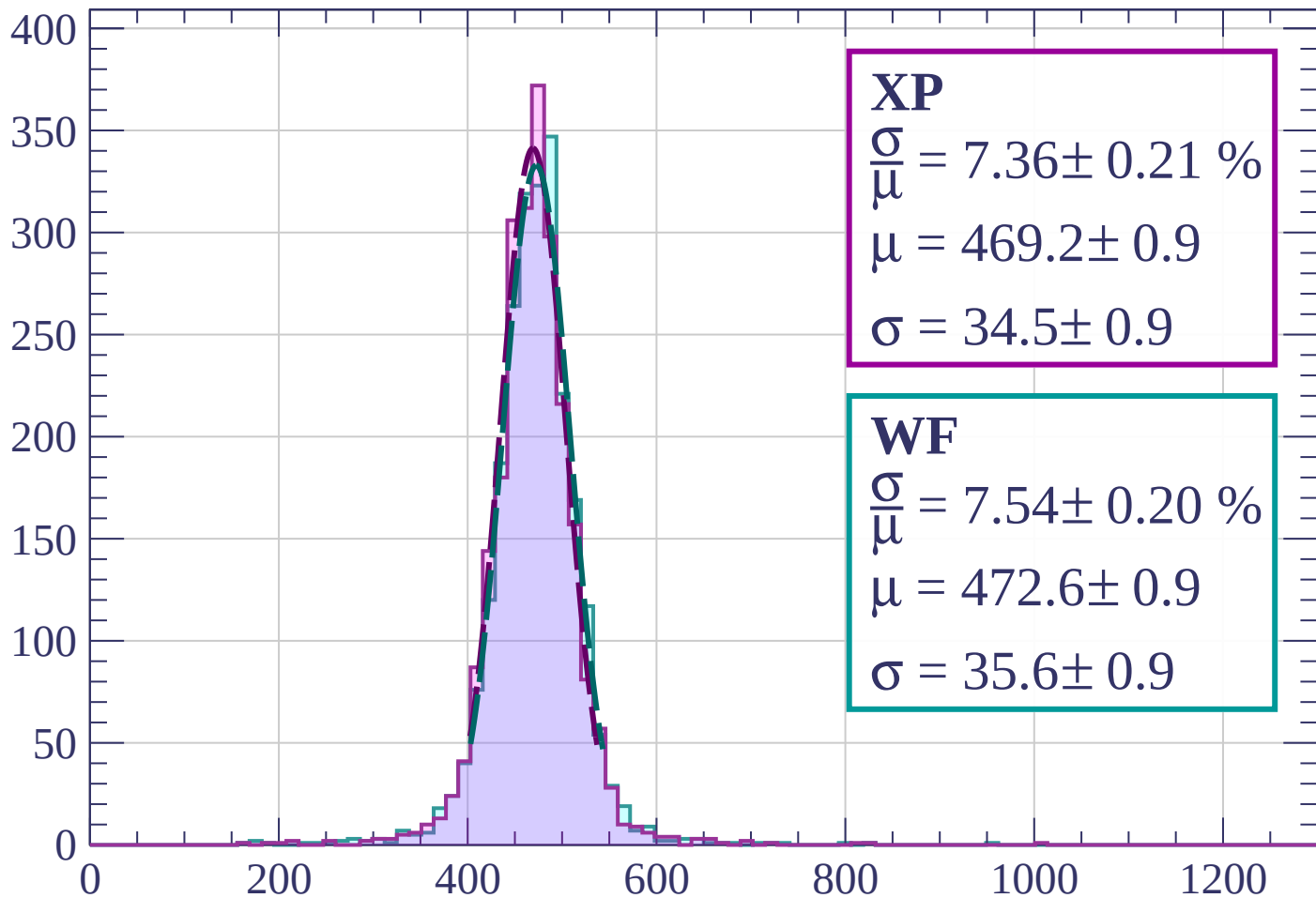
$$\mu = 469.1 \pm 0.8$$

$$\sigma = 33.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 7.36 \pm 0.21 \%$$

$$\mu = 469.2 \pm 0.9$$

$$\sigma = 34.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.54 \pm 0.20 \%$$

$$\mu = 472.6 \pm 0.9$$

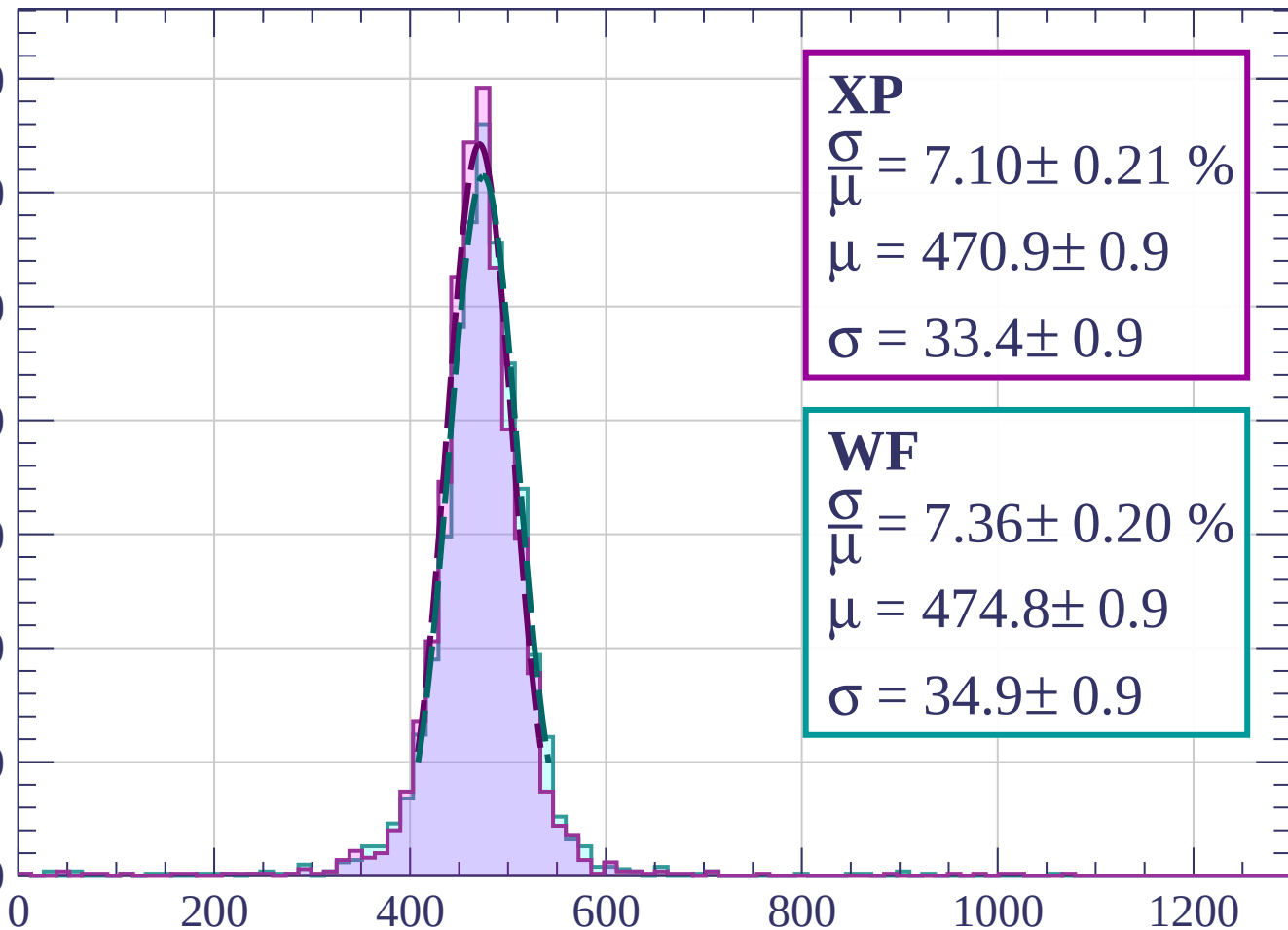
$$\sigma = 35.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1760 < p < 1840$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 7.10 \pm 0.21 \%$$

$$\mu = 470.9 \pm 0.9$$

$$\sigma = 33.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.36 \pm 0.20 \%$$

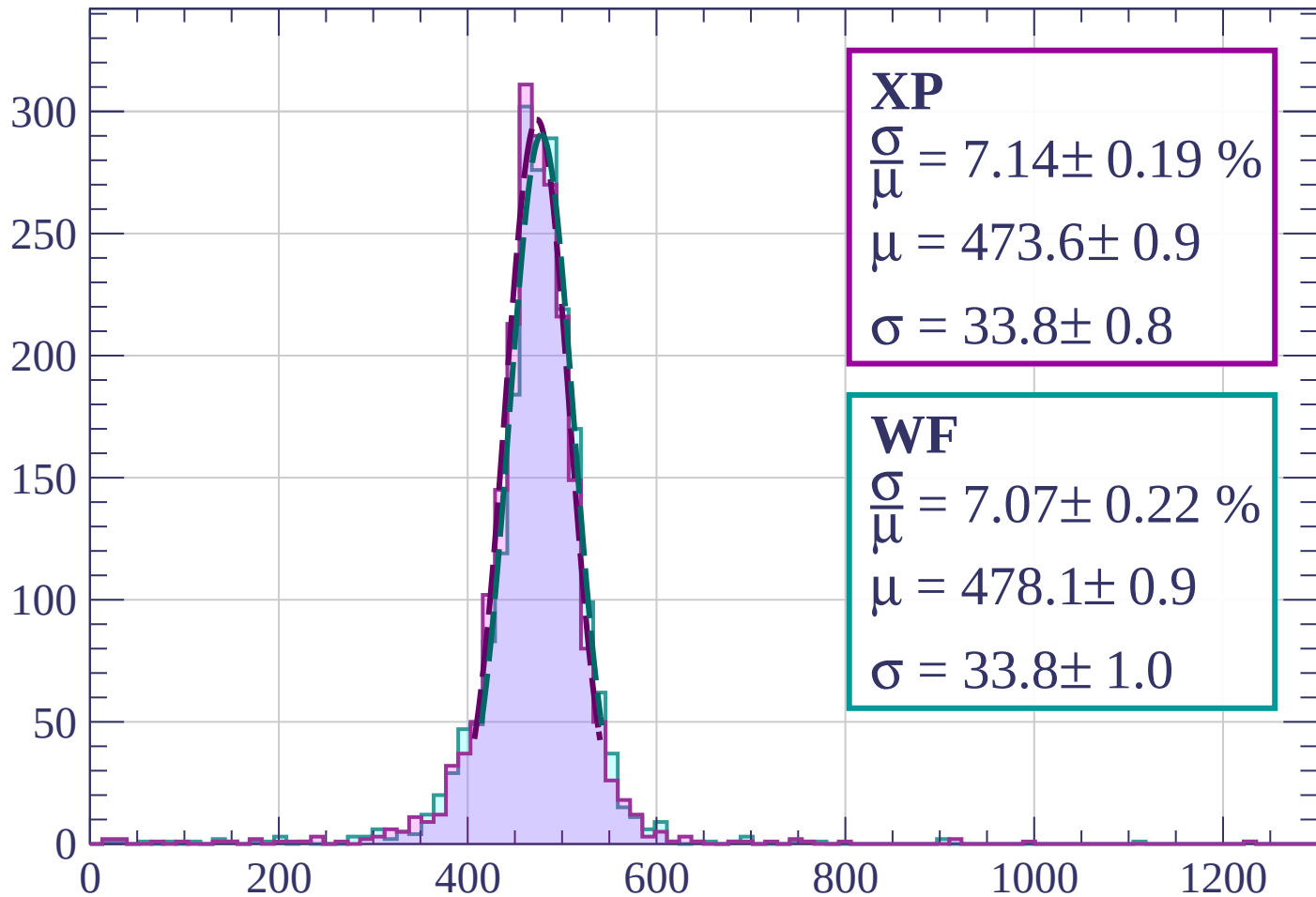
$$\mu = 474.8 \pm 0.9$$

$$\sigma = 34.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 7.14 \pm 0.19 \%$$

$$\mu = 473.6 \pm 0.9$$

$$\sigma = 33.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.07 \pm 0.22 \%$$

$$\mu = 478.1 \pm 0.9$$

$$\sigma = 33.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 2000$

Count

300

250

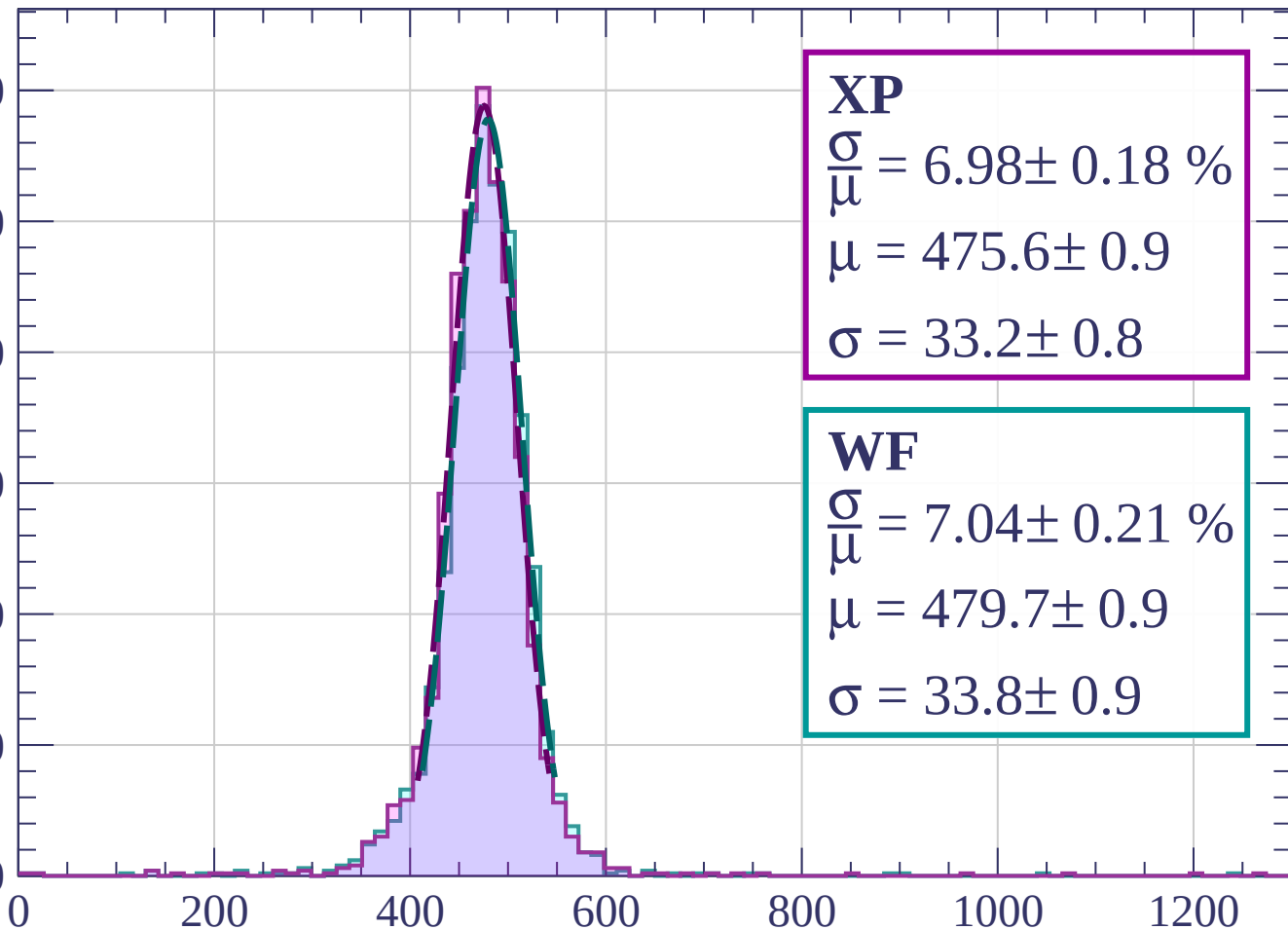
200

150

100

50

0



XP

$$\frac{\sigma}{\mu} = 6.98 \pm 0.18 \%$$

$$\mu = 475.6 \pm 0.9$$

$$\sigma = 33.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.04 \pm 0.21 \%$$

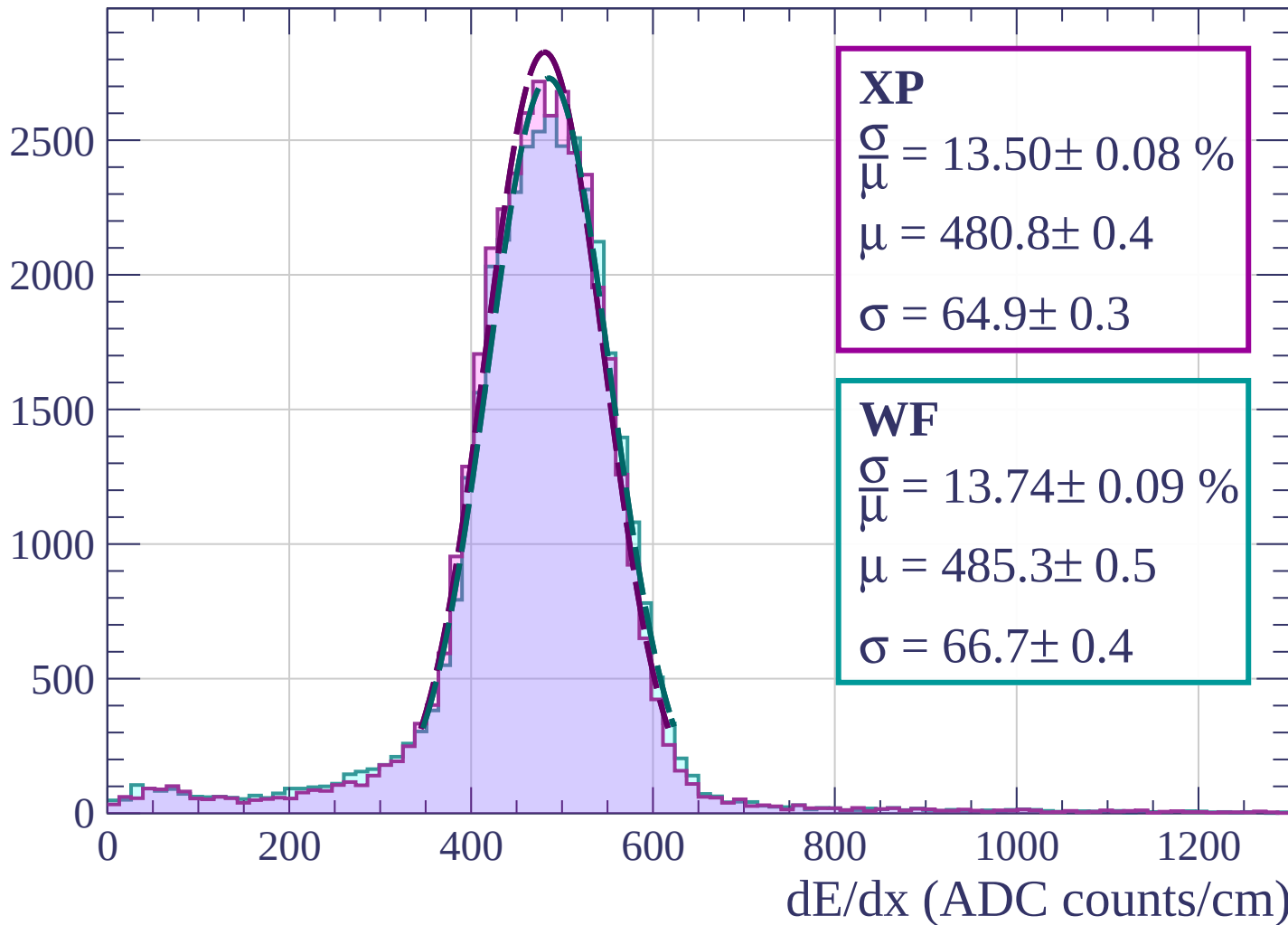
$$\mu = 479.7 \pm 0.9$$

$$\sigma = 33.8 \pm 0.9$$

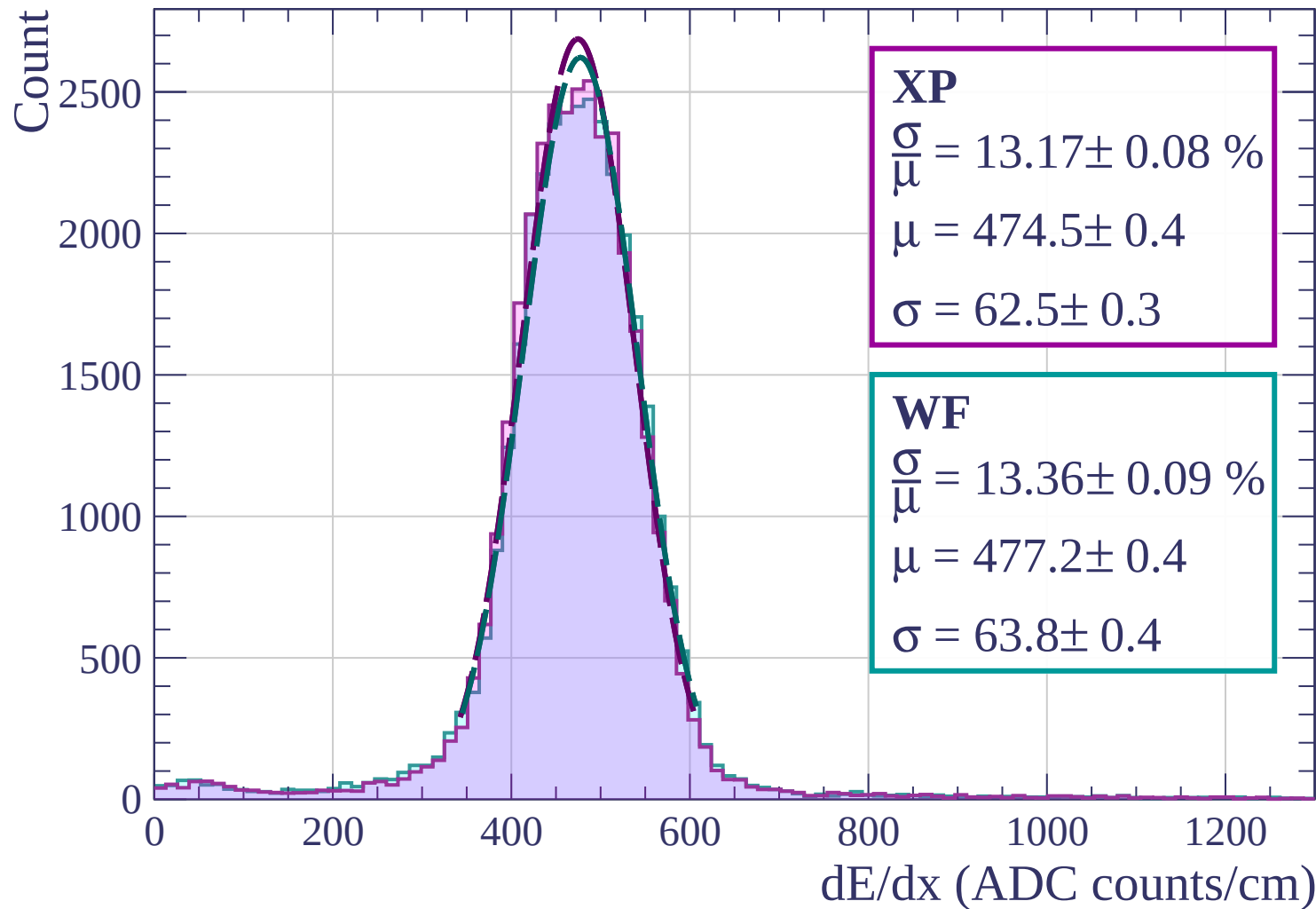
dE/dx (ADC counts/cm)

Energy loss | $0 < \phi < 3$

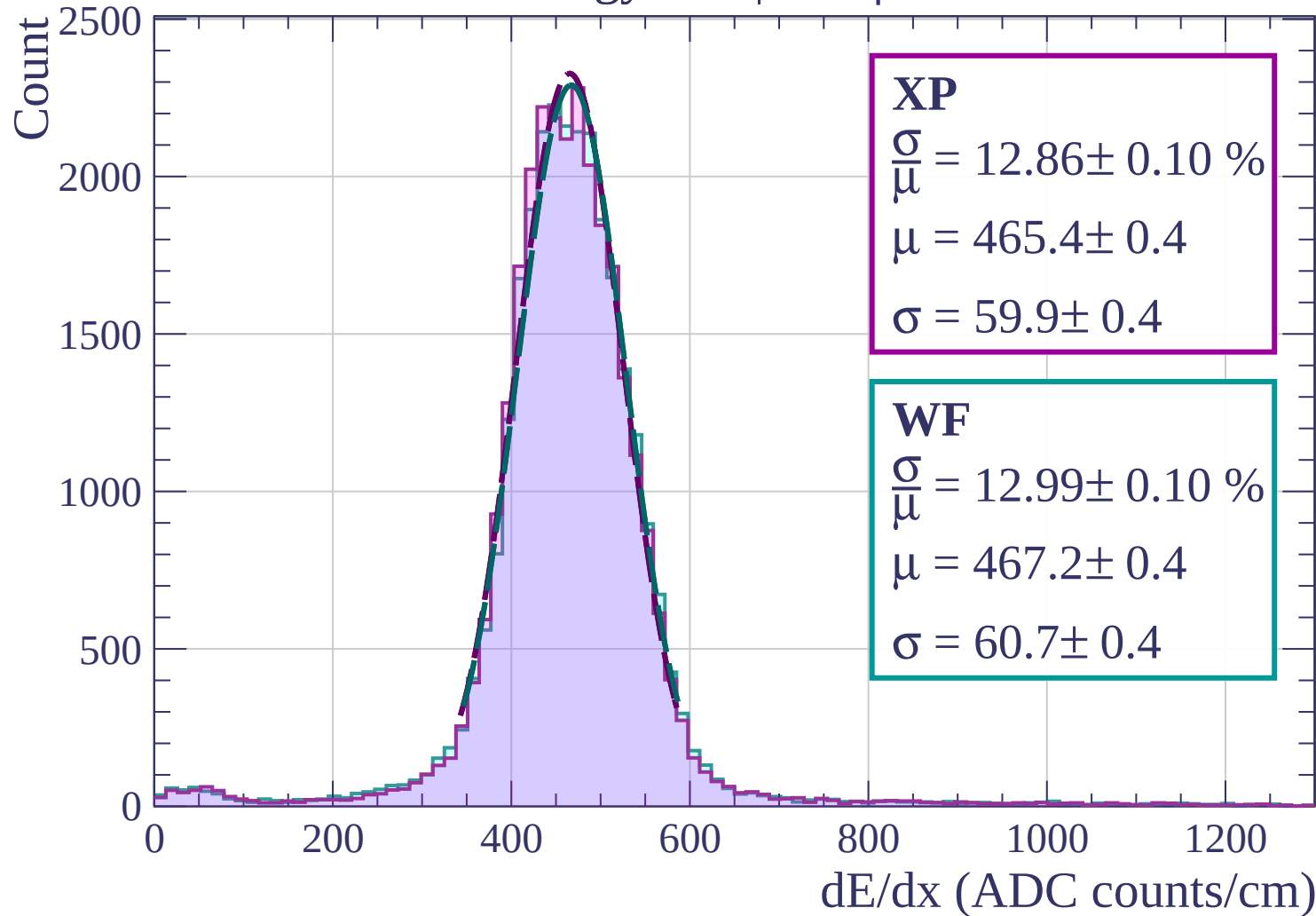
Count



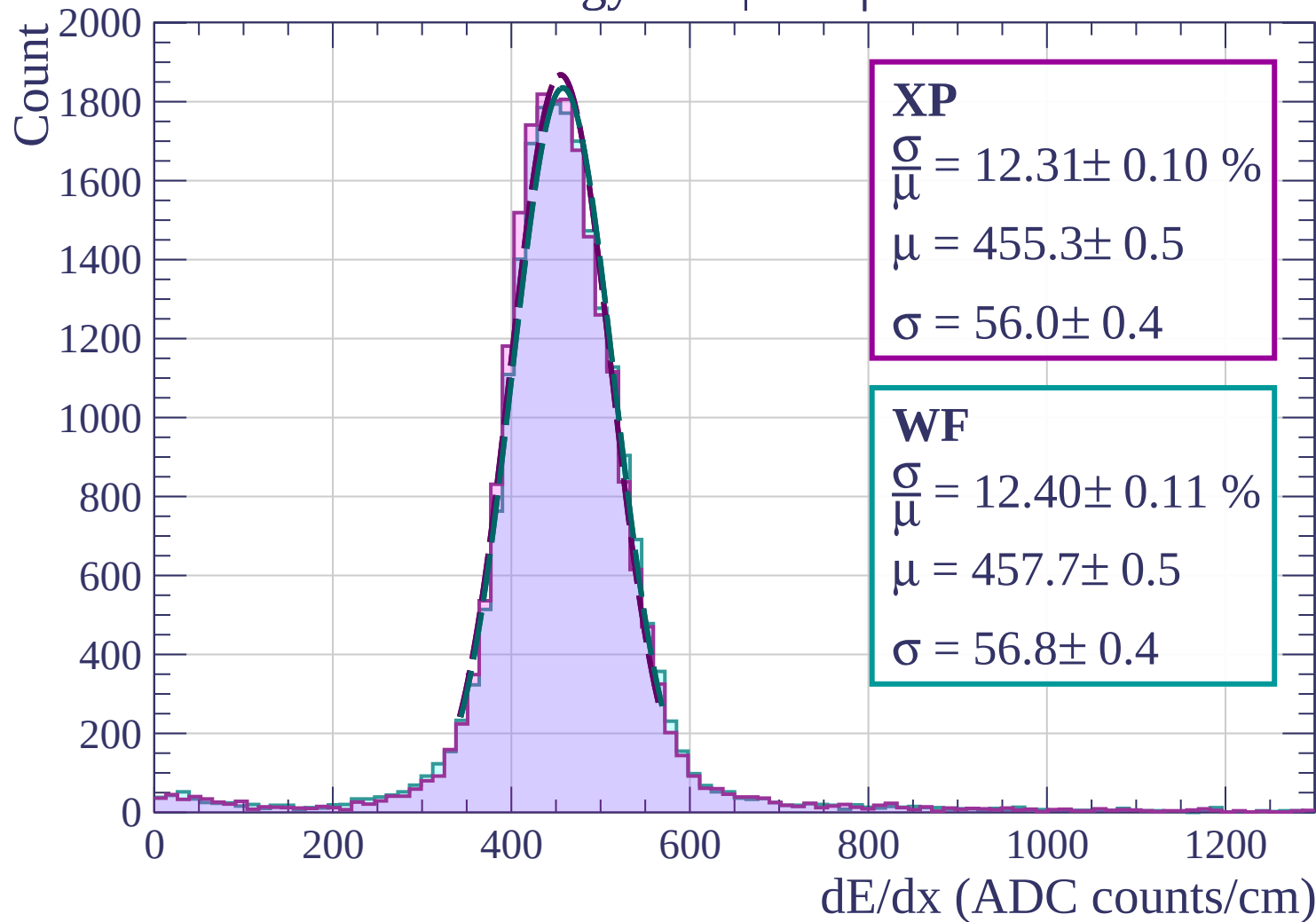
Energy loss | $3 < \phi < 6$



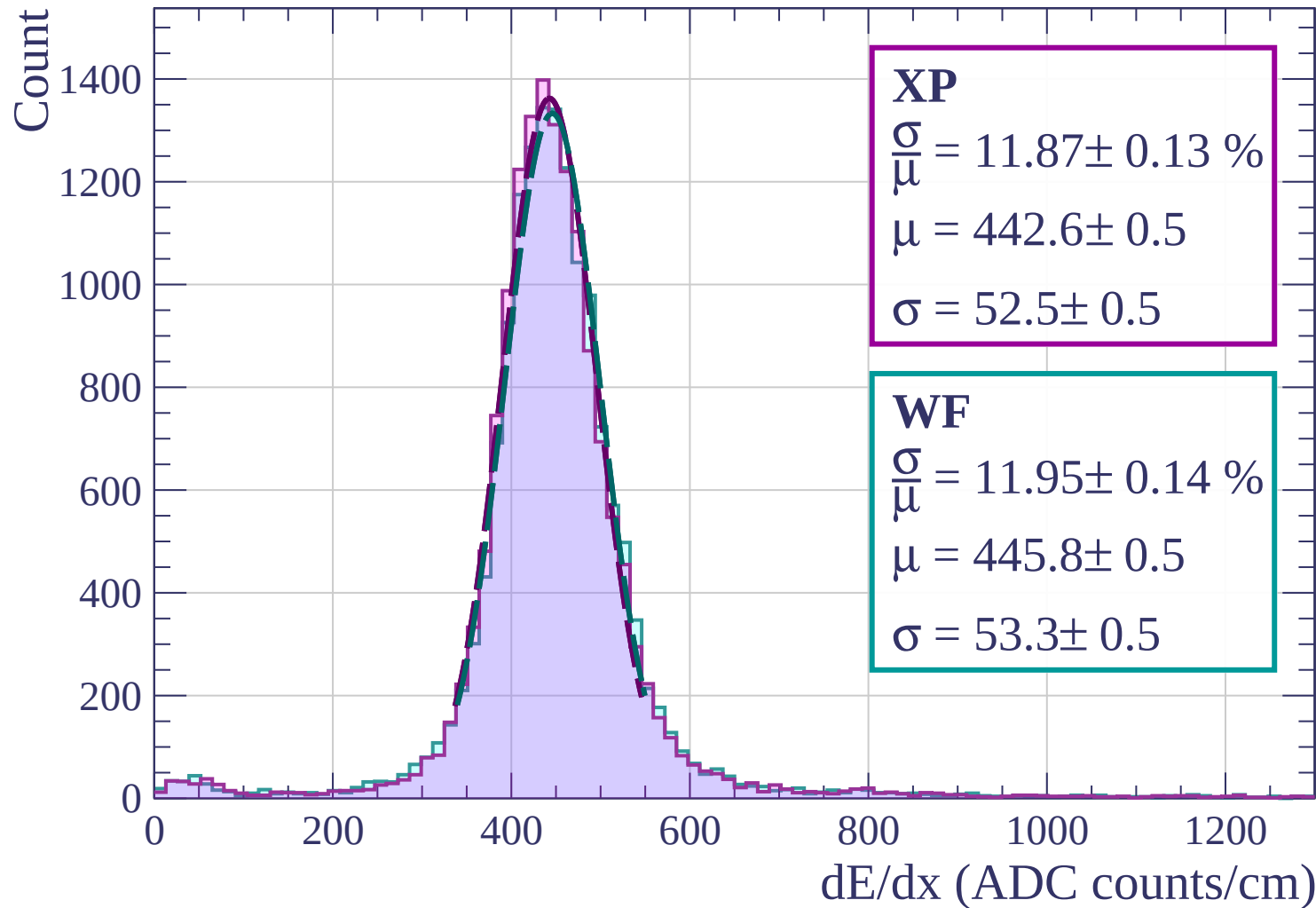
Energy loss | $6 < \phi < 9$



Energy loss | $9 < \phi < 12$

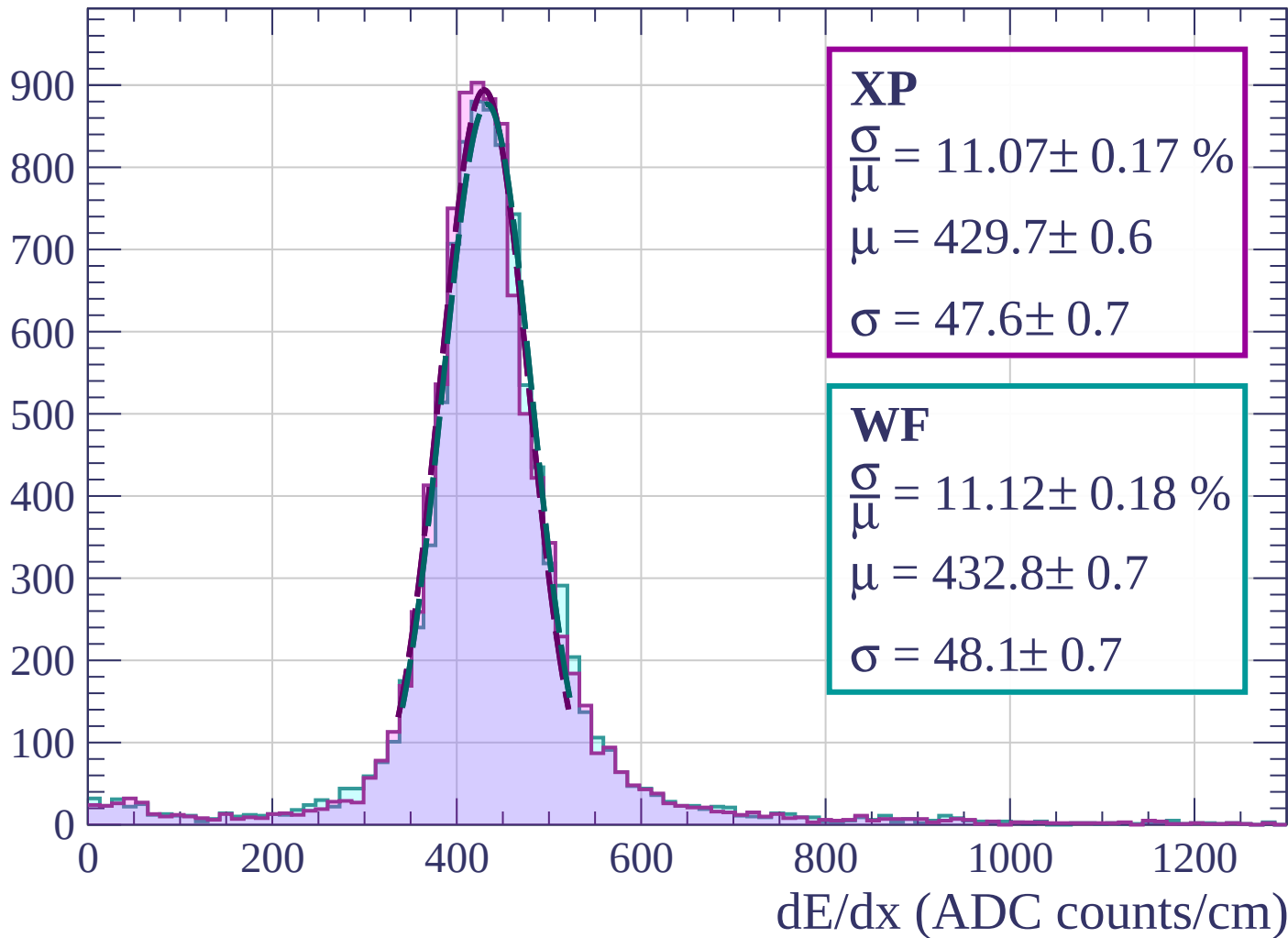


Energy loss | $12 < \phi < 15$



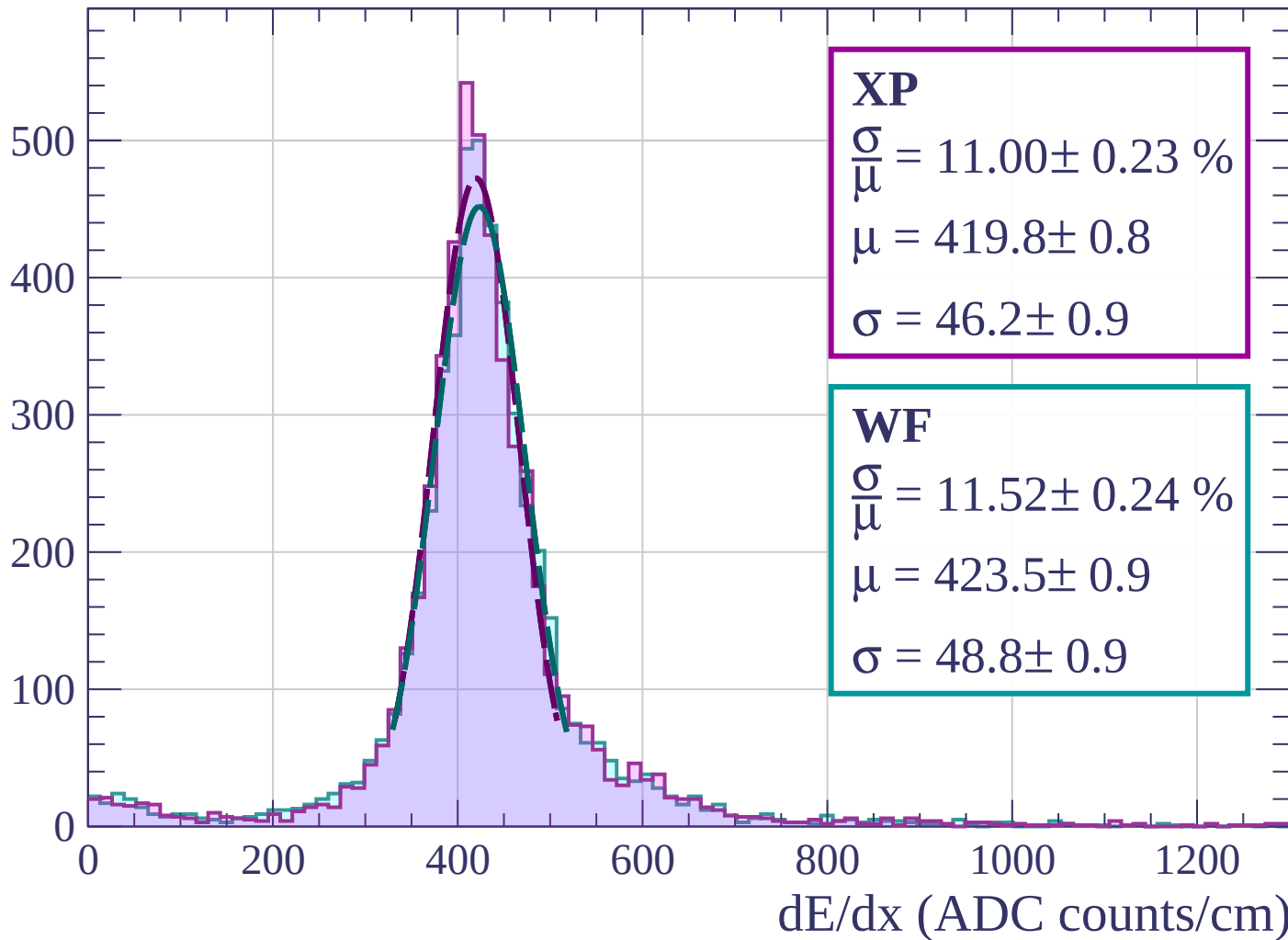
Energy loss | $15 < \phi < 18$

Count



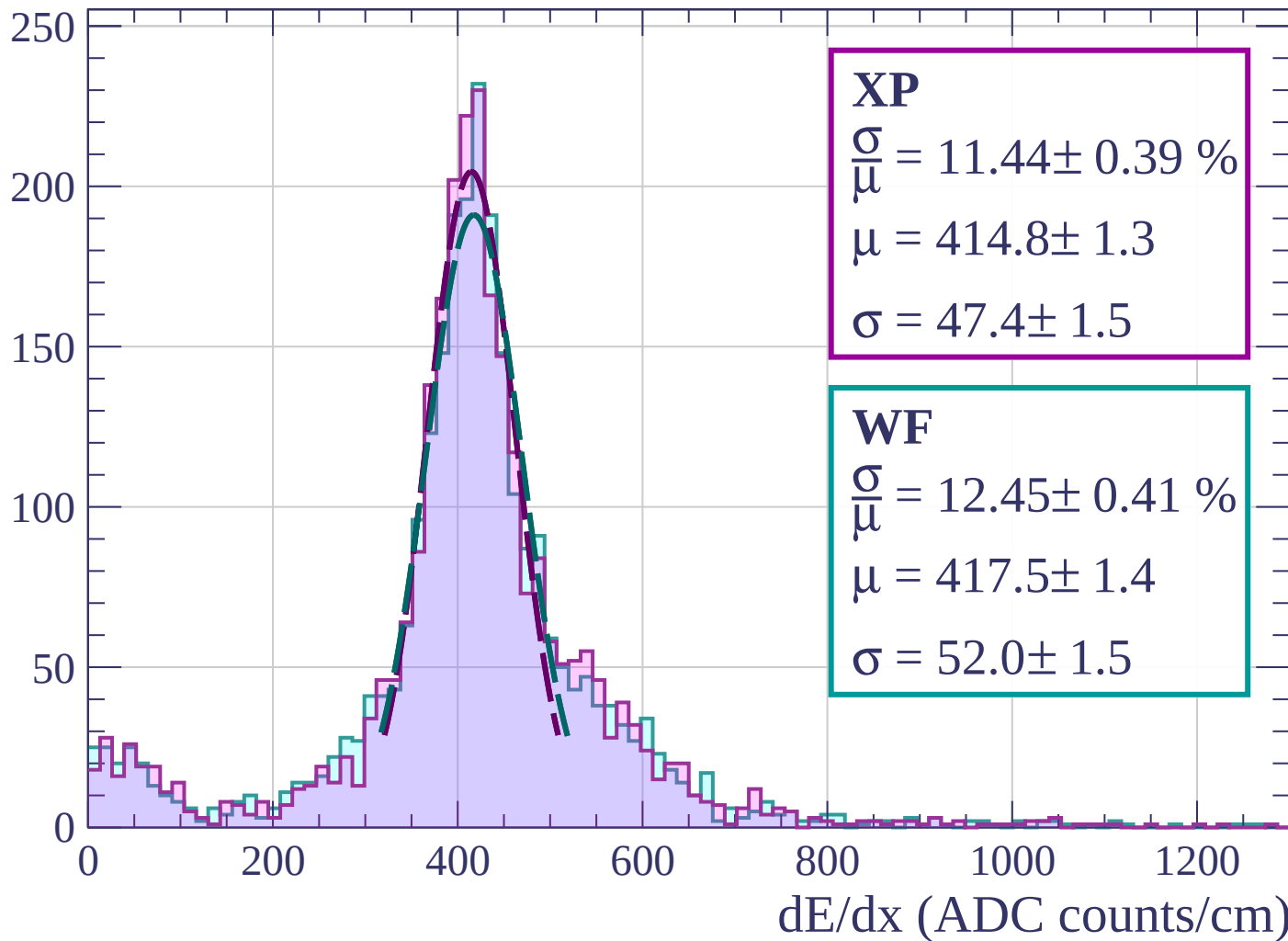
Energy loss | $18 < \phi < 21$

Count



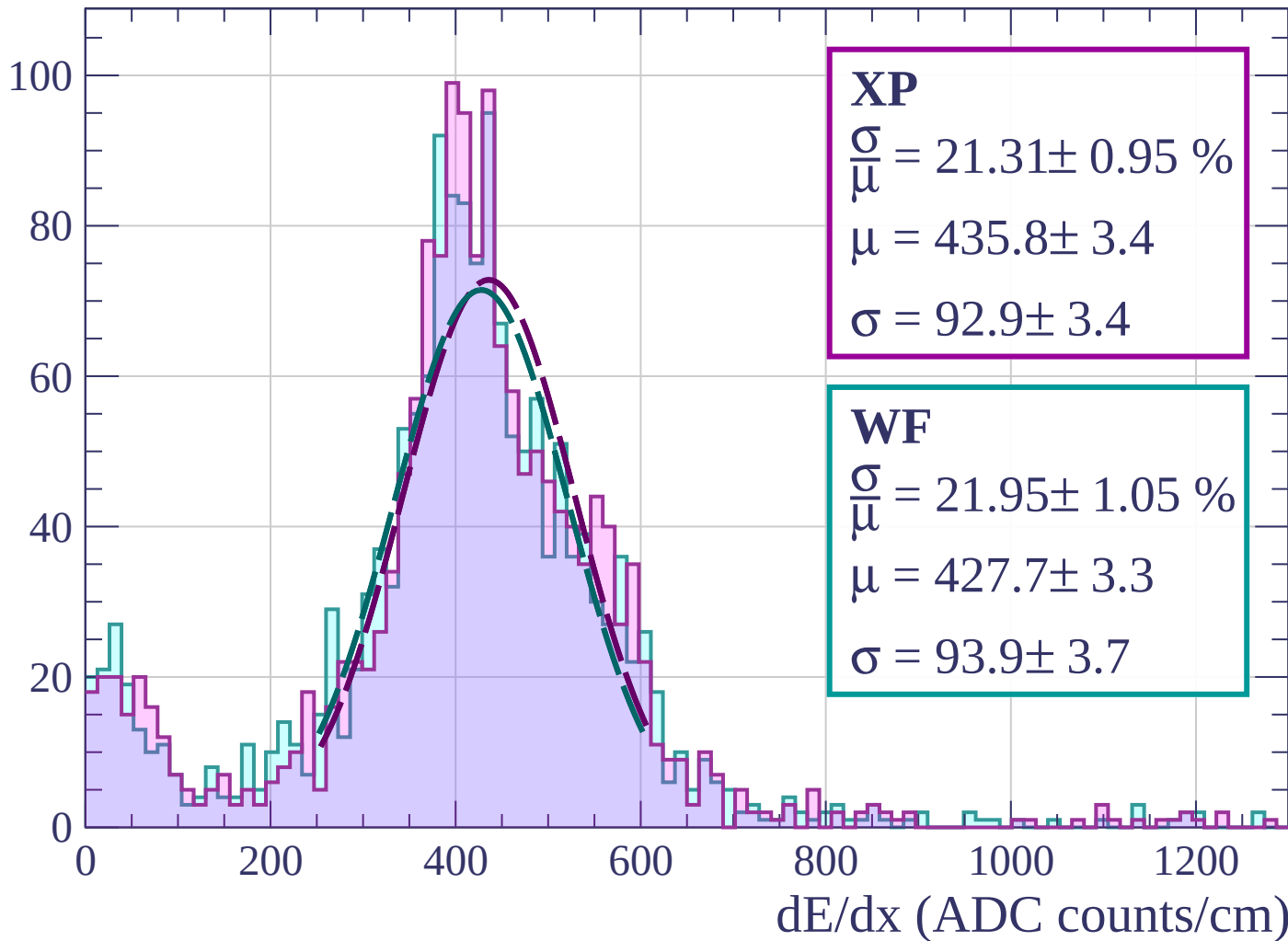
Energy loss | $21 < \phi < 24$

Count



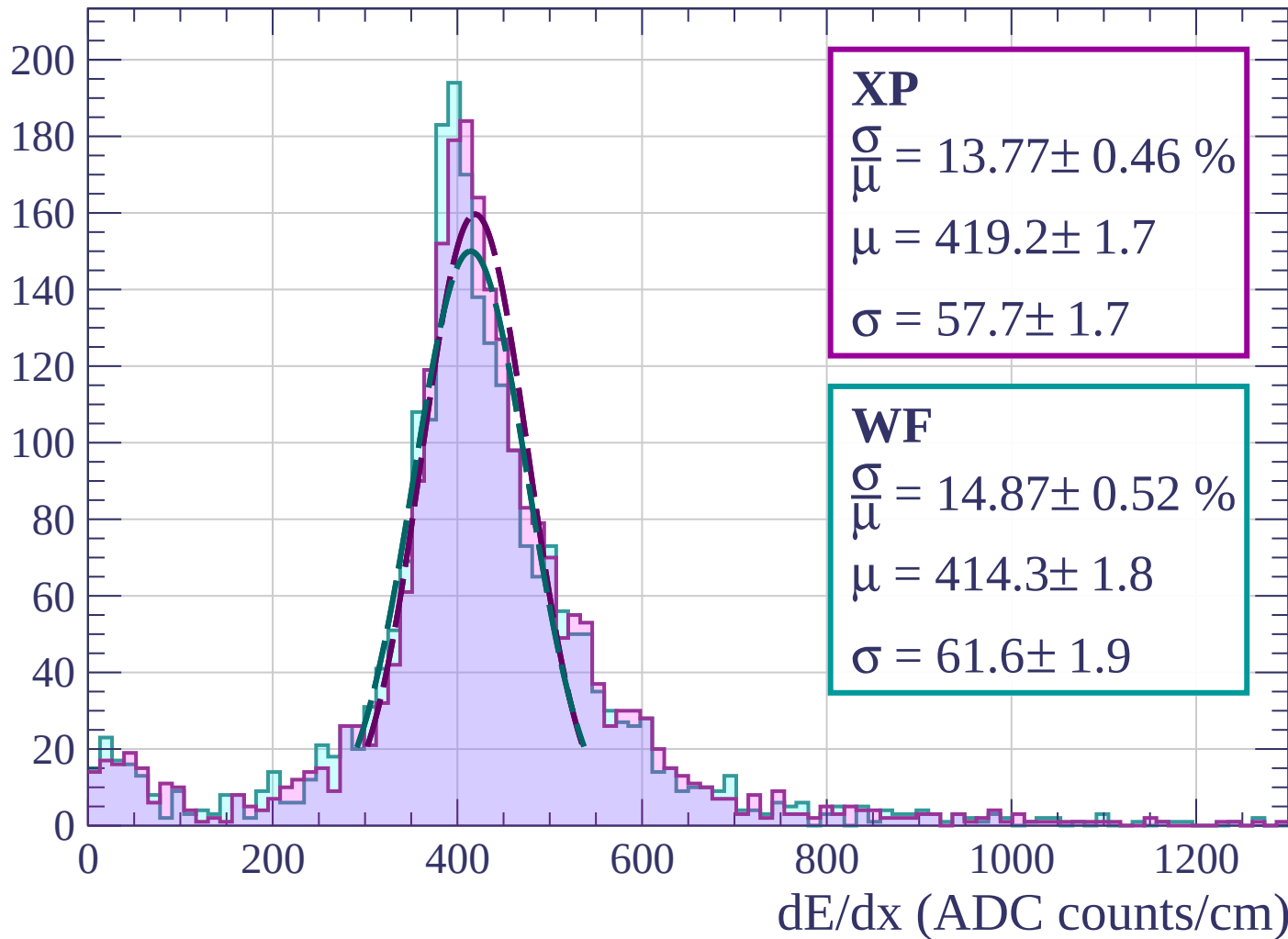
Energy loss | $24 < \phi < 27$

Count



Energy loss | $27 < \phi < 30$

Count



Energy loss | $30 < \phi < 33$

Count

300

250

200

150

100

50

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 12.49 \pm 0.34 \%$$

$$\mu = 416.5 \pm 1.2$$

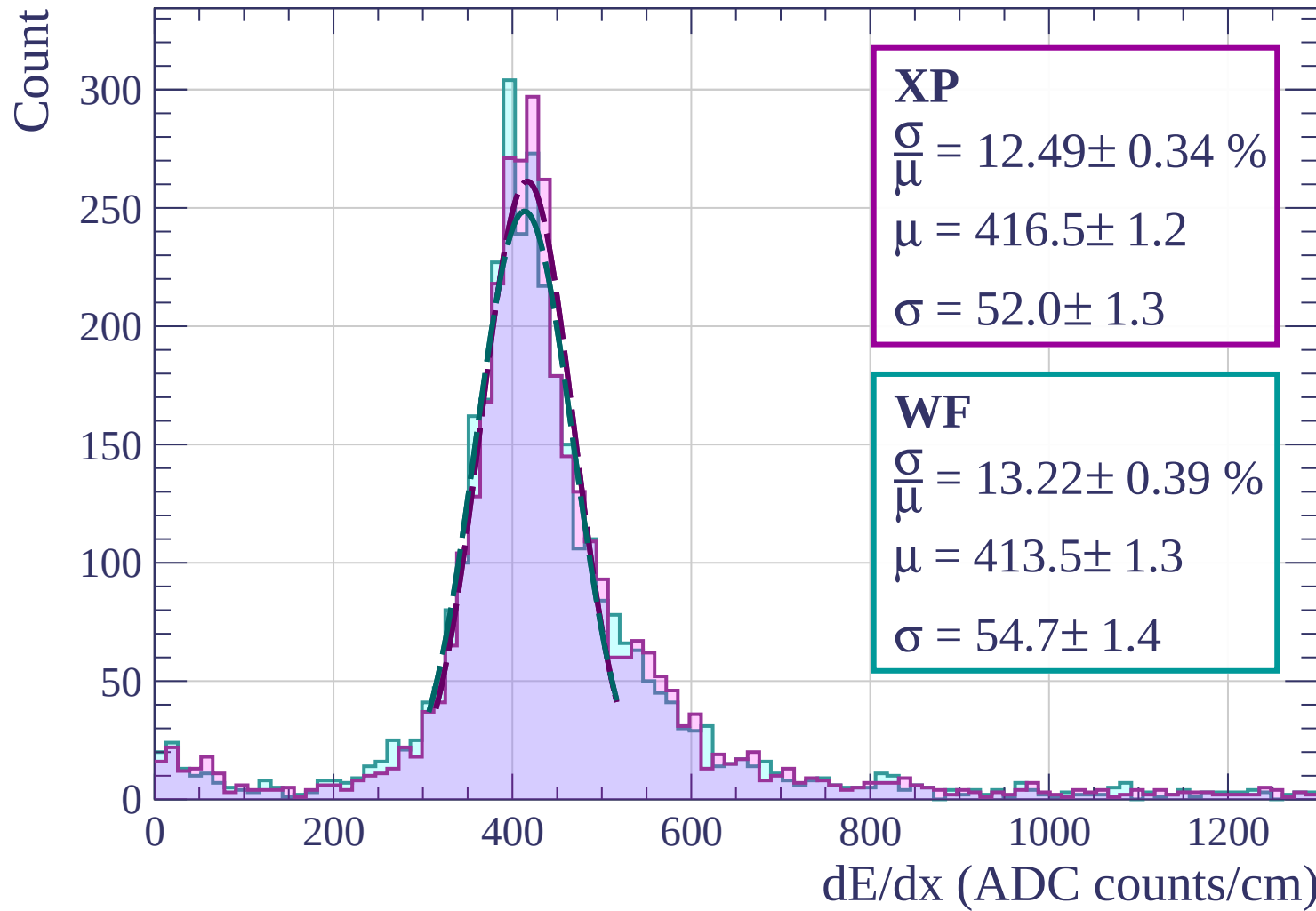
$$\sigma = 52.0 \pm 1.3$$

WF

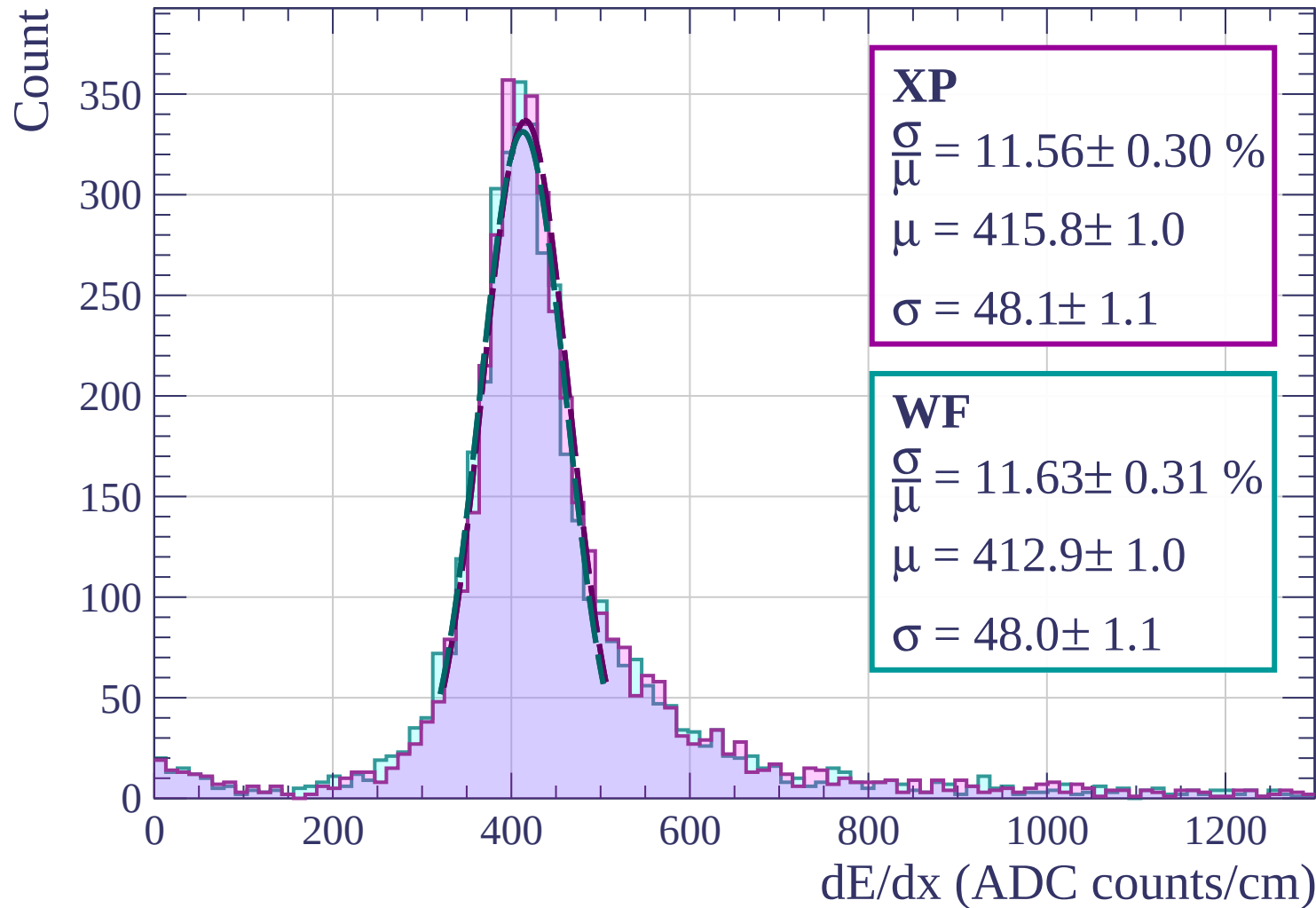
$$\frac{\sigma}{\mu} = 13.22 \pm 0.39 \%$$

$$\mu = 413.5 \pm 1.3$$

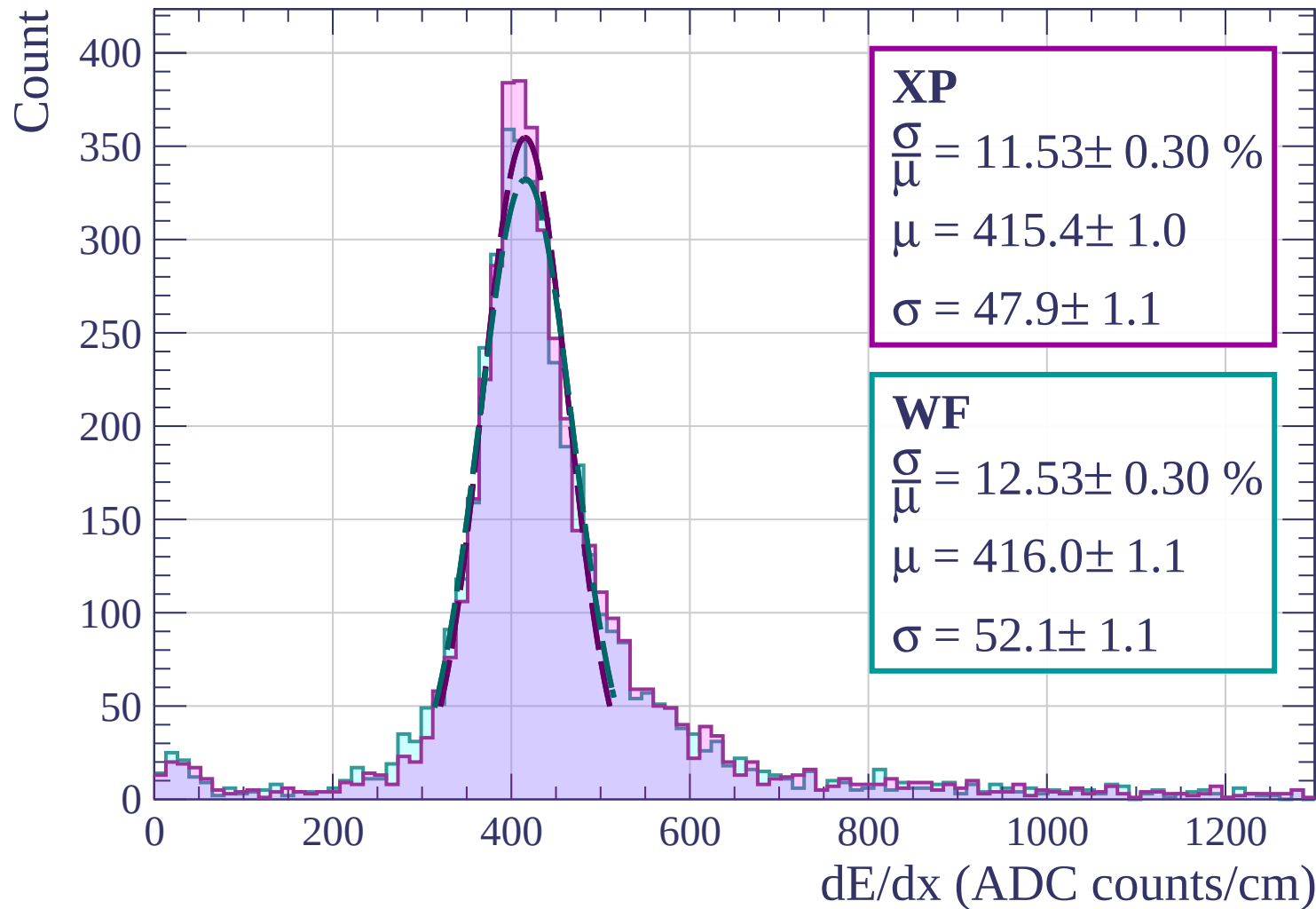
$$\sigma = 54.7 \pm 1.4$$



Energy loss | $33 < \phi < 36$

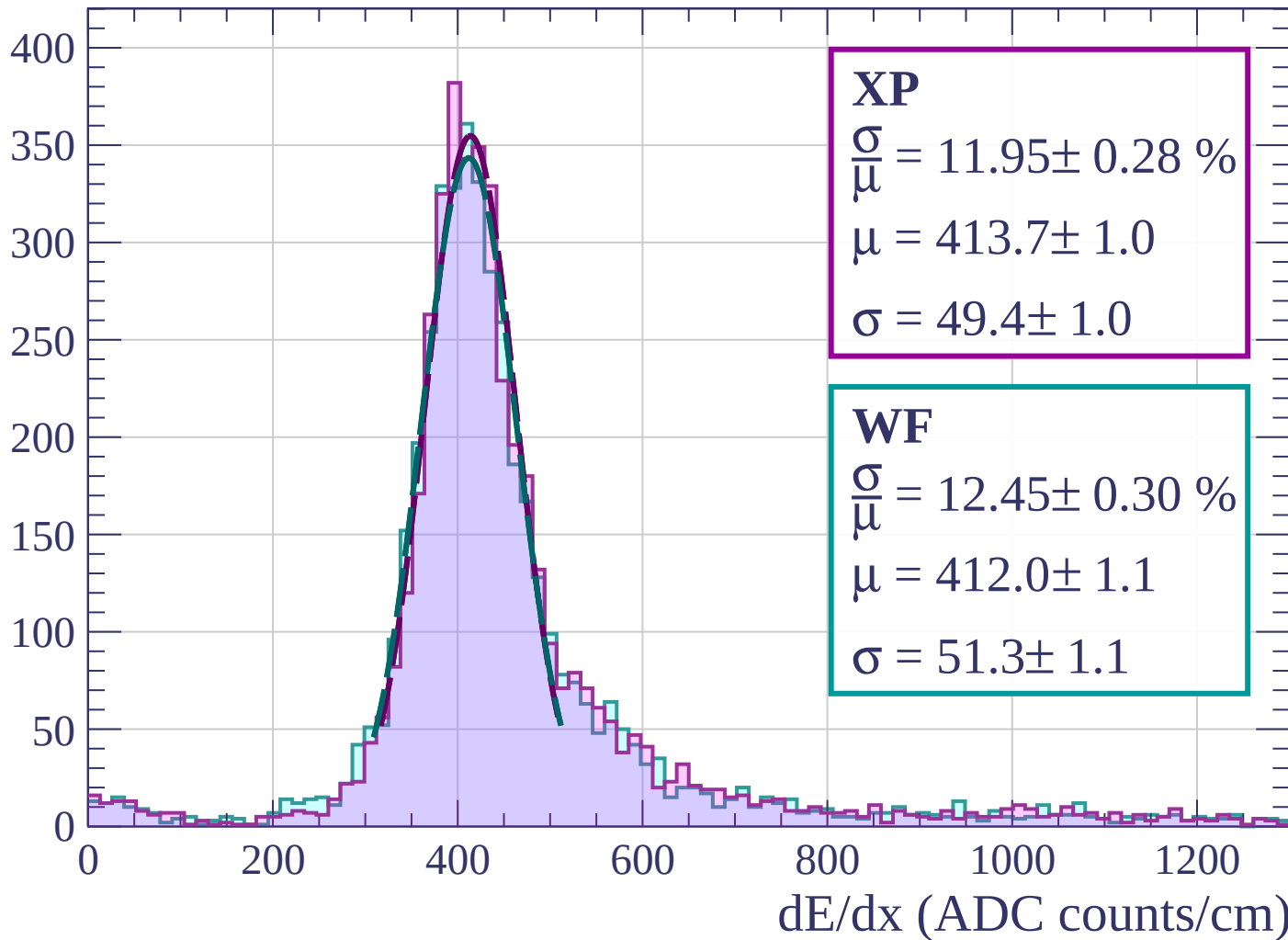


Energy loss | $36 < \phi < 39$



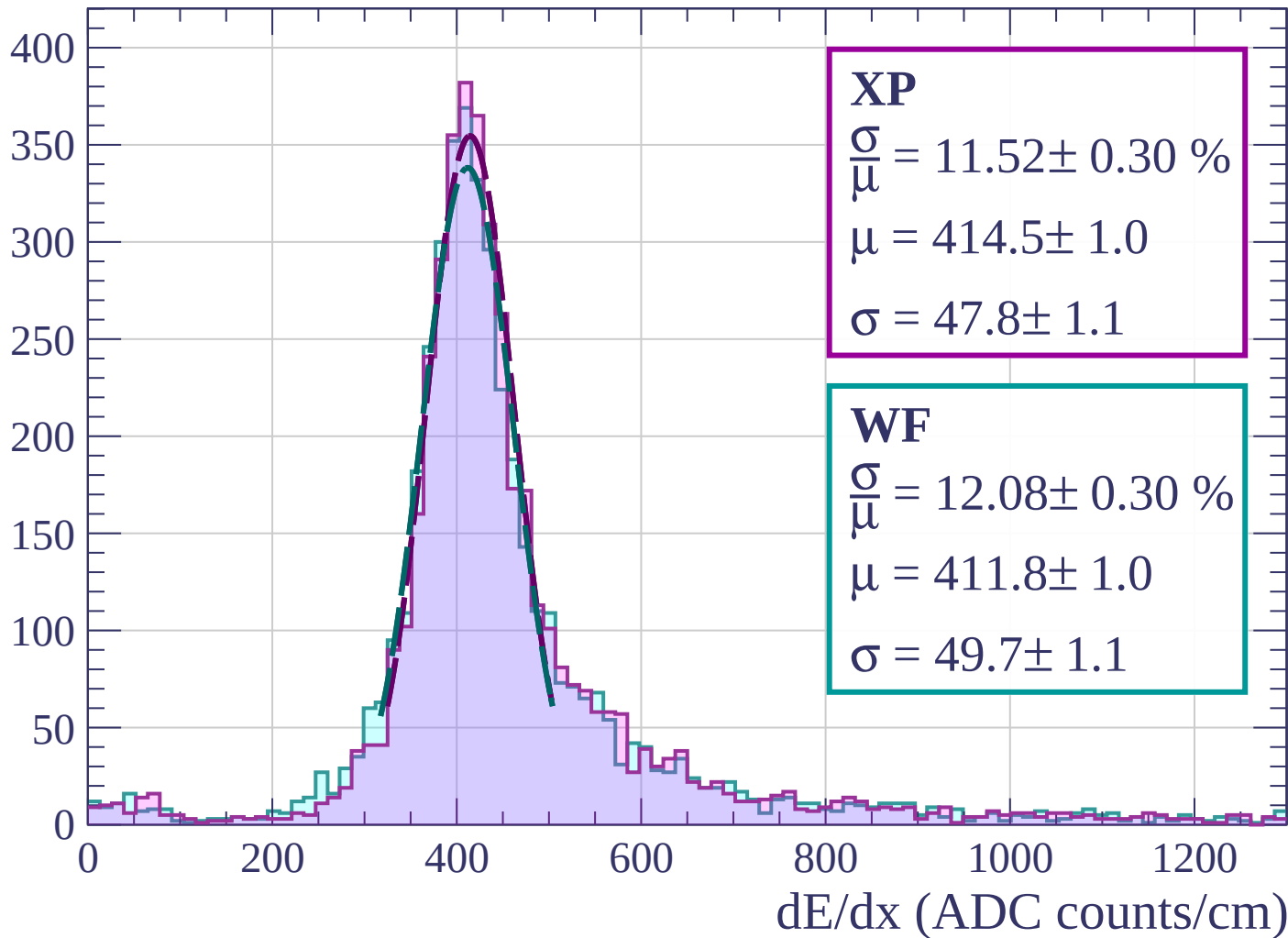
Energy loss | $39 < \phi < 42$

Count

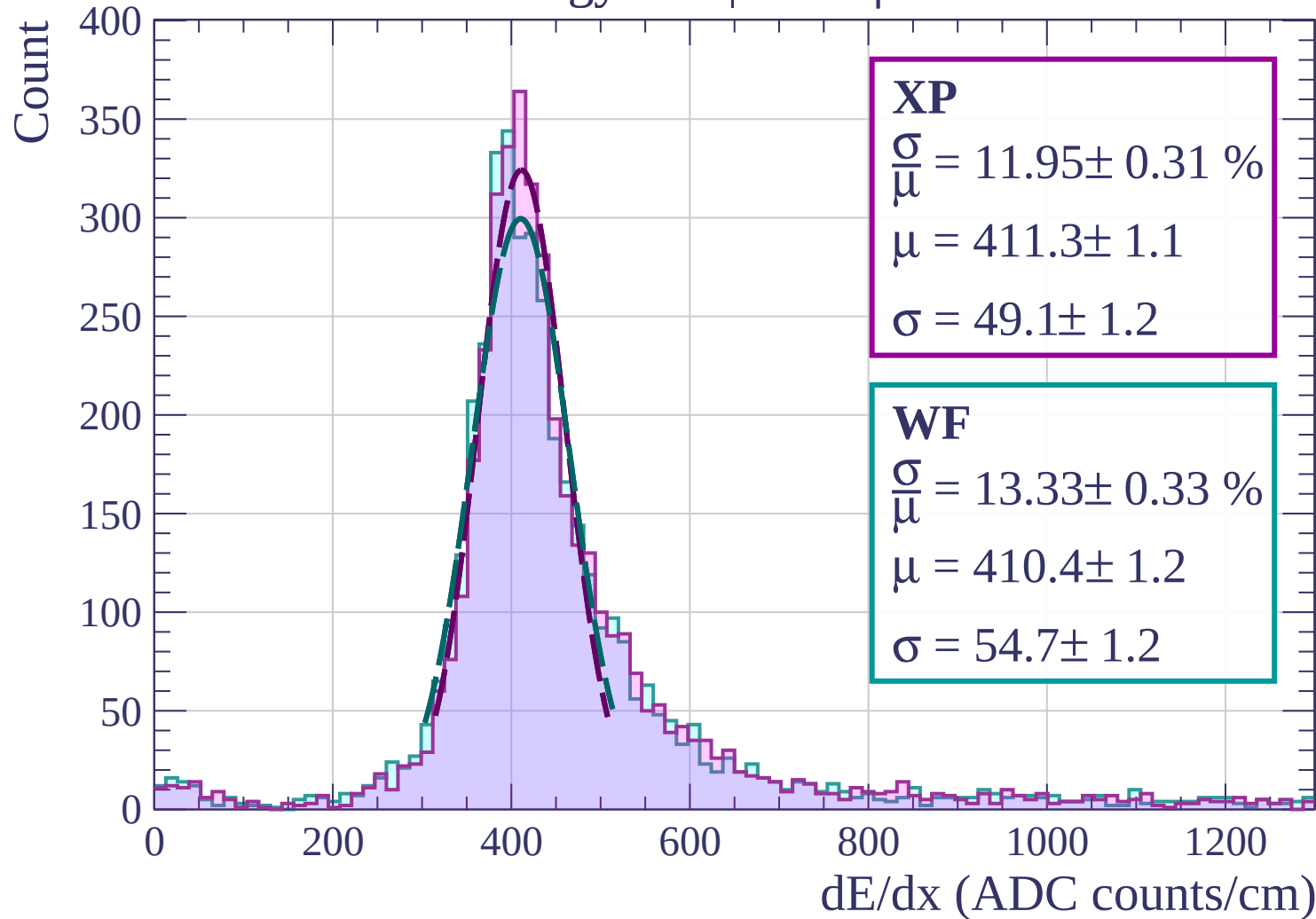


Energy loss | $42 < \phi < 45$

Count

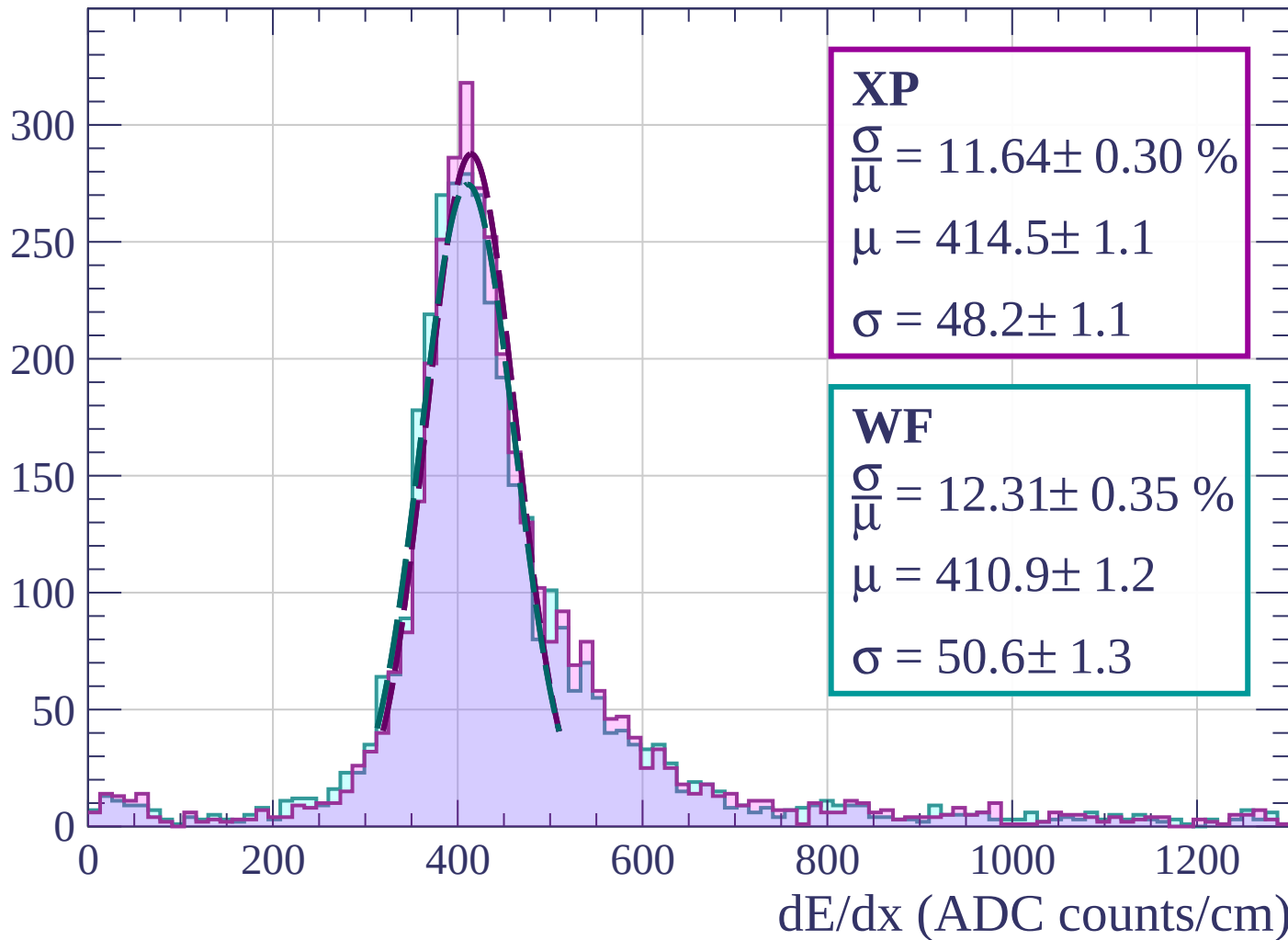


Energy loss | $45 < \phi < 48$



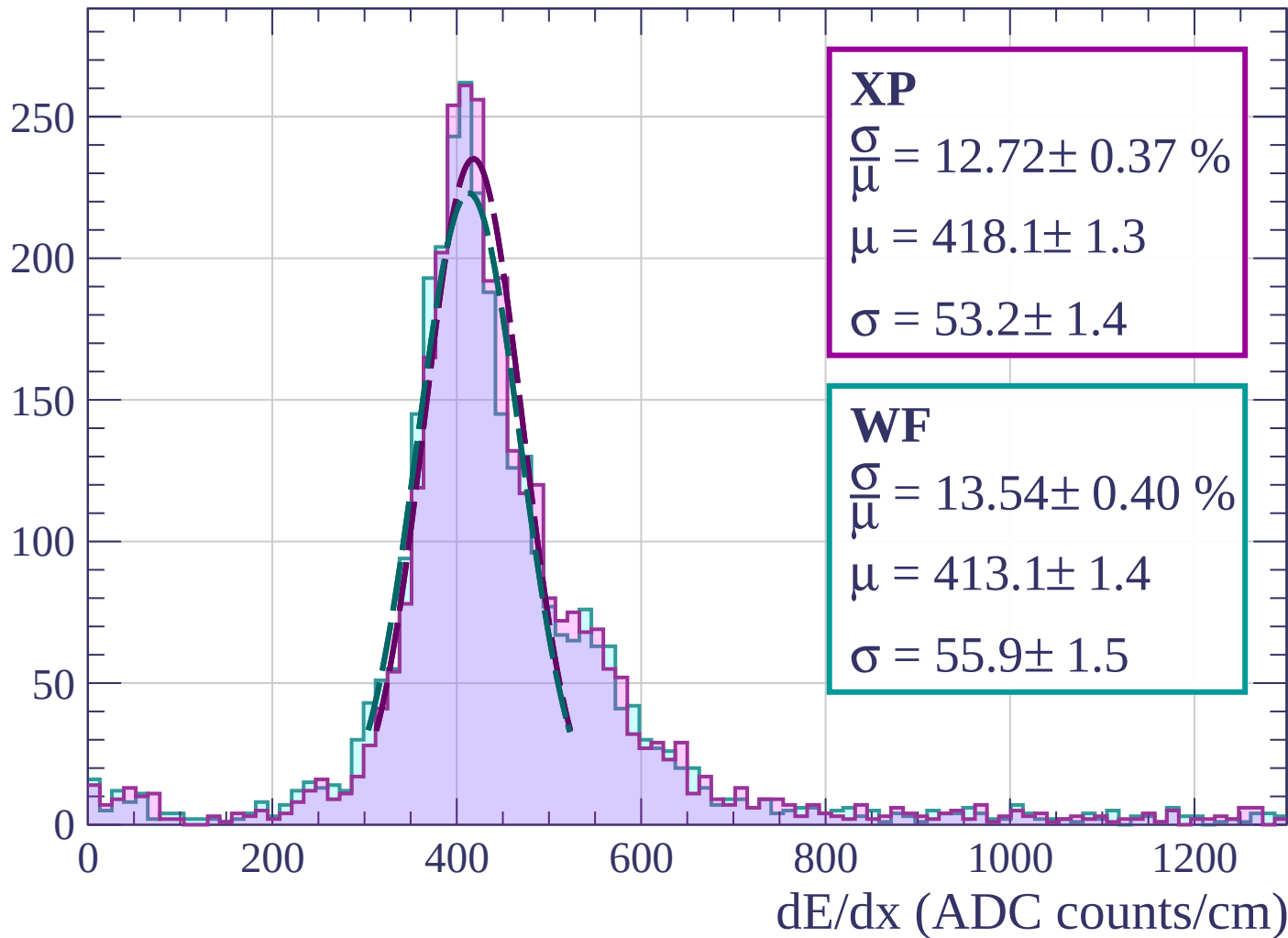
Energy loss | $48 < \phi < 51$

Count



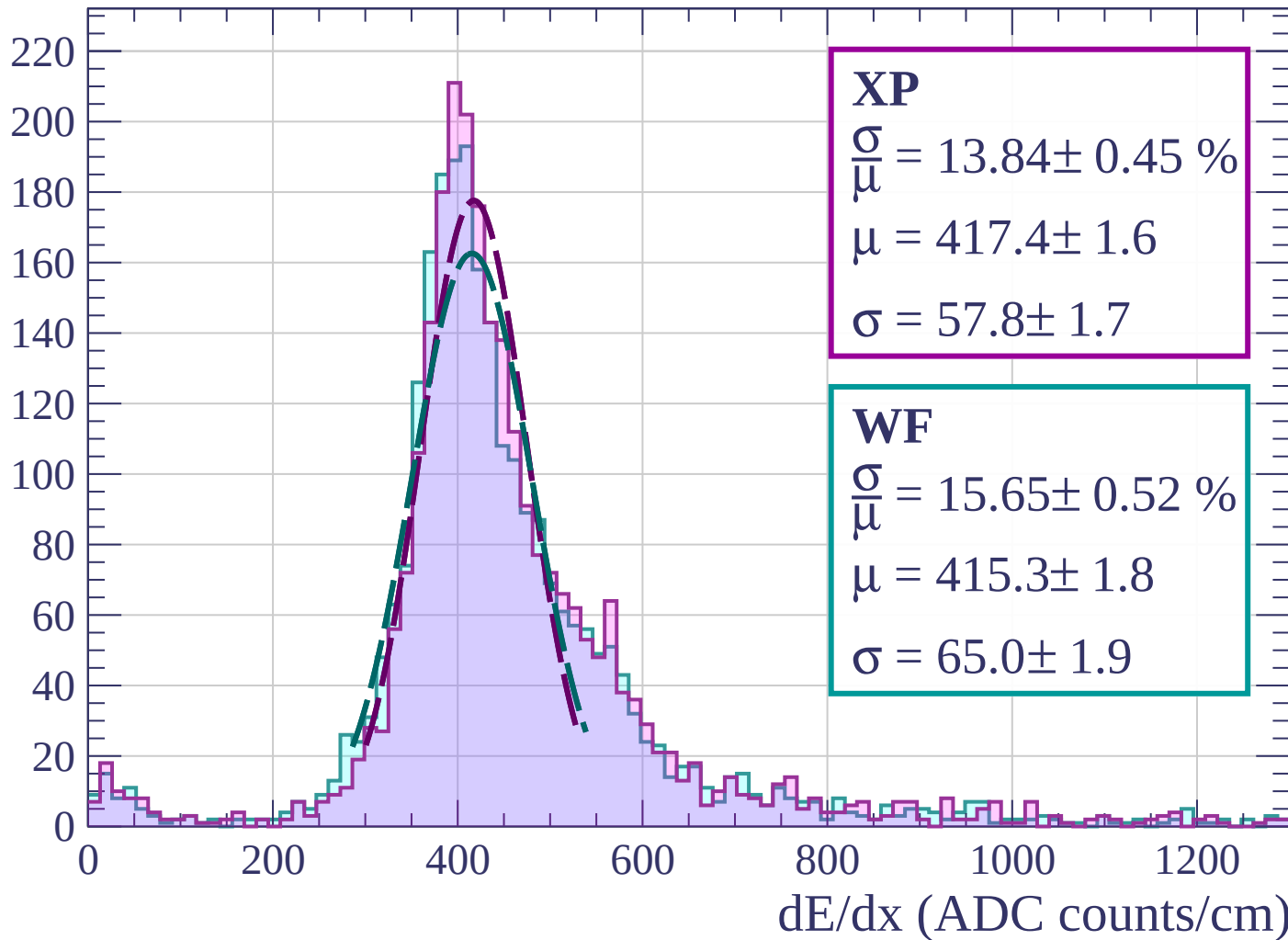
Energy loss | $51 < \phi < 54$

Count



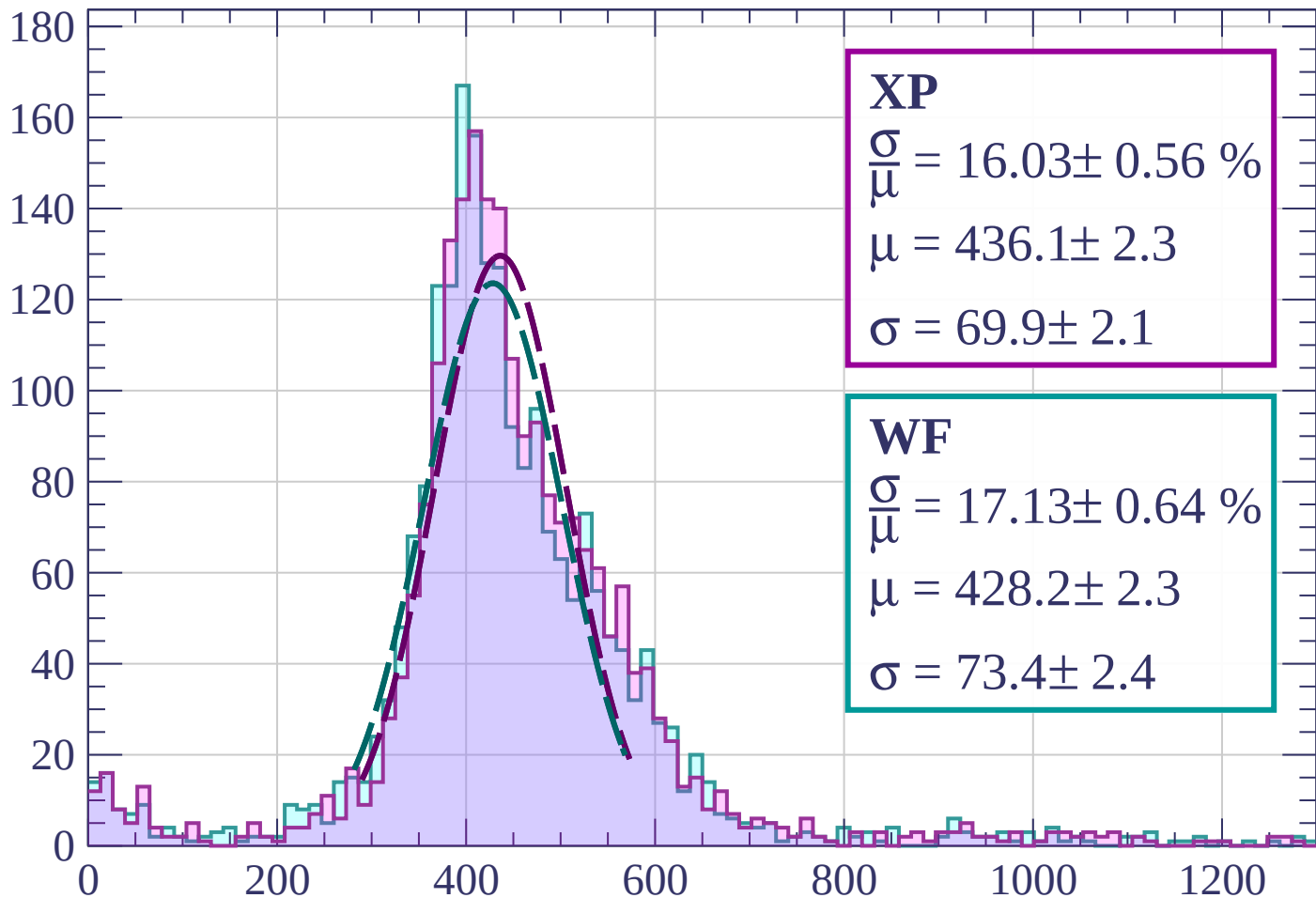
Energy loss | $54 < \phi < 57$

Count



Energy loss | $57 < \phi < 60$

Count



XP

$$\frac{\sigma}{\mu} = 16.03 \pm 0.56 \%$$

$$\mu = 436.1 \pm 2.3$$

$$\sigma = 69.9 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 17.13 \pm 0.64 \%$$

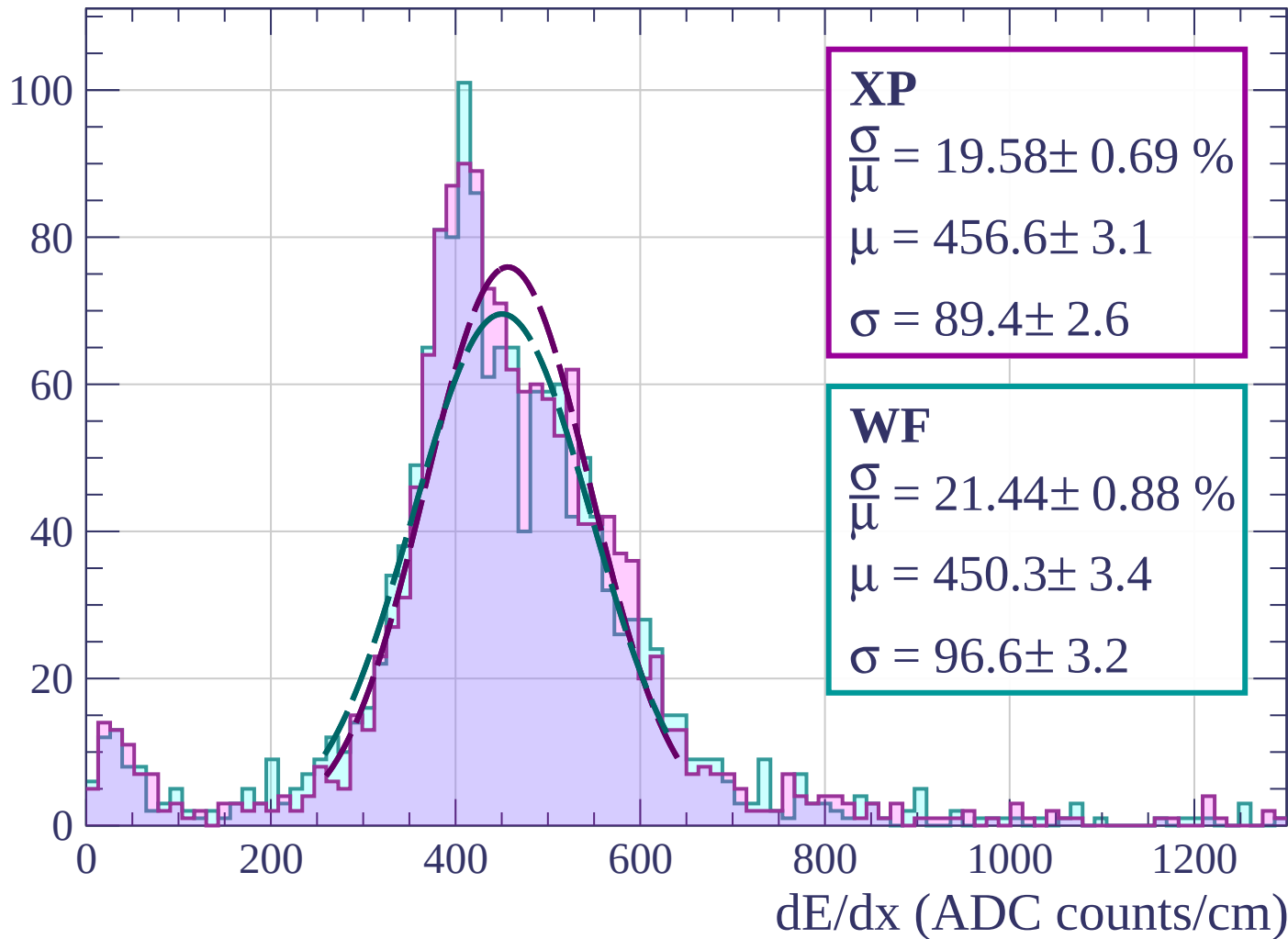
$$\mu = 428.2 \pm 2.3$$

$$\sigma = 73.4 \pm 2.4$$

dE/dx (ADC counts/cm)

Energy loss | $60 < \phi < 63$

Count



Energy loss | $63 < \phi < 66$

Count

XP

$$\frac{\sigma}{\mu} = 19.95 \pm 0.81 \%$$

$$\mu = 469.4 \pm 3.3$$

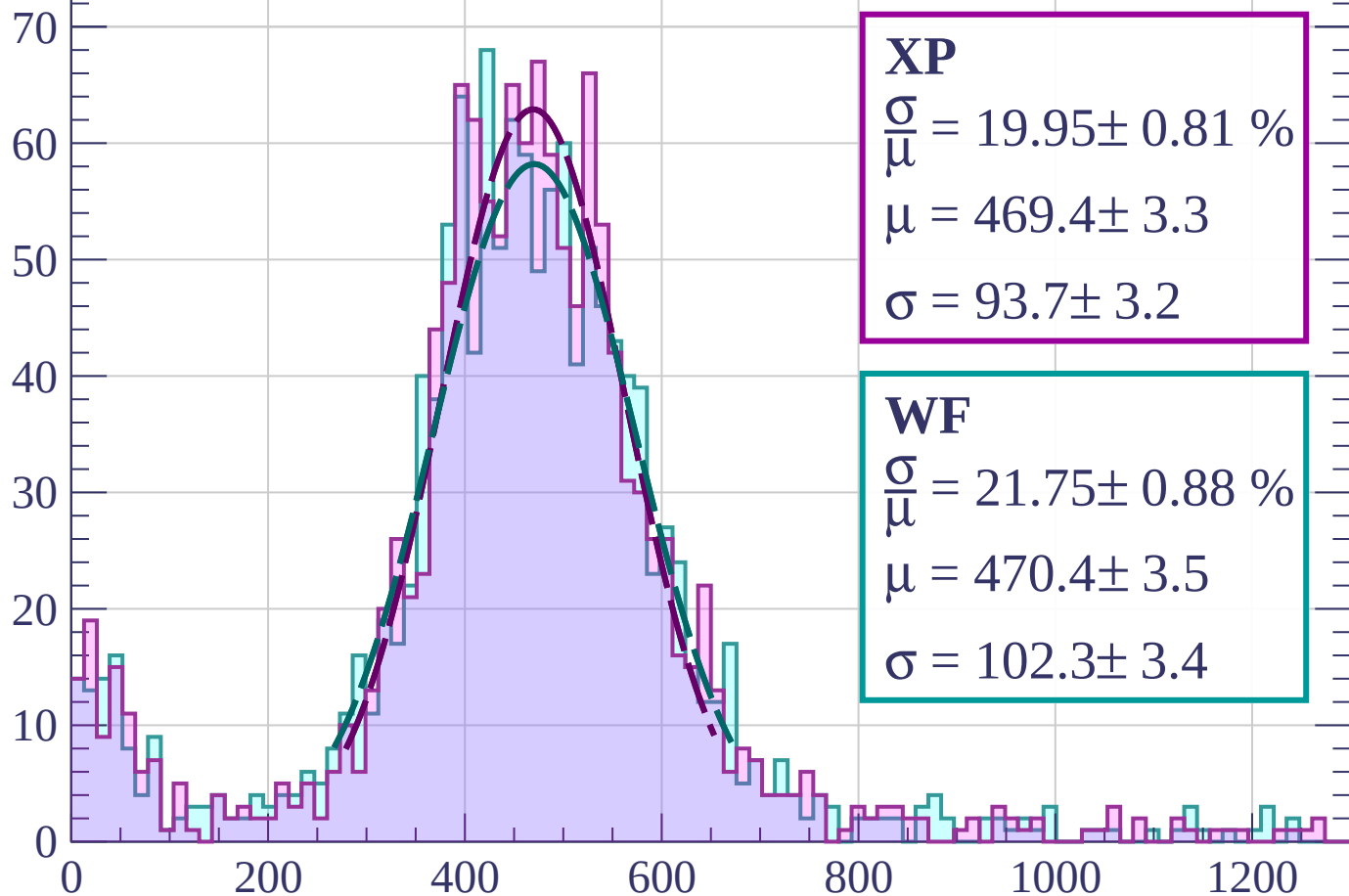
$$\sigma = 93.7 \pm 3.2$$

WF

$$\frac{\sigma}{\mu} = 21.75 \pm 0.88 \%$$

$$\mu = 470.4 \pm 3.5$$

$$\sigma = 102.3 \pm 3.4$$



dE/dx (ADC counts/cm)

Energy loss | $66 < \phi < 69$

Count

XP

$$\frac{\sigma}{\mu} = 19.64 \pm 0.72 \%$$

$$\mu = 466.3 \pm 3.0$$

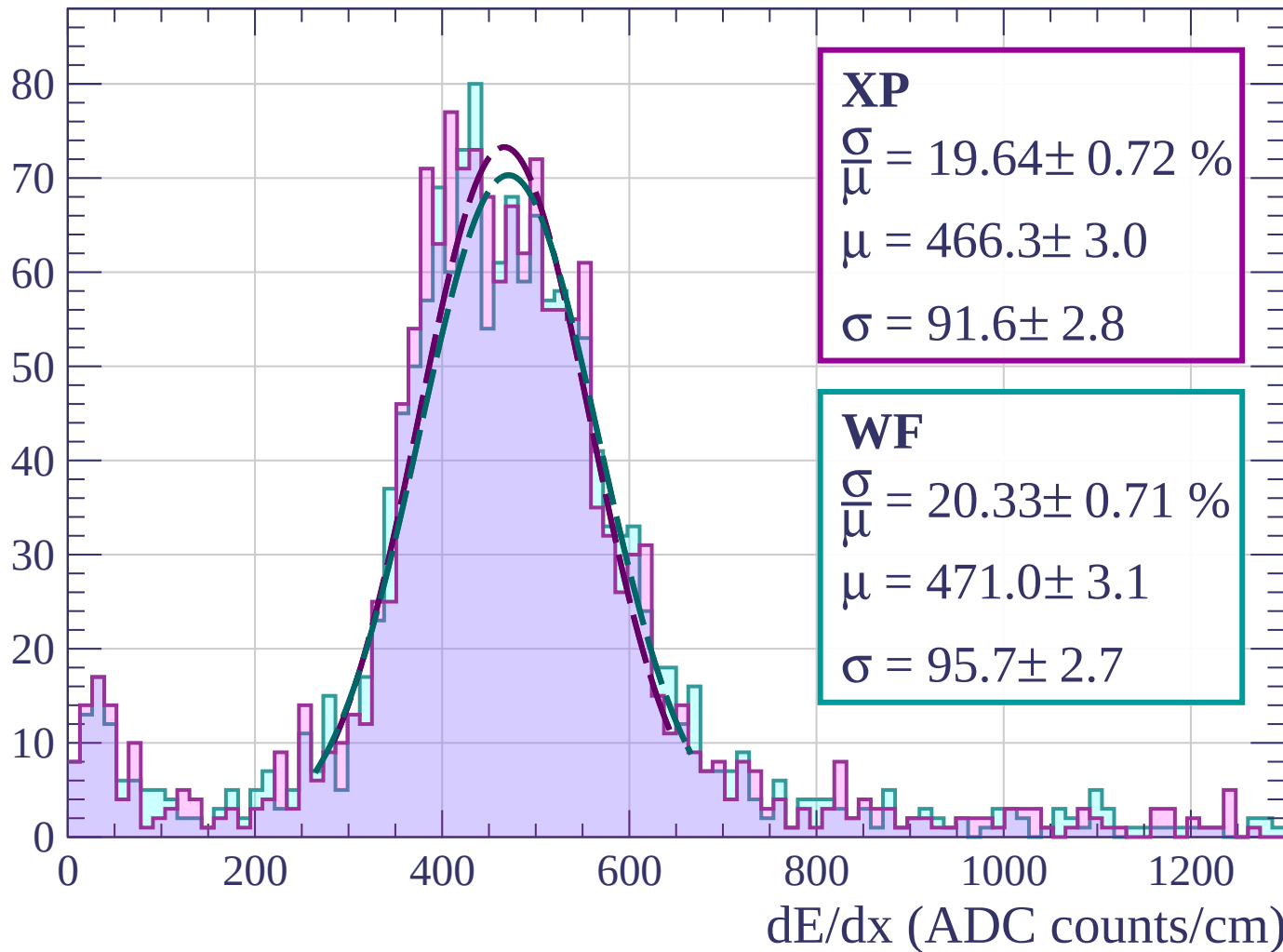
$$\sigma = 91.6 \pm 2.8$$

WF

$$\frac{\sigma}{\mu} = 20.33 \pm 0.71 \%$$

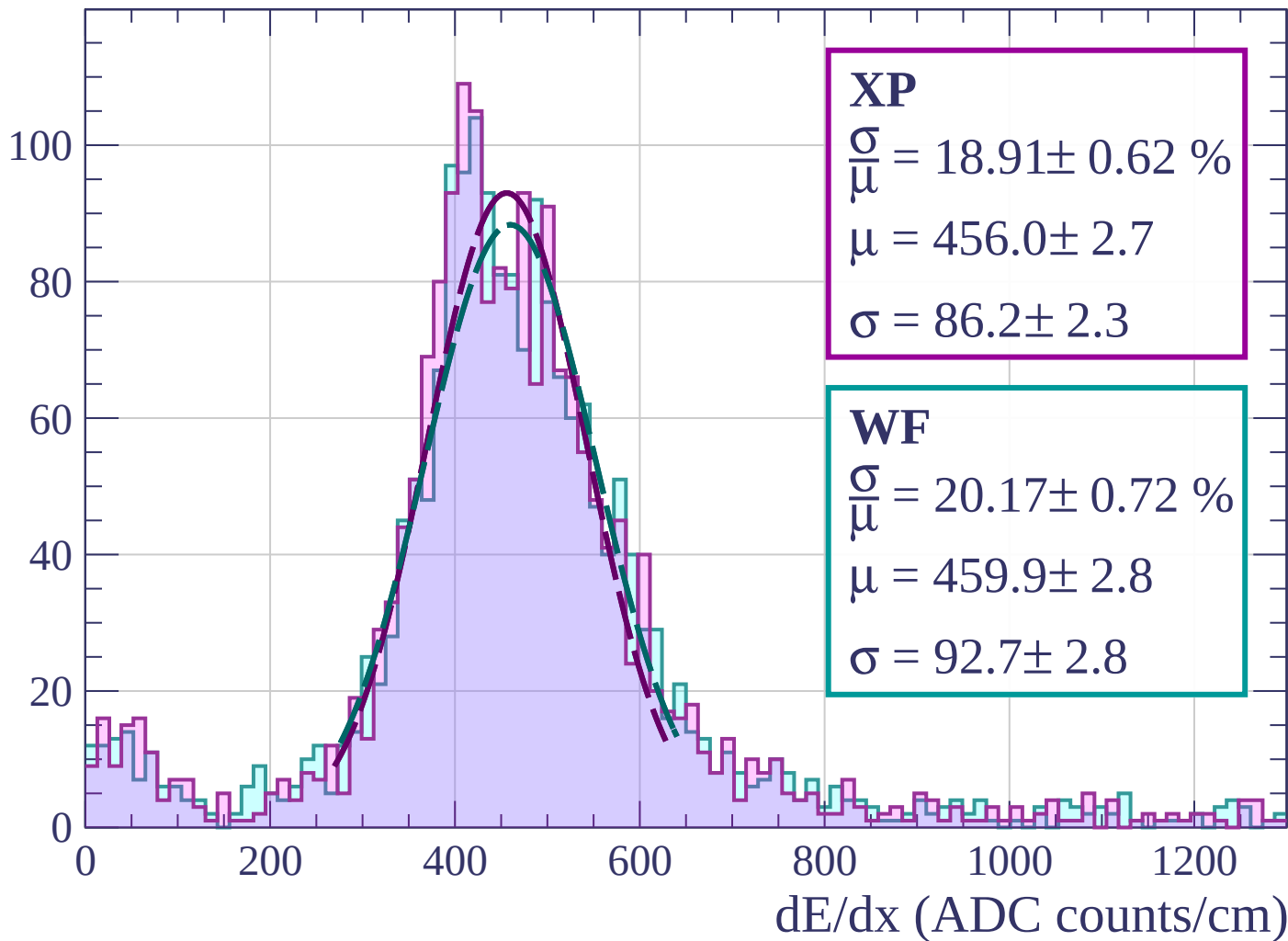
$$\mu = 471.0 \pm 3.1$$

$$\sigma = 95.7 \pm 2.7$$



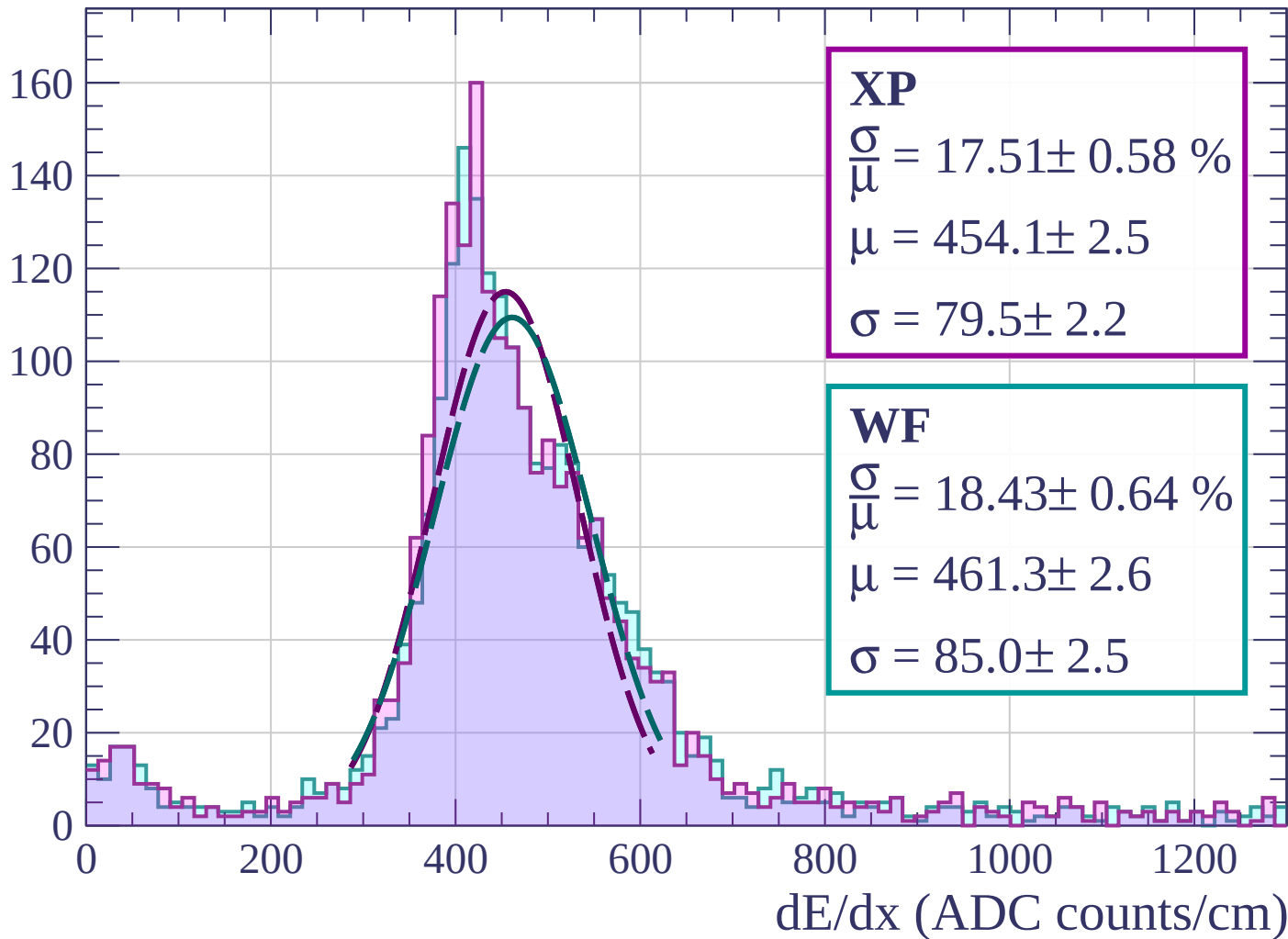
Energy loss | $69 < \phi < 72$

Count



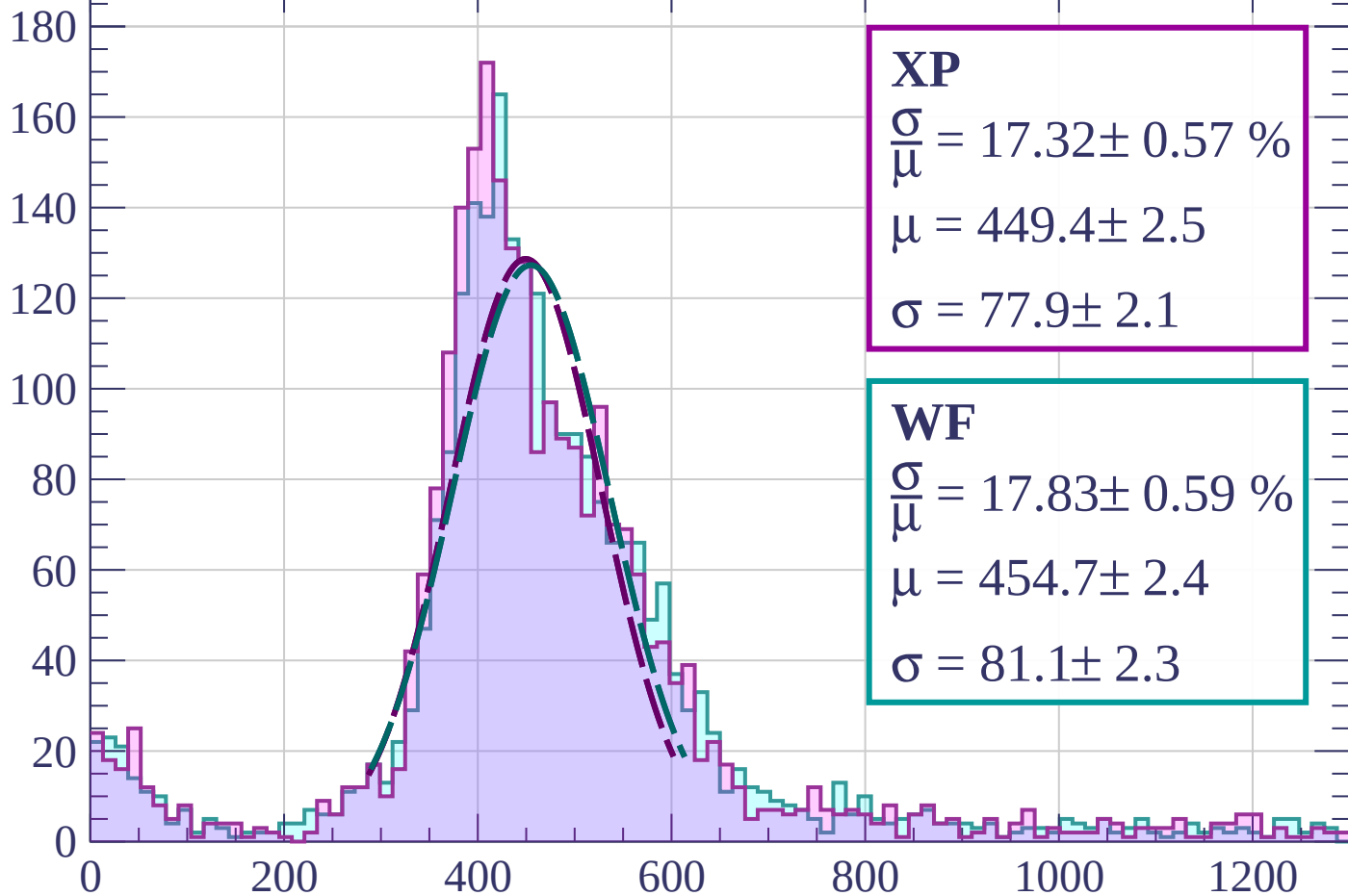
Energy loss | $72 < \phi < 75$

Count



Energy loss | $75 < \phi < 78$

Count



XP

$$\frac{\sigma}{\mu} = 17.32 \pm 0.57 \%$$

$$\mu = 449.4 \pm 2.5$$

$$\sigma = 77.9 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 17.83 \pm 0.59 \%$$

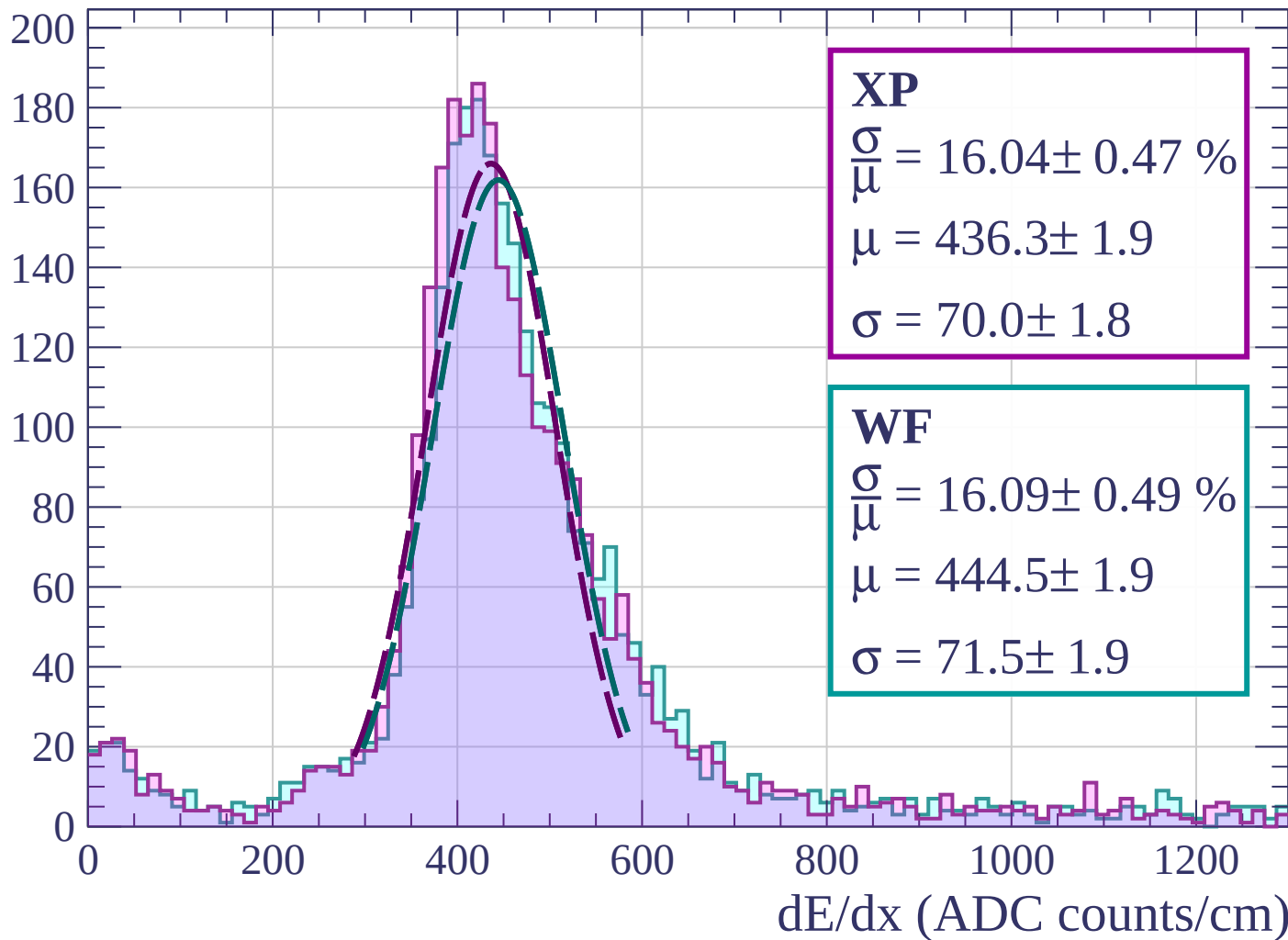
$$\mu = 454.7 \pm 2.4$$

$$\sigma = 81.1 \pm 2.3$$

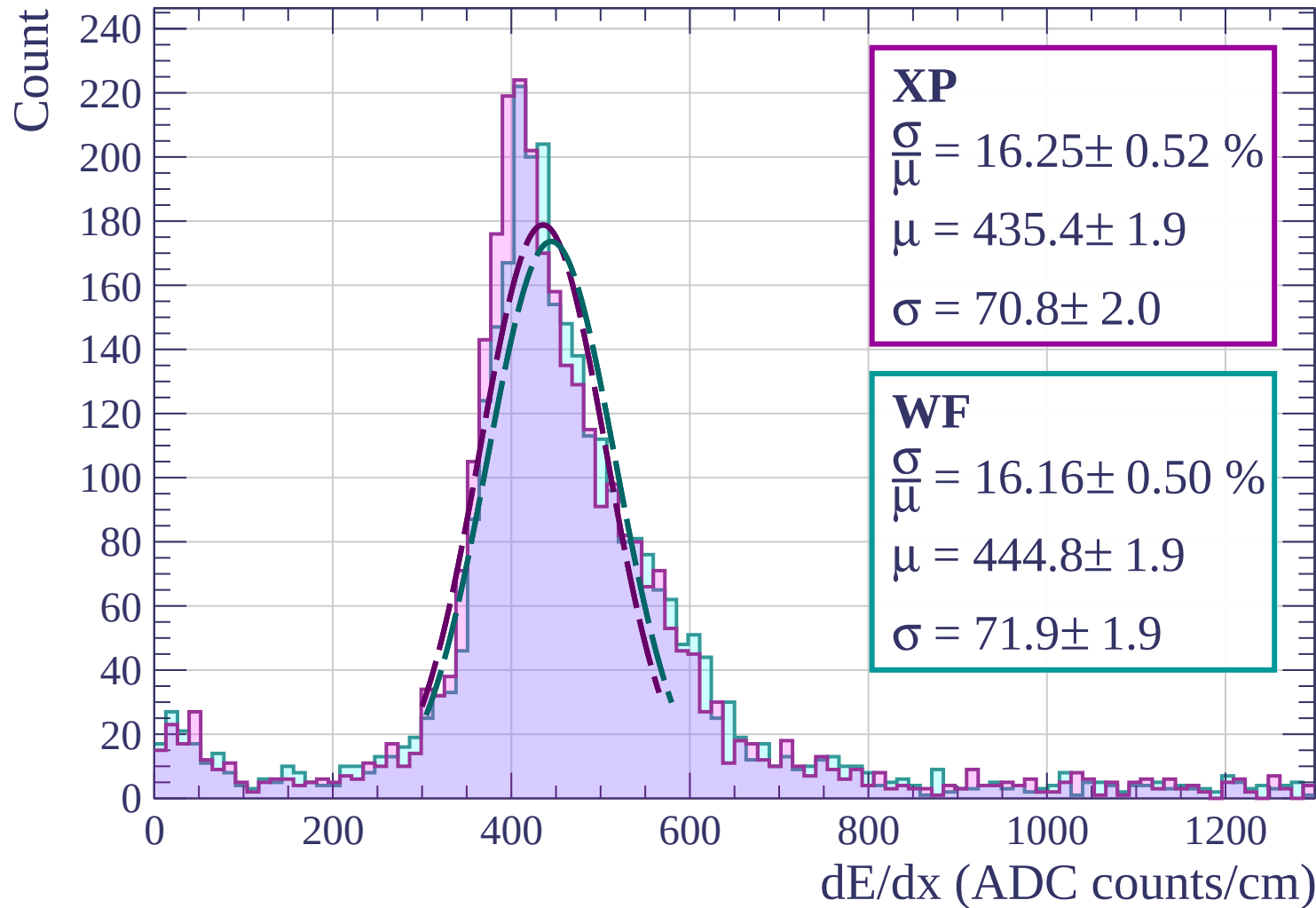
dE/dx (ADC counts/cm)

Energy loss | $78 < \phi < 81$

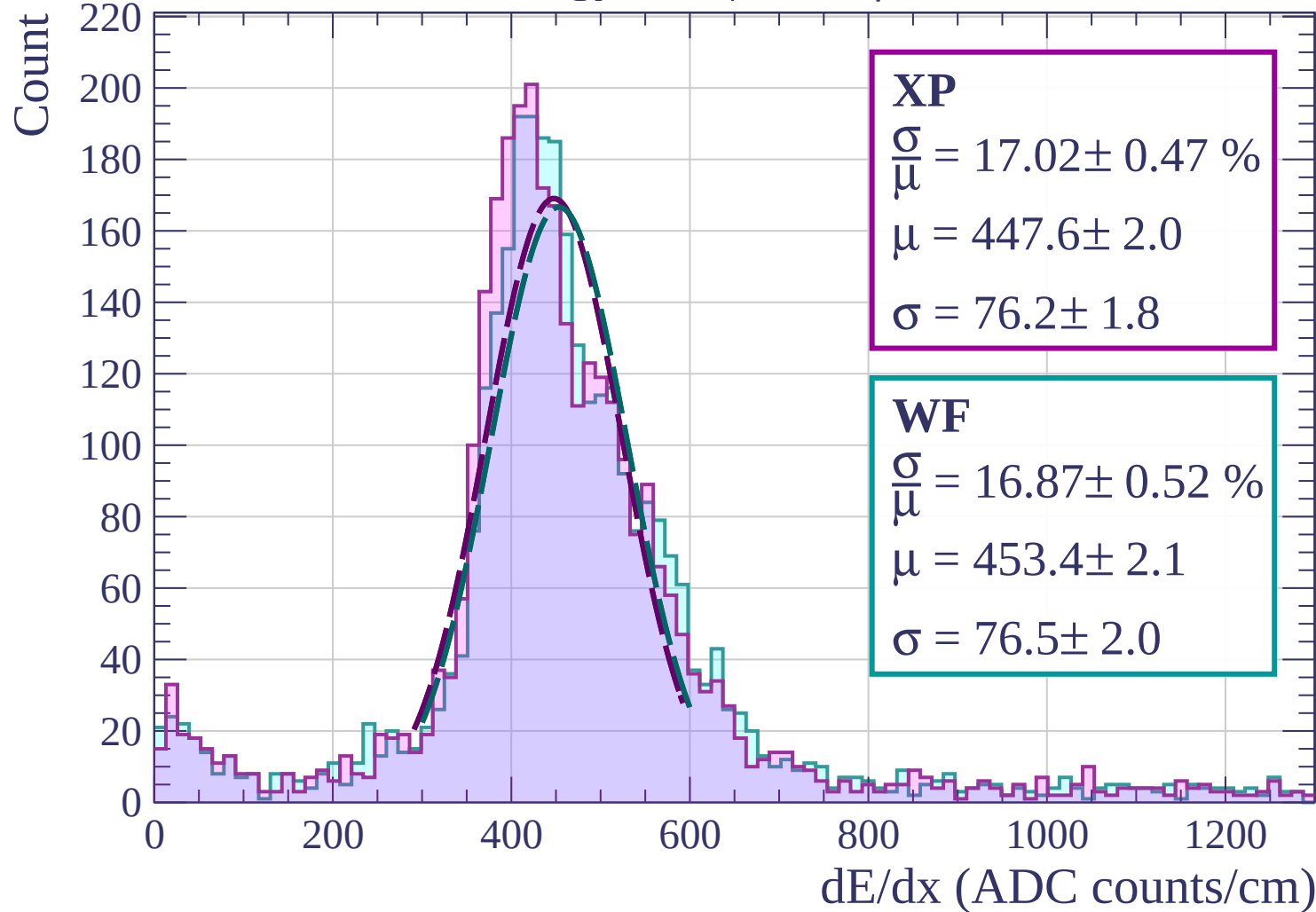
Count



Energy loss | $81 < \phi < 84$

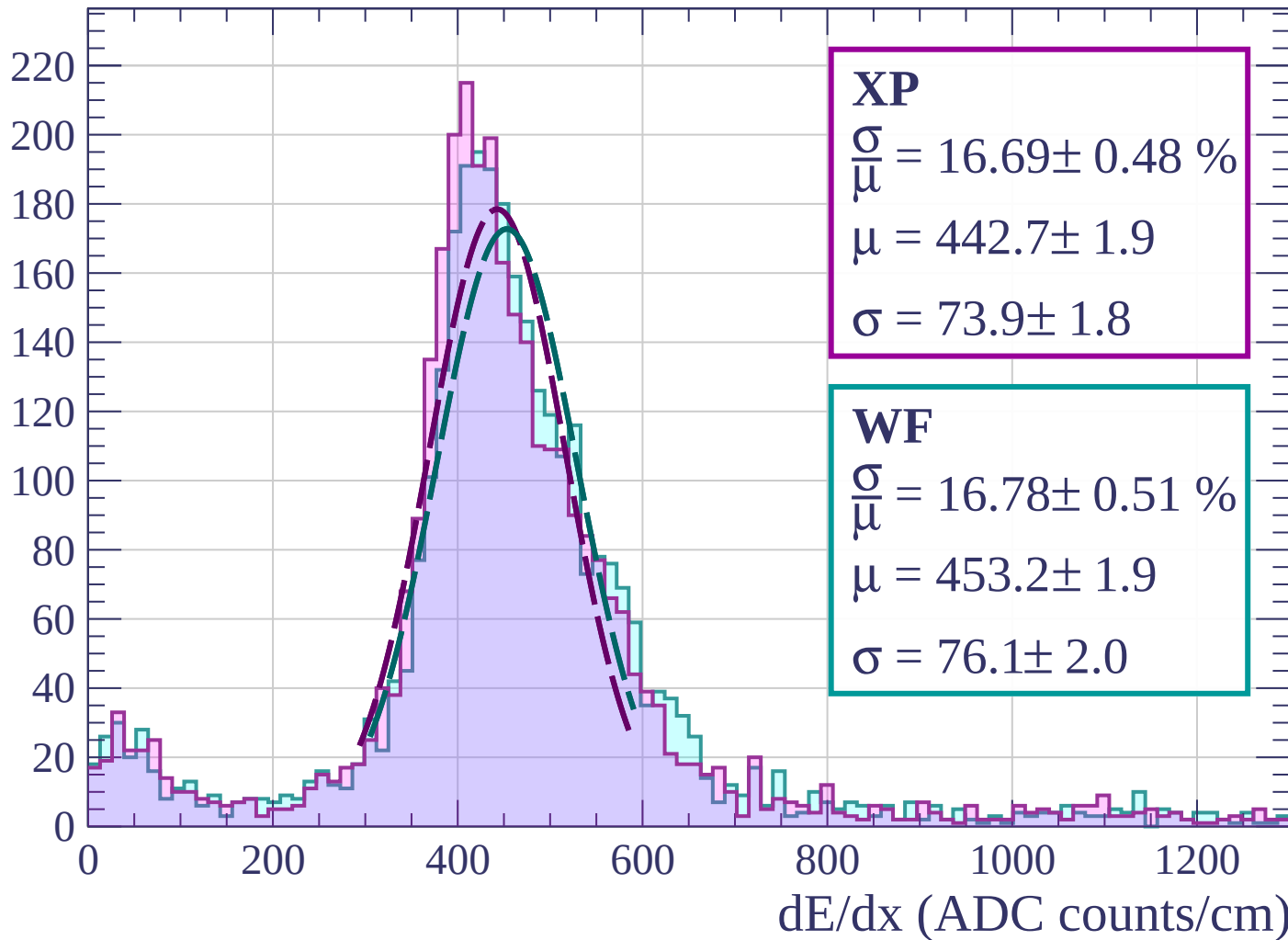


Energy loss | $84 < \phi < 87$

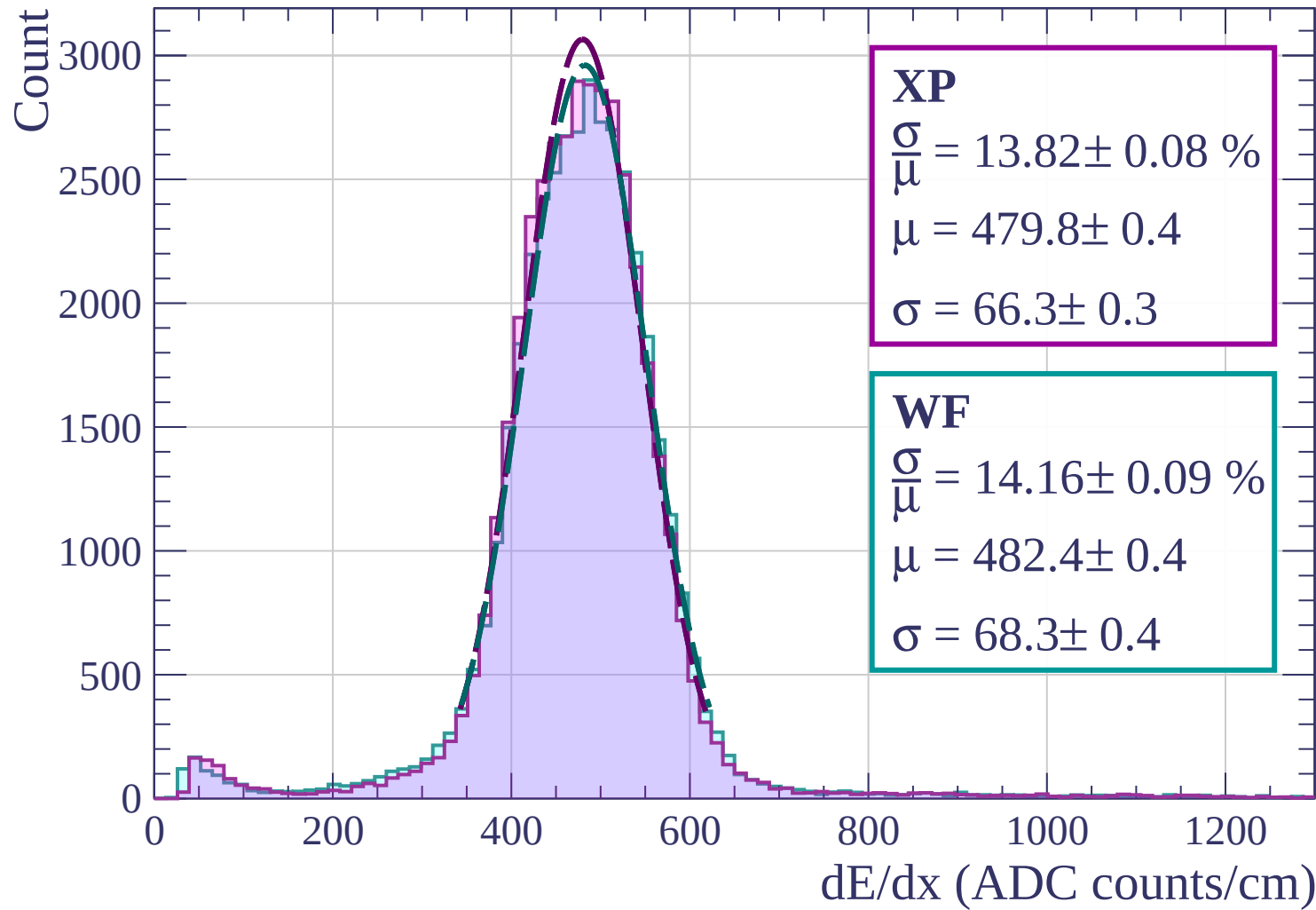


Energy loss | $87 < \phi < 90$

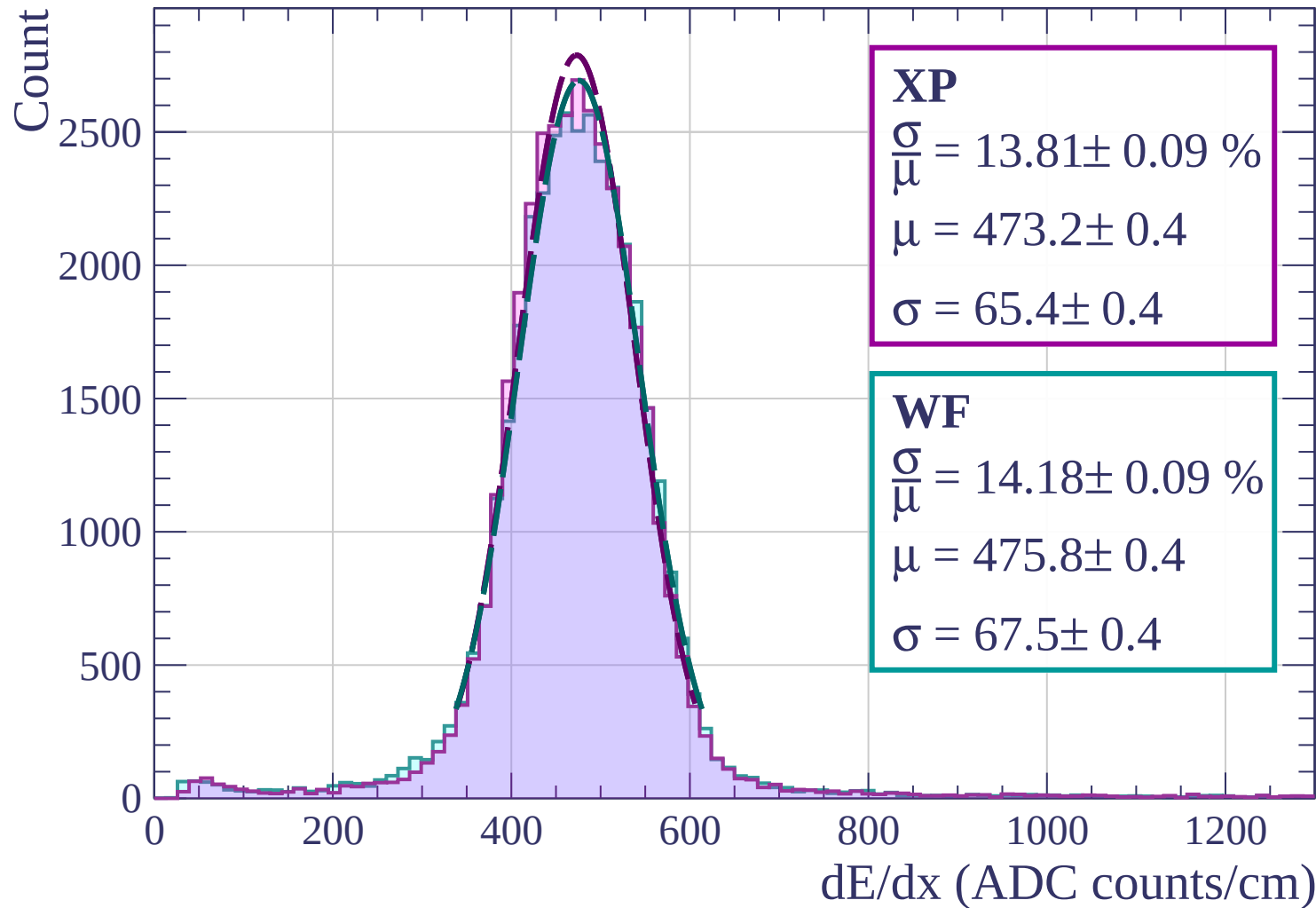
Count



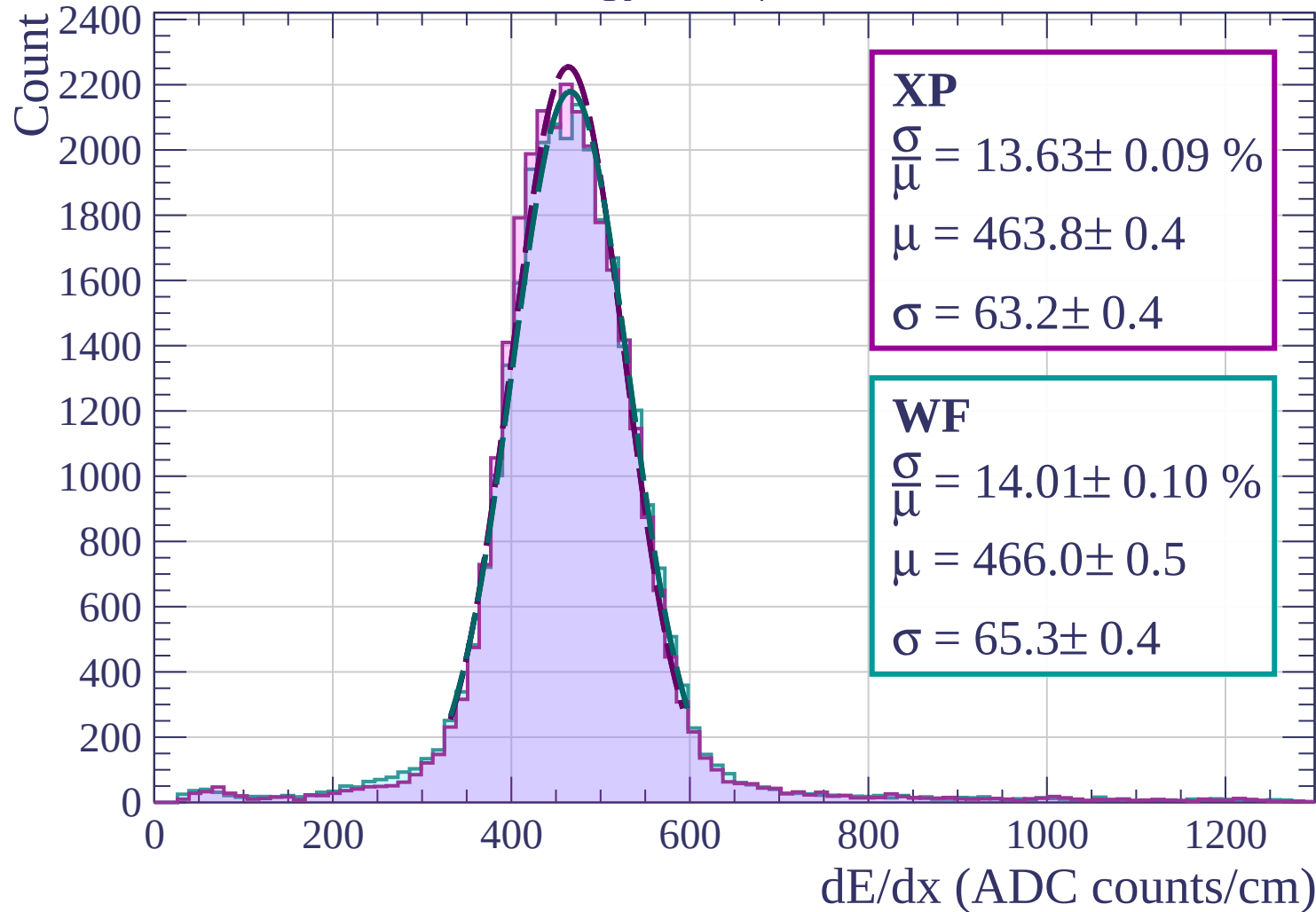
Energy loss | $0 < \theta < 3$



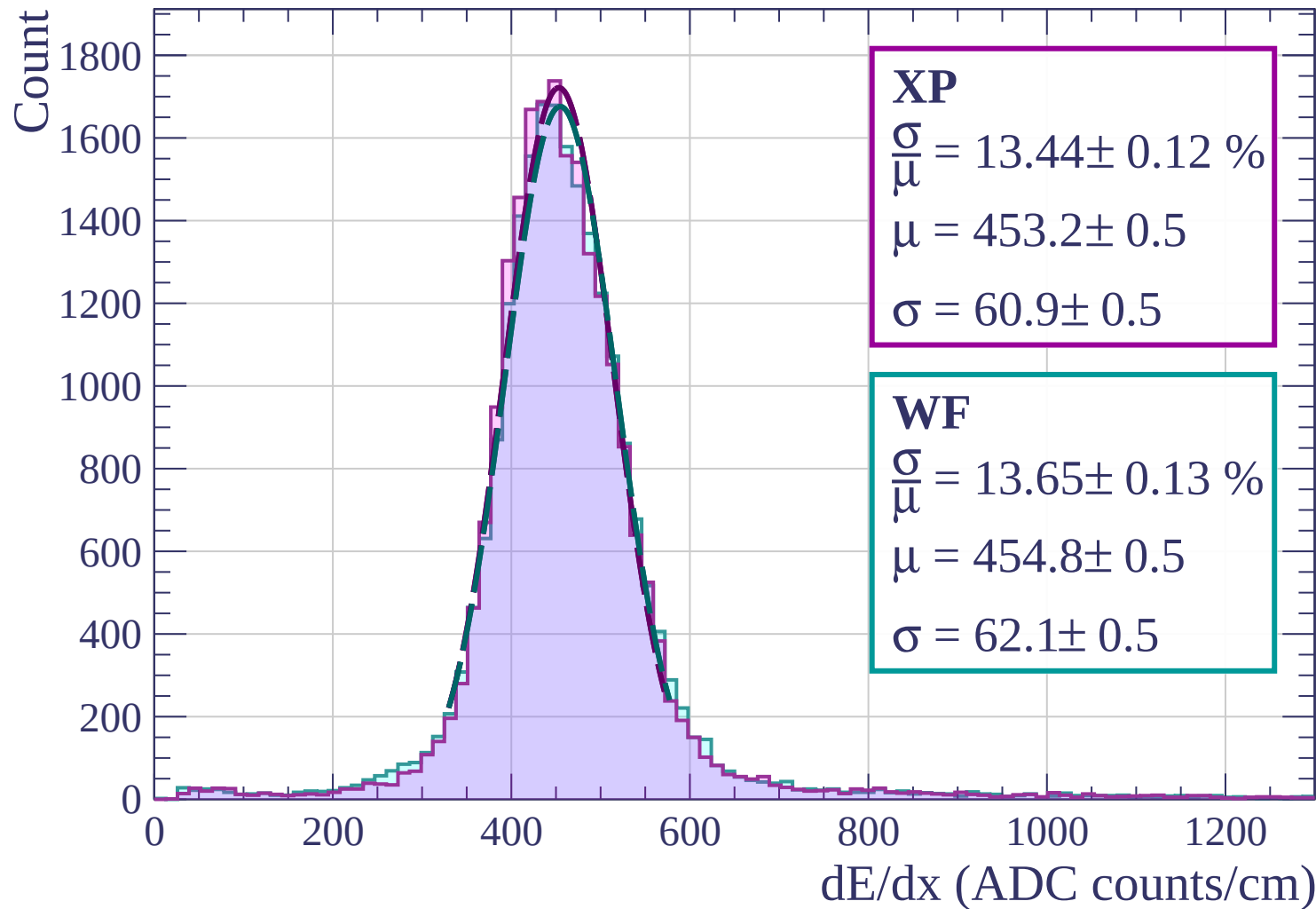
Energy loss | $3 < \theta < 6$



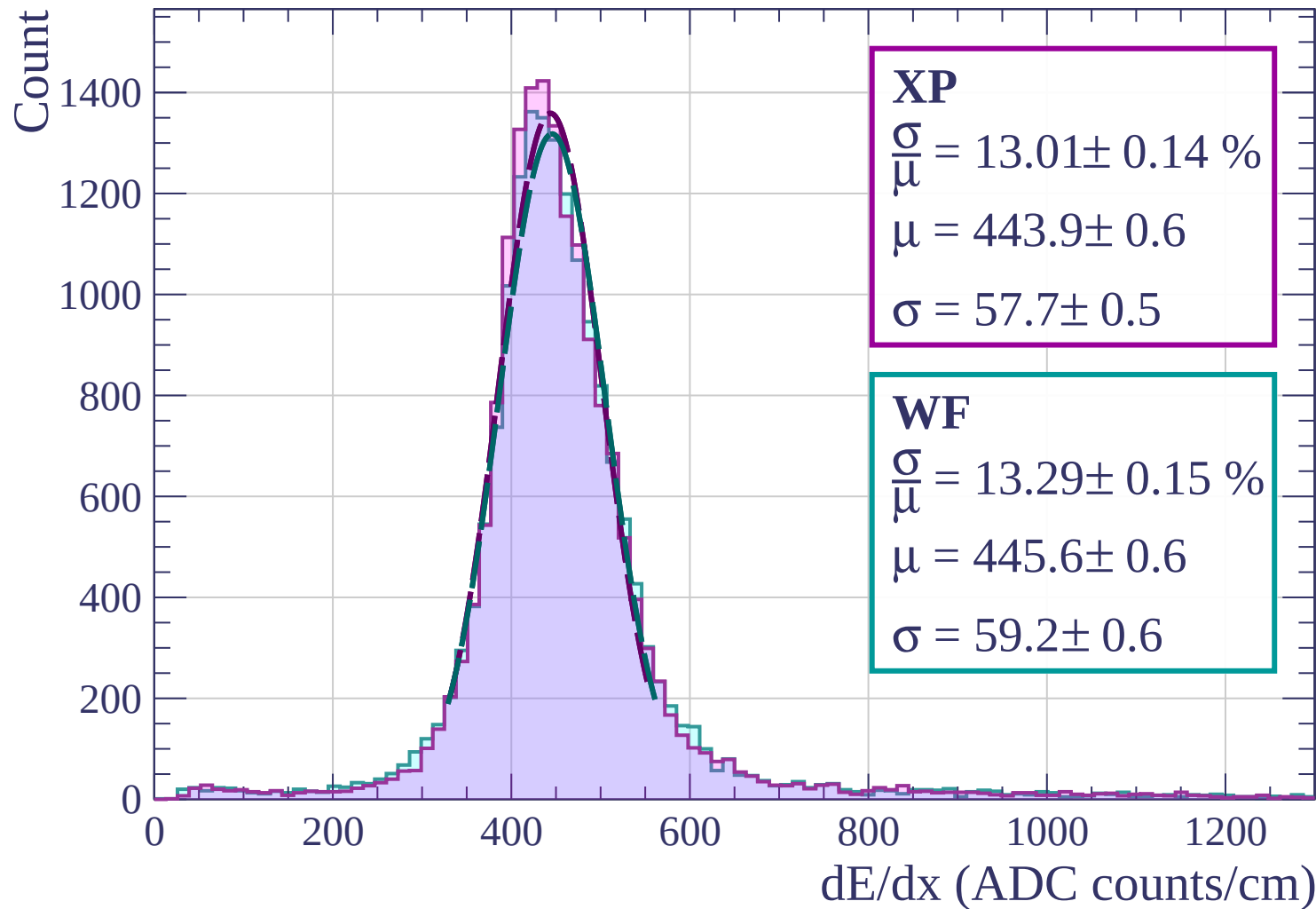
Energy loss | $6 < \theta < 9$



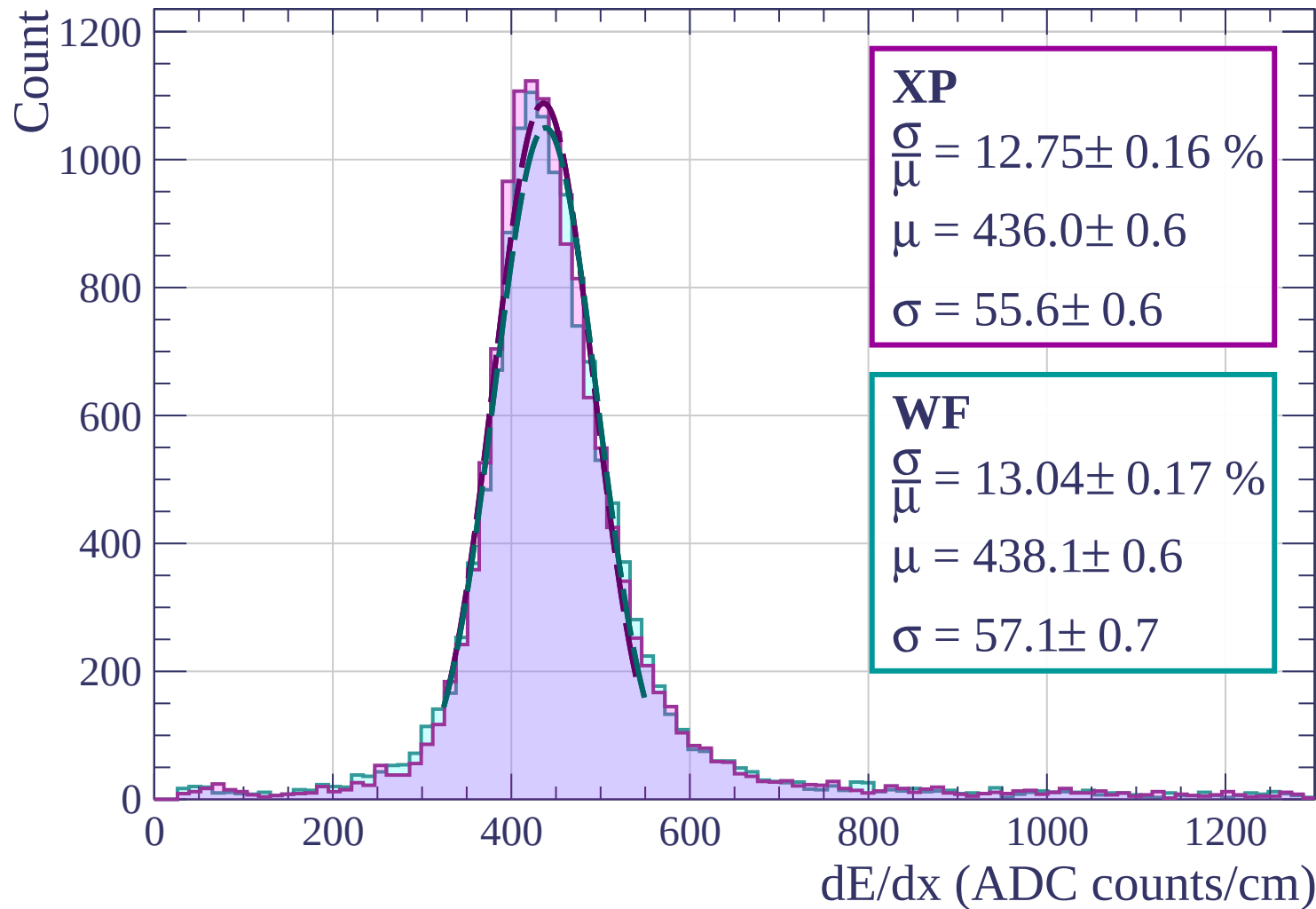
Energy loss | $9 < \theta < 12$



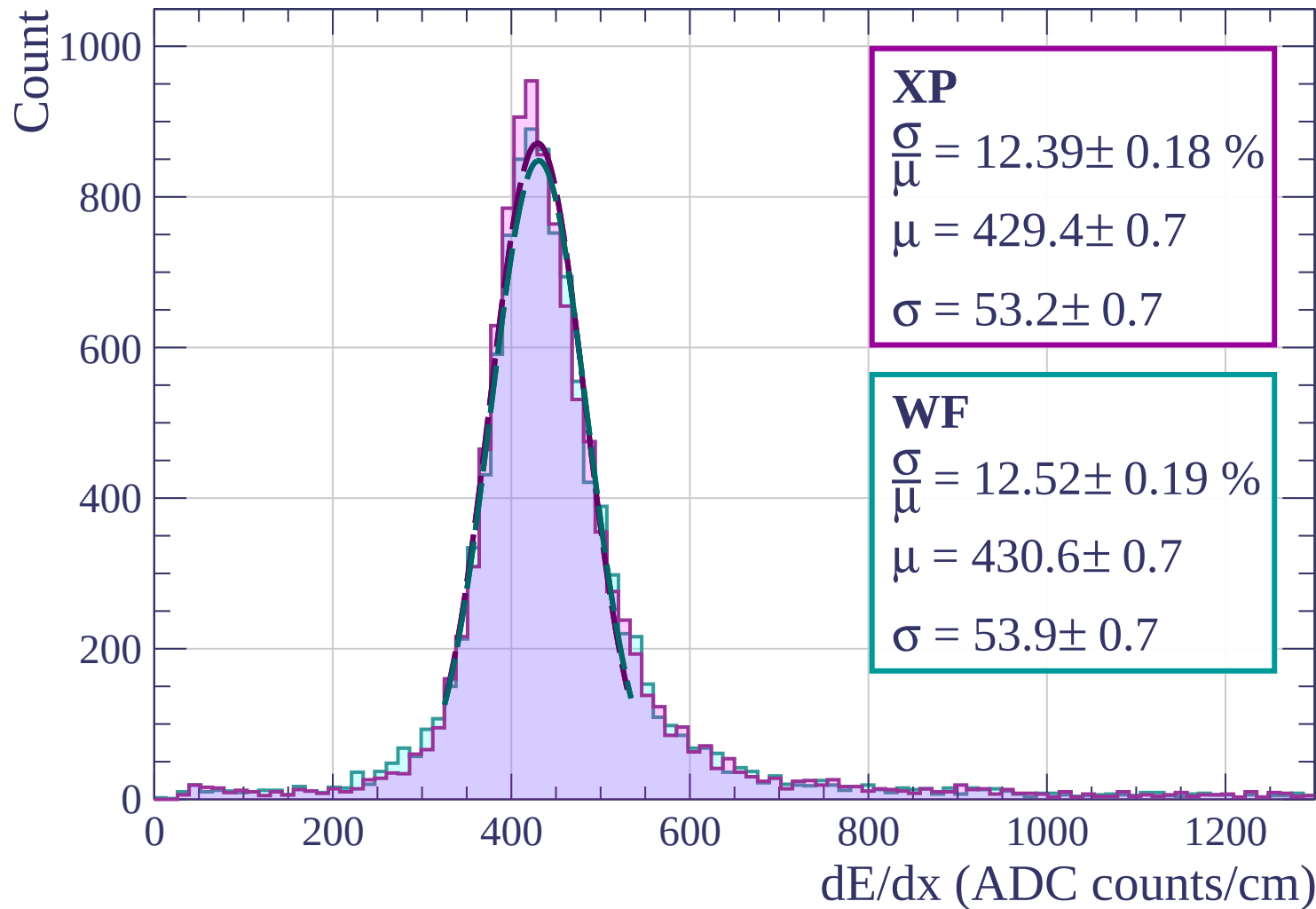
Energy loss | $12 < \theta < 15$



Energy loss | $15 < \theta < 18$

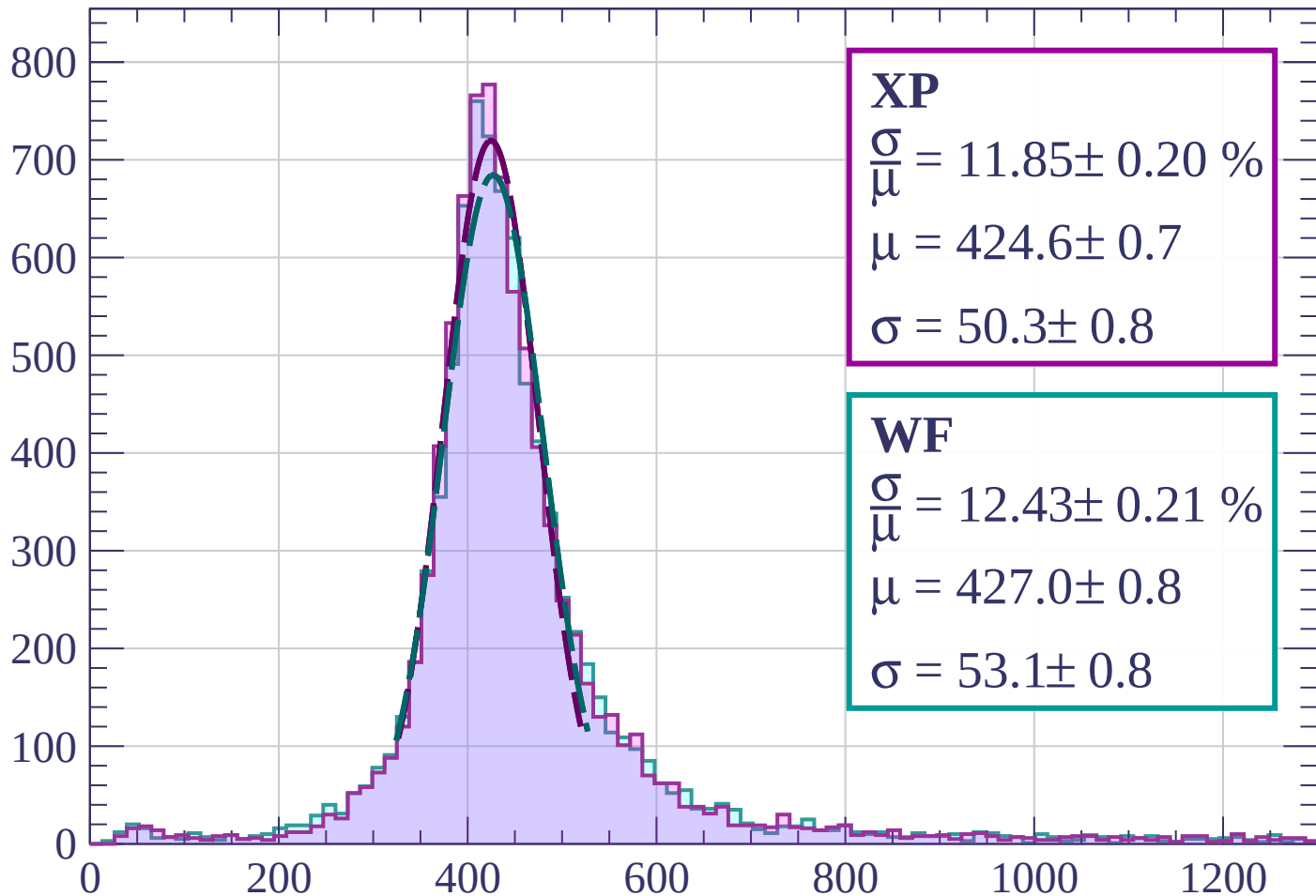


Energy loss | $18 < \theta < 21$



Energy loss | $21 < \theta < 24$

Count



XP

$$\frac{\sigma}{\mu} = 11.85 \pm 0.20 \%$$

$$\mu = 424.6 \pm 0.7$$

$$\sigma = 50.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 12.43 \pm 0.21 \%$$

$$\mu = 427.0 \pm 0.8$$

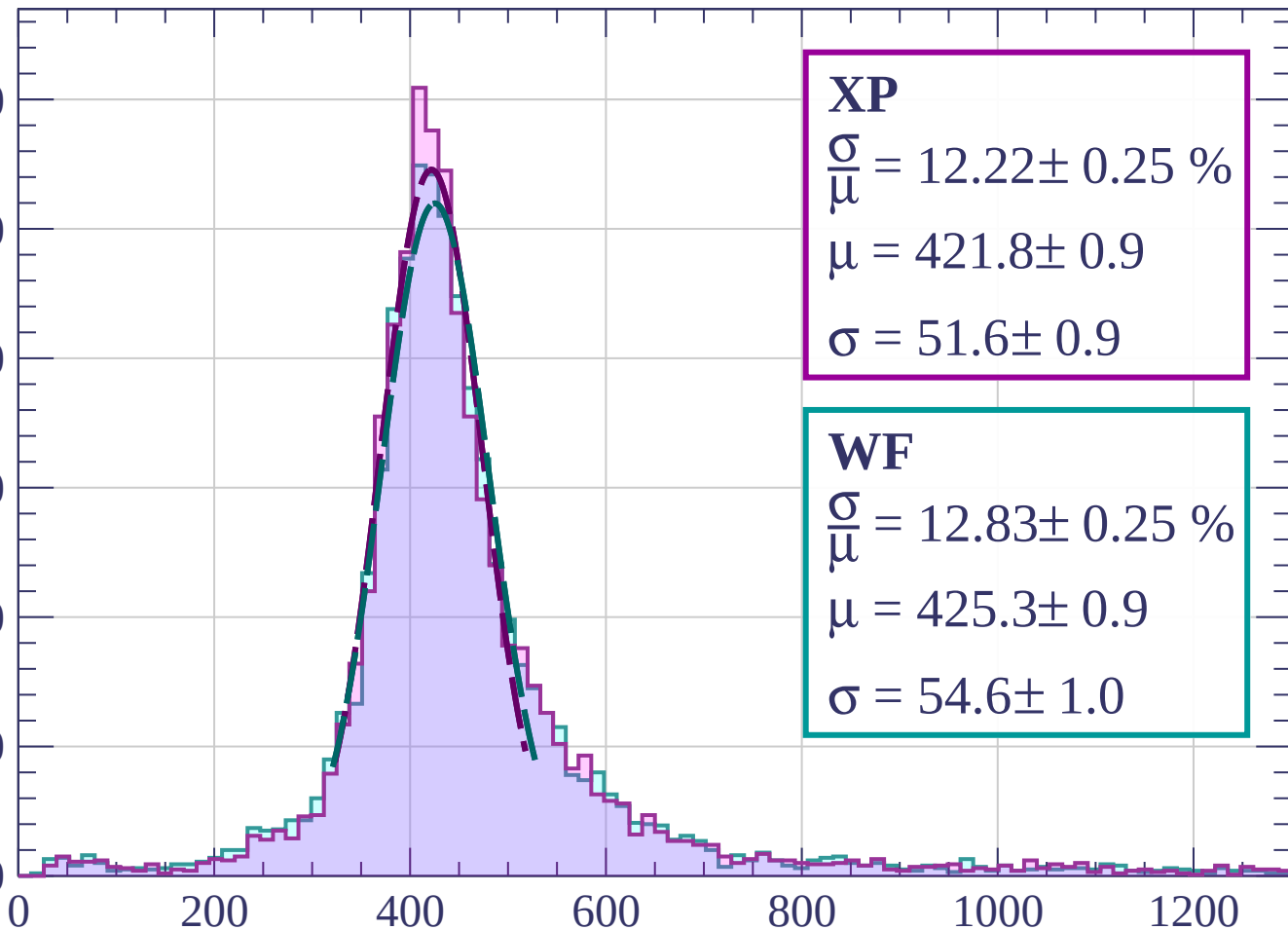
$$\sigma = 53.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $24 < \theta < 27$

Count

600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 12.22 \pm 0.25 \%$$

$$\mu = 421.8 \pm 0.9$$

$$\sigma = 51.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 12.83 \pm 0.25 \%$$

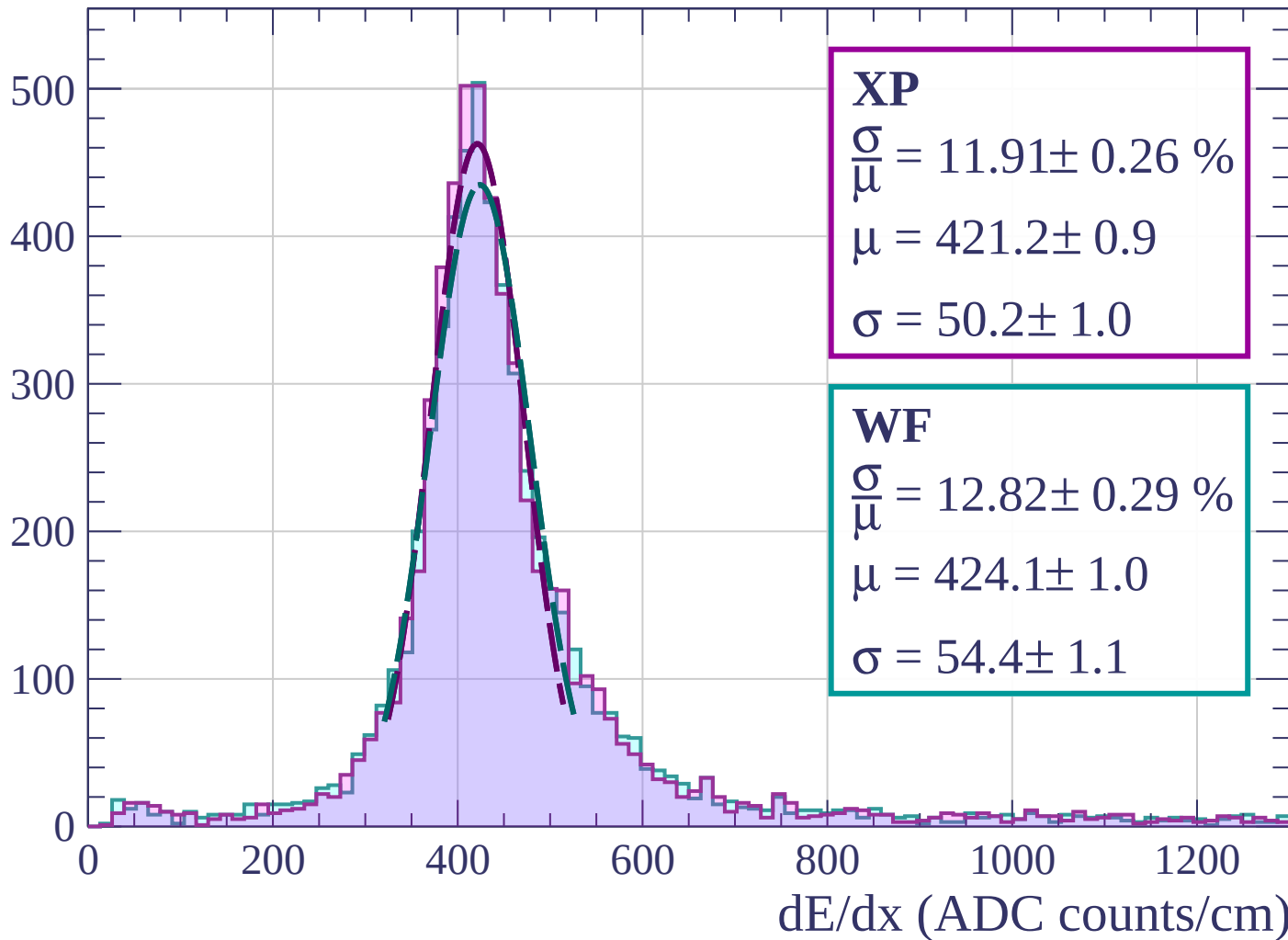
$$\mu = 425.3 \pm 0.9$$

$$\sigma = 54.6 \pm 1.0$$

dE/dx (ADC counts/cm)

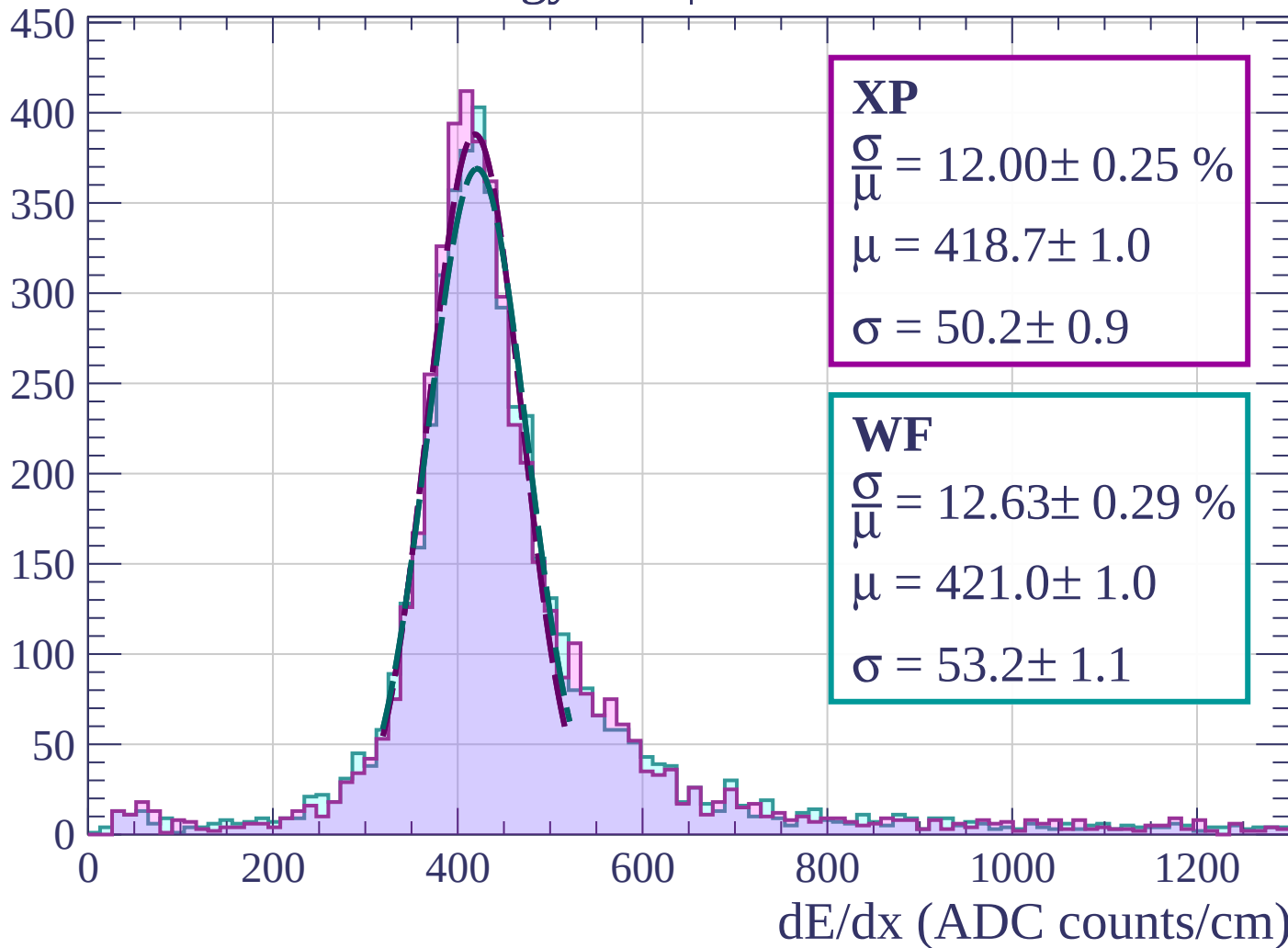
Energy loss | $27 < \theta < 30$

Count



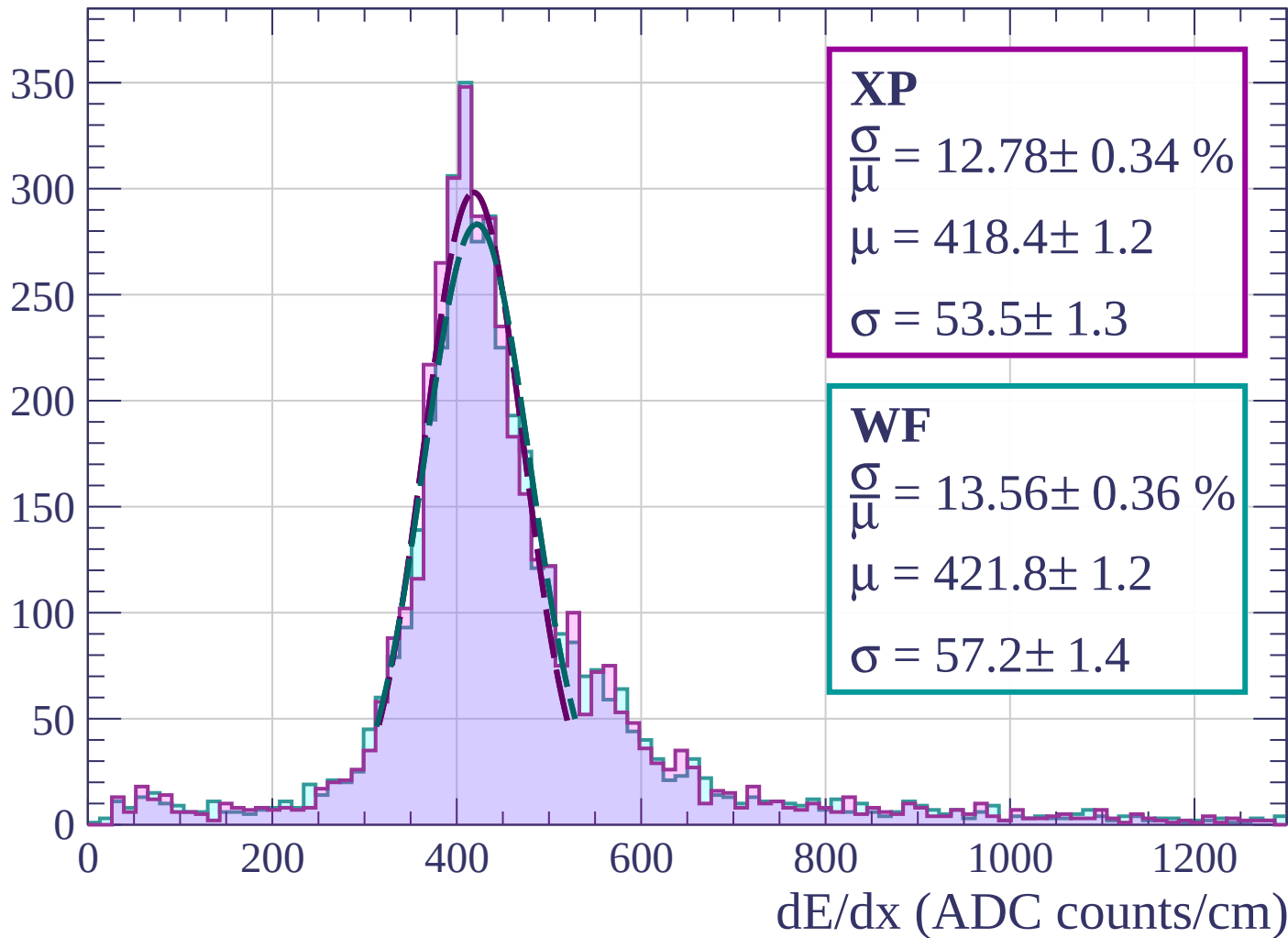
Energy loss | $30 < \theta < 33$

Count

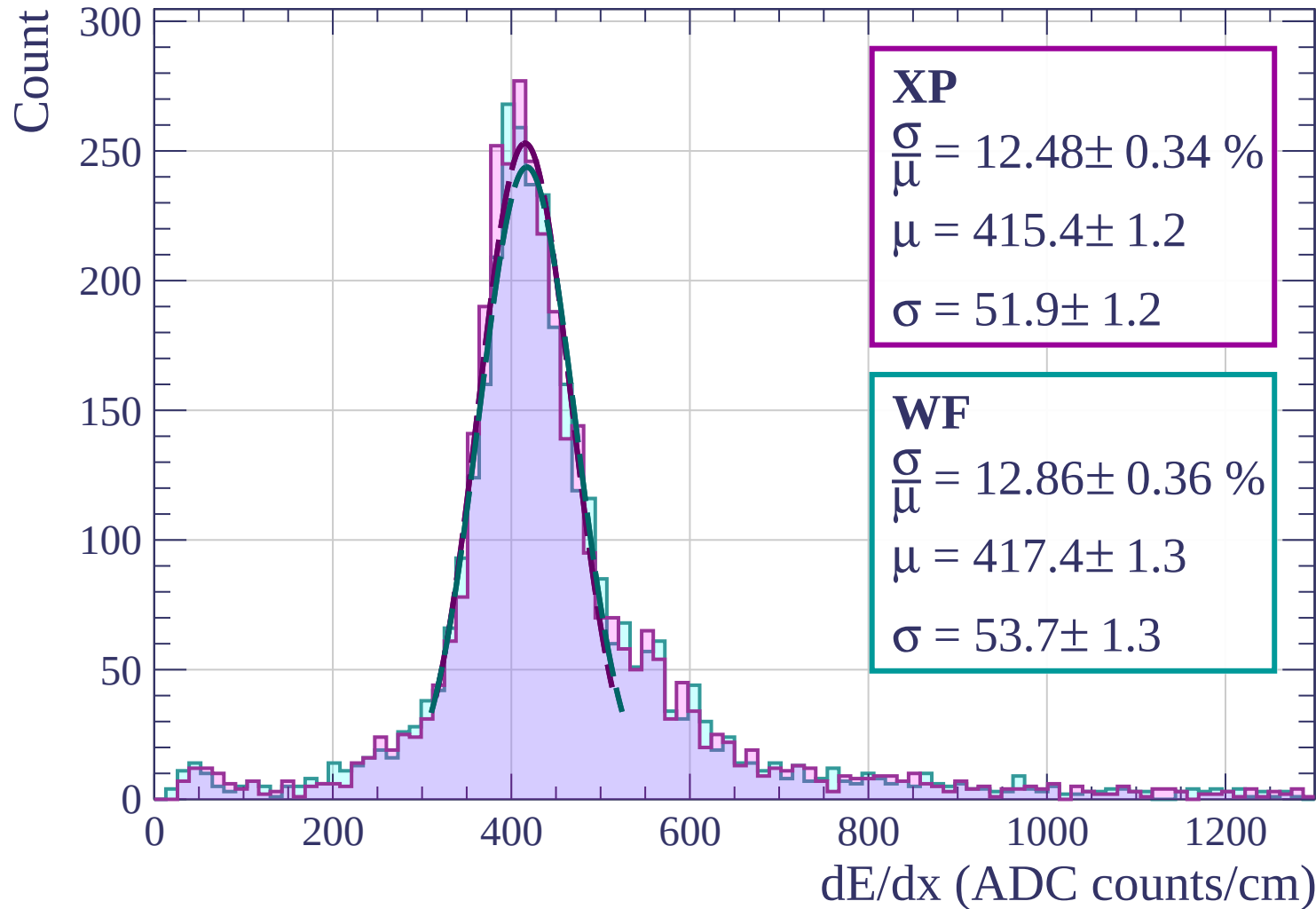


Energy loss | $33 < \theta < 36$

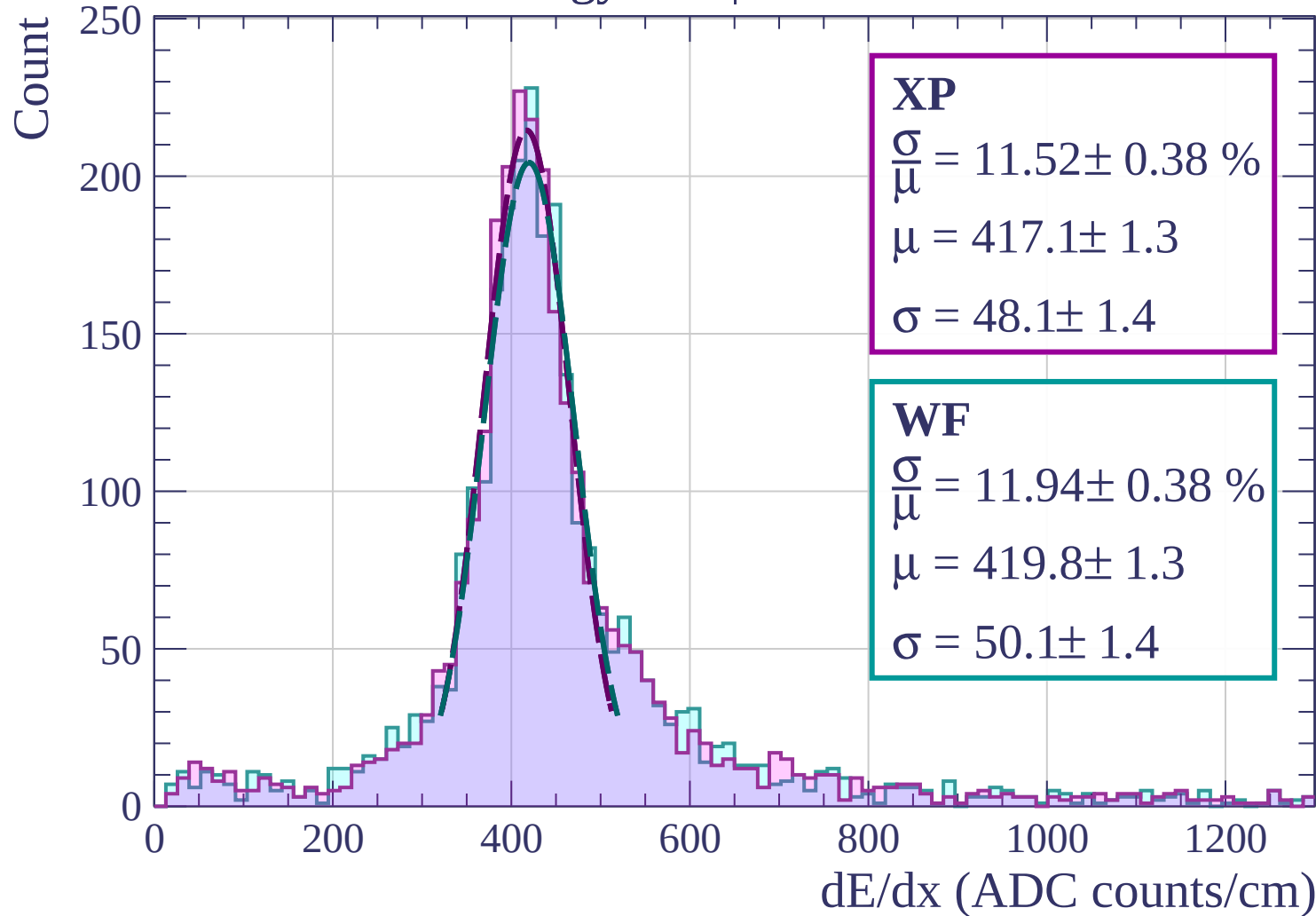
Count



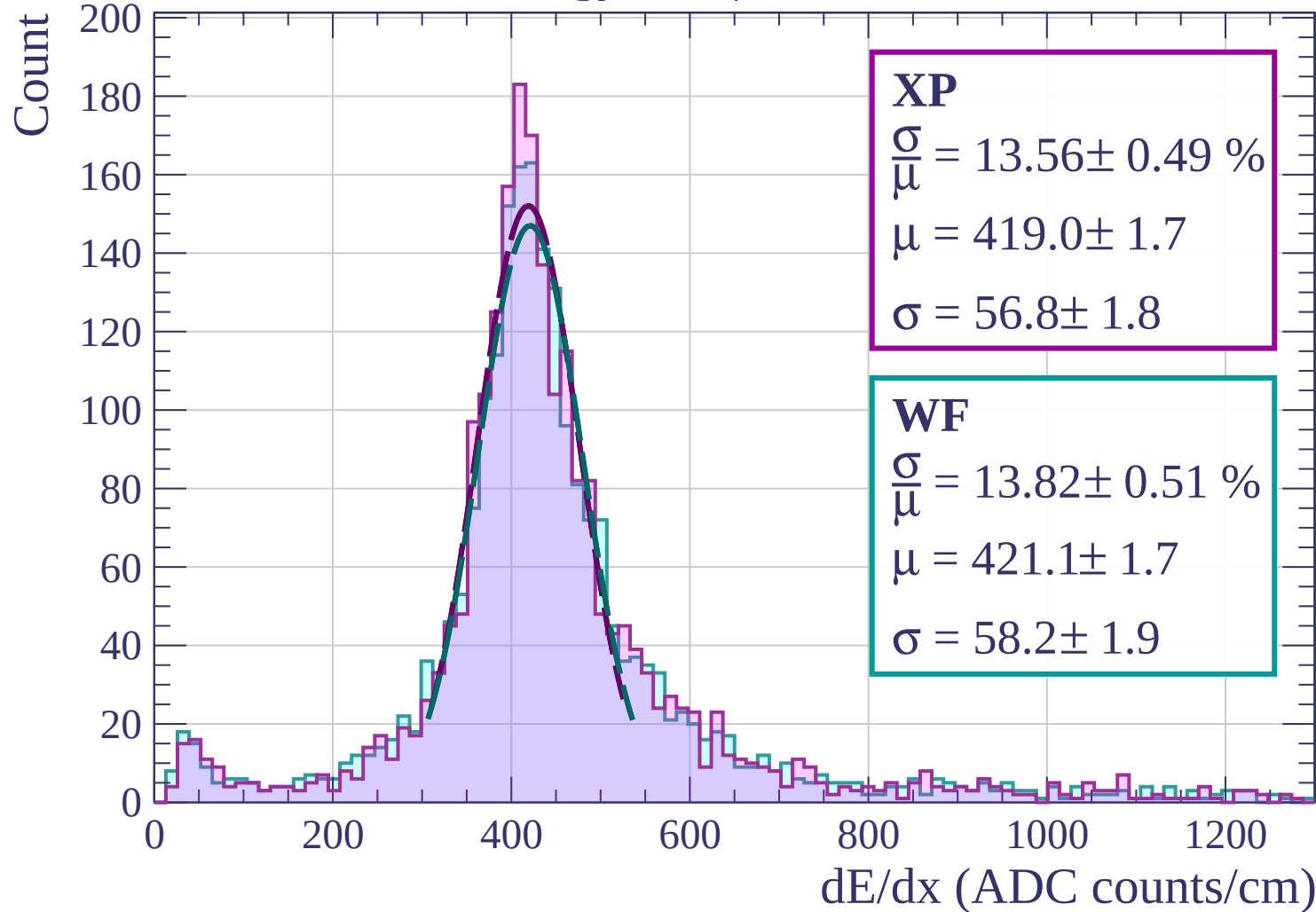
Energy loss | $36 < \theta < 39$



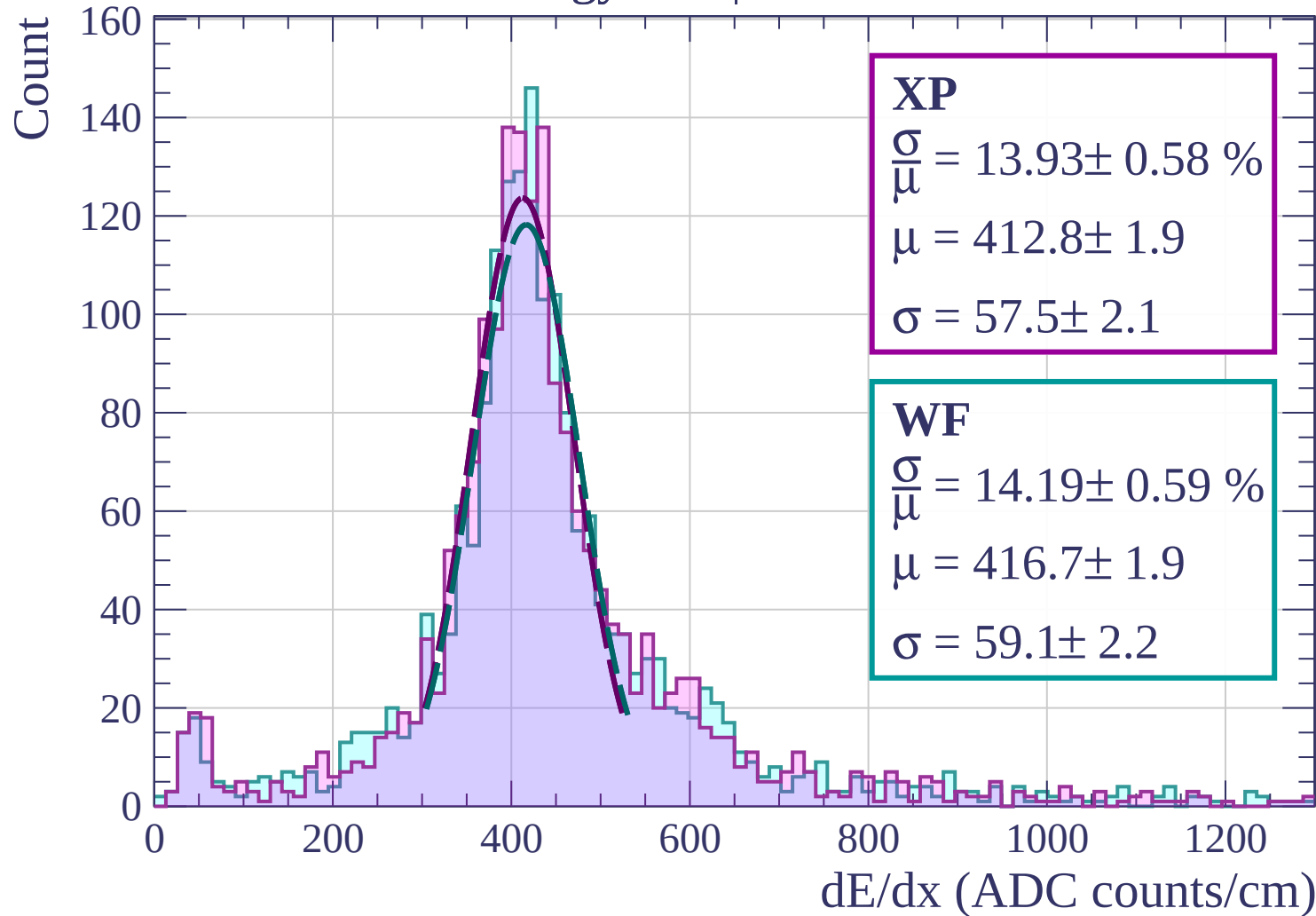
Energy loss | $39 < \theta < 42$



Energy loss | $42 < \theta < 45$

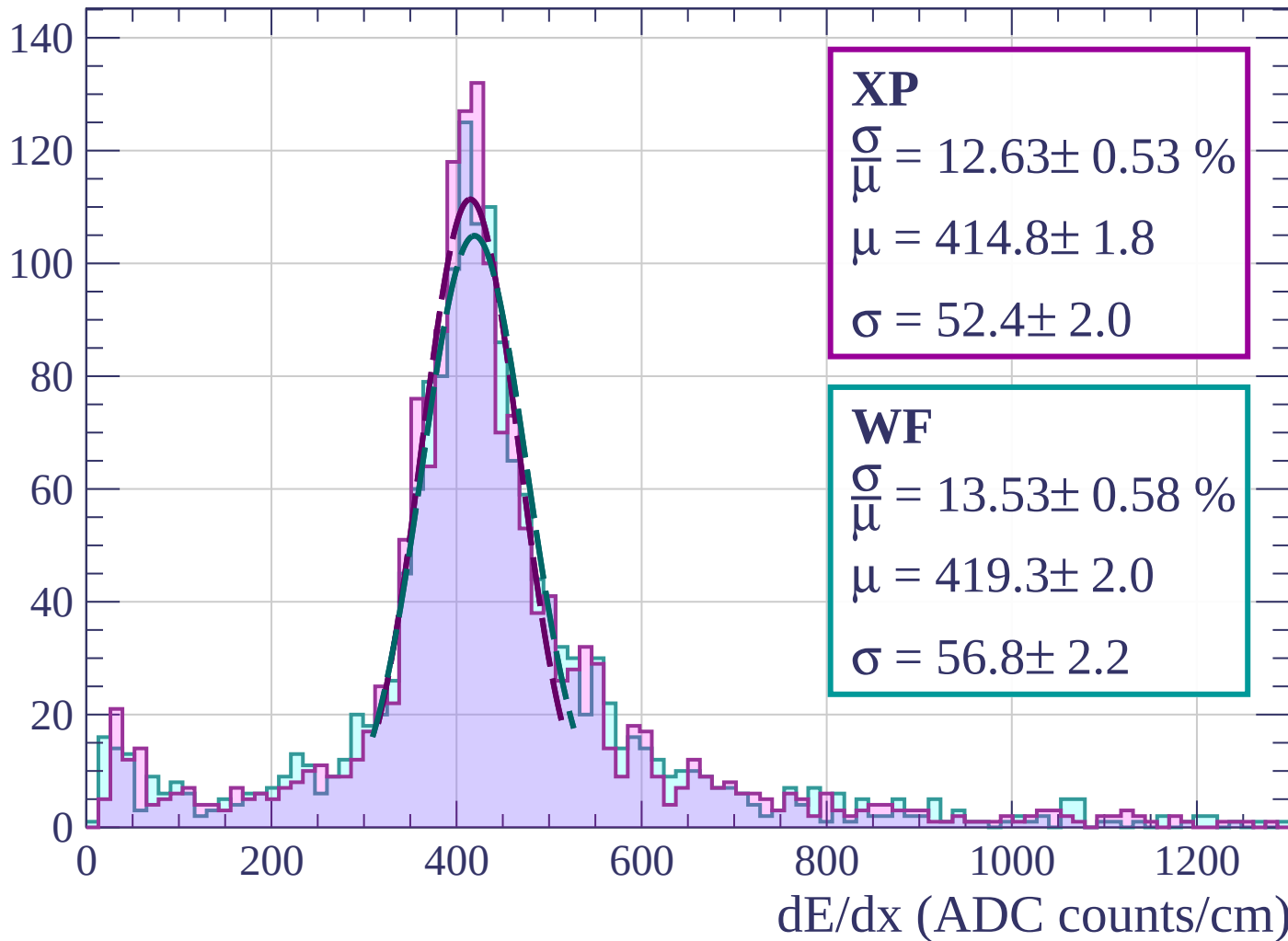


Energy loss | $45 < \theta < 48$



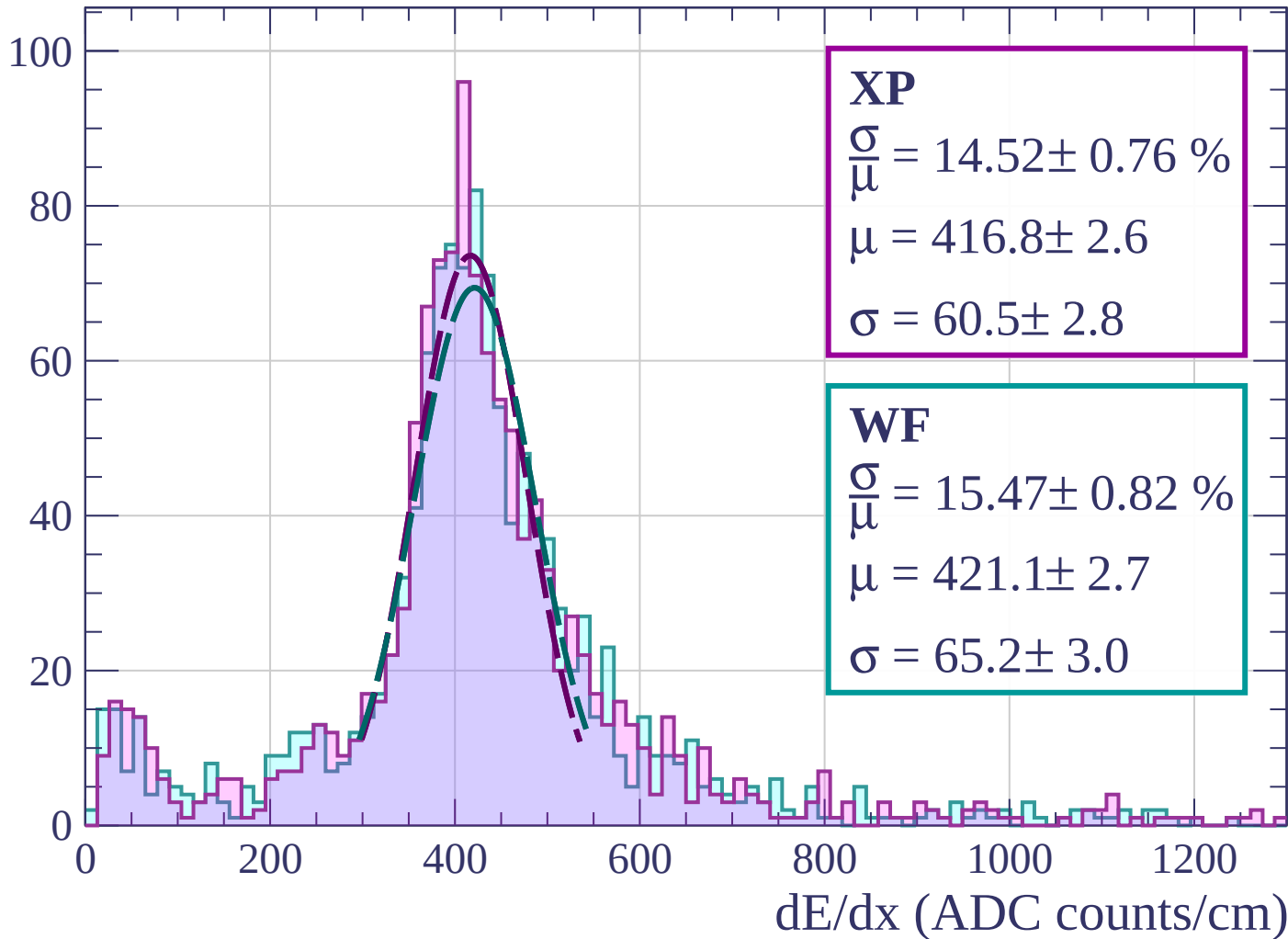
Energy loss | $48 < \theta < 51$

Count



Energy loss | $51 < \theta < 54$

Count



Energy loss | $54 < \theta < 57$

Count

XP

$$\frac{\sigma}{\mu} = 16.04 \pm 1.01 \%$$

$$\mu = 418.9 \pm 3.3$$

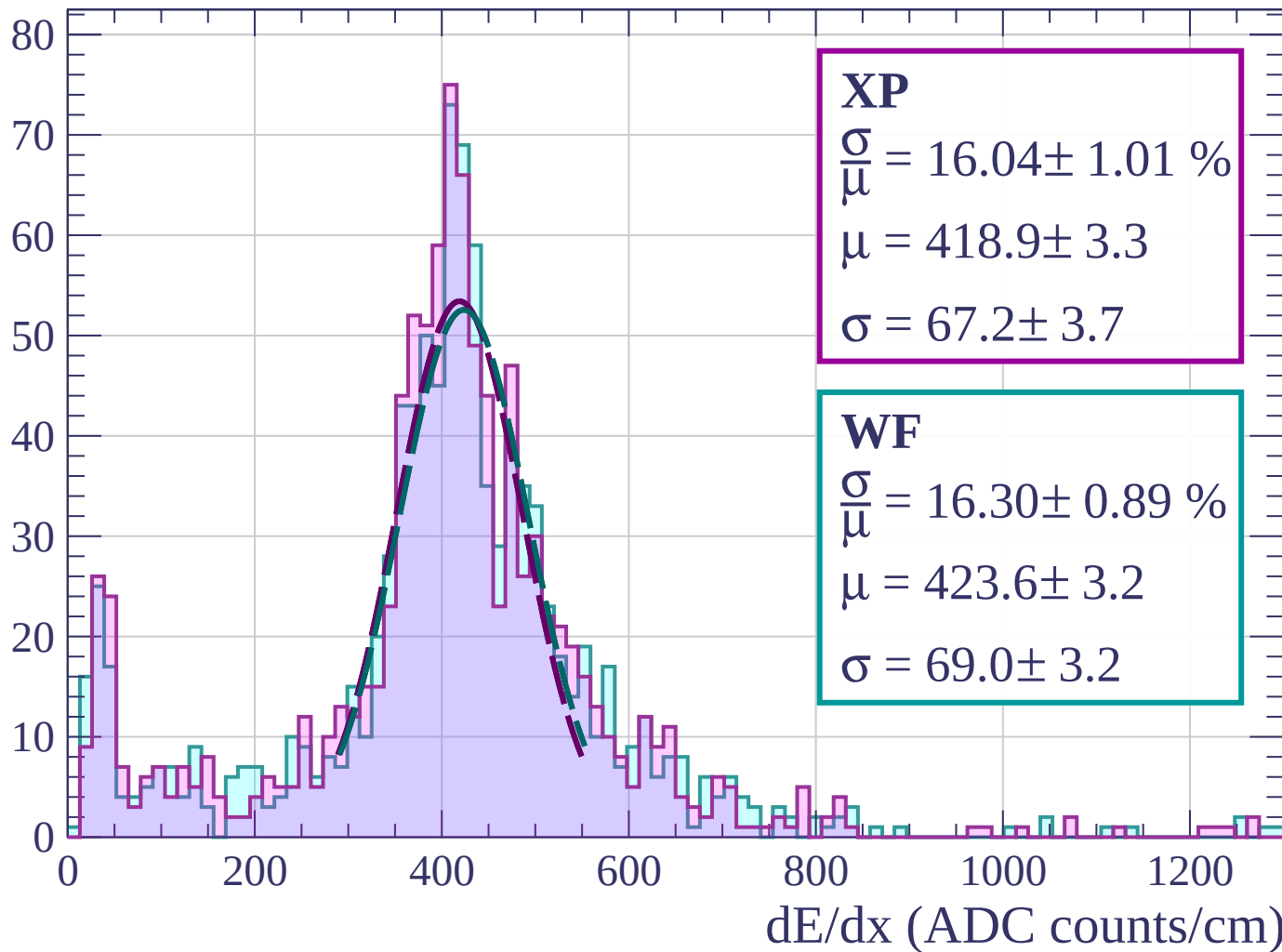
$$\sigma = 67.2 \pm 3.7$$

WF

$$\frac{\sigma}{\mu} = 16.30 \pm 0.89 \%$$

$$\mu = 423.6 \pm 3.2$$

$$\sigma = 69.0 \pm 3.2$$



Energy loss | $57 < \theta < 60$

Count

XP

$$\frac{\sigma}{\mu} = 14.11 \pm 1.05 \%$$

$$\mu = 409.3 \pm 3.2$$

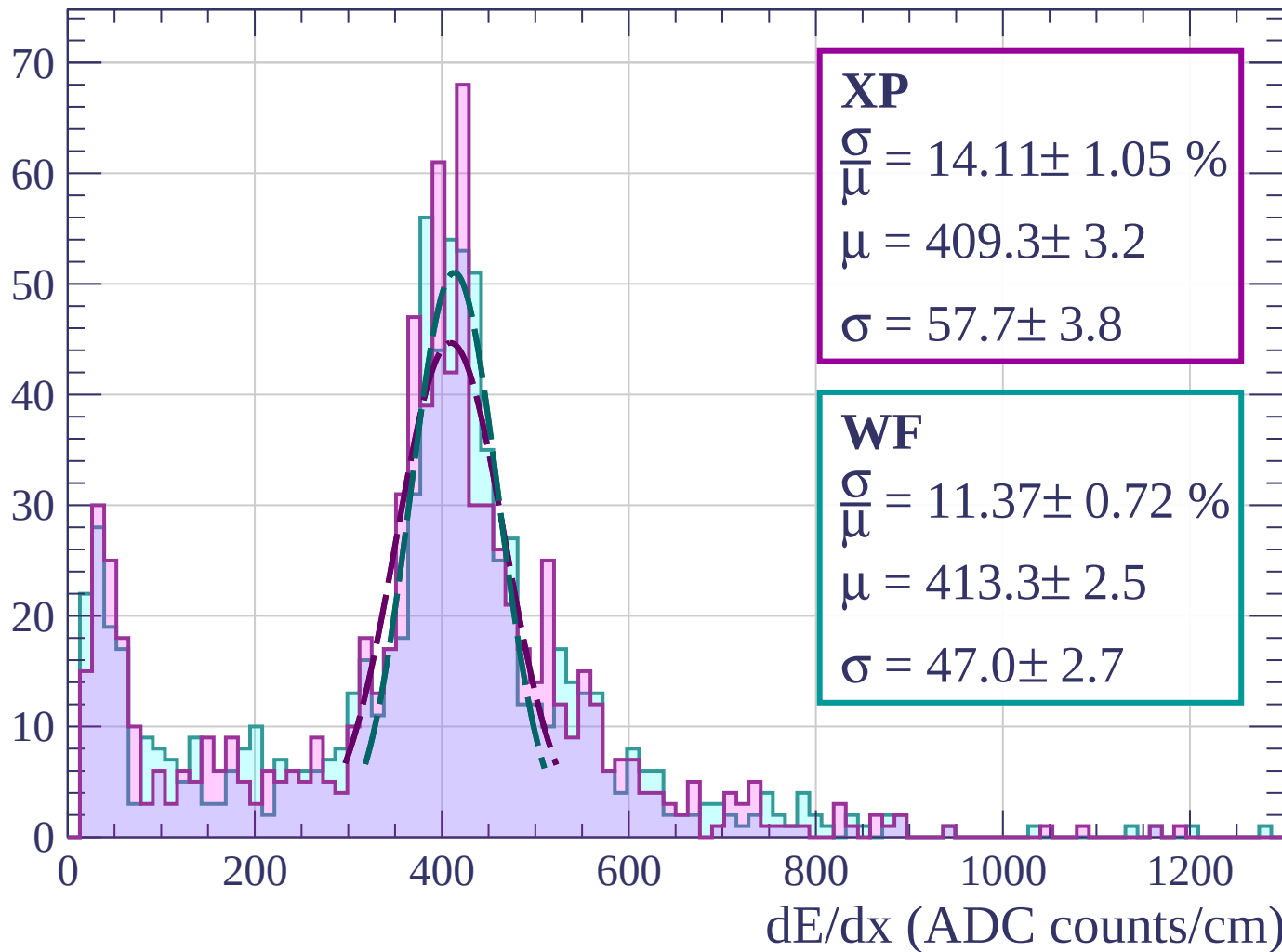
$$\sigma = 57.7 \pm 3.8$$

WF

$$\frac{\sigma}{\mu} = 11.37 \pm 0.72 \%$$

$$\mu = 413.3 \pm 2.5$$

$$\sigma = 47.0 \pm 2.7$$



Energy loss | $60 < \theta < 63$

Count

XP

$$\frac{\sigma}{\mu} = 15.03 \pm 1.22 \%$$

$$\mu = 405.6 \pm 3.7$$

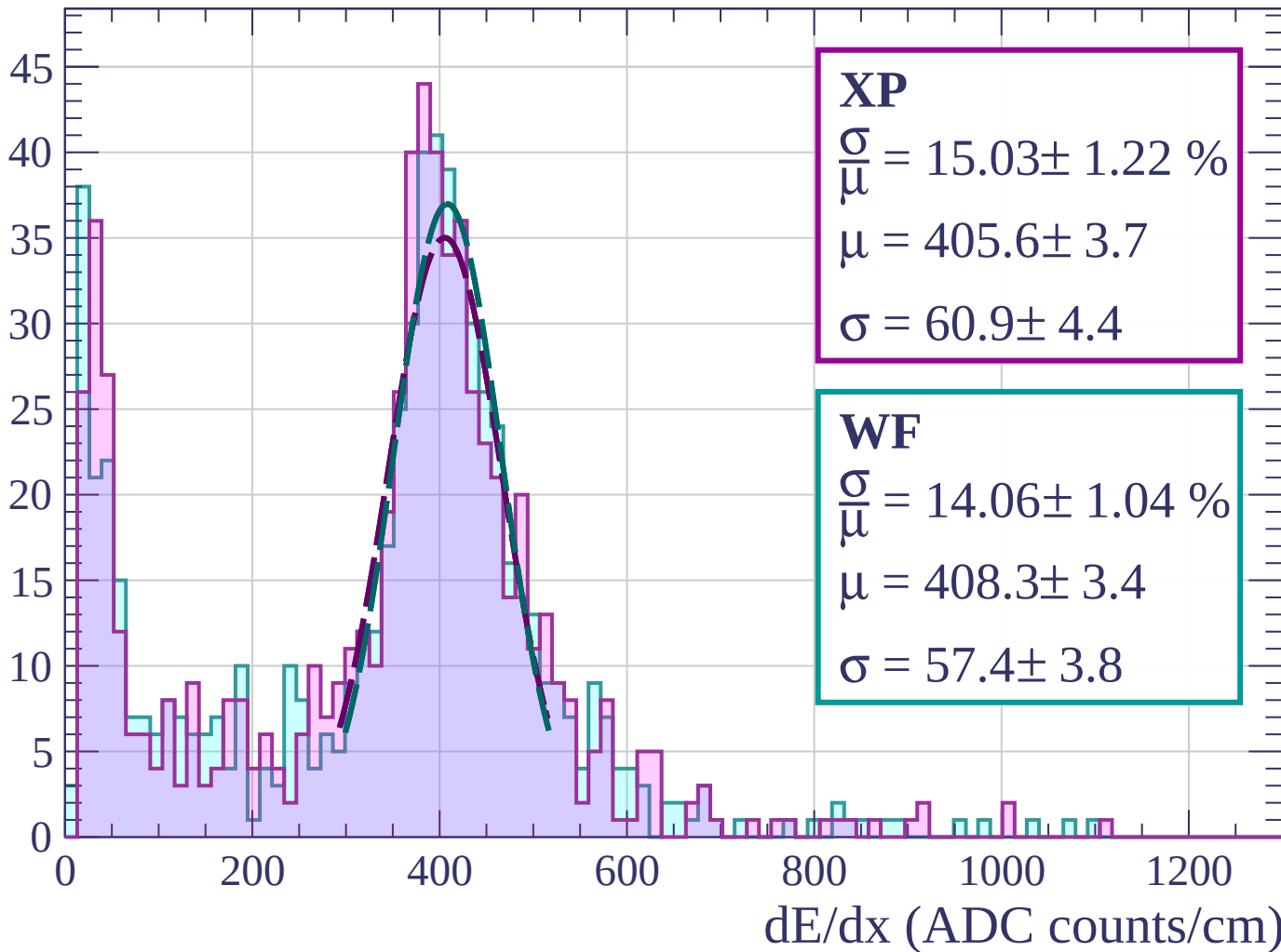
$$\sigma = 60.9 \pm 4.4$$

WF

$$\frac{\sigma}{\mu} = 14.06 \pm 1.04 \%$$

$$\mu = 408.3 \pm 3.4$$

$$\sigma = 57.4 \pm 3.8$$



Energy loss | $63 < \theta < 66$

Count

XP

$$\frac{\sigma}{\mu} = -20.84 \pm 0.87 \%$$

$$\mu = -3712.8 \pm 979.7$$

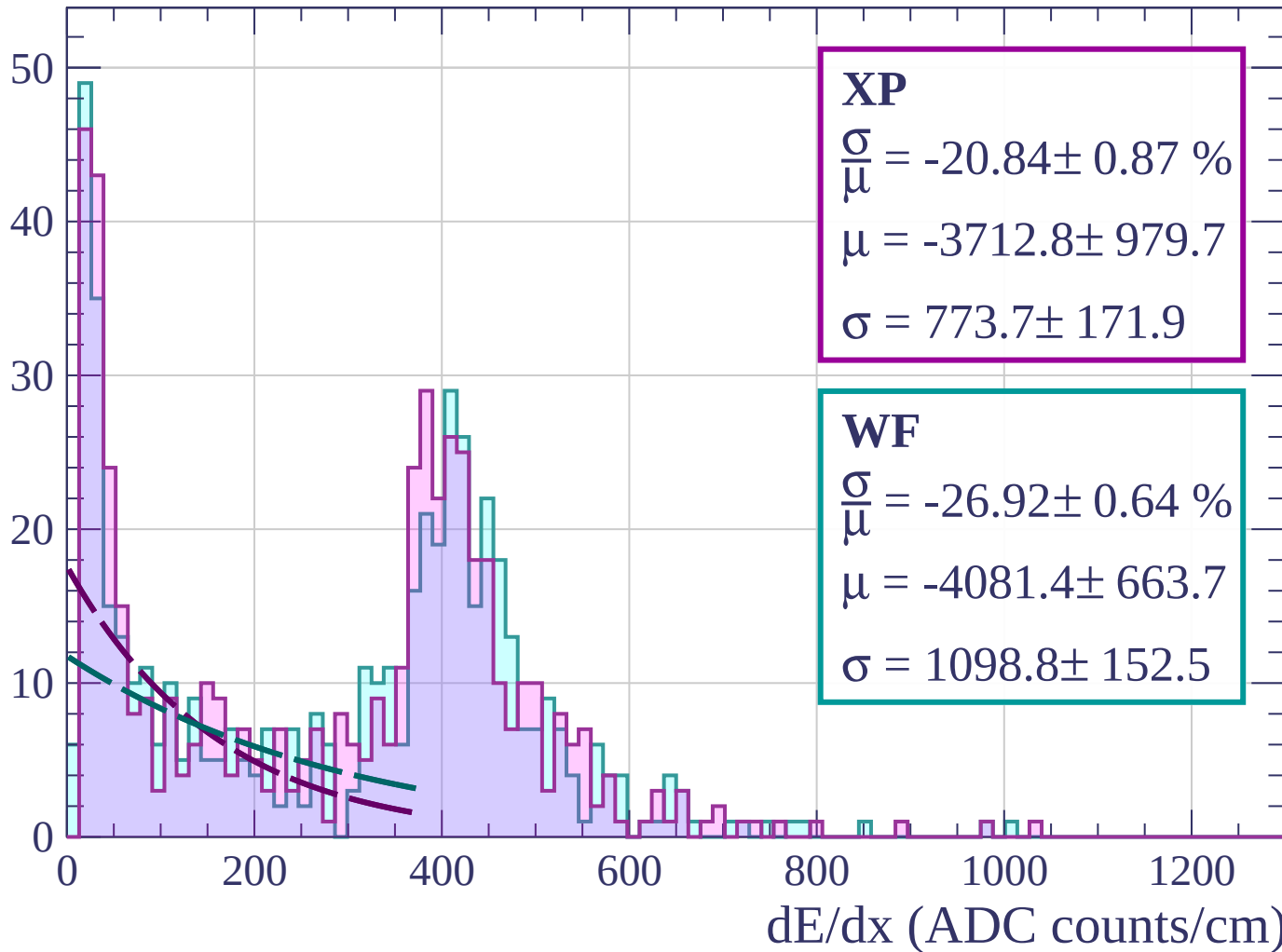
$$\sigma = 773.7 \pm 171.9$$

WF

$$\frac{\sigma}{\mu} = -26.92 \pm 0.64 \%$$

$$\mu = -4081.4 \pm 663.7$$

$$\sigma = 1098.8 \pm 152.5$$



Energy loss | $66 < \theta < 69$

Count

XP

$$\frac{\sigma}{\mu} = 64.84 \pm 14.03 \%$$

$$\mu = 319.4 \pm 23.6$$

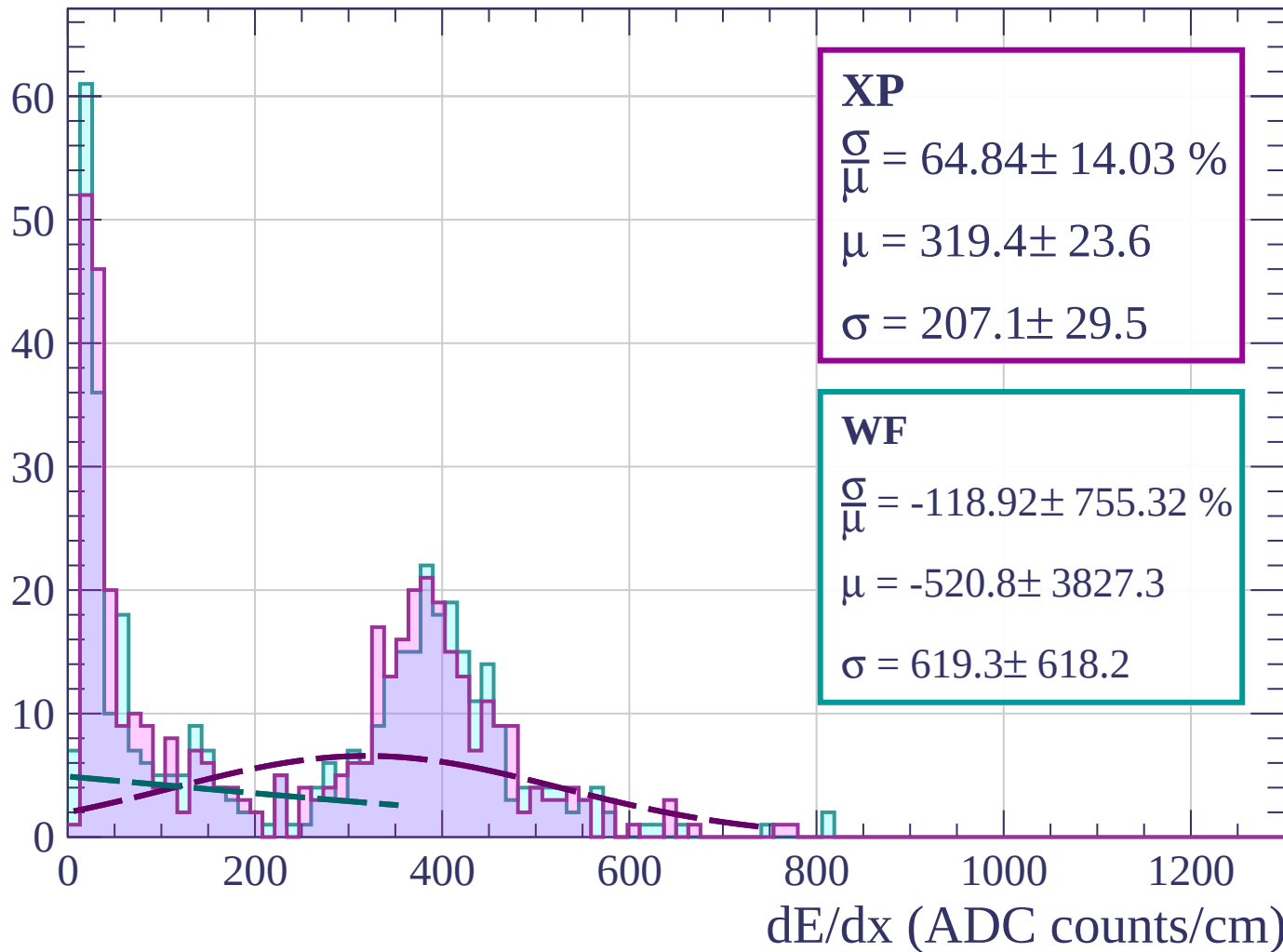
$$\sigma = 207.1 \pm 29.5$$

WF

$$\frac{\sigma}{\mu} = -118.92 \pm 755.32 \%$$

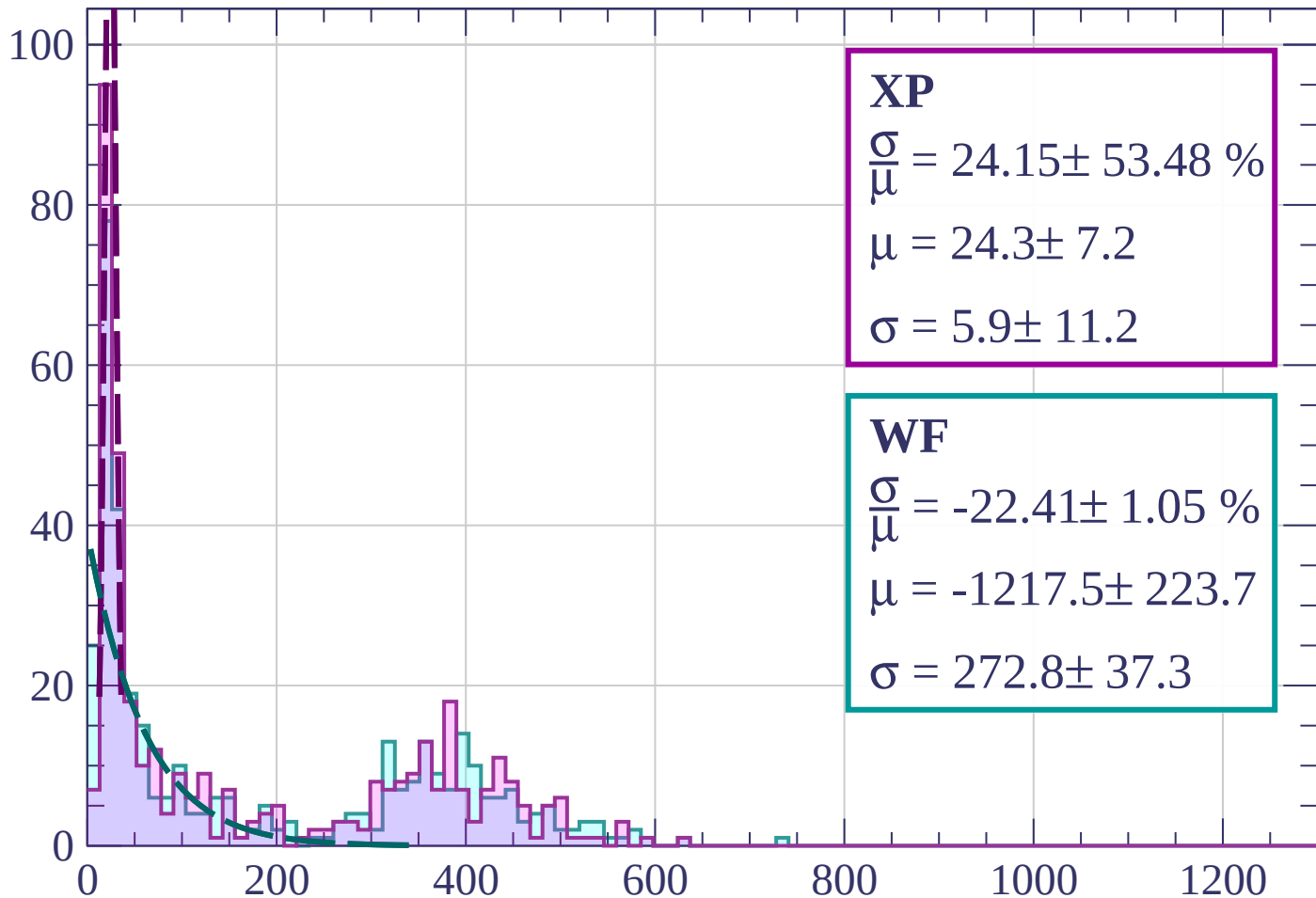
$$\mu = -520.8 \pm 3827.3$$

$$\sigma = 619.3 \pm 618.2$$



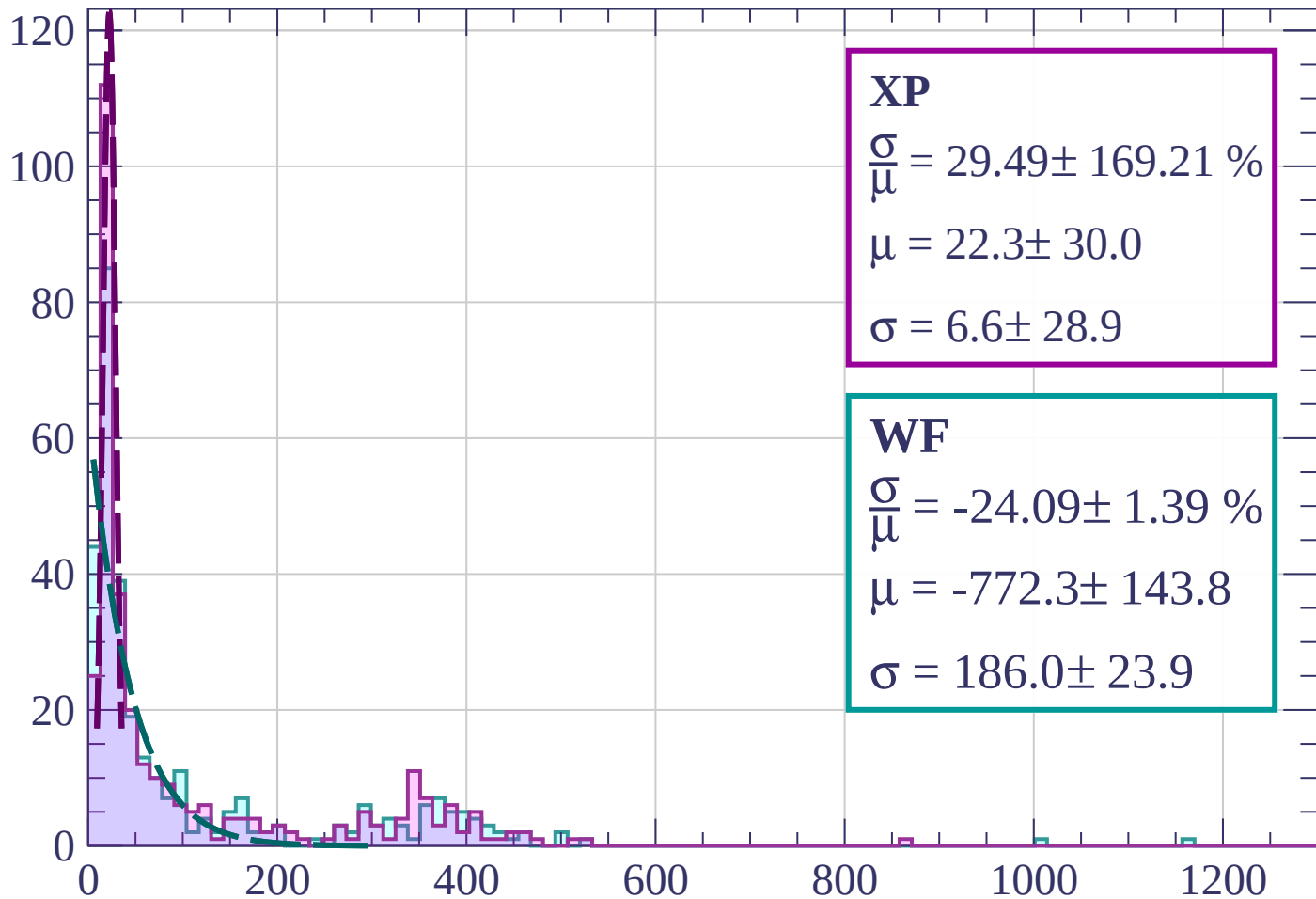
Energy loss | $69 < \theta < 72$

Count



Energy loss | $72 < \theta < 75$

Count



XP

$$\frac{\sigma}{\mu} = 29.49 \pm 169.21 \%$$

$$\mu = 22.3 \pm 30.0$$

$$\sigma = 6.6 \pm 28.9$$

WF

$$\frac{\sigma}{\mu} = -24.09 \pm 1.39 \%$$

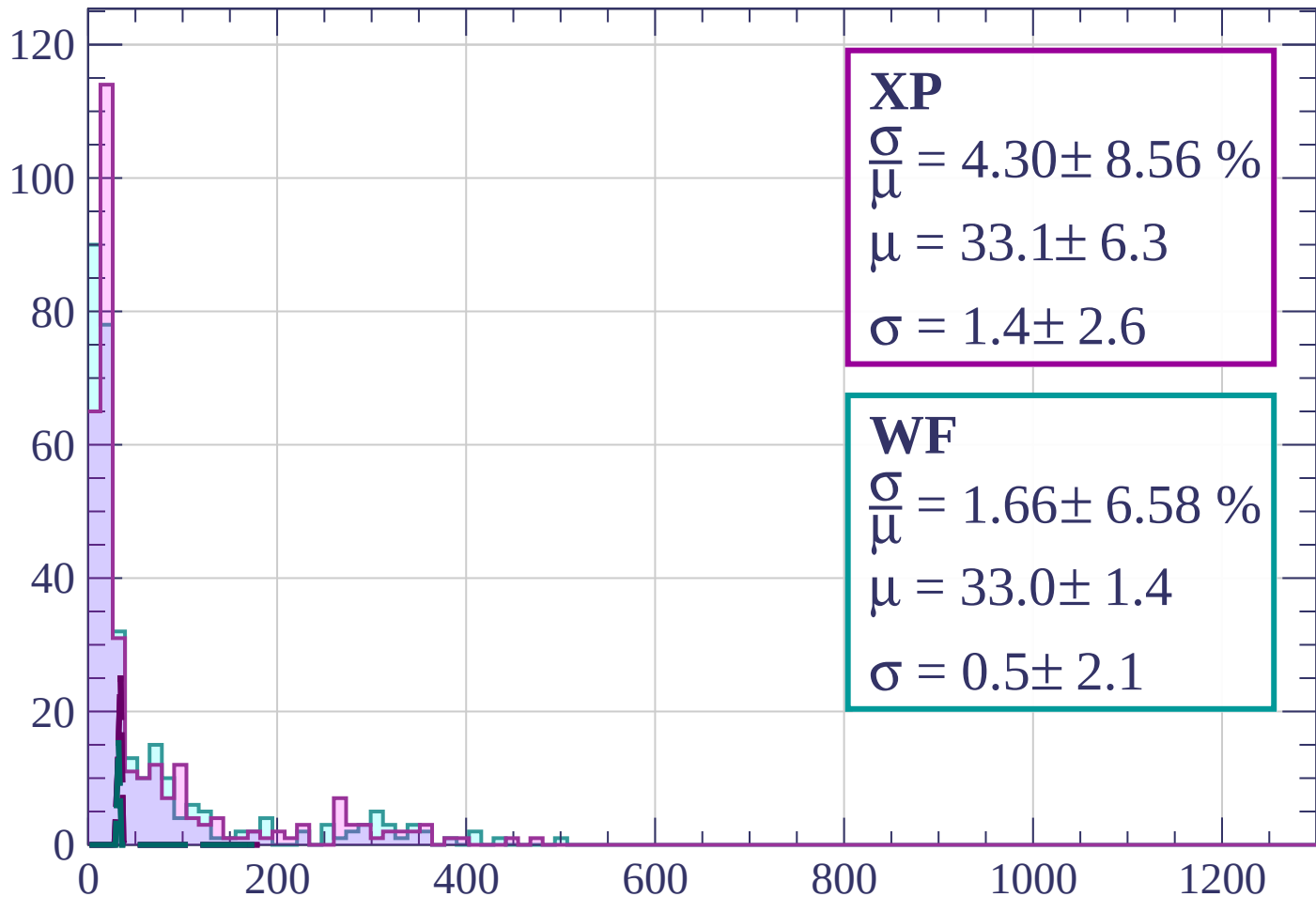
$$\mu = -772.3 \pm 143.8$$

$$\sigma = 186.0 \pm 23.9$$

dE/dx (ADC counts/cm)

Energy loss | $75 < \theta < 78$

Count



XP

$$\frac{\sigma}{\mu} = 4.30 \pm 8.56 \%$$

$$\mu = 33.1 \pm 6.3$$

$$\sigma = 1.4 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 1.66 \pm 6.58 \%$$

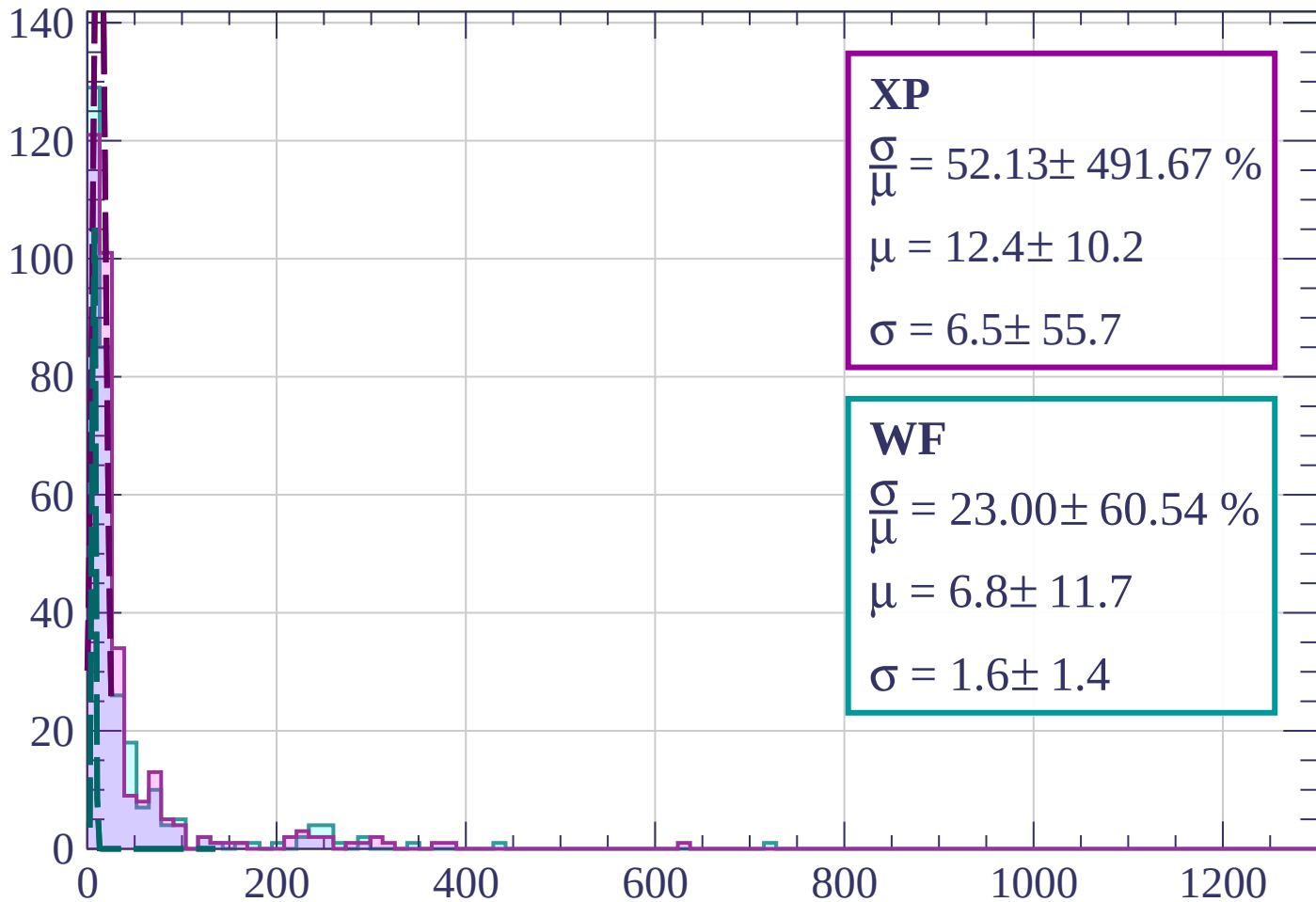
$$\mu = 33.0 \pm 1.4$$

$$\sigma = 0.5 \pm 2.1$$

dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

Count



XP

$$\frac{\sigma}{\mu} = 52.13 \pm 491.67 \%$$

$$\mu = 12.4 \pm 10.2$$

$$\sigma = 6.5 \pm 55.7$$

WF

$$\frac{\sigma}{\mu} = 23.00 \pm 60.54 \%$$

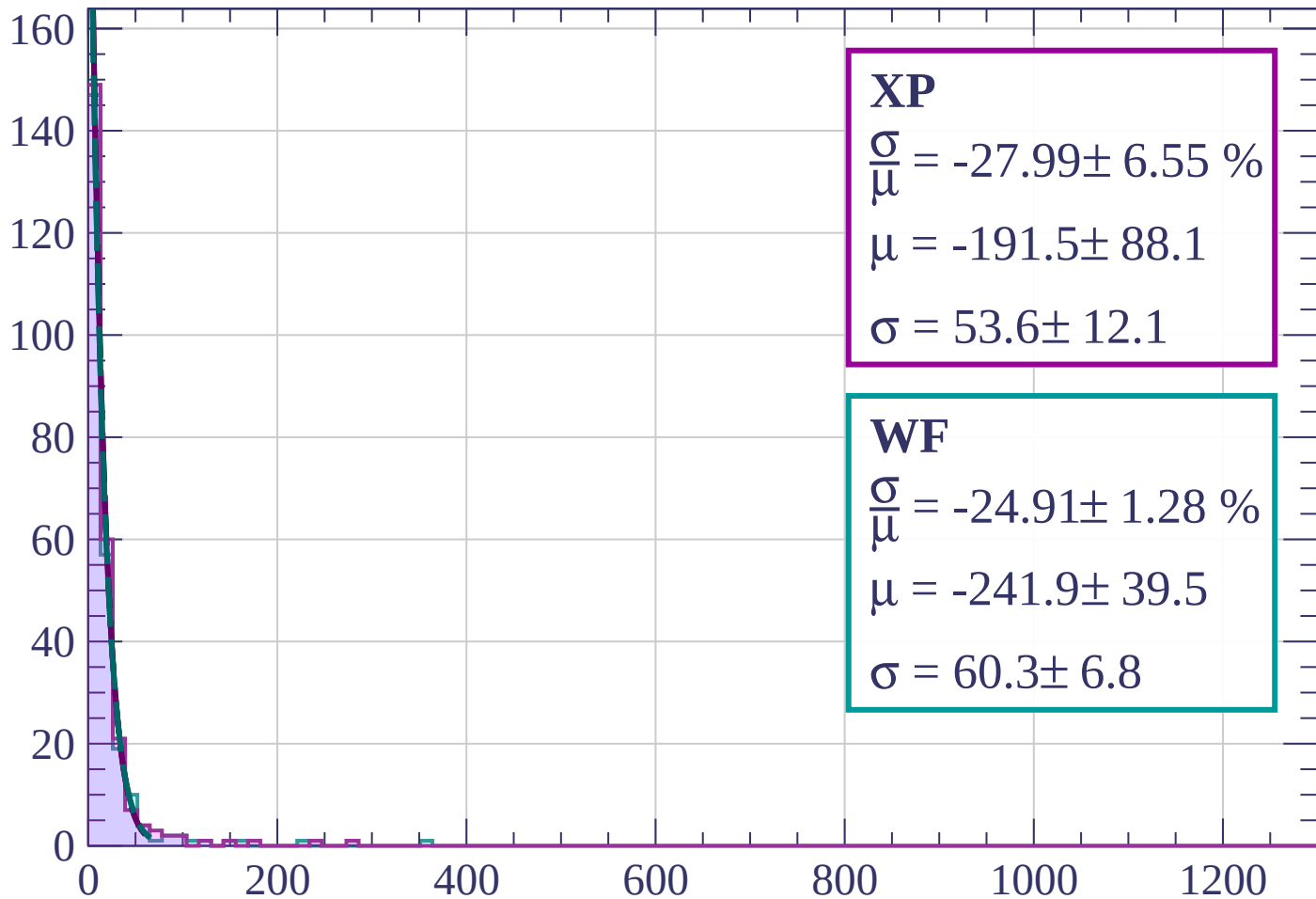
$$\mu = 6.8 \pm 11.7$$

$$\sigma = 1.6 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

Count



XP

$$\frac{\sigma}{\mu} = -27.99 \pm 6.55 \%$$

$$\mu = -191.5 \pm 88.1$$

$$\sigma = 53.6 \pm 12.1$$

WF

$$\frac{\sigma}{\mu} = -24.91 \pm 1.28 \%$$

$$\mu = -241.9 \pm 39.5$$

$$\sigma = 60.3 \pm 6.8$$

dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

Count

120

100

80

60

40

20

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 38.72 \pm 97.98 \%$$

$$\mu = 5.6 \pm 5.8$$

$$\sigma = 2.1 \pm 3.2$$

WF

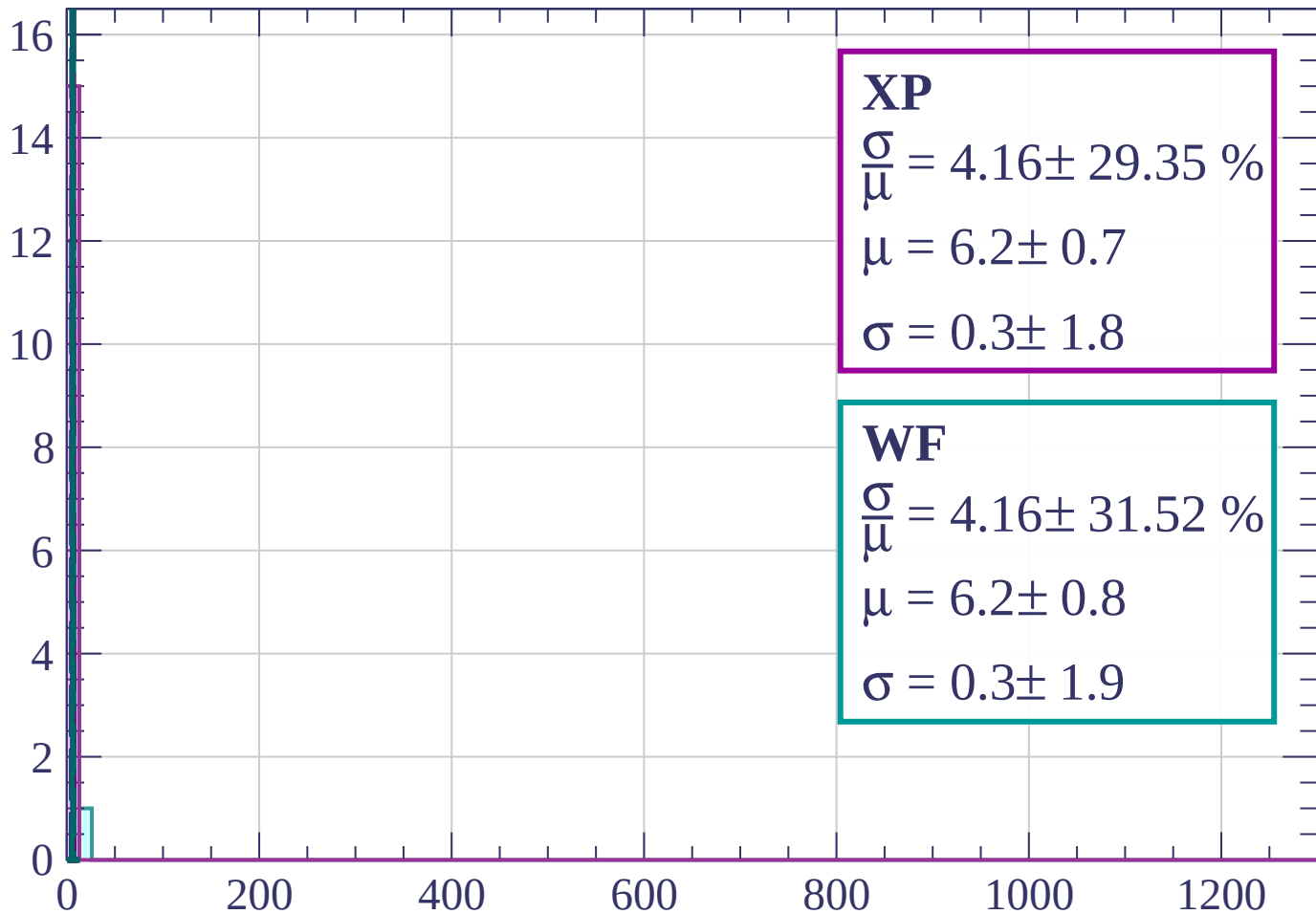
$$\frac{\sigma}{\mu} = 55.22 \pm 105.68 \%$$

$$\mu = 9.8 \pm 7.5$$

$$\sigma = 5.4 \pm 6.2$$

Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

